# Review of Silicon photo-multiplier physics and applications, including a study at low temperature

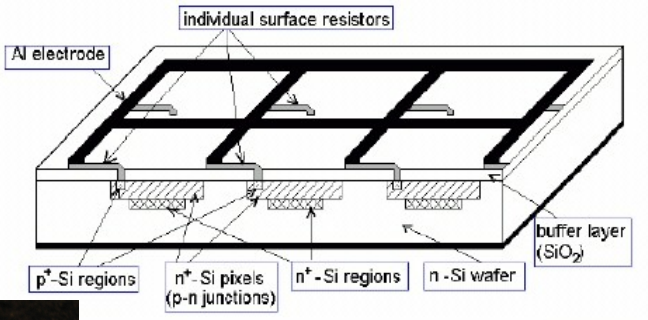
G.Collazuol
Scuola Normale Superiore and INFN - Pisa

## Overview

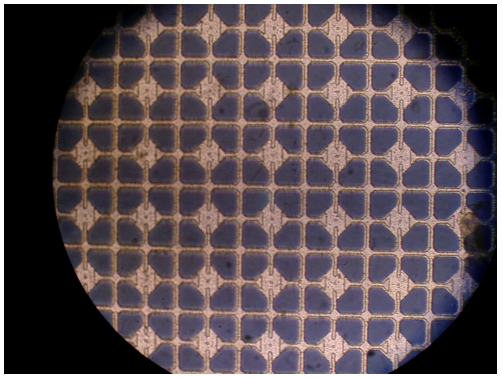
- Introduction
- from device physics
- to detector performance
- and applications
- Conclusions

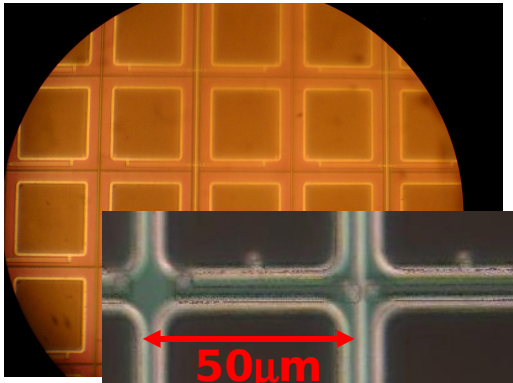
# SiPM = matrix of (hundreds) GM-APD

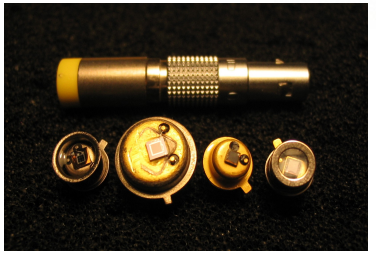
A silicon photo-multiplier consists in a matrix of Single Photon Avalanche Diodes (SPAD) i.e. avalanche diodes operated a few volts above the breakdown voltage: Geiger Mode (GM-) APD

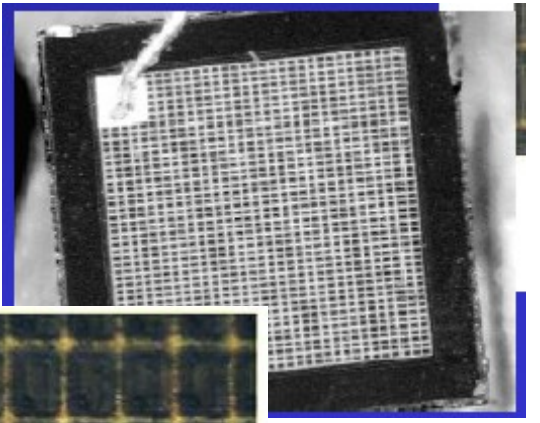


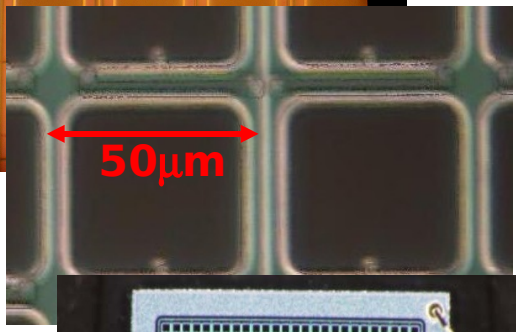

JINR

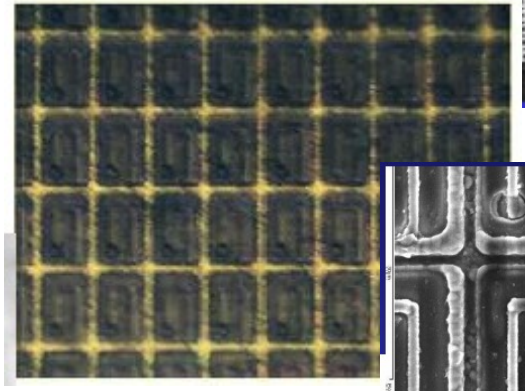


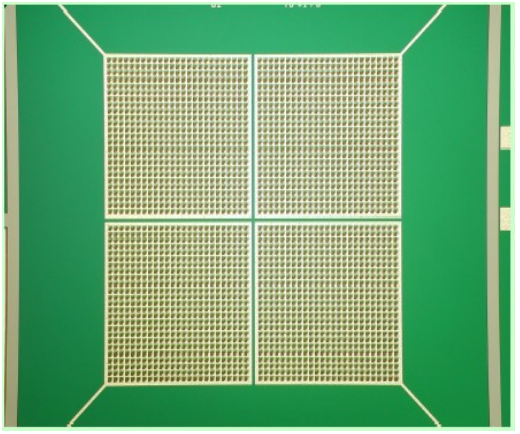




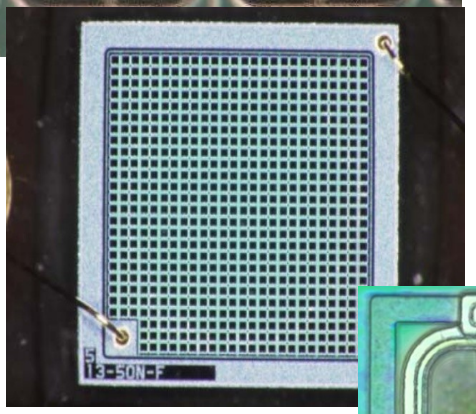




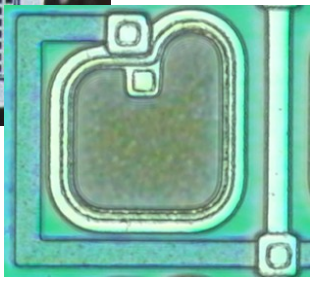



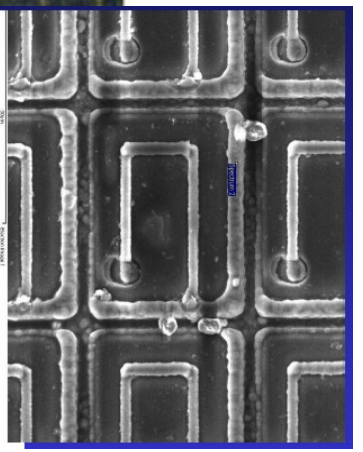


FKB –IRST



HAMAMATSU





CPTA

# Institutes/Companies involved in SiPM

## Pioneering work since late 80-ies at Russian institutes

Investigations of various multi-layer silicon structures with local micro-plasma suppression effect to develop low-cost APD devices

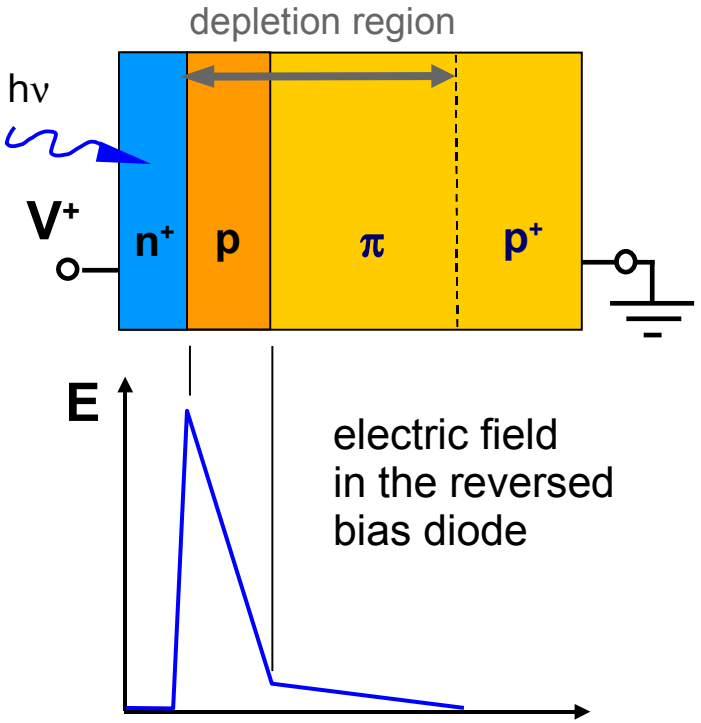
Sadygov - JINR (Dubna)
Golovin - Obninsk/CPTA (Moscow)
Dolgoshein - Mephi/PULSAR (Moscow)

## Today more institutes/companies are involved in SiPM developement/production:

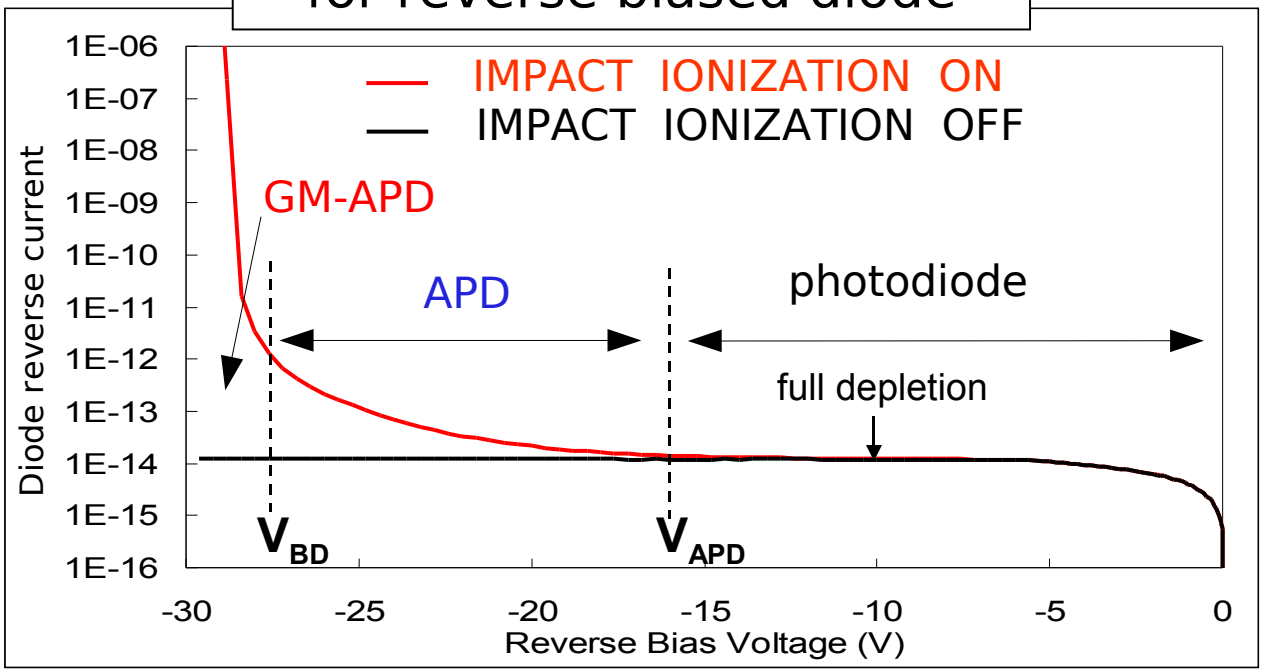
- Hamamatsu, Hamamatsu City, Japan
- FBK-IRST, Trento, Italy $\rightarrow$ collaboration with INFN ("MEMS")
- ST microelectronics
- SensL, Blackrock, Ireland
- MPI, Munich, Germany
- Zecotek, Vancouver, Canada
- RMD, Boston, USA
- JINR/Micron Enterprise, Dubna and Zelenograd, Russia
- MePhi/Pulsar Enterprise, Moscow, Russia
- Center of Perspective Technology and Apparatus (CPTA), Moscow / Photonique, Geneve, Switzerland

# The building block of a SiPM: GM-APD

## Reverse polarized p-n junction




Different working regimes for reverse biased diode




## APD: Linear-Proportional Mode

- Bias BELOW $V_{BD}$ ($V_{APD} < V < V_{BD}$)
- It's an AMPLIFIER
- Gain: limited < 1000 due to fluctuations
- Strong dependence on T and $V_{bias}$
- No single photo-electron resolution

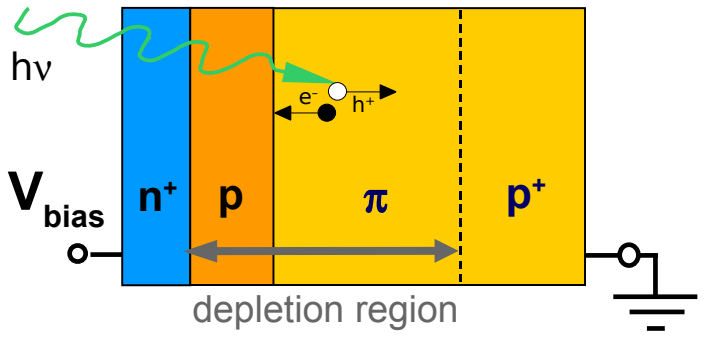
## GM-APD: Geiger Mode

- Bias ABOVE $V_{BD}$ ($V-V_{BD}$ ~a few volts)
- It's a TRIGGER (BINARY) device
- Gain: $\rightarrow\infty$
- Limited by dark count rate
- Single photo-electron resoution

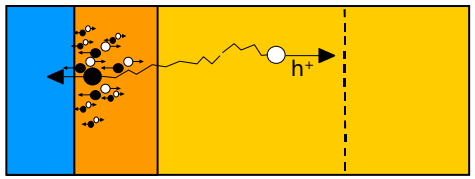
# Operation principle of a GM-APD

## Diode Biased ABOVE $V_{BD}$

- t=0: carrier initiate the avalanche

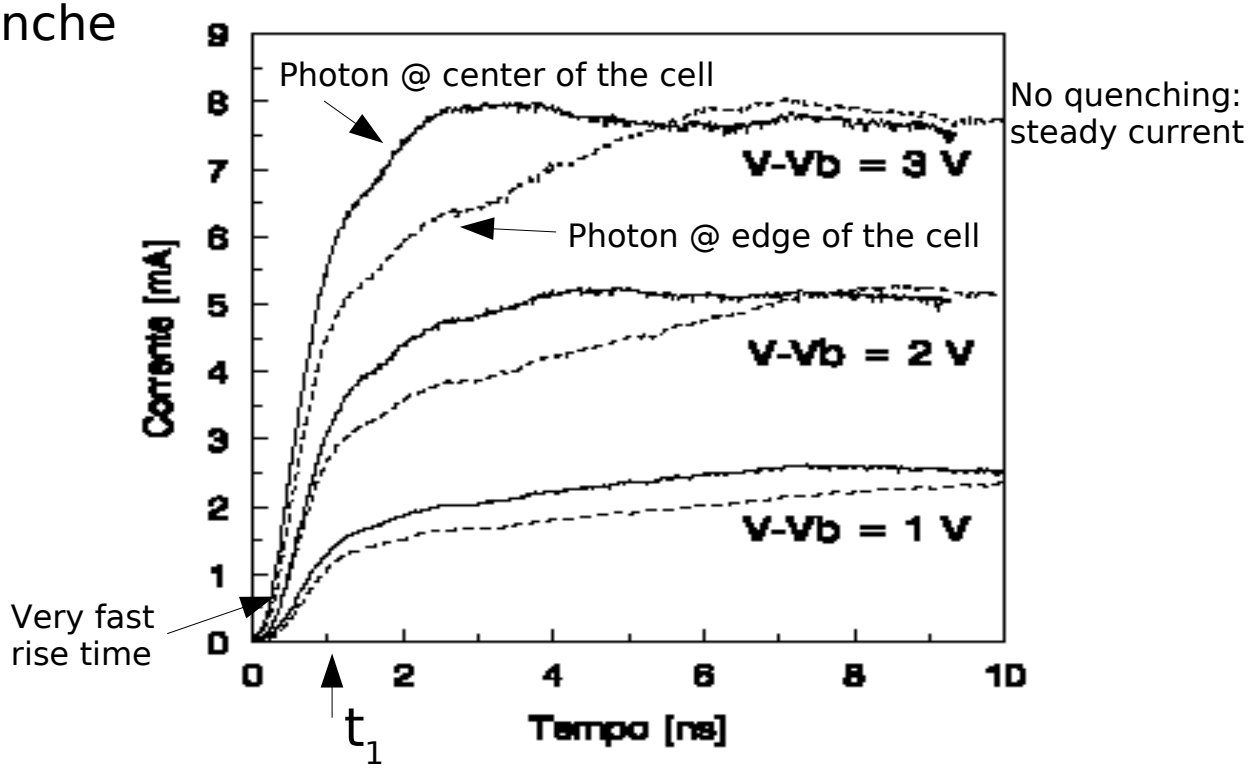


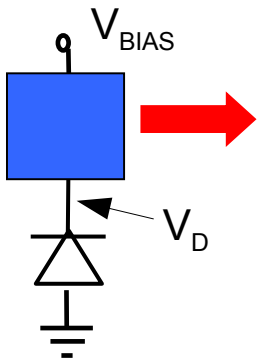

- 0<t<$t_1$: avalanche spreading



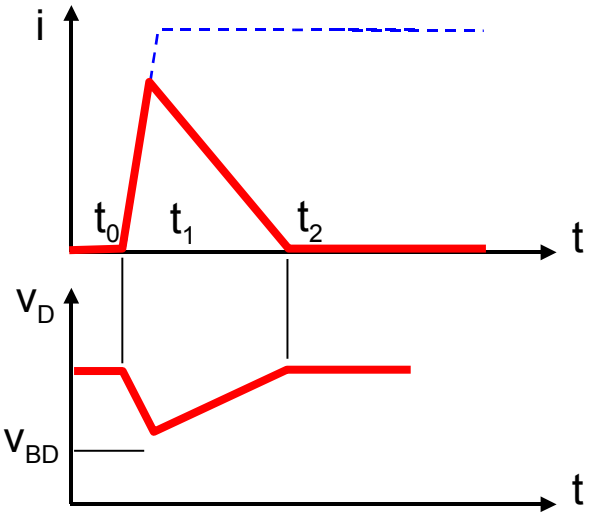

- $t_1$<t: self-sustaining current
  (limited by series resistances )

*A.Spinelli Ph.D thesis (1996)*




To detect another photon need a quenching mechanism. Two solutions:
- large resistance: passive quenching
- analog circuit: active quenching

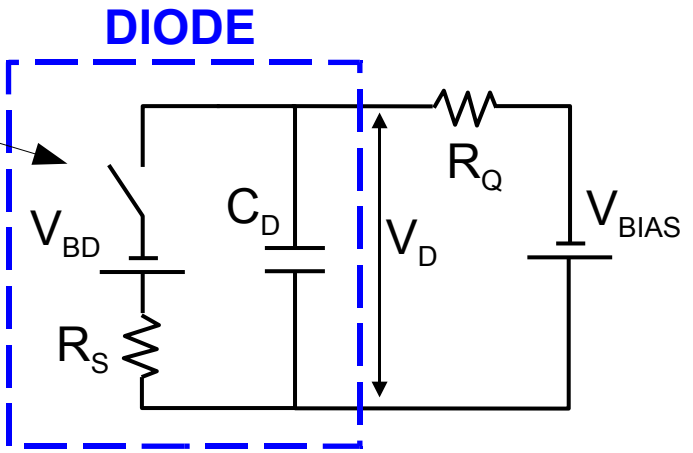


# Electrical model of a GM-APD

Passive quenching studied in detail in the '60 to model micro-plasma instabilities McIntrye JAP 32 (1961), Haitz JAP 35 (1964)

The Geiger-Mode APD can be modeled with an electrical circuit and two probabilities:

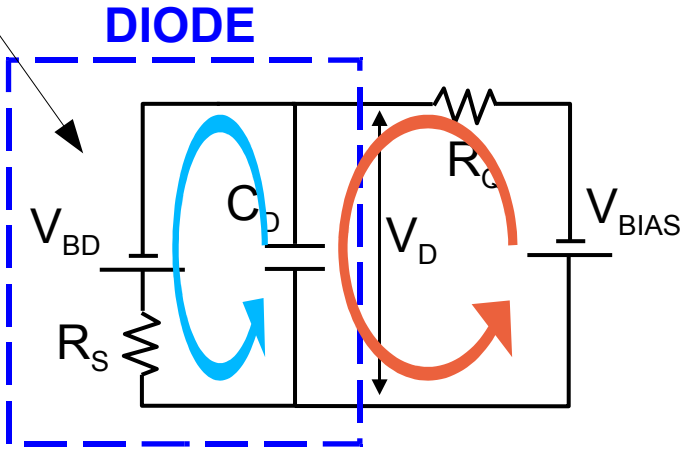
- Switch OFF = micro-plasma non-conducting
- Switch ON = micro-plasma conducting




- $C_d$ diode capacitance (some 10fF)
- $R_s$ series resistance (~1kΩ)
- $R_q$ quenching resistance (> 300kΩ)
- $V_{bd} < V_{bias}$ (few % relative)

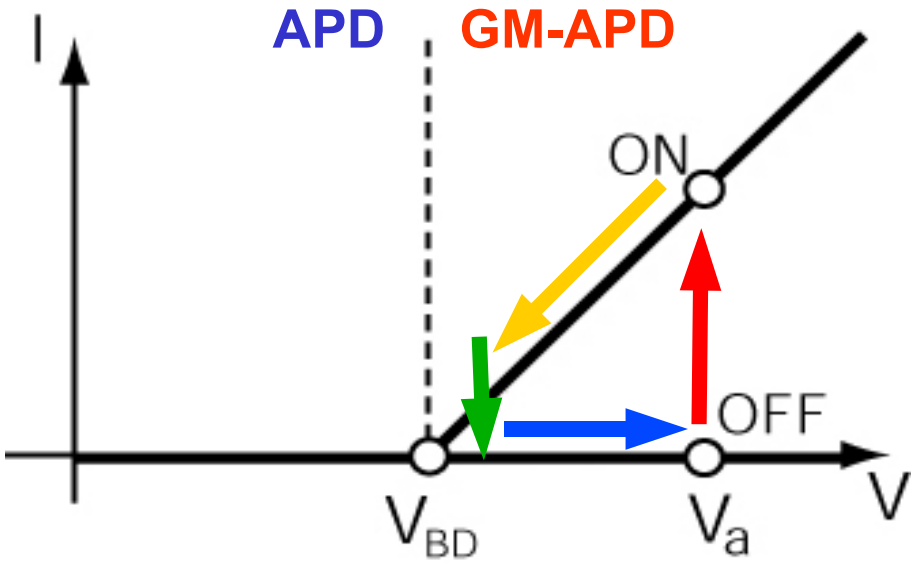
- $P_{01}$ turn-ON
  Probability that a carrier traversing the high field region trigger an avalanche

- $P_{10}$ turn-OFF
  Probability that number of carriers in the high field region fluctuates to 0




Internal/external currents

# Operation principle of a GM-APD




Haitz JAP 35 (1964)

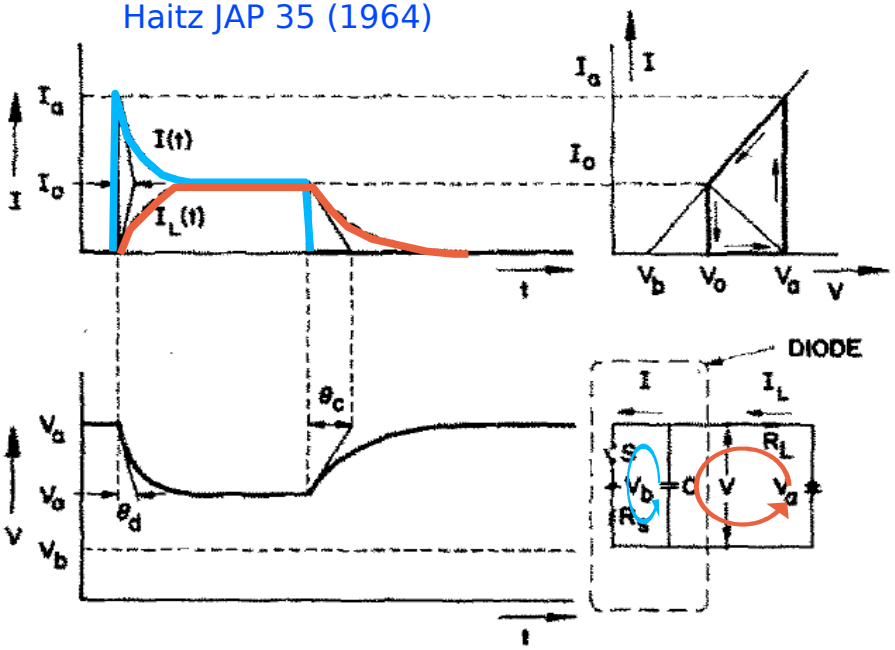



FIG. 3. Shape of current pulse for $\theta_d \ll \tau_1(I_0)$.

OFF condition: avalanche quenched, switch open, capacitance charged until no current flowing from $V_{BD}$ to $V_{BIAS}$ with time constant $R_q x C_D = \tau_{Quenching}$ ($\rightarrow$ recovery time)

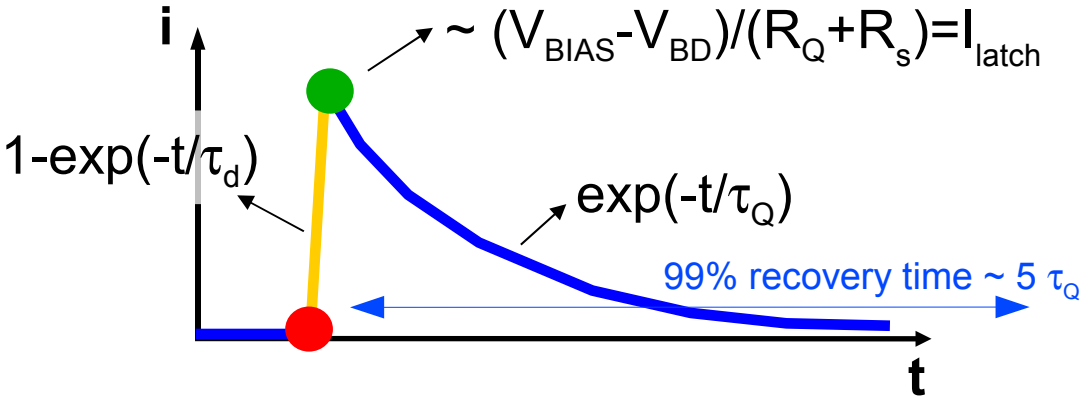
$P_{01}$ = turn-on probability
probability that a carrier traversing the high-field region triggers the avalanche

$P_{10}$ = turn-off probability
probability that the number of carriers traversing the high-field region fluctuates to 0

ON condition: avalanche triggered, switch closed
$C_D$ discharges to $V_{BD}$ with a time constant $R_S x C_D = \tau_{discharge}$,
at the same time the external current asymptotic grows to $(V_{BIAS}-V_{BD})/(R_Q+R_s)$

# Operation principle of a GM-APD

If $R_Q$ is high enough the internal current is so low that statistical fluctuations may quench the avalanche




The leading hedge of the signal is much faster than trailing edge:

1. $\tau_d = R_S \bullet C_D << R_Q \bullet C_D = \tau_Q$
2. turn-off mean time is very short (if $R_{Q\ is}$ sufficiently high, $I_{latch} \sim 10\mu A$)

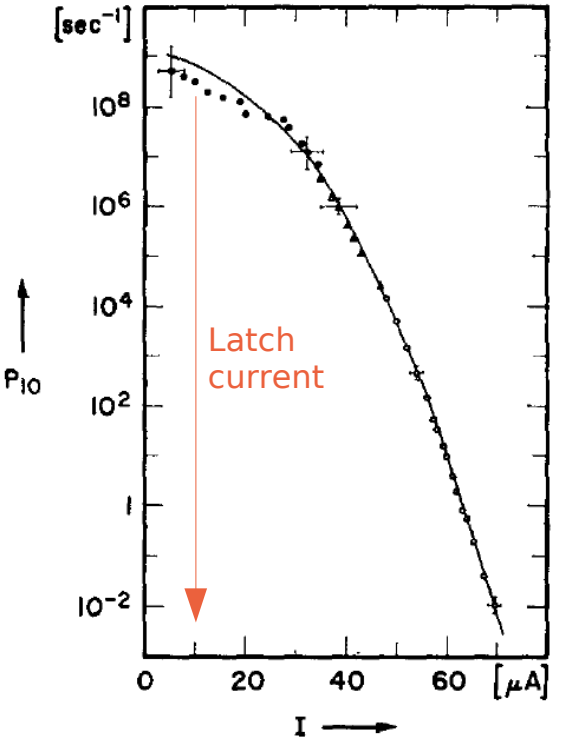



FIG. 2. Turnoff probability per second as function of pulse current.

Haitz JAP 35 (1964)

The charge collected per event is the area under the exponential which is determined by circuital elements and bias.

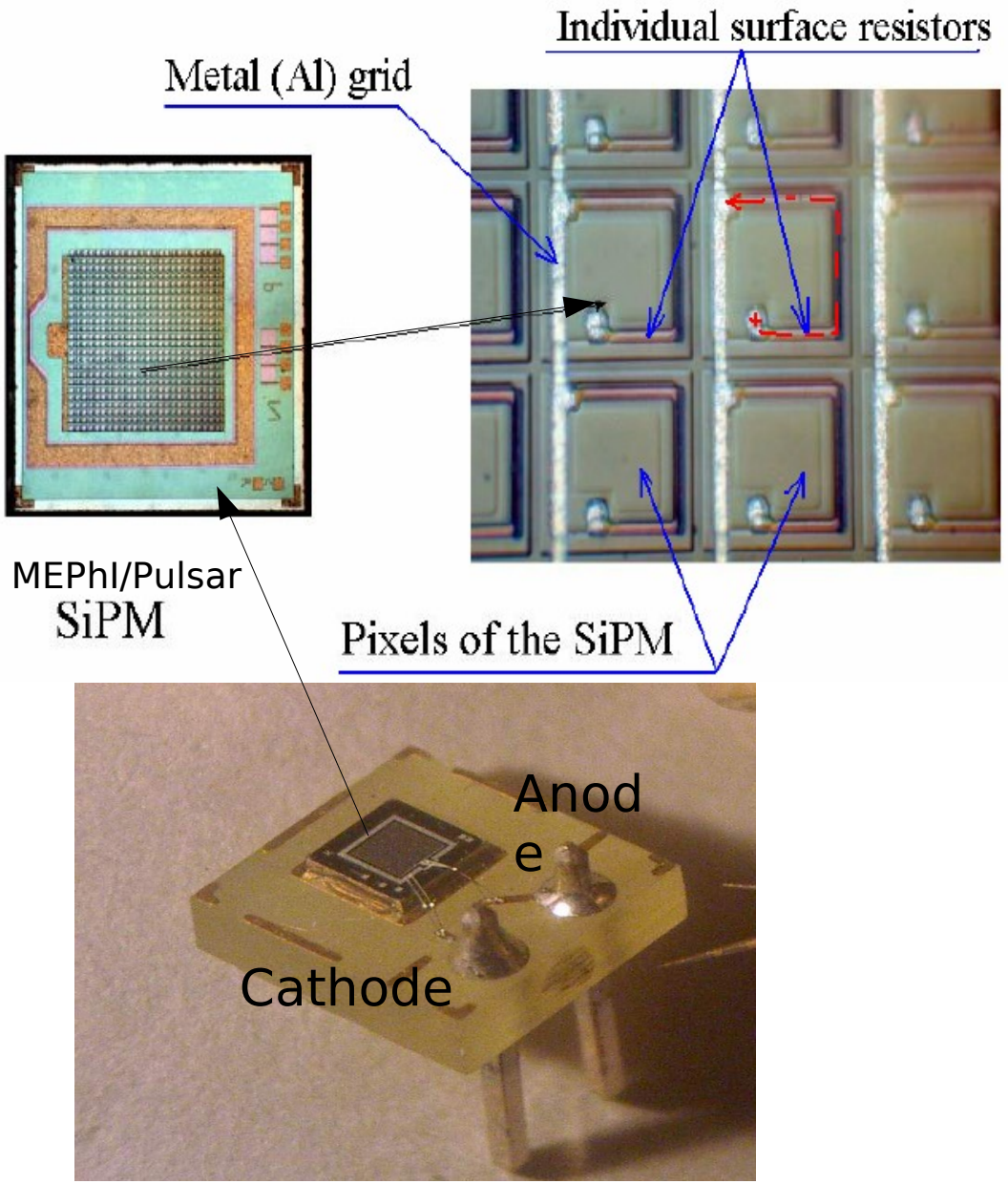
➡ It is possible to define a GAIN (discharge of a capacitor !!!)

$$G = \frac{I_{max} \cdot \tau_Q}{q_e} = \frac{(V_{bias} - V_{BD}) \cdot \tau_Q}{(R_Q + R_S) \cdot q_e} = \frac{(V_{bias} - V_{BD}) \cdot C_D}{q_e}$$

➡ Gain fluctuations in GM-APD are smaller than in APD because electrons and holes give the same signal !!!

# The silicon photomultiplier

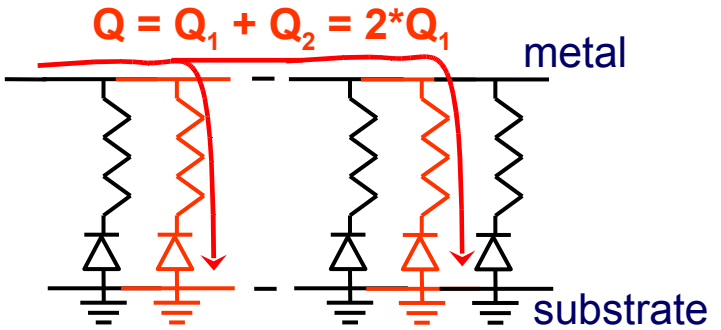
Single GM-APD gives no information on light intensity → MATRIX structure first proposed by Golovin and Sadygov in the'90s




A single GM-APD is segmented in tiny micro GM-APD cells with one passive quenching $R_Q$ each and connected in parallel.

Each element is independent and gives the same signal when fired by a photon

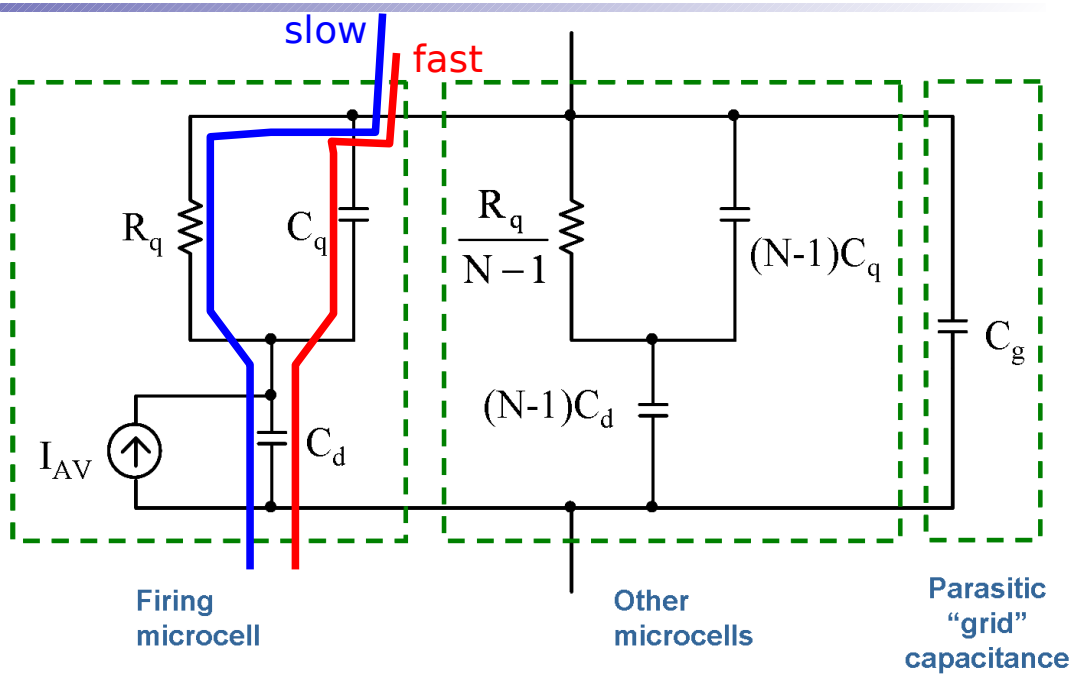
Output charge is proportional to the number of triggered cells that is the number of incident photons (if efficiency =1)

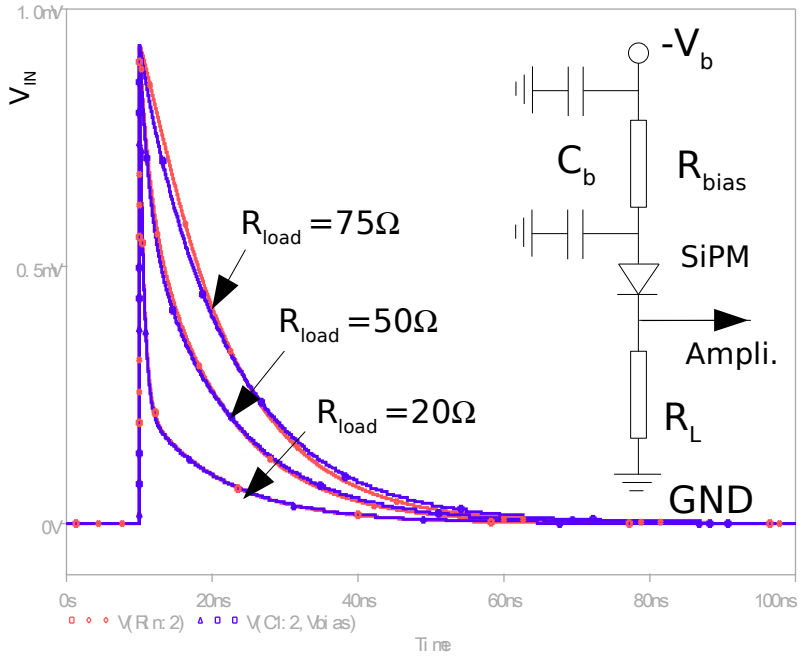



Σ digital signals = analogic signal !!!

# Electrical model of a SiPM

$R_q$: quenching resistor
(hundreds of kΩ)
$C_d$: junction capacitance
(few tens of fF)
$C_q$: parasitic capacitance in parallel to $R_q$ (few tens of fF, $C_q < C_d$)
$I_{AV}$: SiPM ~ ideal current source
current source modeling the total charge delivered by a cell during the avalanche $Q=\Delta V(C_d+C_q)$
$C_g$: parasitic capacitance due to the routing of Vbias to the cells (metal grid, few tens of pF)




1) the peak of $V_{IN}$ is independent of $R_s$

A constant fraction $Q_{IN}$ of the charge Q delivered during the avalanche is instantly collected on $C_{tot}=C_g+C_{eq}$.

2) The circuit has two time constants:

- $\tau_{IN}= R_L\ C_{tot}$ (fast)
- $\tau_r = R_q\,(C_d+C_q)$ (slow)

Decreasing $R_s$, the time constant $\tau_{IN}$ decreases, the current on $R_s$ increases and the collection of Q is faster

F. Corsi, C. Mazzocca et al.

# SiPM physics and characterization

**Gain**

related to the recharge of the diode capacitance from $V_{BD}$ to $V_{bias}$ during the avalanche quenching time after $I_{latch}$ is reached.

$G=(V_{BIAS}-V_{BD}) * C_D/q$

valid for few volts above $V_{BD}$

**Noise: dark count**

pulses triggered by non-photo-generated carriers (thermal / tunneling generation in the bulk or in the surface depleted region around the junction)

**afterpulse**

carriers can be trapped during an avalanche and then released triggering another avalanche

**optical cross-talk**

photo-generation during the avalanche discharge. Some of the photons can be absorbed in the adjacent cell possibly triggering new discharges

**Photo-detection efficiency**

$PDE = QE * P_{01} * \varepsilon$

QE = quantum efficiency

$P_{01}$ = avalanche triggering prob.

$\varepsilon$ = geometrical fill factor

**Dynamic Range**

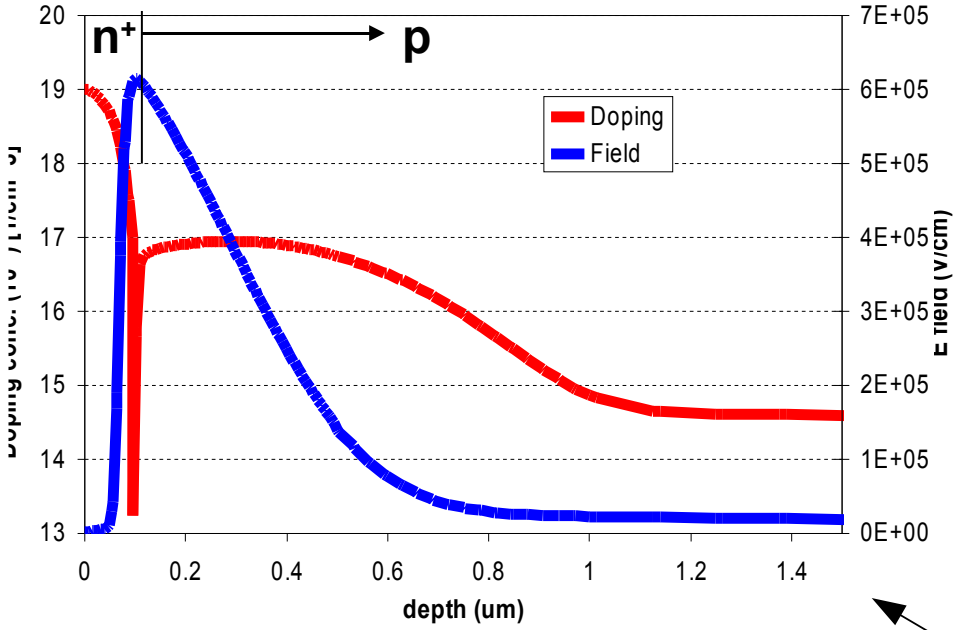
Related to the density of cells and recovery time

**Time resolution**

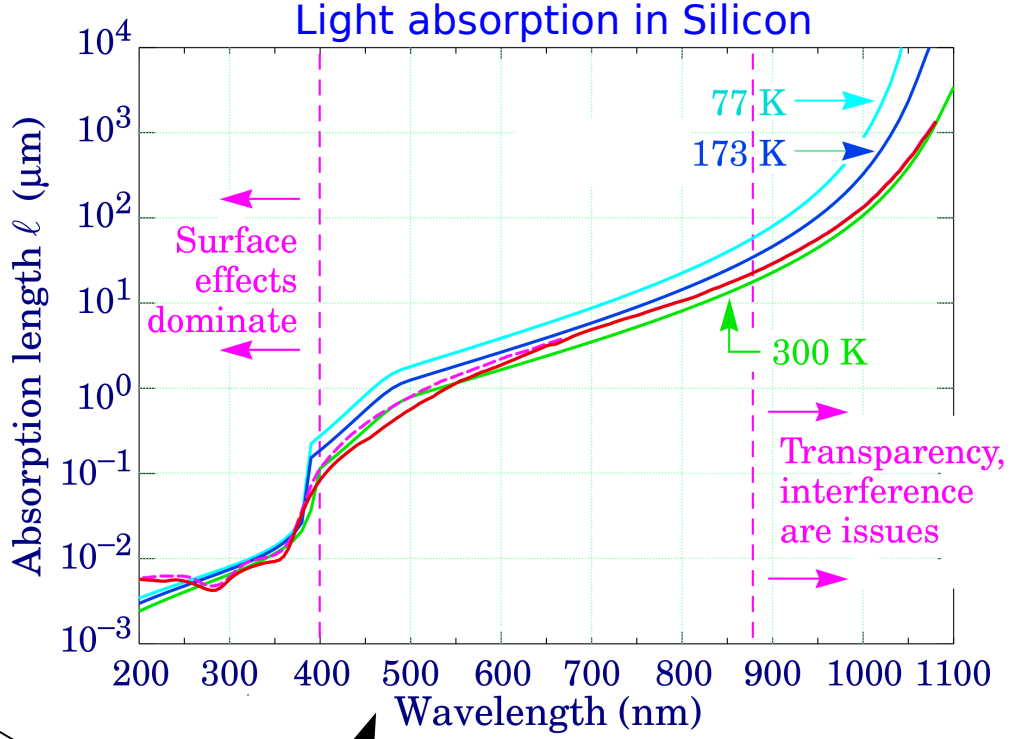
Related to the photo-generation and to the avalanche propagation

# Close up of a cell

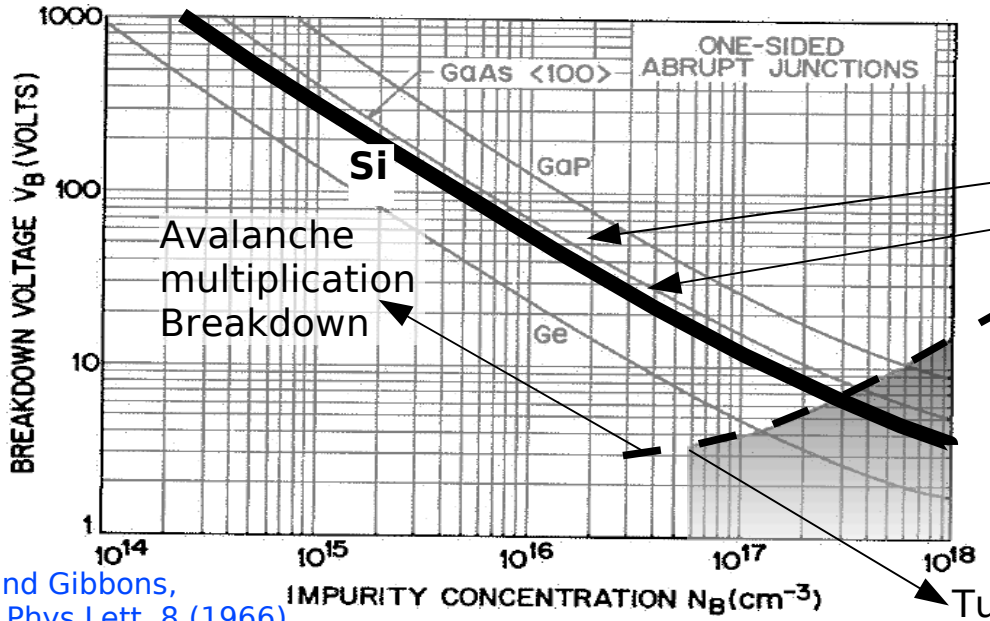
## Doping and Field profiles (IRST)



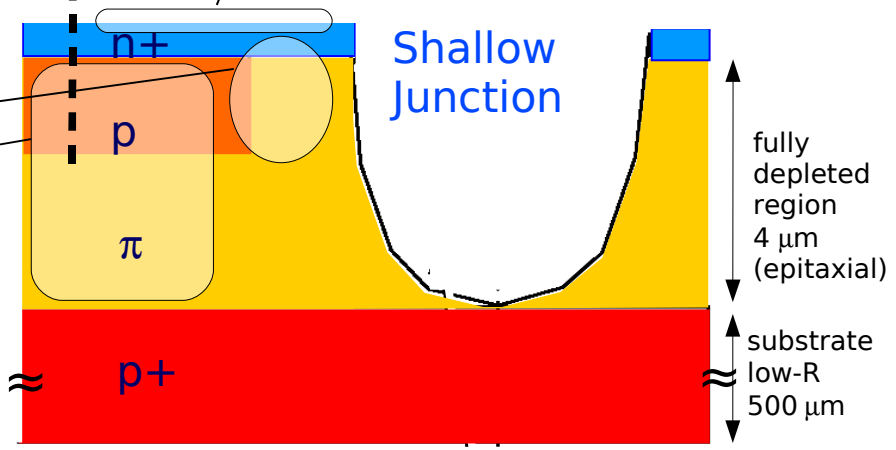

## Light absorption in Silicon




## $V_{BD}$ versus doping concentration




Sze and Gibbons, Appl. Phys Lett. 8 (1966)

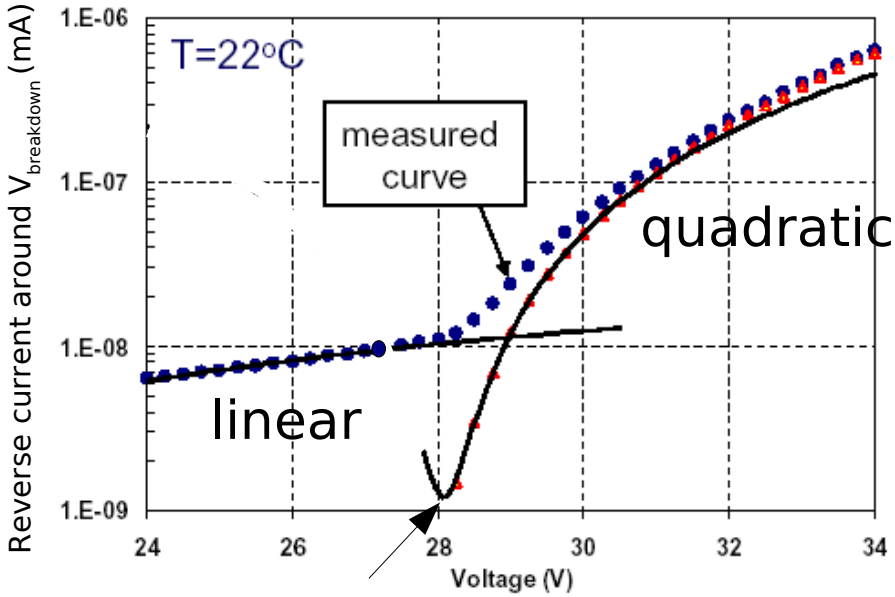


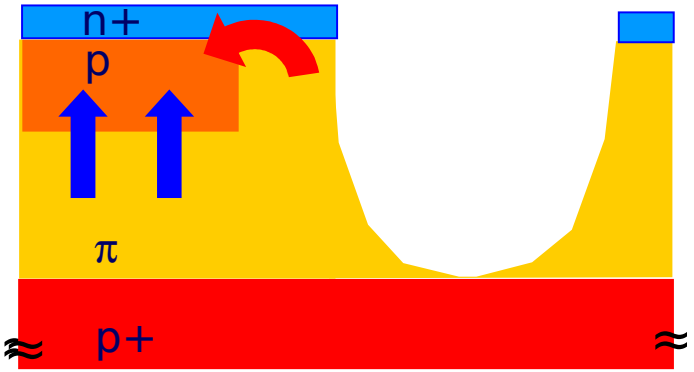

Tunneling effect Breakdown

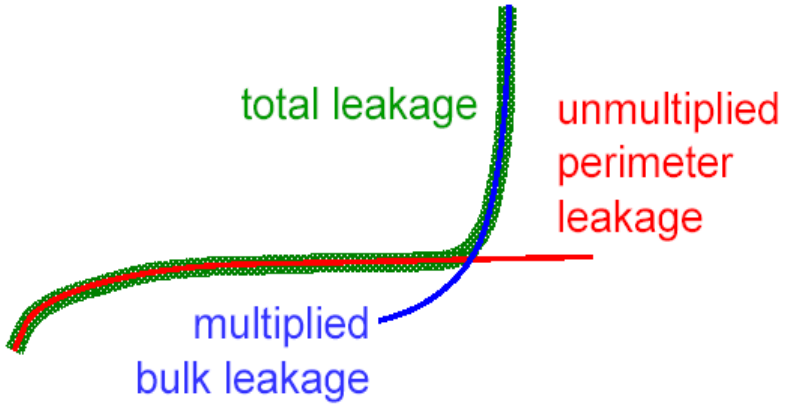
# Static characteristics (I-V measurement)

First tests to verify the functionality of the device:

- Breakdown voltage ($V_{bd}$)
- Dark count rate ($N_{dark}$)
- Quenching resistor value $R_Q$




Pre-breakdown: current mainly due to generation in the surface region around diode:

$I_{pre\text{-}BD} \sim V_{bias}$ (linear)

Post-breakdown: up to few V current due dark events is:

$I_{post\text{-}BD} = q \cdot G \cdot N_{dark} \sim q \cdot V_{bias} \cdot V_{bias}$ (quadratic)

Technological issues:

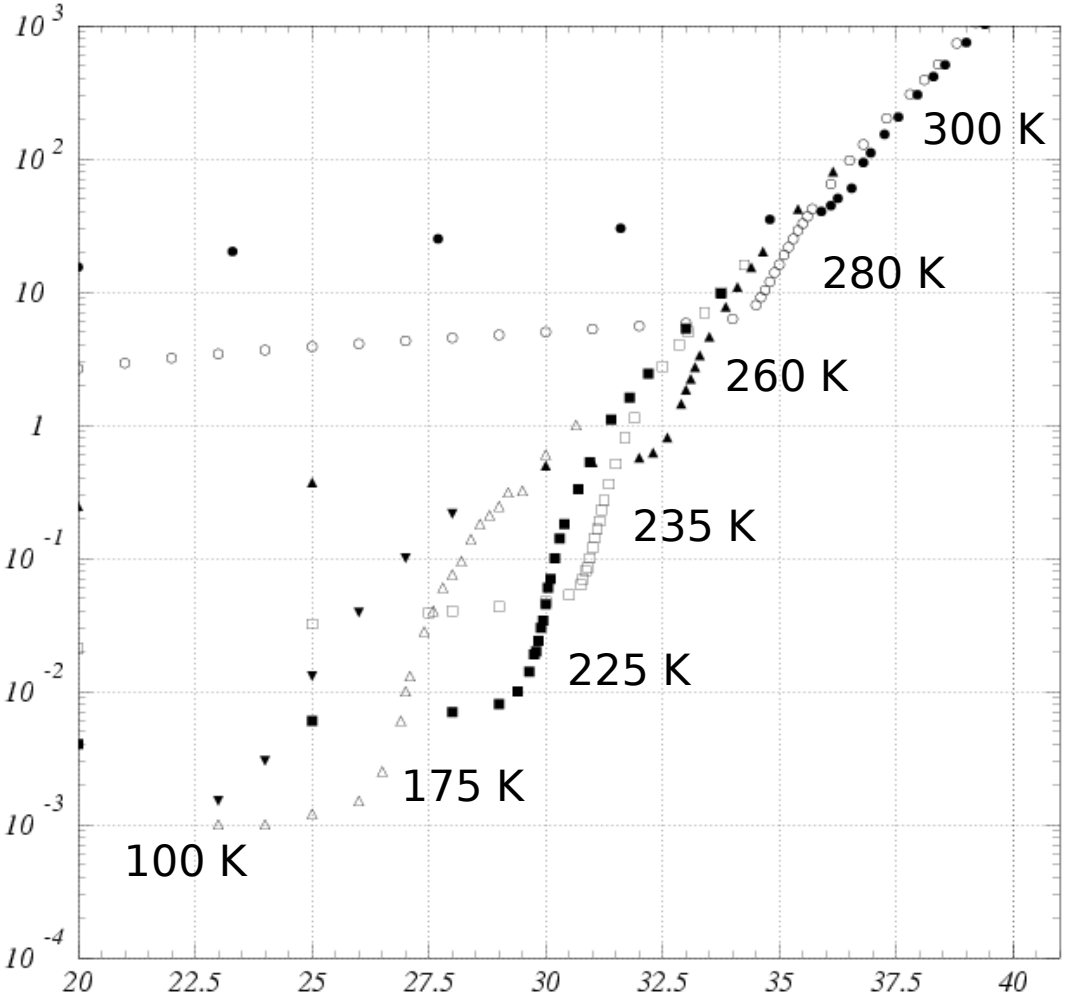
- Uniformity of layout
  check spread of $V_{BD}$ in a wafer (typ. < 1%)
- Uniformity of dark count rate
  check spread of post-breakdown current
- Yield of good devices
  Fraction of devices showing anomalous current behavior (e.g. at IRST <20%)
  Due to rare defects. Crucial parameter for matrices on the same substrate.

# T dependence: breakdown voltage $V_{BD}$

## $V_{BD}$ increases with T

At higher T carriers loose more to lattice
→ lower mobility, shorter mean free path (λ)
→ carriers need higher V to impact-ionize
→ rate of $\Delta v_{bd}/\Delta T$ depending on doping concentration

SiPM - IRST
(1x1mm$^2$ , fill factor 50%)

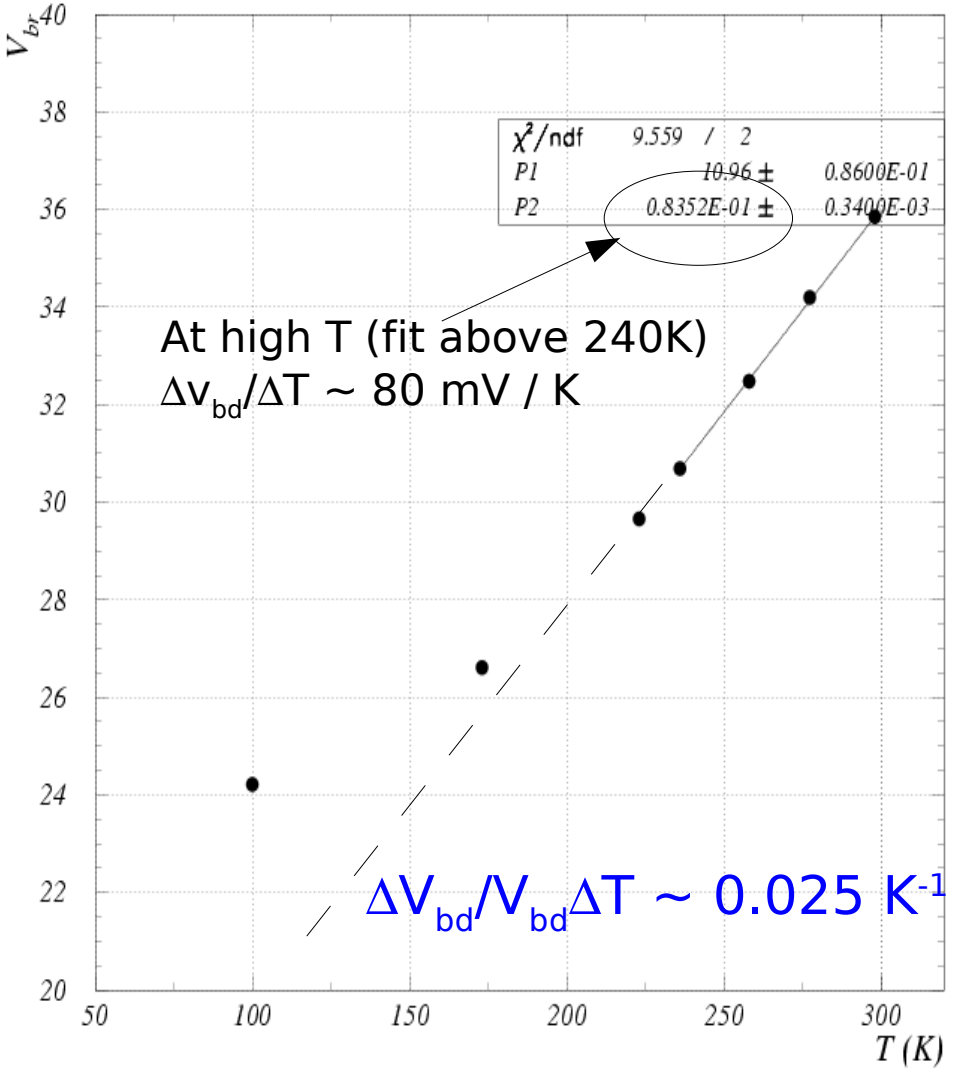



T characterization in vacuum with cryo-cooler system

G.Collazuol (unpublished)
Acknowledgments: A.Baldini, A.Brez

# $V_{BD}$ vs T

## Data: IRST devices




G.Collazuol (unpublished)

## Model: Baraff PR 128 (1962) 2507

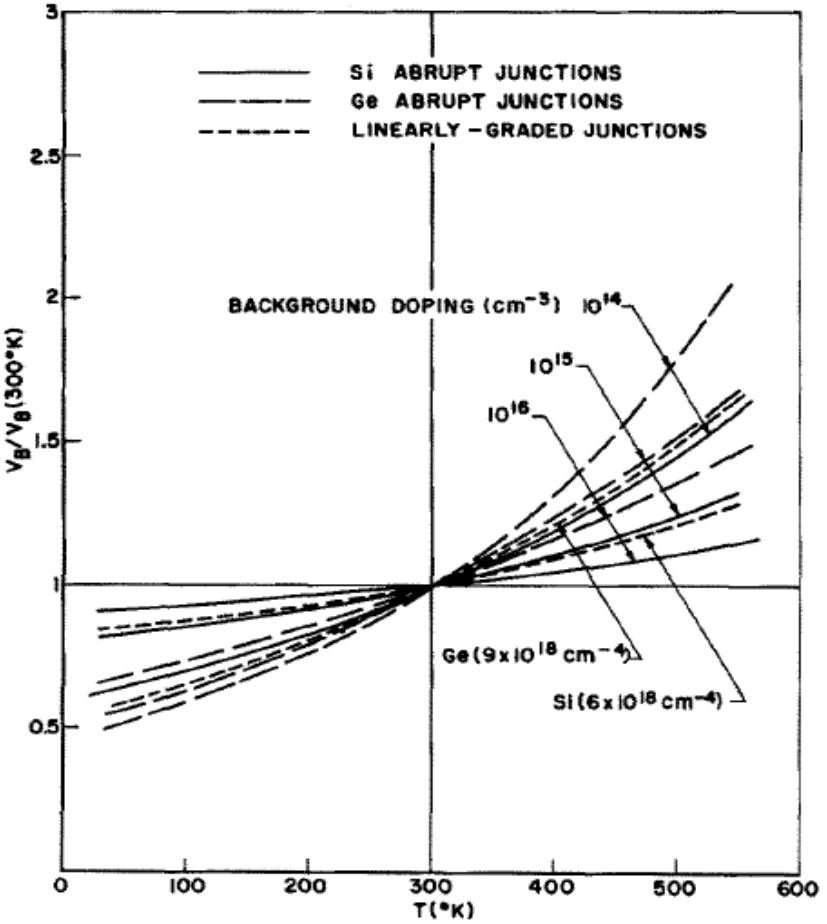
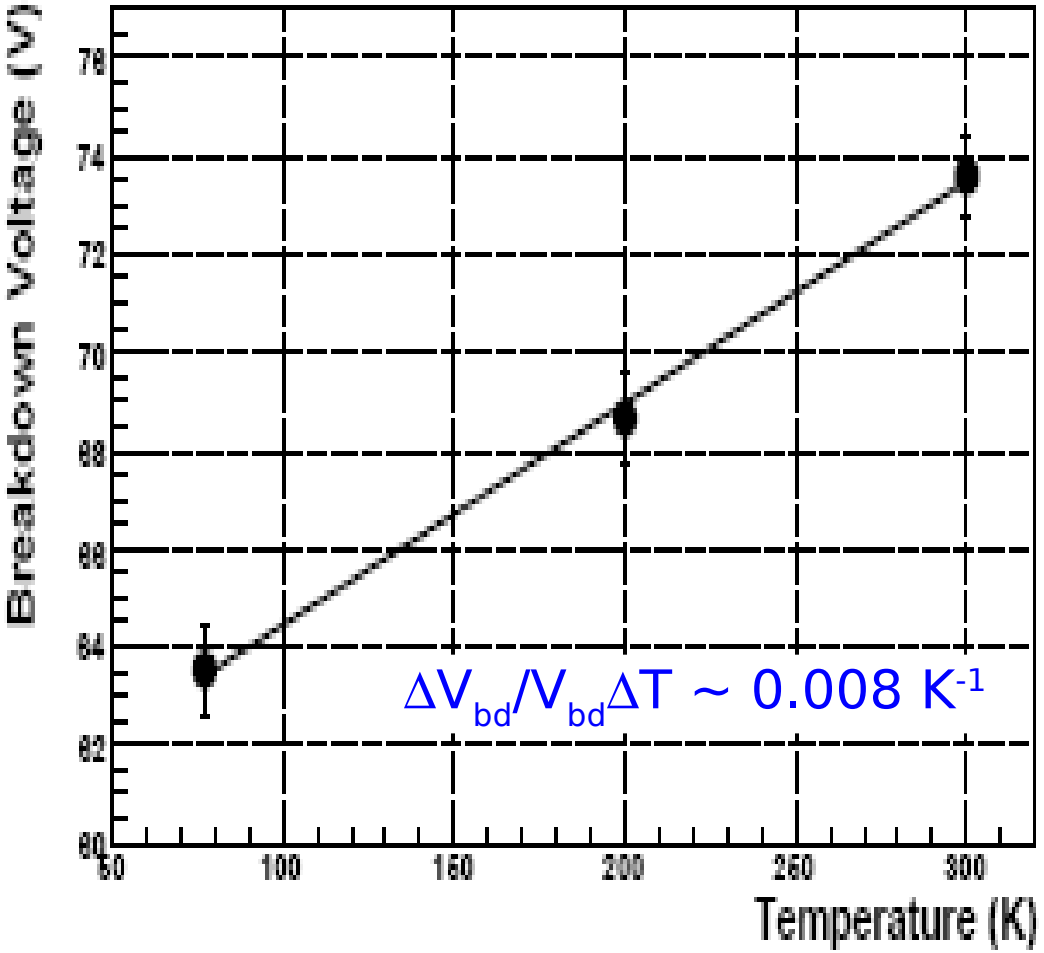



**Fig. 4. Breakdown voltage vs temperature for Si and Ge *p-n* junctions. $V_B$(300°K) is 2000, 330, and 60 V for Si and 950, 150, and 25 V for Ge for dopings of $10^{14}$, $10^{15}$, and $10^{16}$ cm$^{-3}$ respectively. The linear-graded junctions have $V_B$(300°K) the same as those for doping of $10^{15}$ cm$^{-3}$.**

Crowell and Sze, APL 9 (1966) 242

# $V_{BD}$ vs T

## Data: HPK devices




Otono et al – PD07

## Model: Baraff PR 128 (1962) 2507

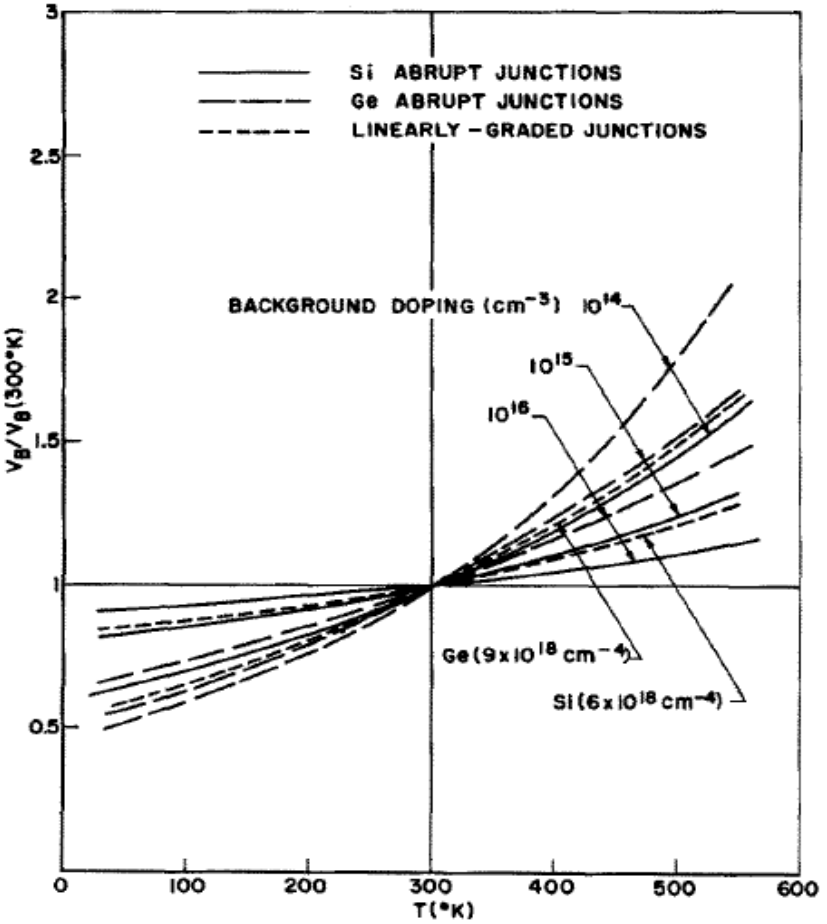



**Fig. 4. Breakdown voltage vs temperature for Si and Ge *p-n* junctions. $V_B(300°K)$ is 2000, 330, and 60 V for Si and 950, 150, and 25 V for Ge for dopings of $10^{14}$, $10^{15}$, and $10^{16}$ $cm^{-3}$ respectively. The linear-graded junctions have $V_B(300°K)$ the same as those for doping of $10^{15}$ $cm^{-3}$.**

Crowell and Sze, APL 9 (1966) 242

# SiPM physics and characterization

related to the recharge of the diode capacitance from $V_{BD}$ to $V_{bias}$ during the avalanche quenching time after $I_{latch}$ is reached.

$G=(V_{BIAS}-V_{BD}) * C_D/q$

valid for few volts above $V_{BD}$

Gain

pulses triggered by non-photo-generated carriers (thermal / tunneling generation in the bulk or in the surface depleted region around the junction)

Noise: dark count
afterpulse
optical cross-talk

carriers can be trapped during an avalanche and then released triggering another avalanche

photo-generation during the avalanche discharge. Some of the photons can be absorbed in the adjacent cell possibly triggering new discharges

**$PDE = QE * P_{01} * \varepsilon$**

QE = quantum efficiency

$P_{01}$ = avalanche triggering prob.

$\varepsilon$ = geometrical fill factor

Photo-detection efficiency

Dynamic Range

Related to the density of cells and recovery time

Related to the photo-generation and to the avalanche propagation

Time resolution

# Dynamic characteristics

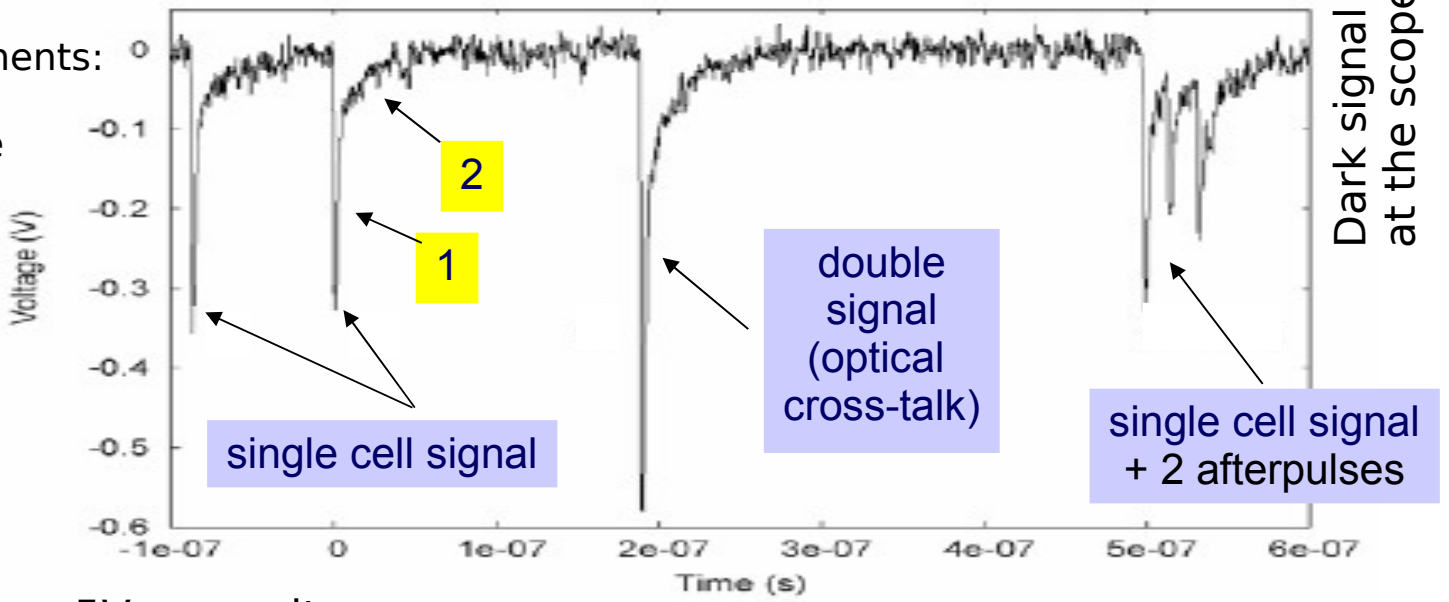
Complete characterization of signal and noise:

•Signal shape • Gain • Dark noise rate • Optical cross-talk • After-pulse

The signal presents 2 components:

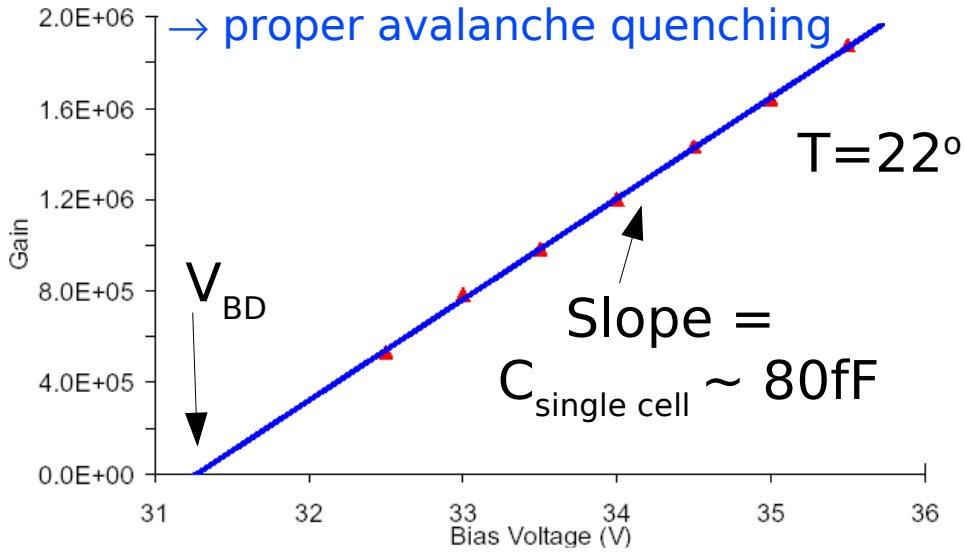
1. fast component: avalanche current reproduced at the output by parasitic capacitor

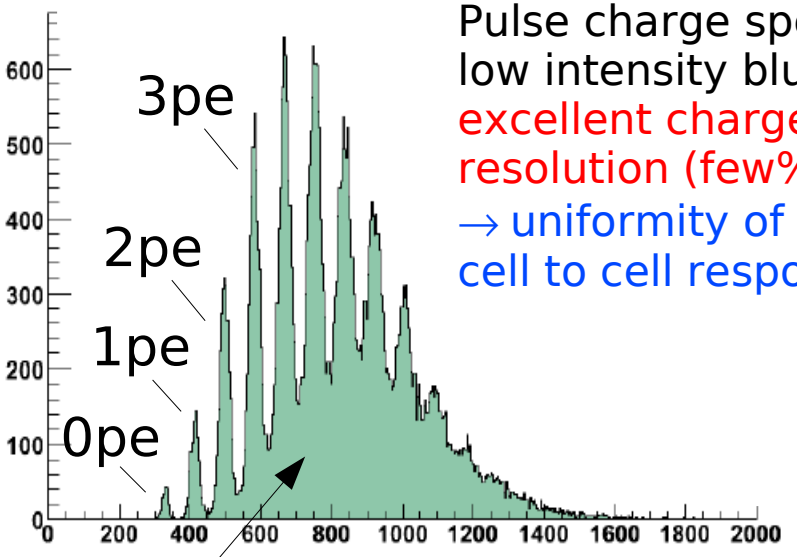
2. slow component due to the recharge of the diode capacitance (99% recovery time ~100ns)




Gain is linear up to ~ 5V overvoltage

→ proper avalanche quenching




Pulse charge spectrum low intensity blue LED

excellent charge resolution (few%)

→ uniformity of cell to cell response

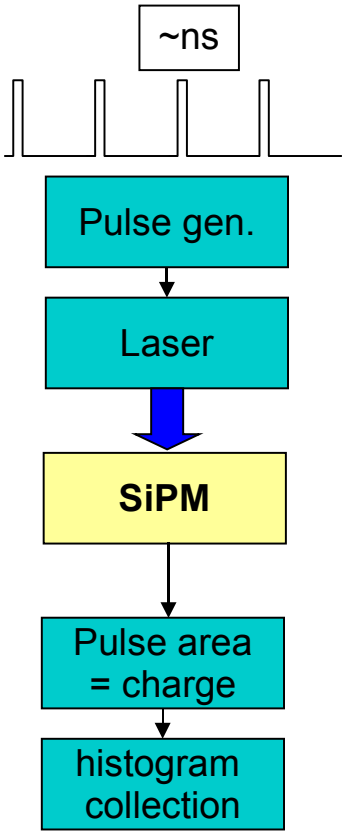
NOTE: very easy to measure the gain (wrt PMT !!!)

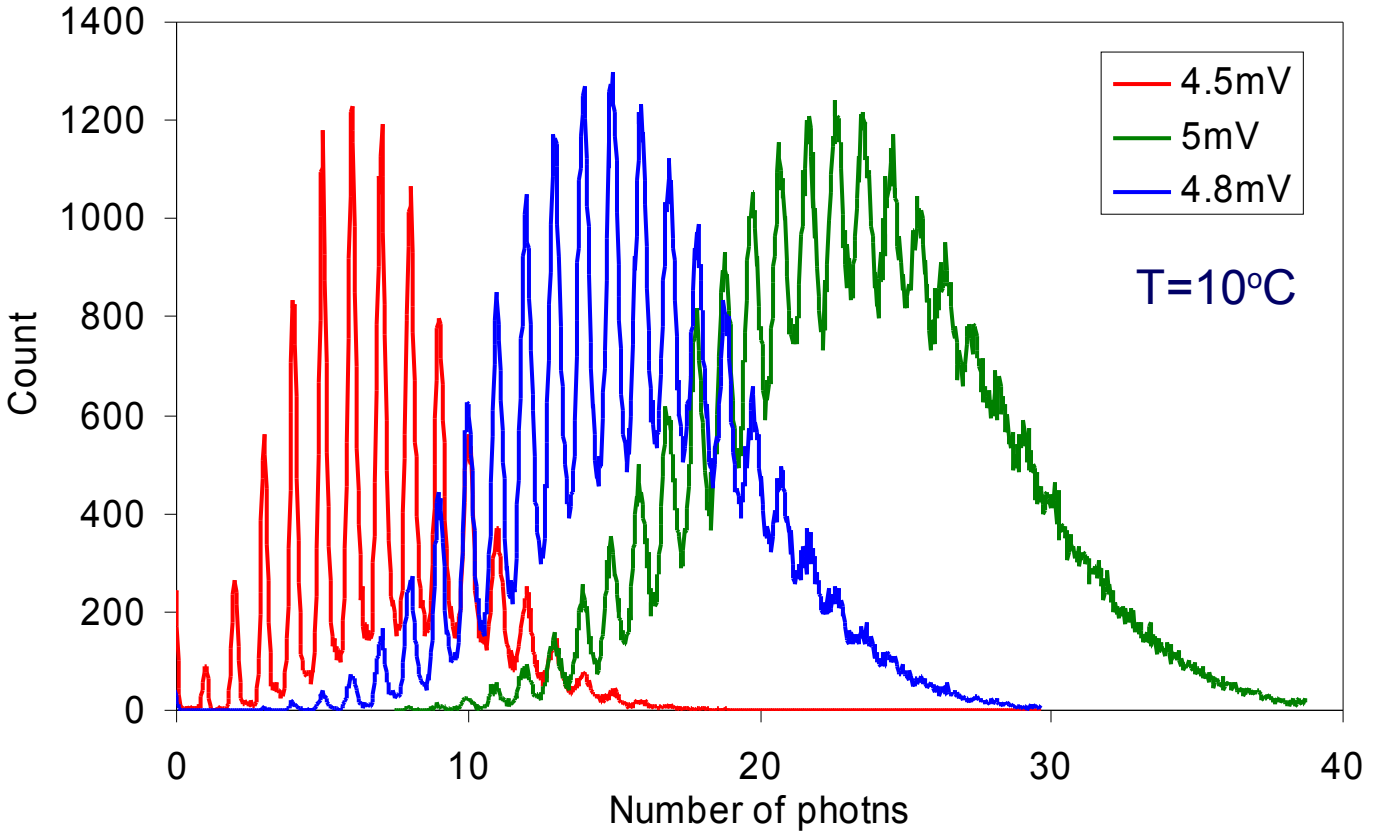
(IRST devices)

# Single cell charge resolution

Device illuminated with short weak light pulses from a blue LED.
Device biased at 3V over-voltage.

Effective quenching and cell-to-cell uniformity !



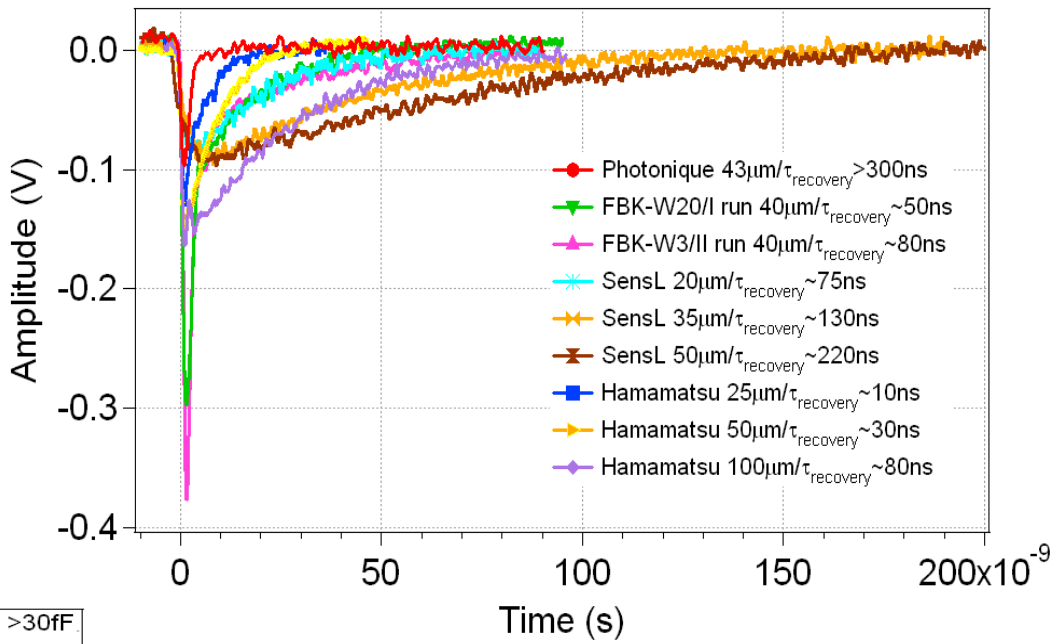

NOTE: resolution limited by electronic noise



C.Piemonte, workshop on "Photon Detection" - Perugia 13-14/6/2007

# Dynamic characteristics ($C_d$ and $R_q$)

| Producer | SiPM ID | Nr. μ cells | μ cell size (μ m) | ε geom (%) |
|---|---|---|---|---|
| Photonique | SSPM-0701-BG | 556 | 43 x 43 | 70 |
| FBK-irst | W20-B10-T3V2PD/ I run | 625 | 40 x 40 | 20 |
| FBK-irst | W3-B3-T6V1PD/ II run | 625 | 40 x 40 | 16 |
| SensL | SPM-20 | 848 | 29 x 32 | 43 |
| SensL | SPM-35 | 400 | 44 x 47 | 59 |
| SensL | SPM-50 | 216 | 59x 62 | 68 |
| HPK | S10362-11-25 | 1600 | 25 x 25 | 31 |
| HPK | S10362-11-50 | 400 | 50 x 50 | 61.6 |
| HPK | S10362-11-100 | 100 | 100x 100 | 78.5 |




g.2 Single-μcell dark signals shape of the SiPM

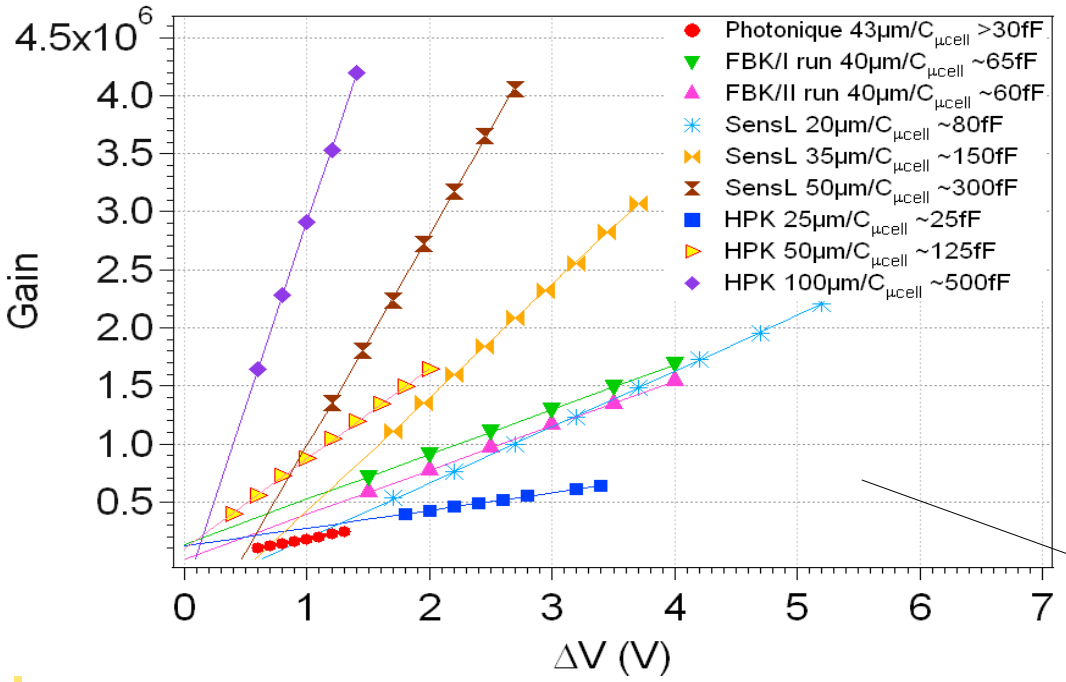



Fig. 3 The gain of the SiPM prototypes vs. Δ V

Electro-optical characterization of SiPM: a comparative study N.Dinu et al. NIM A (2008) – to be publshed

Integration window 40ns

# T dependence: $\tau_Q$ and Gain

## $V_{BD}$ breakdown voltage: $V_{BD}$ increases with T:

At higher T carriers loose more to lattice
→ lower mobility, shorter mean free path (λ)
→ carriers need higher V to impact-ionize
(temp. coefficient $\Delta V_{BD}/\Delta T \sim$ 20mV/K – 80mV/K depending on doping concentration)




Haitz JAP 34 (1963)

## Recovery time: $\tau_Q$ decreases with T

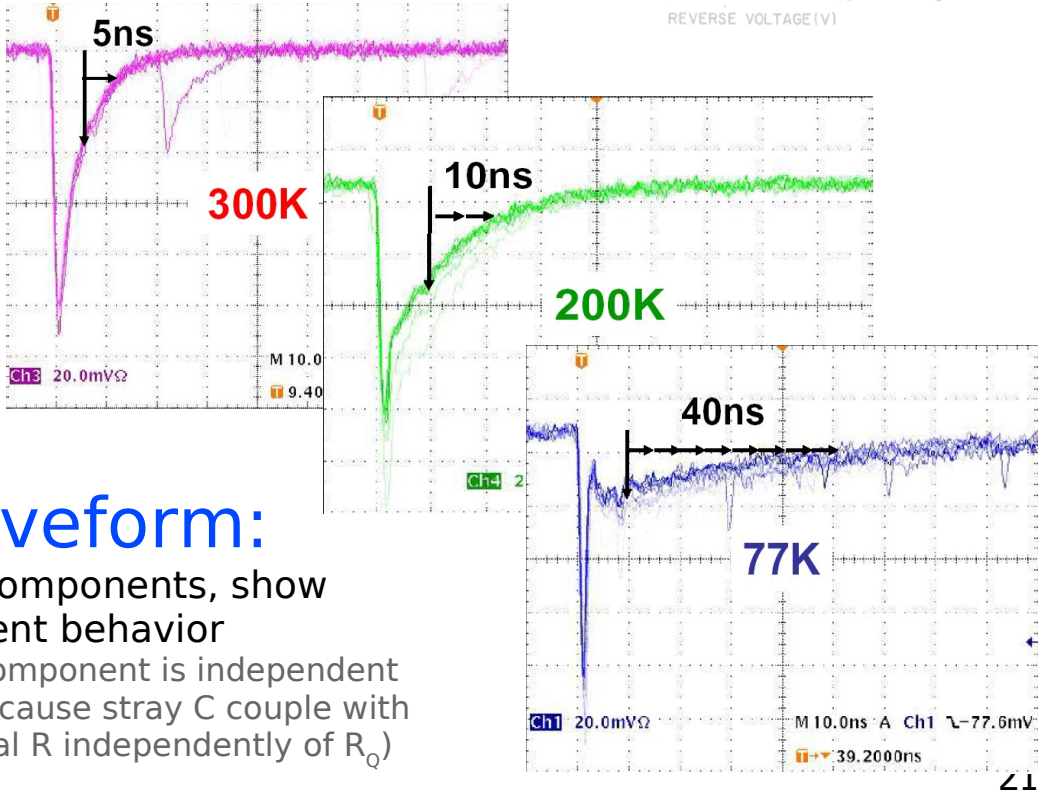
Due polysilicon $R_Q$ properties

| T (K) | Rq (MΩ) |
|---|---|
| 300 | 0.2 |
| 200 | 0.4 |
| 77 | 1.7 |

H.Otono – PD07
Characterization
of SiPM by HKP at low T

Cell Capacity does not vary with T

## Gain independent of T
at fixed Over-Voltage




## Waveform:

Two components, show different behavior
(fast component is independent of T because stray C couple with external R independently of $R_Q$)

# SiPM physics and characterization

related to the recharge of the diode capacitance from $V_{BD}$ to $V_{bias}$ during the avalanche quenching time after $I_{latch}$ is reached.

$G=(V_{BIAS}-V_{BD}) * C_D/q$

valid for few volts above $V_{BD}$

Gain

Noise: dark count
after-pulse
optical cross-talk

pulses triggered by non-photo-generated carriers (thermal / tunneling generation in the bulk or in the surface depleted region around the junction)

carriers can be trapped during an avalanche and then released triggering another avalanche

photo-generation during the avalanche discharge. Some of the photons can be absorbed in the adjacent cell possibly triggering new discharges

**$PDE = QE * P_{trigg} * \varepsilon_{geom}$**

QE = quantum efficiency

$P_{trigg}$ = avalanche triggering prob.

$\varepsilon_{geom}$ = geometrical fill factor

Photo-detection efficiency

Dynamic Range

Related to the density of cells and recovery time

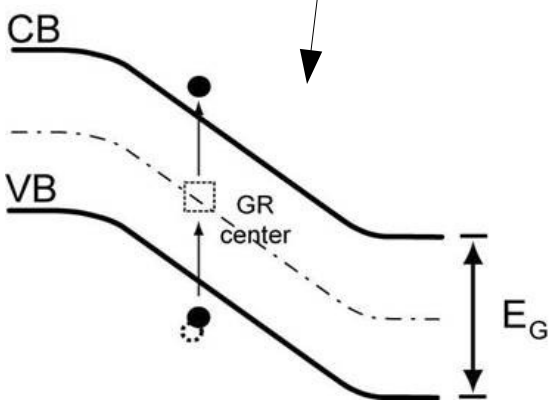
Related to the photogeneration and to the avalanche propagation

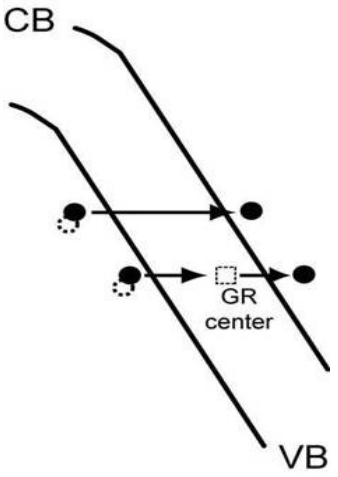
Time resolution

# Dark count rate: free carrier generation

## Two main mechanisms

(2) Field-Assisted Generation: tunneling
(trap-assisted and band to band)




(1) Generation - Recombination Centers
SHR generation (Field Enhanced)
in the depletion region

$$\frac{dN_{emiss}}{dt} \equiv \frac{n_i}{2\tau_g} \quad [1/cm^3 s]$$

$n_i$: intrinsic carrier concentration

$\tau_g$: minority carrier lifetime $\sim 1/N_{generation\ centers}$

Example:

- effective volume $V_{eff} = A_{eff} \cdot W_{depletion}$
  $V_{eff} \sim 1mm^2 \cdot 50\% \cdot 4\mu m$
- $\tau_g \sim 10ms$ (good quality technology)
- Prob. to trigger avalanche $P_{01}$
  $P^e_{01} \sim 100\%$ for electrons
  $P^h_{01} \sim 1/2\ P^e_{01} \sim 50\%$ for holes

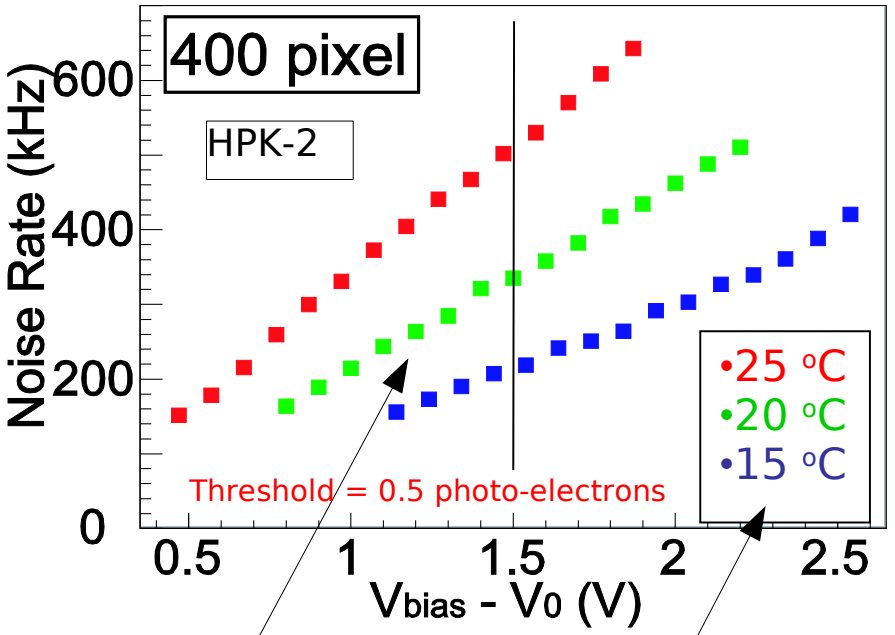
$\rightarrow$ Dark rate $\sim V_{eff} P_{01} / \tau_g \sim$ 2MHz (n+/p: e trigger the avalanche in depl. region)
$\sim$ 1MHz (p+/n: h trigger the avalanche in depl. region)

# Dark count rate

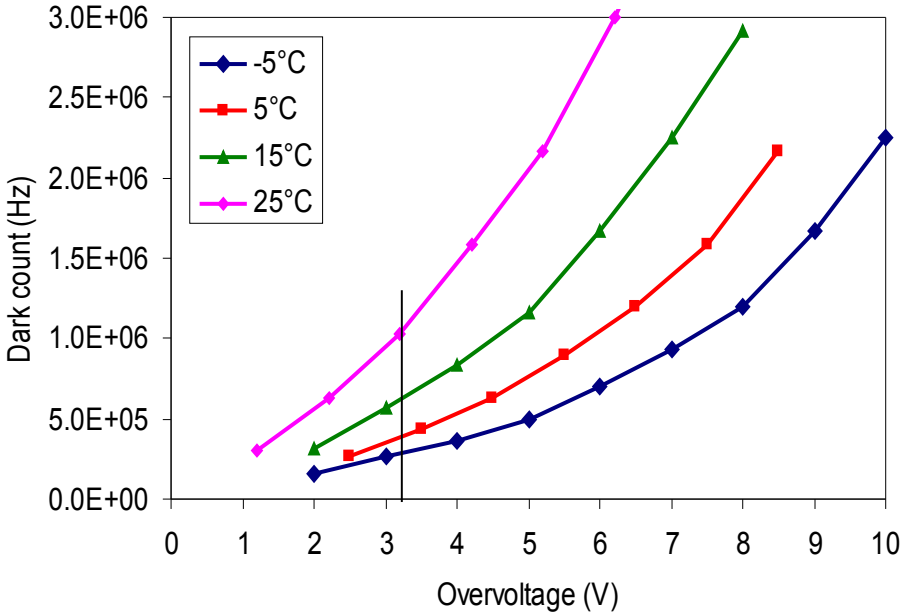
Critical issues: • quality of epitaxial layer
• gettering techniques

## Hamamatsu device (1mm²)

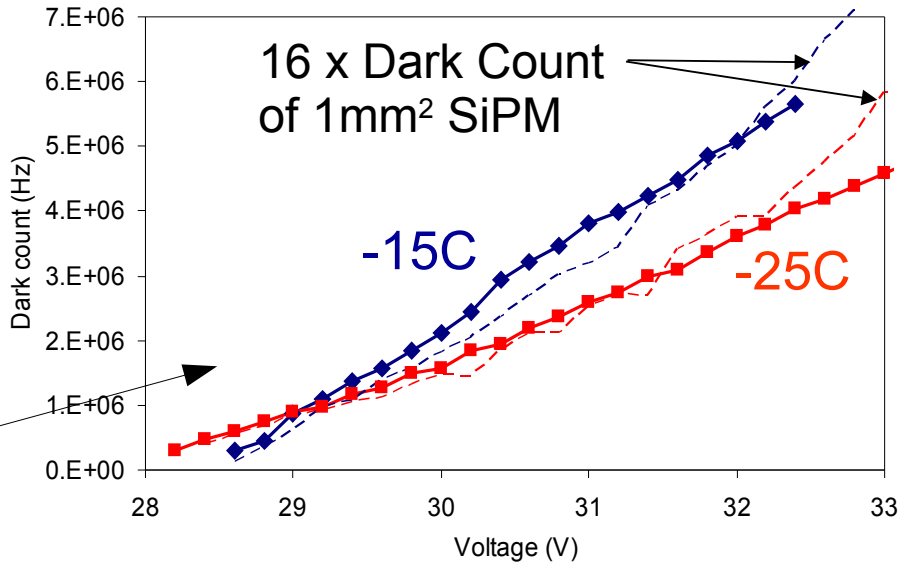
S.Uozumi – VCI07




## IRST device (1x1 mm²)




## IRST device (4x4 mm²)




NOTE:
T dependence

NOTE:
~ linear dependence due to $P_{01} \propto \Delta V$
~ scales with active surface
~ non-linear at high $\Delta V$ due to cross-talk and after-pulsing ($\propto \Delta V^2$)

# Dark count rate

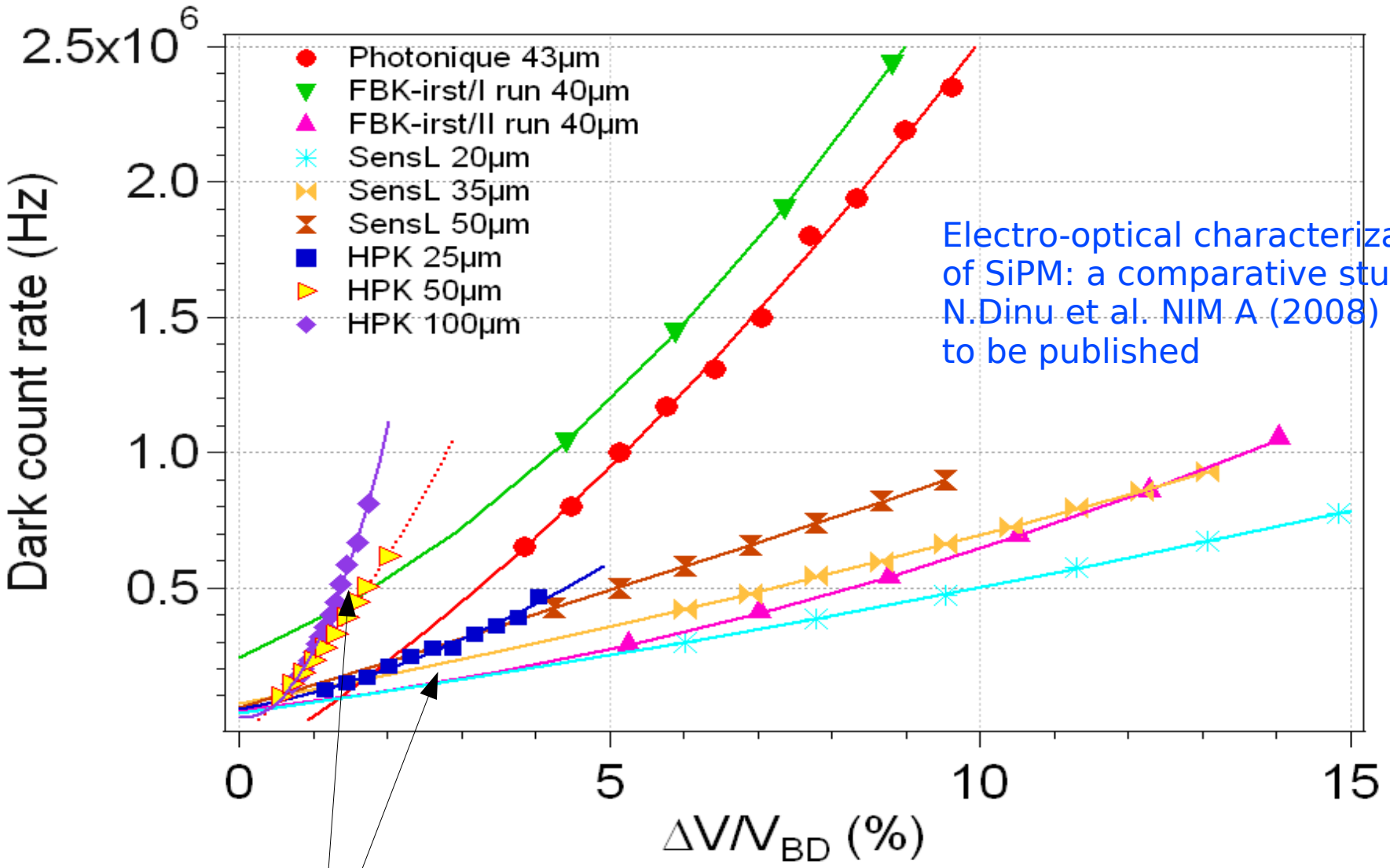


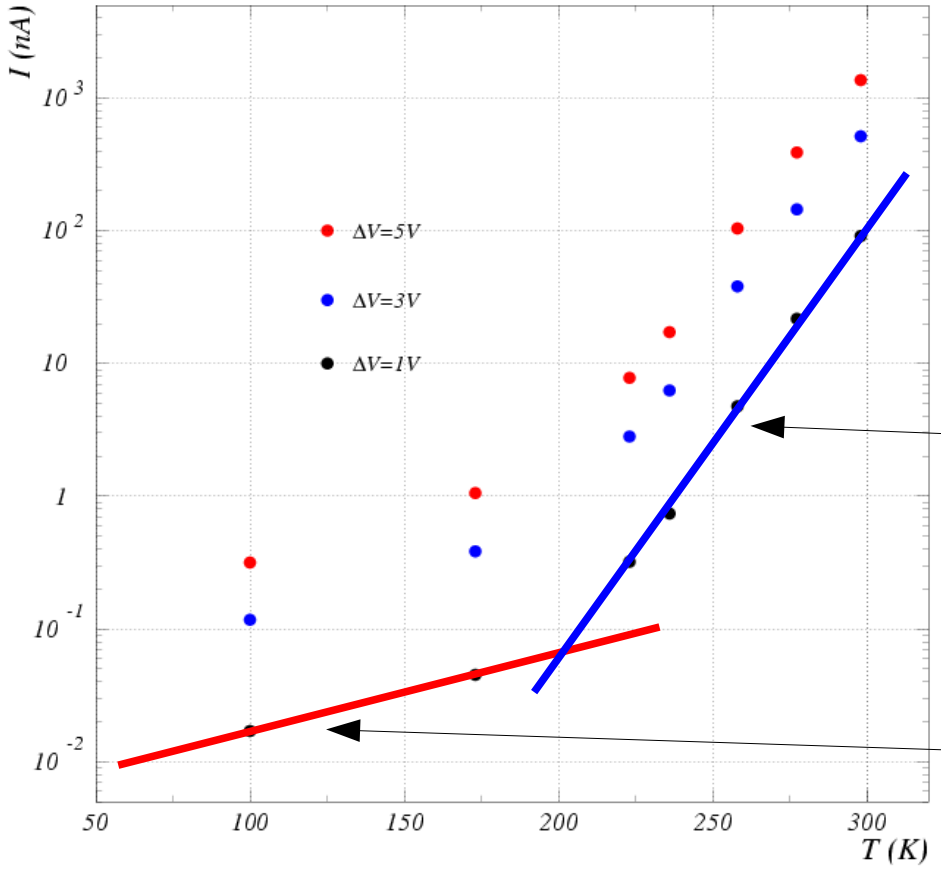

Fig.4 The DCR of the SiPM prototypes vs. $\Delta V/V_{BD}$

The slope (at low $\Delta V/V$) is depending on the slopw of breakdown probability ($P_{01}$)
(see later comments on PDE)

# T dependence: Dark Current

## Data (IRST devices)

(fixed over-voltage)




## Dark rate sources:

### 1. Diffusion

$I_{reverse}$ by minority carriers: negligible ay T room

### 2. SHR (Field Enhanced)

Rule of thumb: factor x2/10K (at fixed Over-voltage)
Dominating at T room

$$I_{reverse} \propto T^2 \exp\frac{-E_g}{2K_B T}$$

### 3. Band to band Tunnel

Strong dependence on the Electric field profile
Dominating at low T

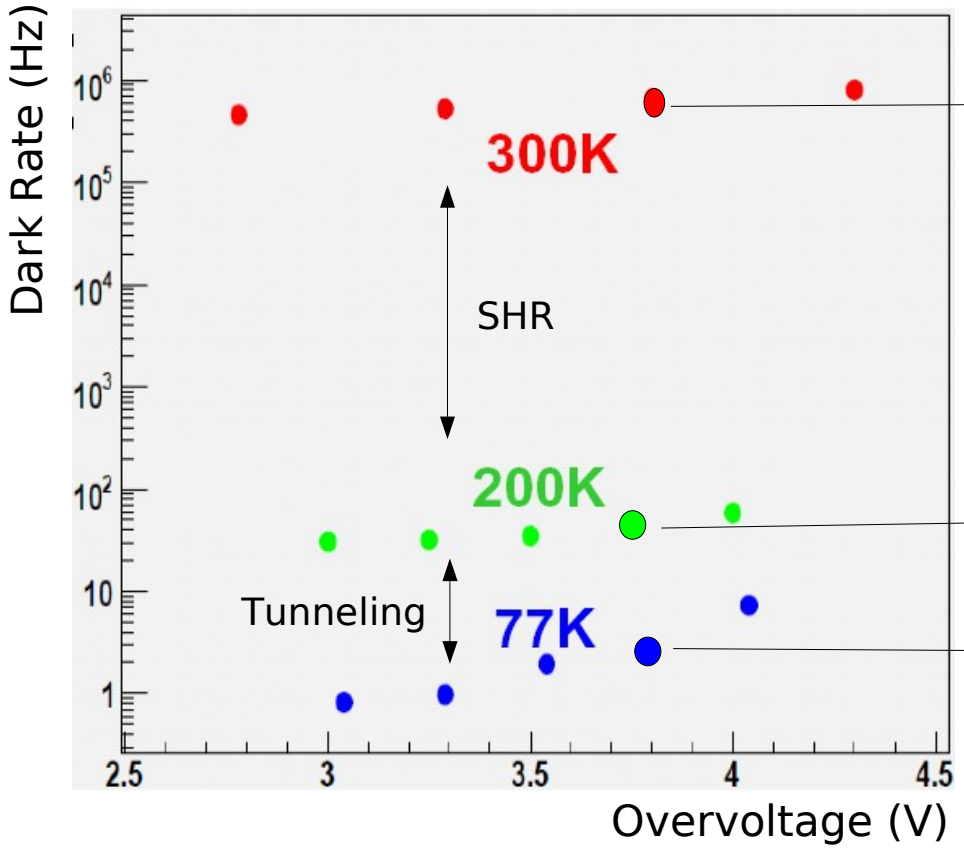
Tunneling current dominates
at higher doping and lower temperatures

G.Collazuol (unpublished)

Acknowledgments: A.Baldini, A.Brez – INFN Pisa

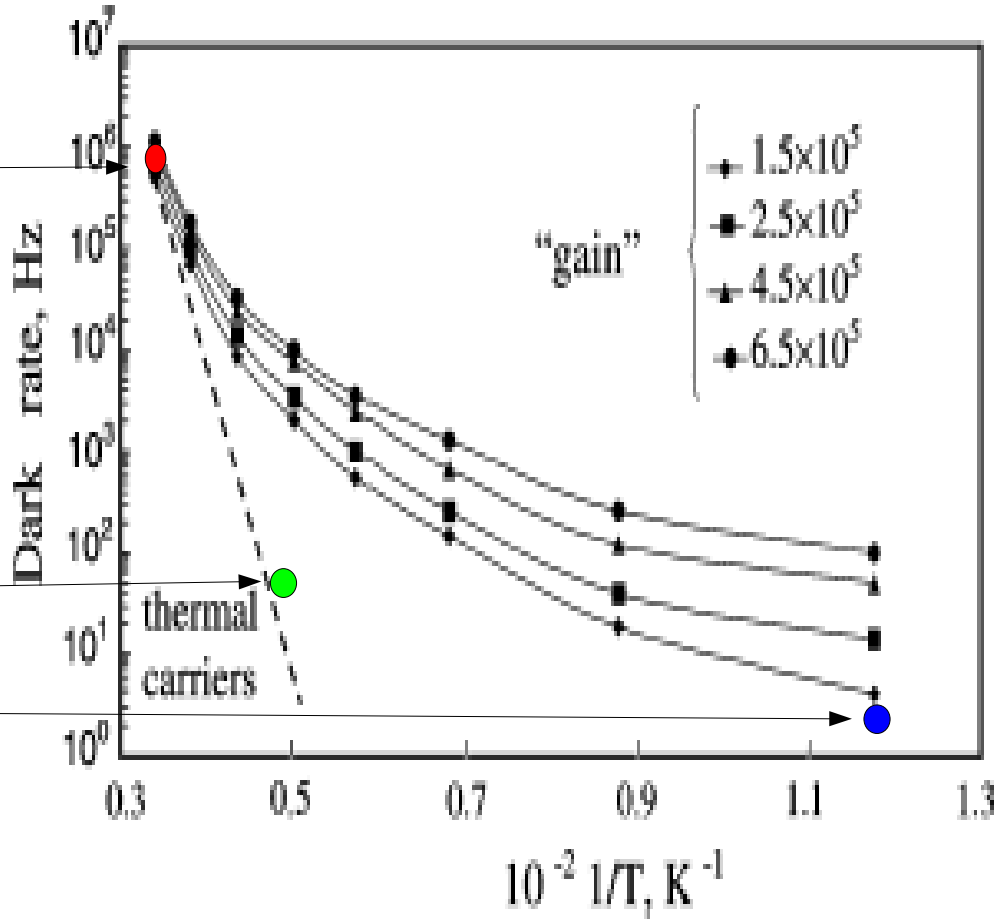
# T dependence: Dark Rate

Hamamatsu




H.Otono – PD07

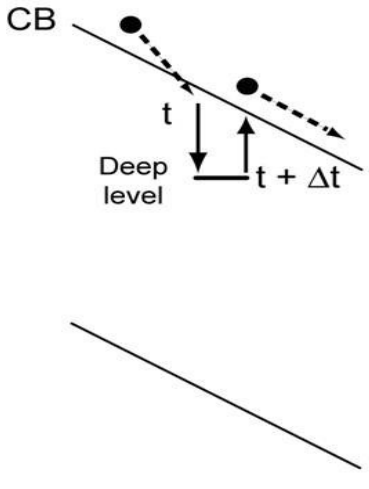
MePHY/Pulsar




Dolgoshein et al, NIM A 442 (2000)

Electric field engineering and silicon quality
make huge differences in dark noise as a function of T

# After-pulsing




## i.e. Carrier trapping and delayed release

$$P_{after-pulse}(t) = P_c \cdot \frac{\exp(-t/\tau)}{\tau} \cdot P_{01} \propto \Delta V^2$$

$P_{01}$ : trigger probability
∝ ΔV(t) (over-voltage, recovery)

τ : trap lifetime
depends on trap level position

quadratic dependence on ΔV

$P_c$ : trap capture probability
∝ carrier flux (current) during avalanche ∝ ΔV (over-voltage)
∝ N traps

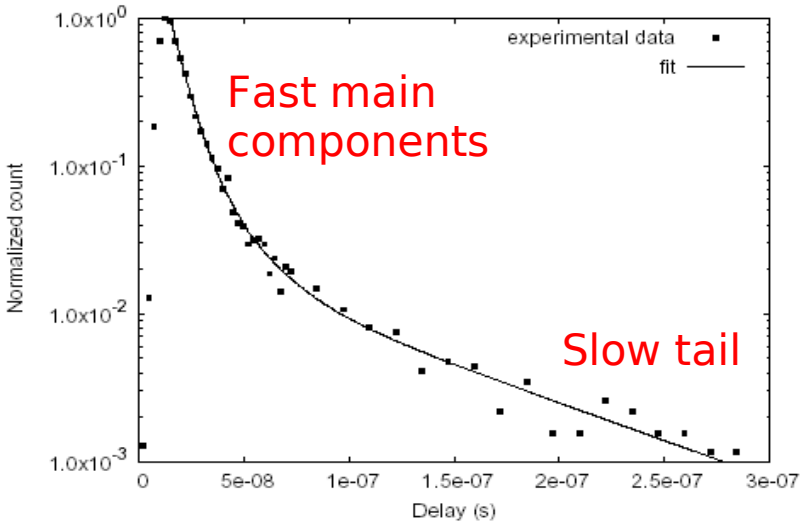
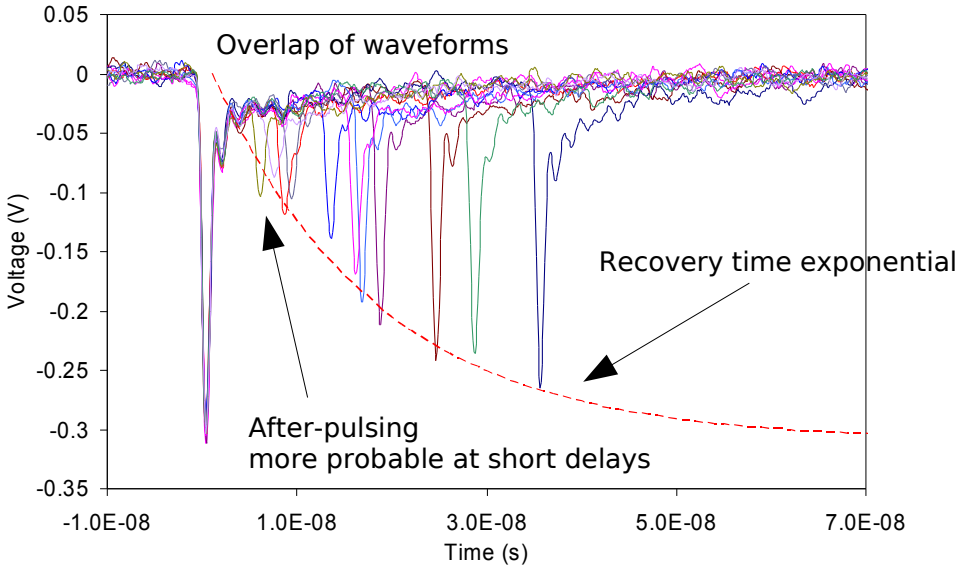



Fig. 10. Spectrum of the delay time from the primary pulse to the after-pulse.

C.Piemonte et al. IEEE TNS 54 (1) (2007) 236

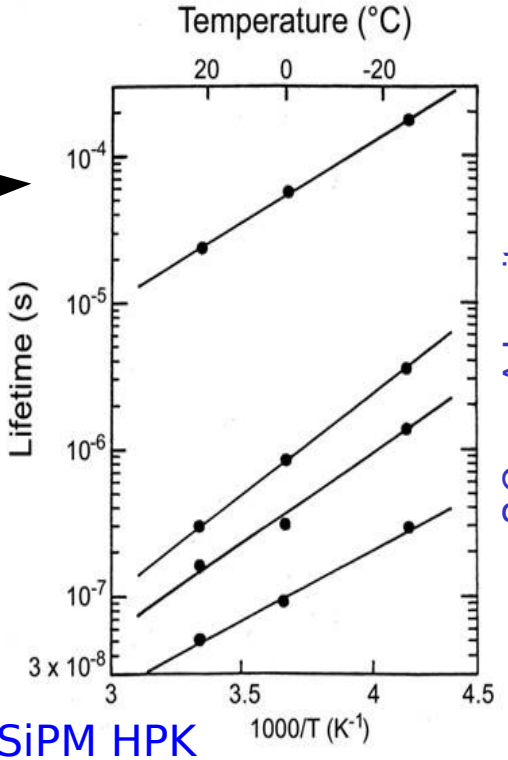



It can be reduced to % in a wide ΔV range

# T dependence: after-pulse, cross-talk

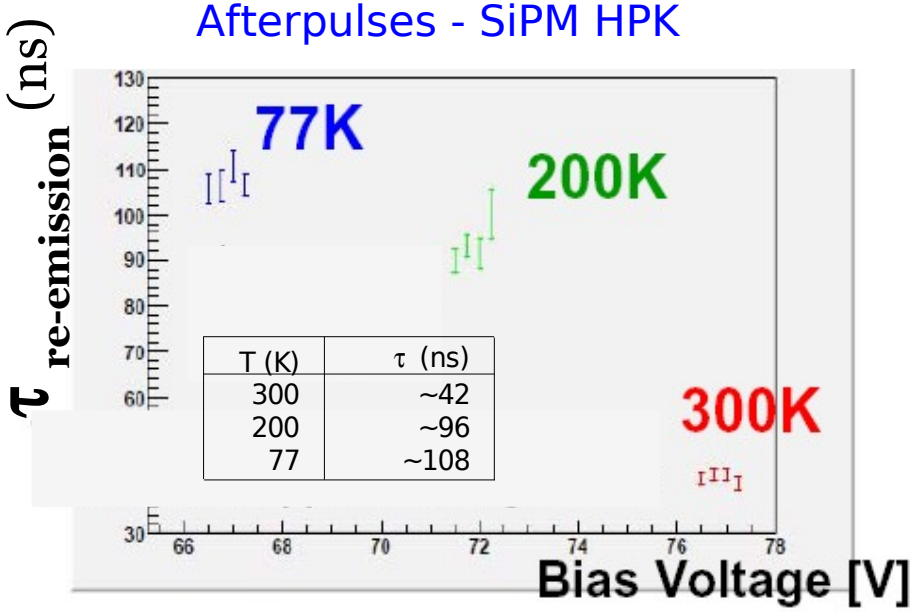
## After-pulses: increases at low T

Trap lifetime decreases as T increases



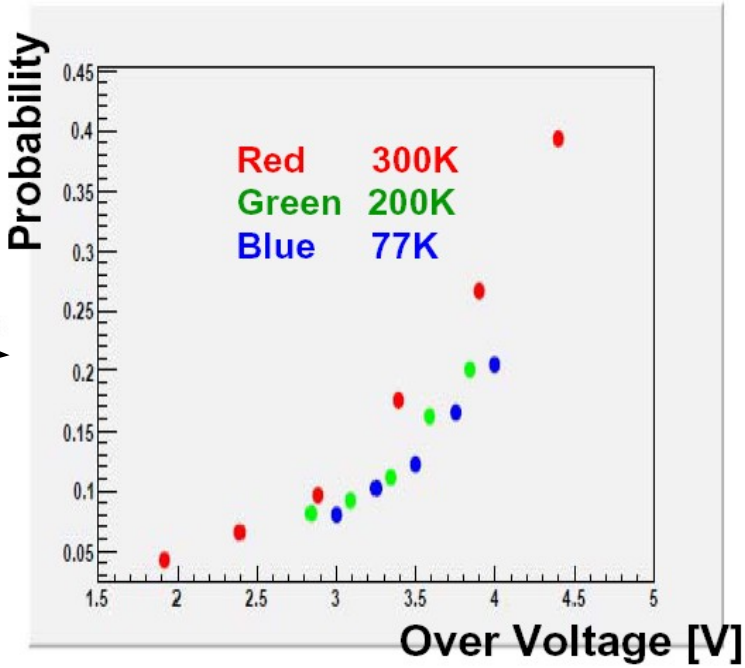

S.Cova, A.Lacaita, G.Ripamonti, IEEE EDL (1991)

Afterpulses - SiPM HPK




H.Otono – PD07

Crosstalk - SiPM HPK




## Cross-talk: decreases at low T

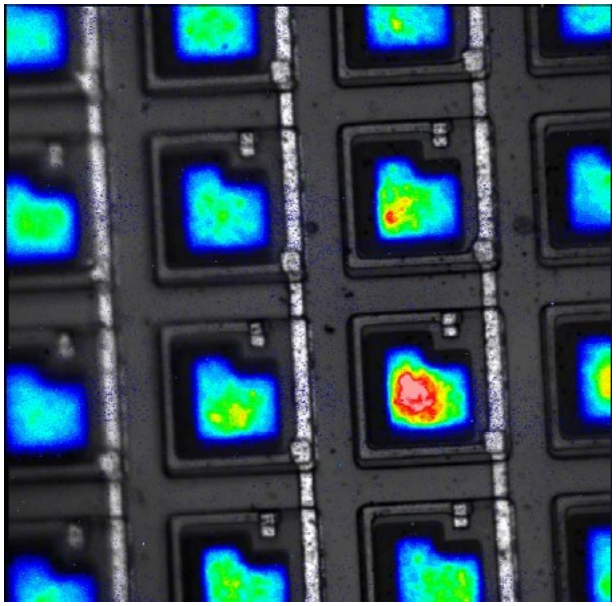
Observed slight reduction at low T

(I guess: due to lower PDE for long wavelength photons which dominate the carrier luminescence spectrum)

# Optical cross-talk

Carrier luminescence (spontaneous direct relaxation in the conduction band) during the avalanche: probability $3 \cdot 10^{-5}$ per carrier (crossing the junction) to emit photons with E> 1.14 eV

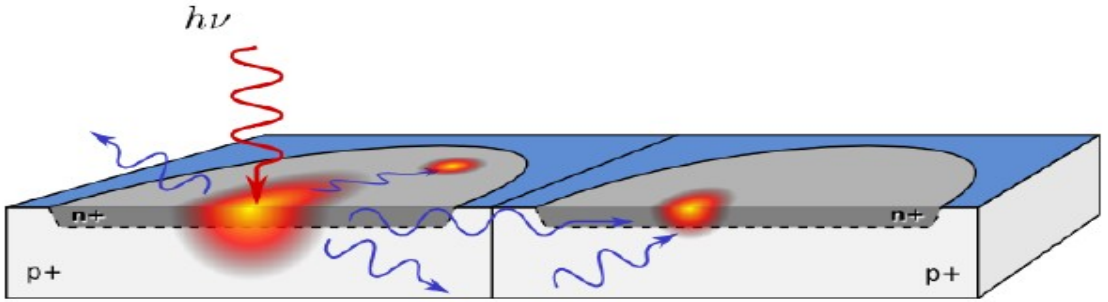
A.Lacaita et al. IEEE TED (1993)



N.Otte, SNIC 2006

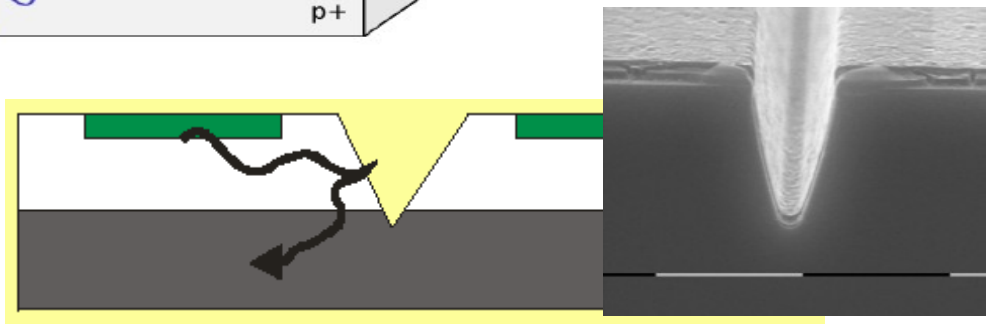
Photons can induce avalanches in neighboring cells.
Depends on distance between high-field regions
Quadratic dependence on over-voltage:

- carrier flux (current) during avalanche ∝ ΔV
- gain ∝ ΔV




Counteract:

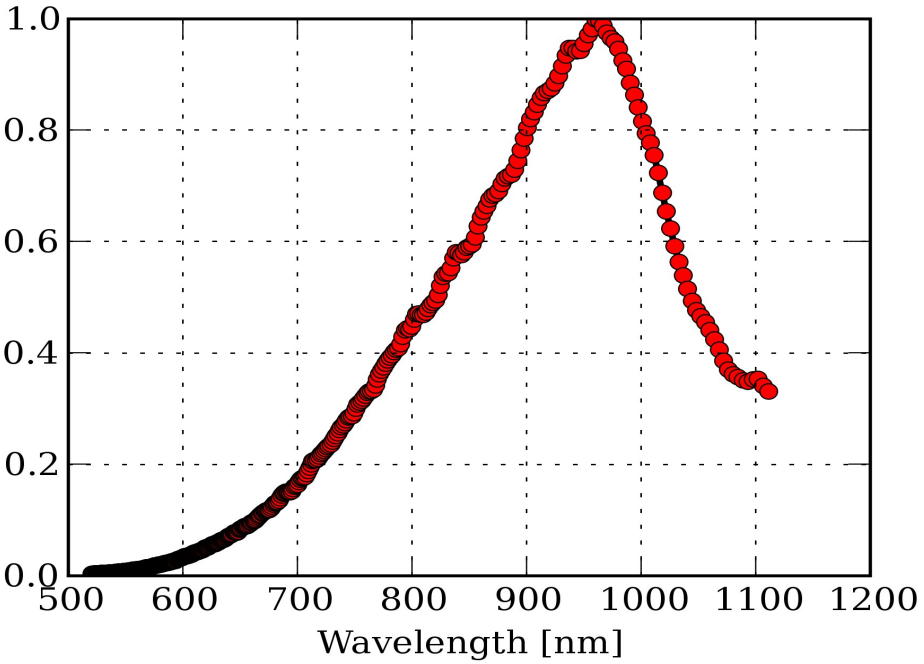
- optical isolation between cells by trenches filled with opaque material
- low over-voltage operation helps



It can be reduced below % in a wide ΔV range

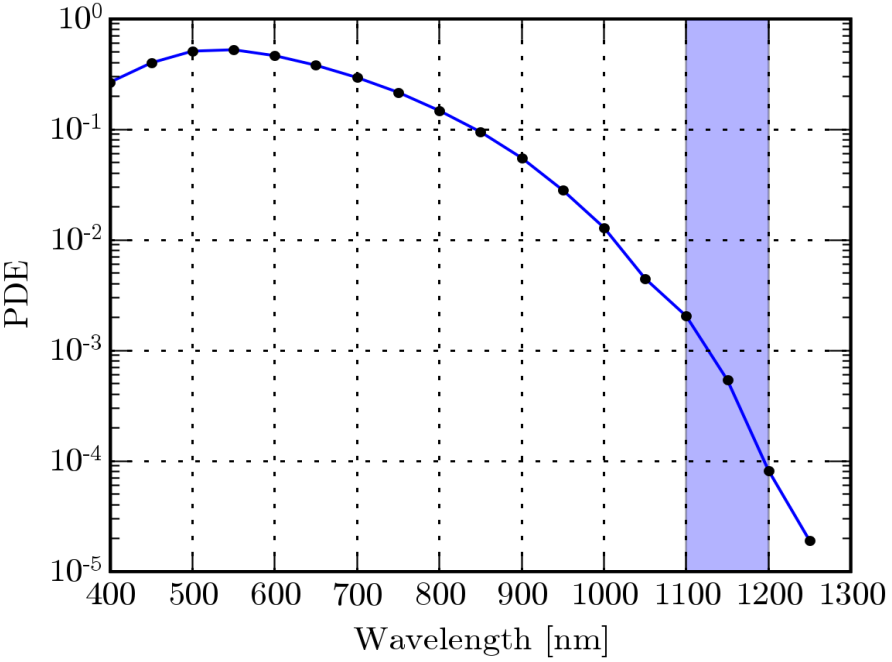
# Optical cross-talk: wavelength range

Measured Emission spectrum




⊗

PDE

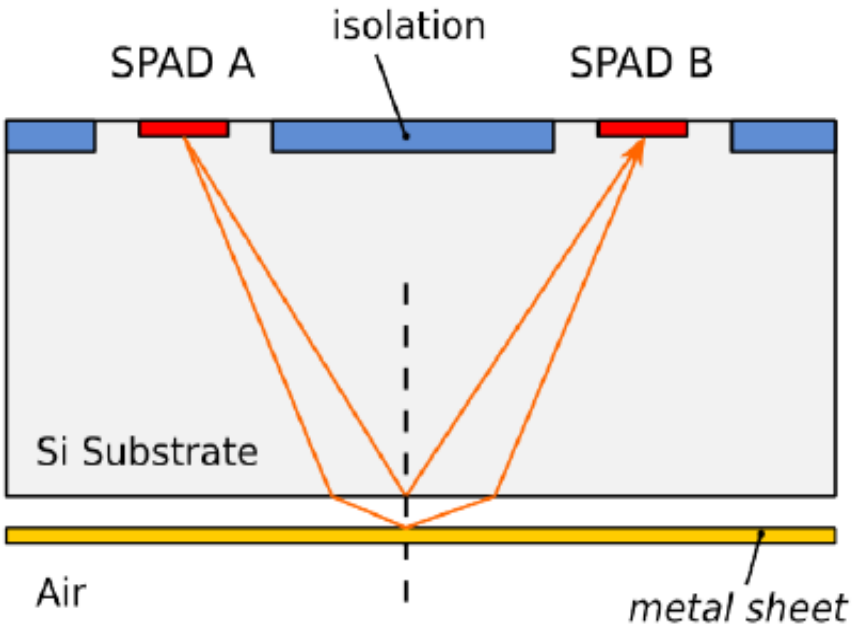



**Crosstalk is due to a *narrow* wavelength range** *(< 100 nm)*

A.Ingargiola - NDIP08

# Optical cross-talk: reflections off the bottom

A.Ingargiola - NDIP08




## Hypothesis

- Is there the contribution due to the bottom reflection *dominant*?

## Proof

- Measurements with a reflective layer on the backside show a *marked increase in the crosstalk intensity*

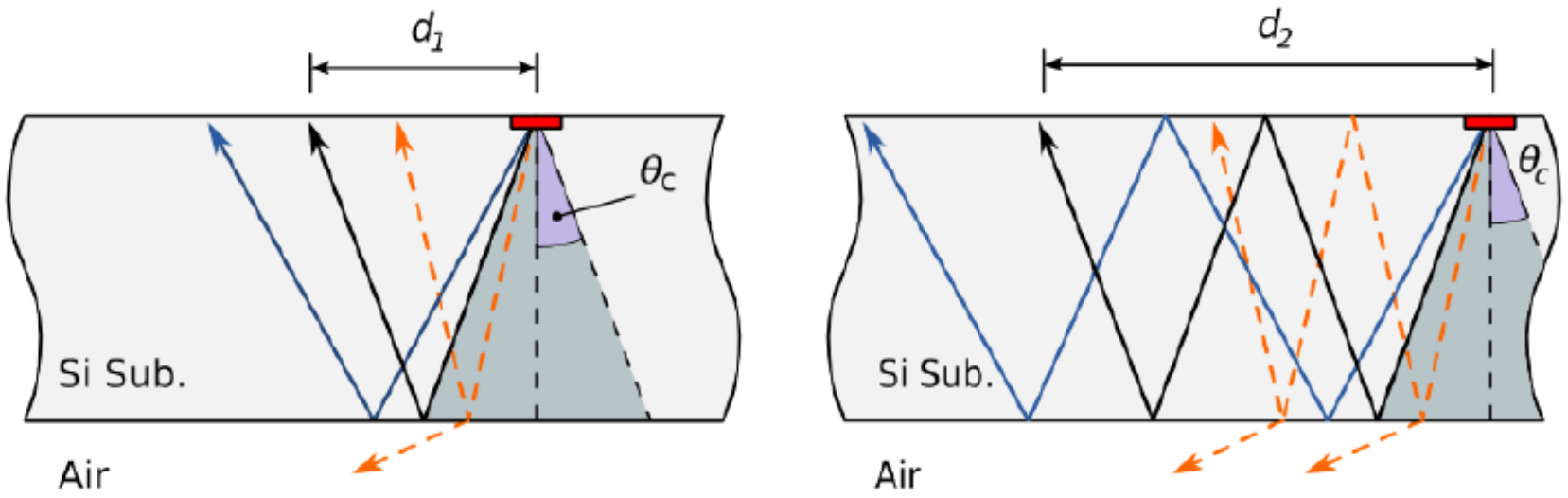
Notwithstanding the small fraction of re-injected photons we recorded a crosstalk increase of 10–20%

**Optical paths reflected off the bottom give a dominant contribution**

# Optical cross-talk: reflections off the bottom



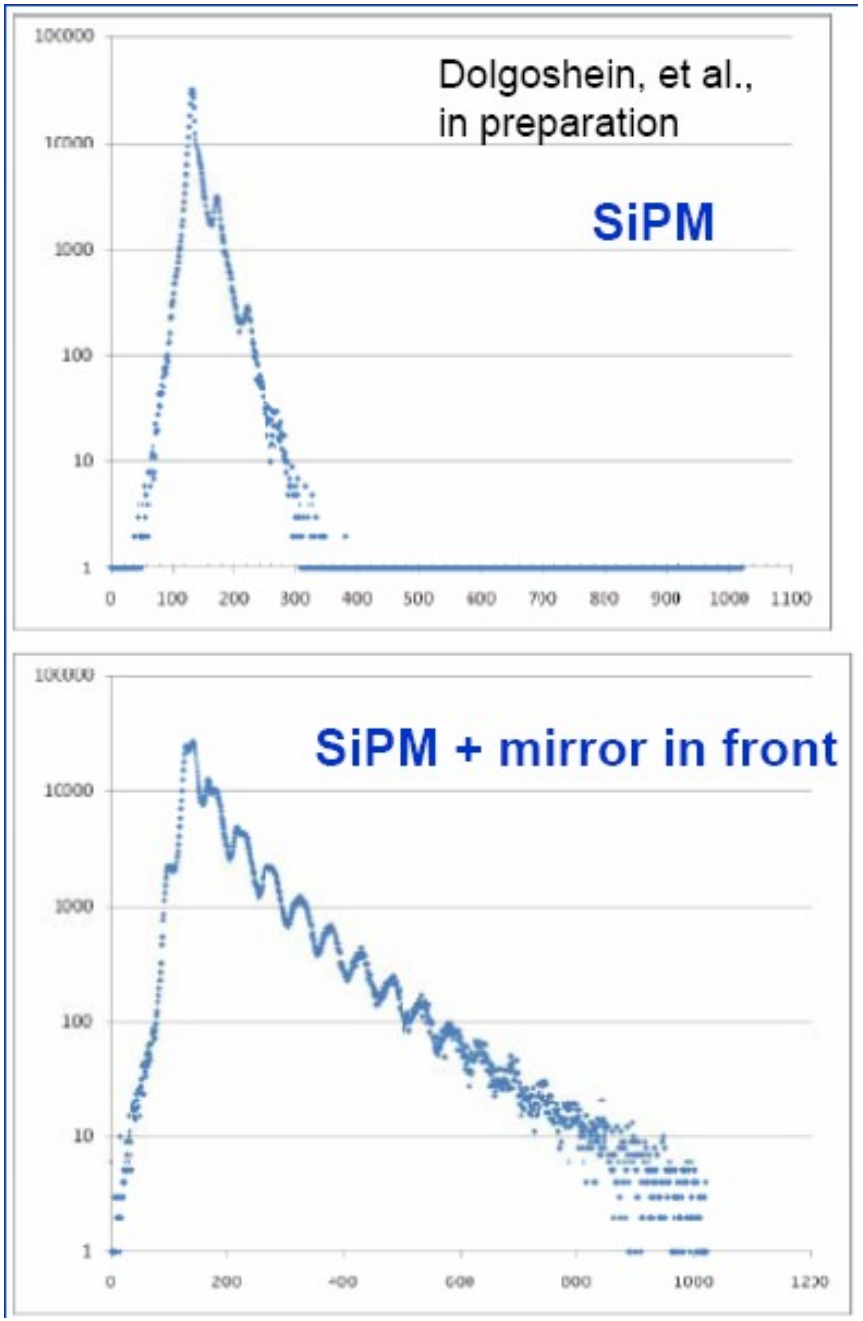
*Due to total internal reflection ...*




- Single reflection → peak at distance $d_1$
- Double reflection → peak at distance $d_2$

→ ***We observe a non monotonous dependence !***

# External Cross-talk: SiPM in front of a mirror




R.Myrzoyan
NDIP08

The emitted light bounces back inducing a kind of 'external' cross-talk effect

Amplitude resolution could be degraded when SiPMs are coupled to scintillators

# SiPM physics and characterization

related to the recharge of the diode capacitance from $V_{BD}$ to $V_{bias}$ during the avalanche quenching time after $I_{latch}$ is reached.

$G=(V_{BIAS}-V_{BD}) * C_D/q$

valid for few volts above $V_{BD}$

Gain

pulses triggered by non-photogenerated carriers (thermal / tunneling generation in the bulk or in the surface depleted region around the junction)

carriers can be trapped during an avalanche and then released triggering another avalanche

Noise: dark count
afterpulse
optical cross-talk

photoegeneration during the avalanche discharge. Some of the photons can be absorbed in the adjacent cell possibly triggering new discharges

**$PDE = QE * P_{01} * \varepsilon$**

QE = quantum efficiency

$P_{01}$ = avalanche triggering prob.

$\varepsilon$ = geometrical fill factor

Photo-detection efficiency

Related to the density of cells and recovery time

Dynamic Range

Related to the photogeneration and to the avalanche propagation

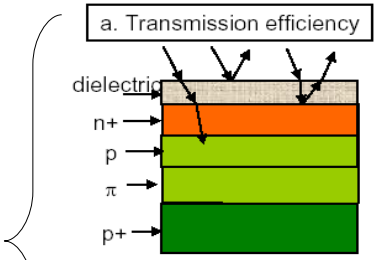
Time resolution

# Photo-detection efficiency (PDE)

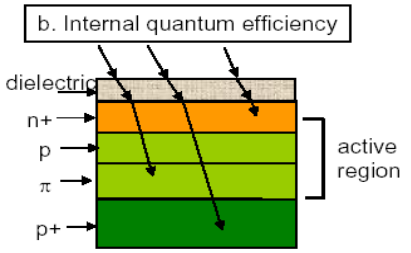
$$PDE = N_{pulses} / N_{photons} = QE \bullet P_{01} \bullet \varepsilon_{geom}$$

## Carrier Photo-generation

(QE = probability for a photon to generate a carrier that reaches the high field region)




"External QE"




"Internal QE"

*

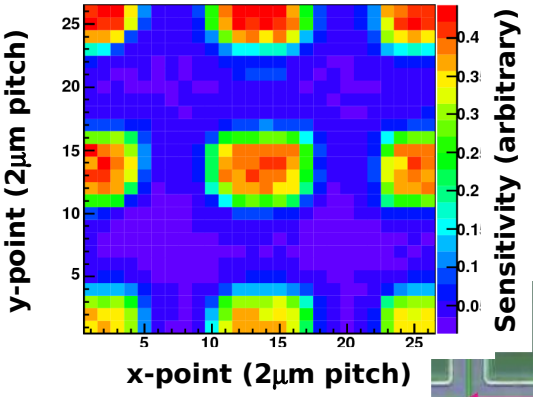
## Avalanche triggering

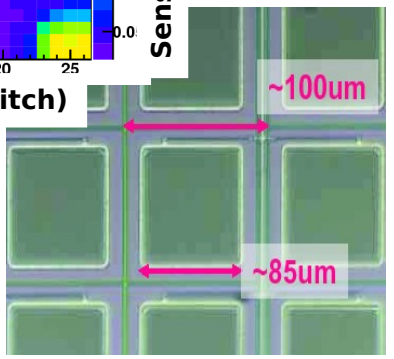
($P_{01}$ = probability for a carrier traversing the high-field to generate the avalanche)

*

## Geometrical fill factor

(ε = fraction of dead area due to structures between the cells, eg. guard rings, trenches)



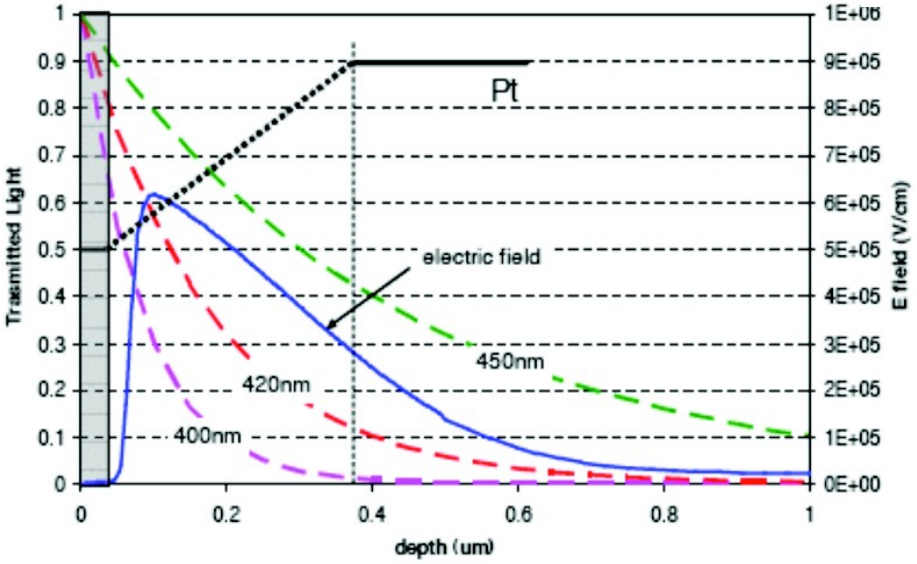

Hamamatsu SiPM close up

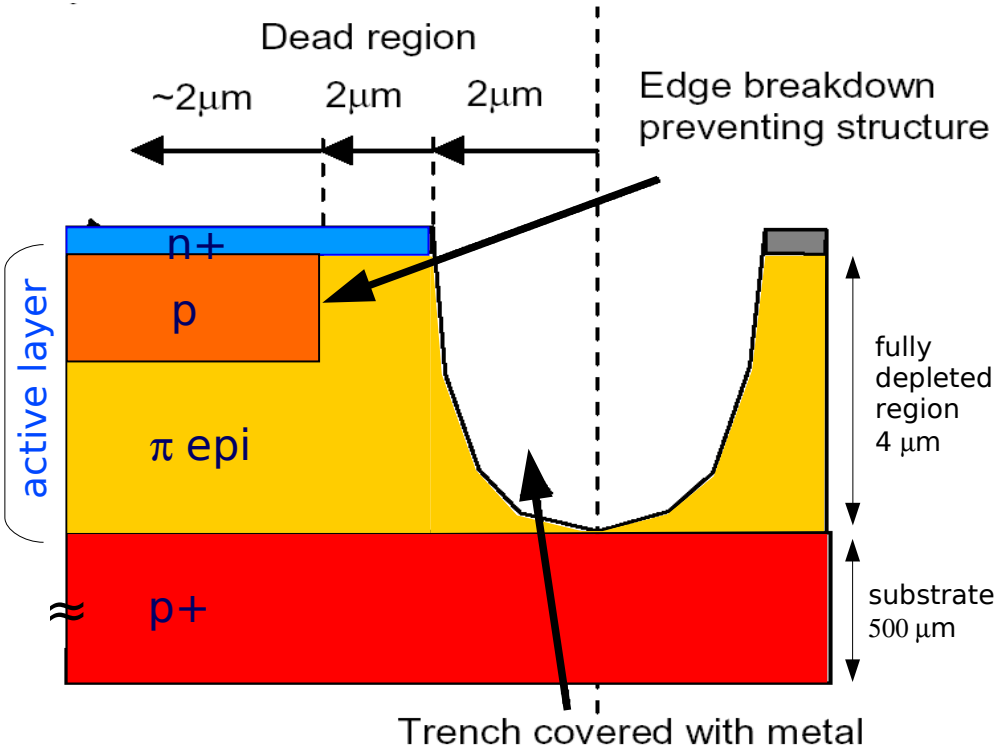
W.Oldham et al. IEEE TED (1972)

# QE: Efficiency of a single cell

Two factors in QE:
(1) transmittance of the entrance window (dielctric on top of silicon surface)
(2) probability of a photon inside to generate a e-h pair in the active layer (internal quantum efficiency)

Only the depleted region is fully active to efficiently photo-generate because of high recombination probability in the un-depleted regions.
Only a small layer ($\lambda_{diffsion} \sim \sqrt{D\tau_{recomb}}$) at the edge of un-depleted regions contributes to the photo-generation (critical for UV light)



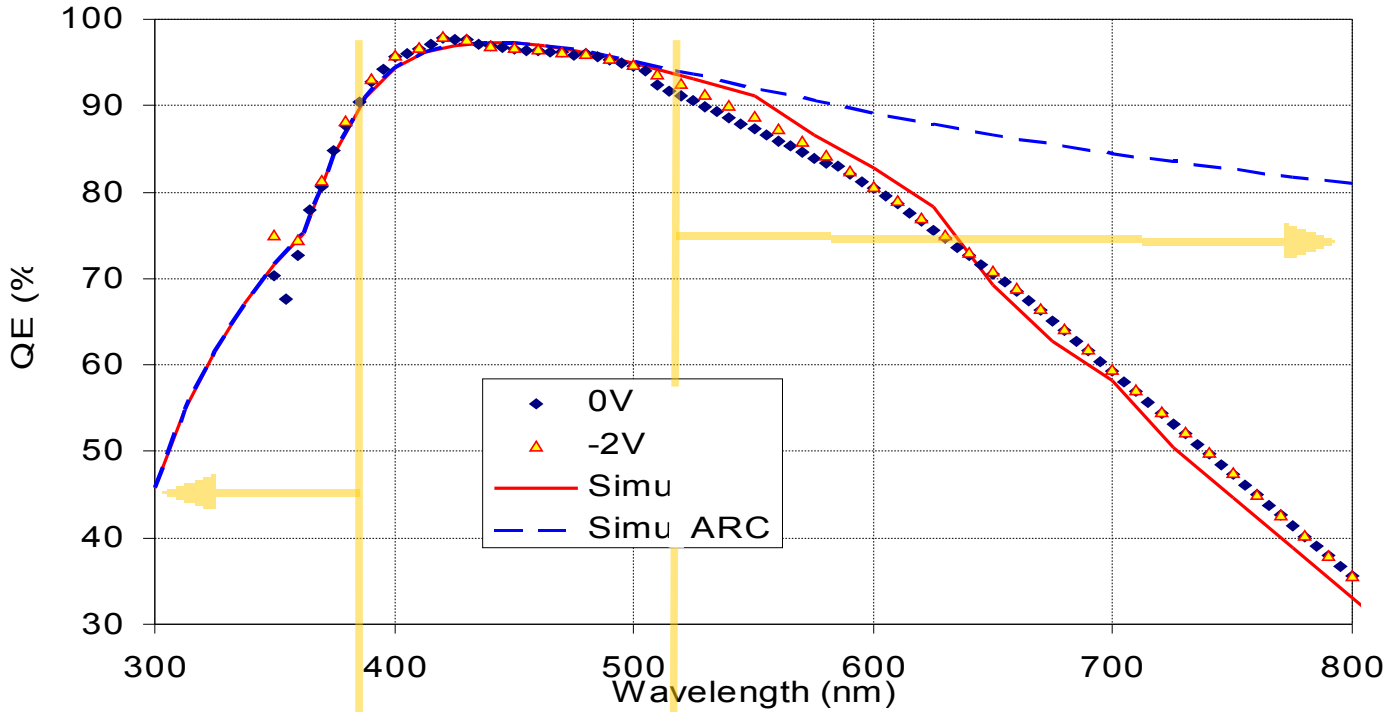




## QE optimization

- Anti-reflective coating (ARC)
- Shallow junctions for short λ
- Thick epi layers for long λ

# QE: Efficiency of a single cell

Direct access to internal QE and transmittance through ARC
by measuring photo-voltaic regime ($V_{bias}$~ 0 V)
the photon detection efficiency of a
diode with the same $n^+/p$ junction structure and same ARC

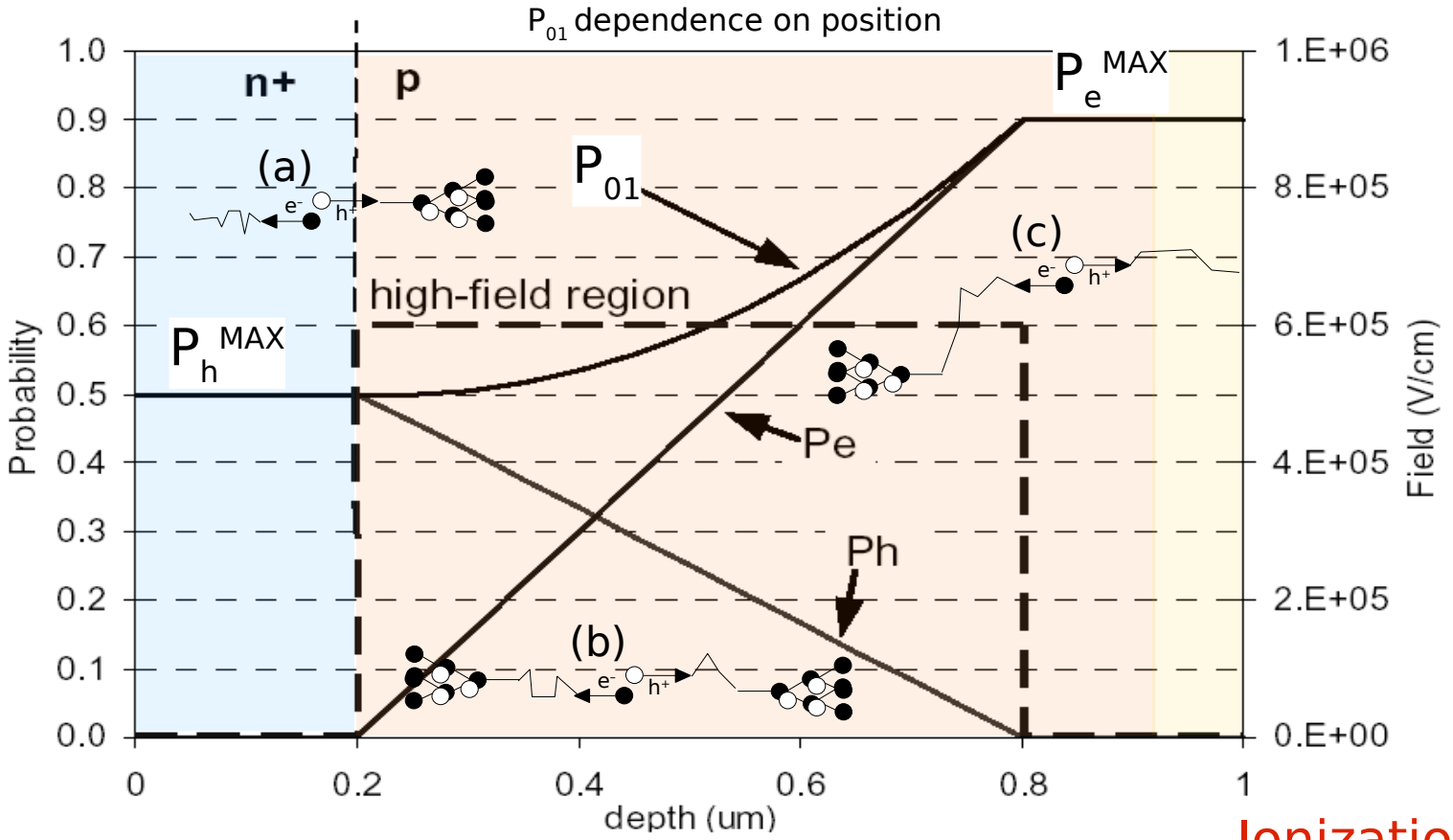



Reduced by
ARC Transmittance

Reduced by the
small π layer thickness

IRST devices

# Avalanche trigger probability ($P_{01}$)



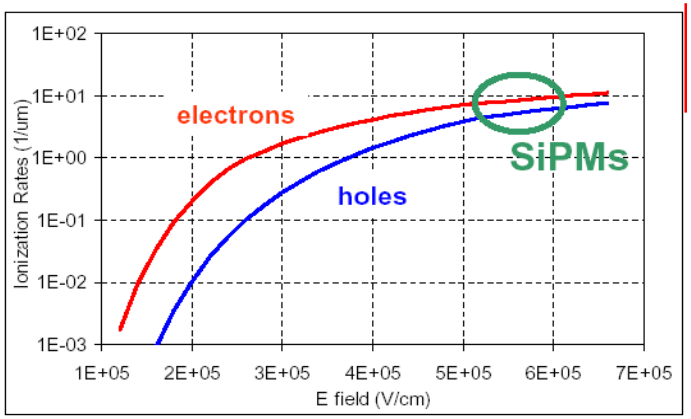

C.Piemonte
NIM A 568
(2006) 224

Example with constant high-field:
(a) only holes may trigger the avalanche
(b) both electrons and holes may trigger
(but in afraction of the high-field region)
(c) only electrons may trigger

$P_{01}$ optimization

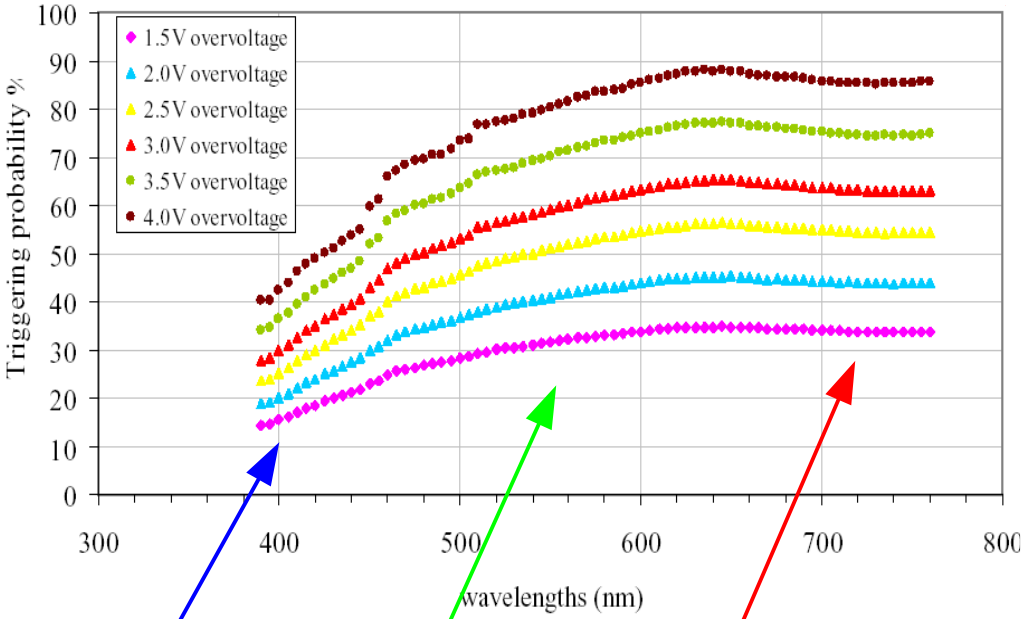
- high over-voltage
- photo-generation in the p-side of the junction

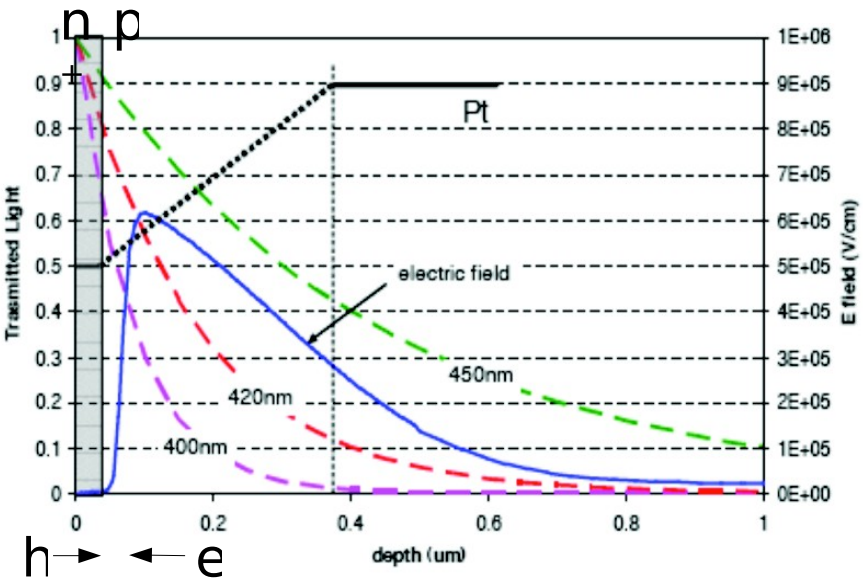
Ionization rate in Silicon



# Avalanche trigger probability ($P_{01}$)

$P_{01}$ = PDE / QE / $\varepsilon_{geom.}$



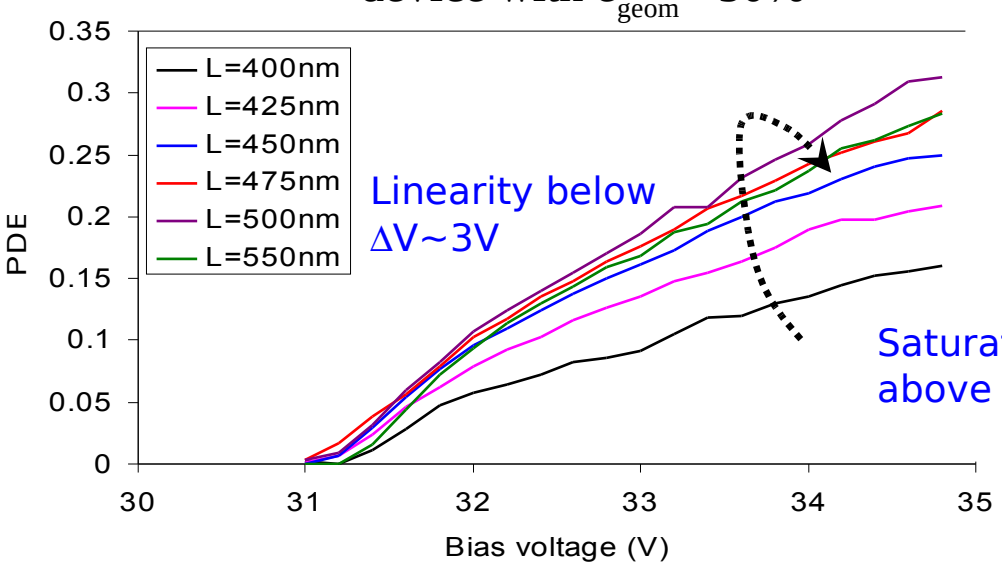

Only $e^-$ cross the high E field region and trigger the avalanche

Both $h^+$ and $e^-$ might trigger the avalanche (but cross only a fraction of high field region)

Only $h^+$ cross the high E field trigger the avalanche

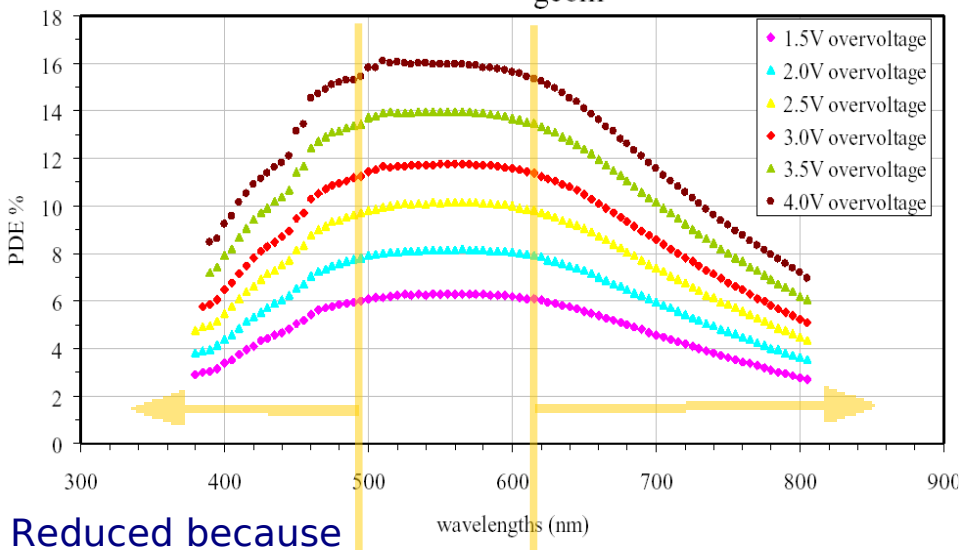
# IRST devices – PDE vs ΔV and λ

device with $e_{geom}$ ~50%




PDE dependence on ΔV (at different λ)

device with $\varepsilon_{geom}$ ~ 22%

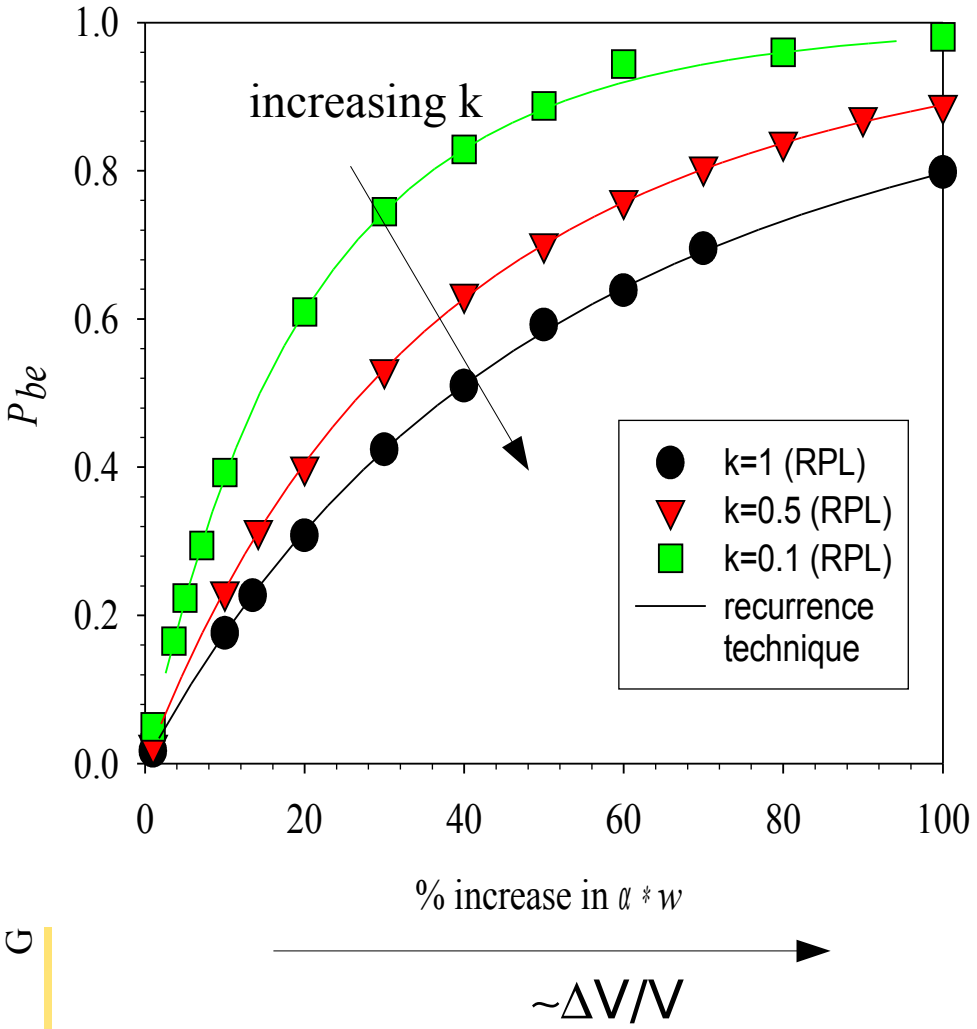



PDE dependence on λ (at different DV)

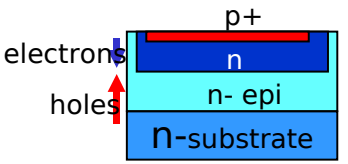
# PDE vs ΔV

"Statistics of Avalanche Current Buildup Time in Single-Photon Avalanche diodes"
C.H.Tan, J.S.Ng, G.J.Rees, J.P.R.David (Sheffield U.)
IEEE J.Quantum Electronics 13 (4) (2007) 906

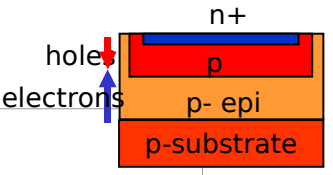
RPL model Courtesy of C.H.Tan

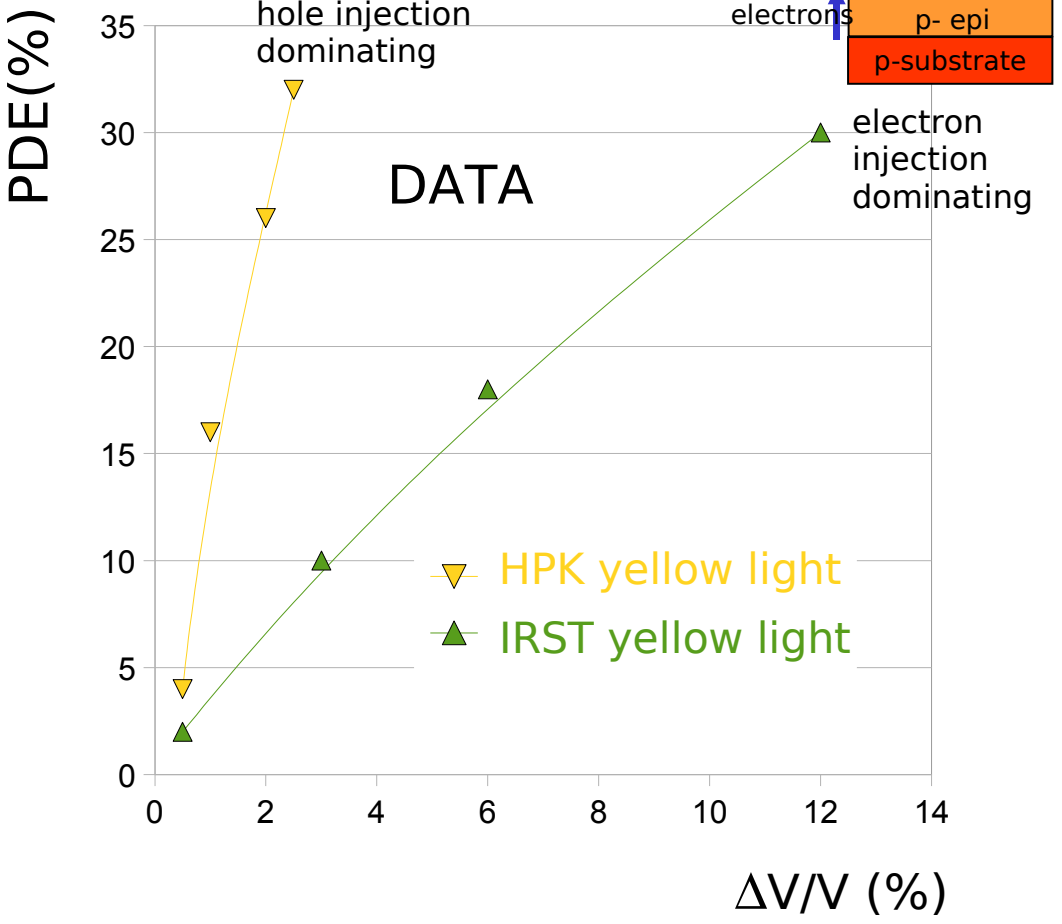



p-on-n structure



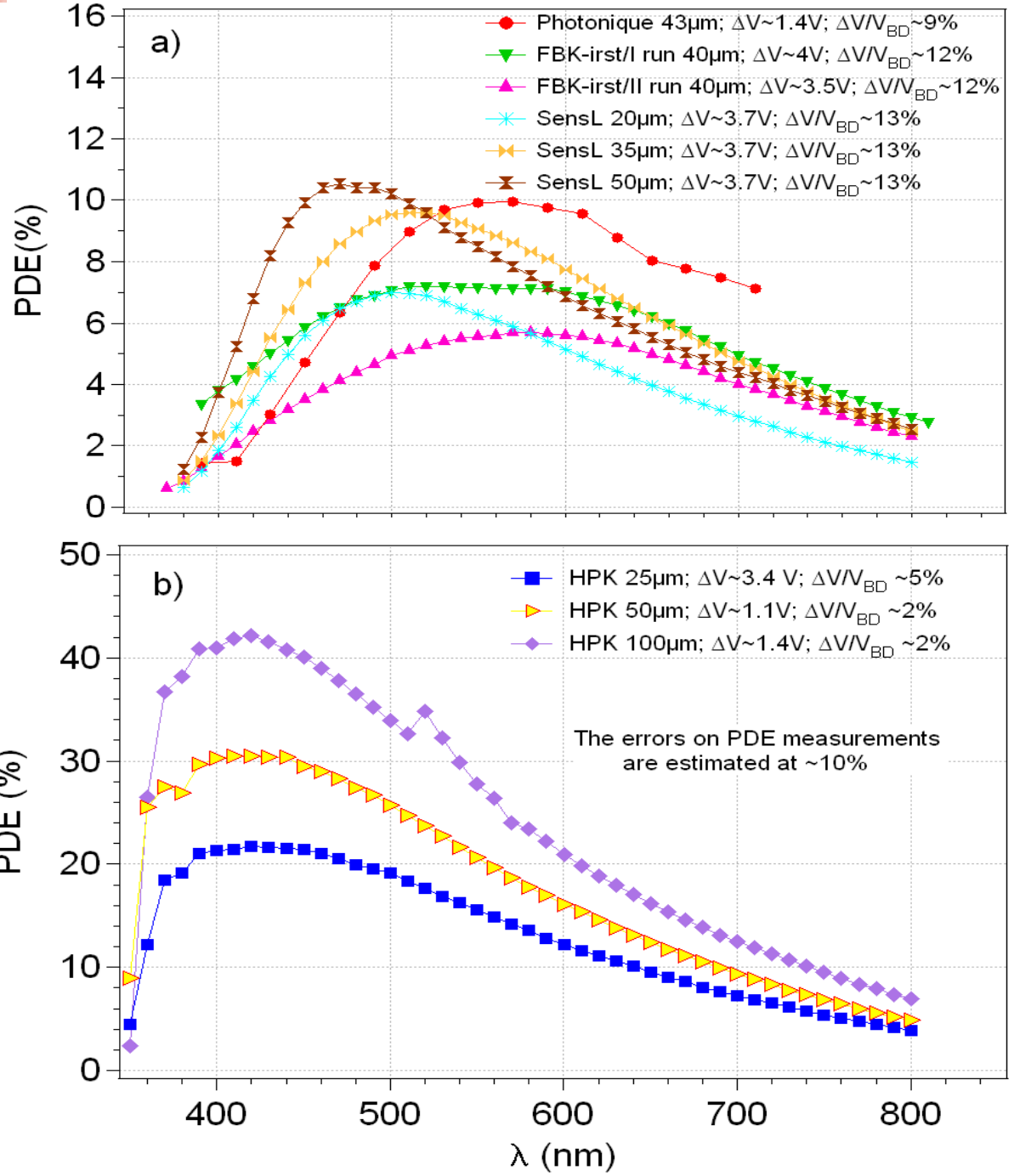

n-on-p structure



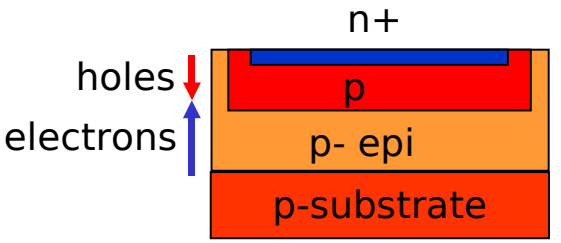



# PDE VS λ

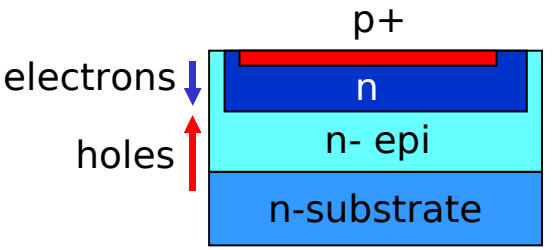
N.Dinu et al. NIM A (2008)
to be published




## n-on-p structures




## p-on-n structure



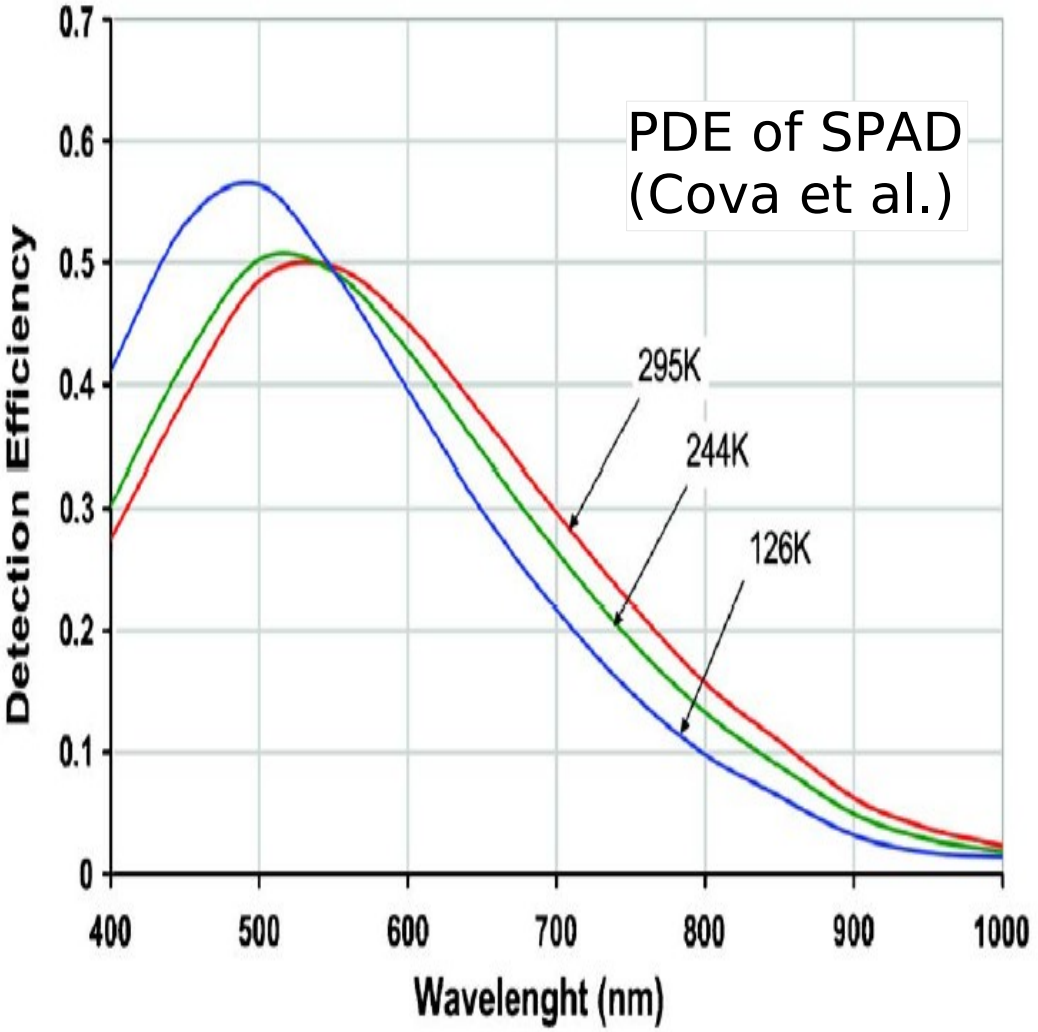

Note:
1) geometrical fill factor included
2) PDE in HPK catalog is 20%-30% more than the real PDE due to after-pulses and cross-talk

# T dependence: PDE (SPAD devices)

## PDE dependence on T
(Over-voltage fixed)

Combination of two effects:
- $P_{01}$ increases at low T
  because of increased impact ioniz.
- Energy gap increases at low T

Similar effect expected also for SiPM




I.Rech el al, Rev.Sci.Instr. 78 (2007)

# SiPM physics and characterization

related to the recharge of the diode capacitance from $V_{BD}$ to $V_{bias}$ during the avalanche quenching time after $I_{latch}$ is reached.

$G=(V_{BIAS}-V_{BD}) * C_D/q$

valid for few volts above $V_{BD}$

Gain

pulses triggered by non-photogenerated carriers (thermal / tunneling generation in the bulk or in the surface depleted region around the junction)

Noise: dark count
afterpulse
optical cross-talk

carriers can be trapped during an avalanche and then released triggering another avalanche

photoegeneration during the avalanche discharge. Some of the photons can be absorbed in the adjacent cell possibly triggering new discharges

**$PDE = QE * P_{01} * \varepsilon$**

QE = quantum efficiency
$P_{01}$ = avalanche triggering prob.
$\varepsilon$ = geometrical fill factor

Photo-detection efficiency

Related to the density of cells and recovery time

Dynamic Range

Related to the photogeneration and to the avalanche propagation

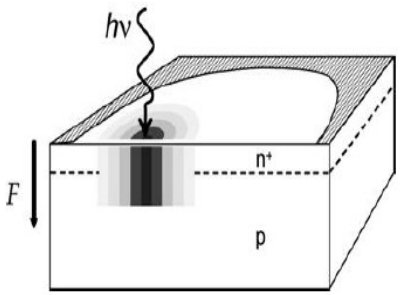
Time resolution

# GM-APD timing: fast and slow components

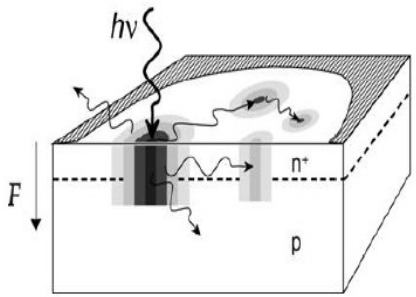
## 1) Fast component: gaussian with time scale O(10ps)

Statistical fluctuations in the avalanche:

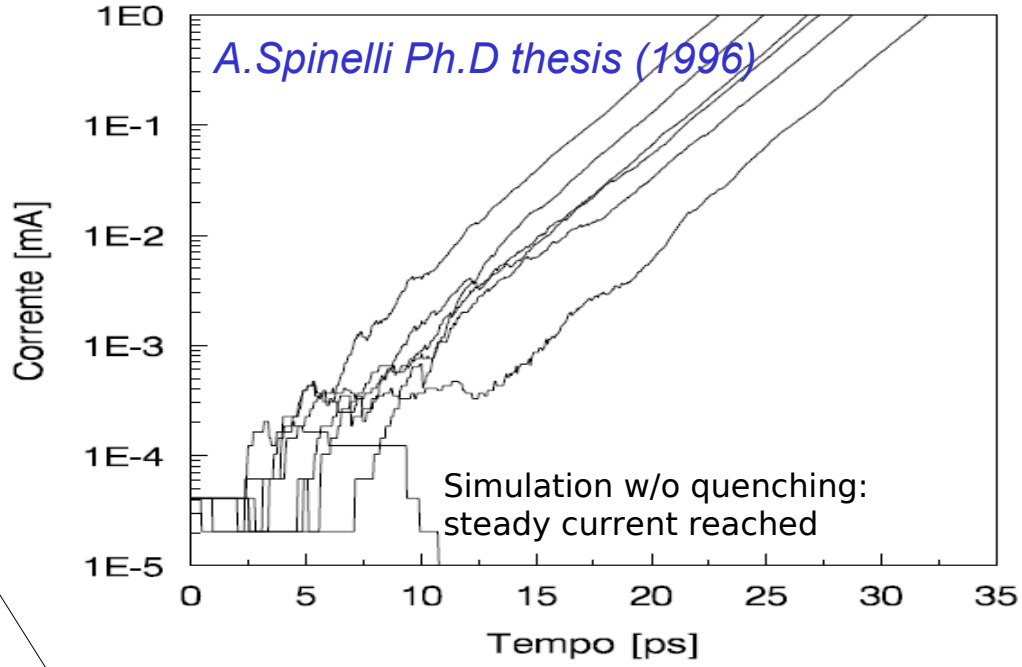
- Longitudinal build-up (minor contribution)
- Transversal propagation (main contribution):
  - via Multiplication assisted diffusion (dominating)
    A.Lacaita et al. APL and El.Lett. 1990
  - via Photon assisted propagation
    PP.Webb, R.J. McIntyre RCA Eng. 1982
    A.Lacaita et al. APL 1992

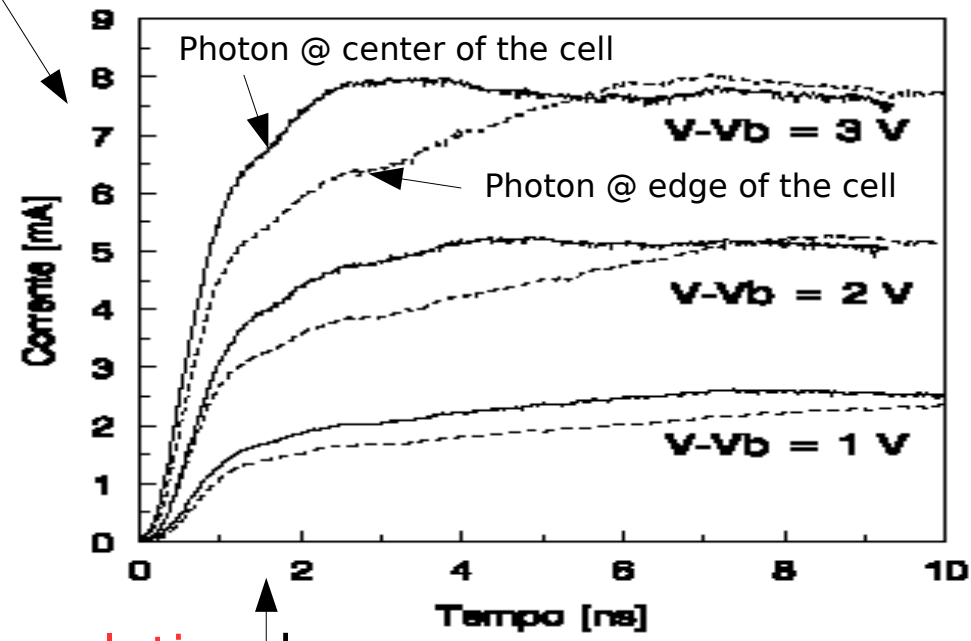


Multiplication assisted diffusion



Photon assisted propagation




Dependence of avalanche build-up rate on the impact position (→ cell size)

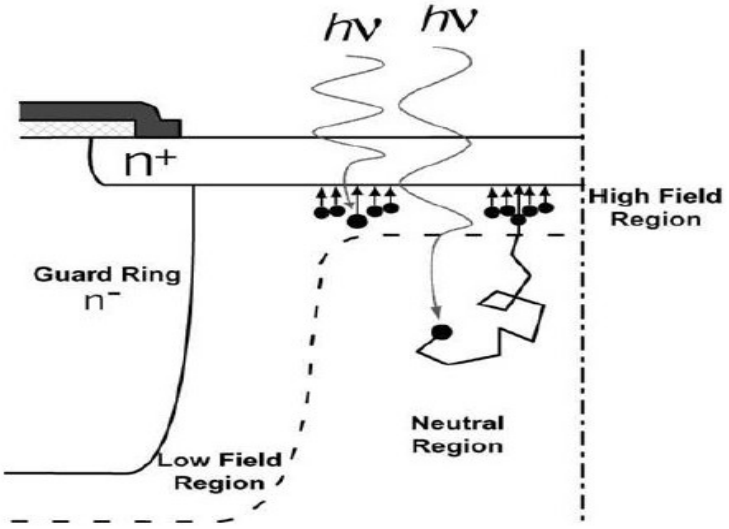
Higher overvoltage → improved time resolution

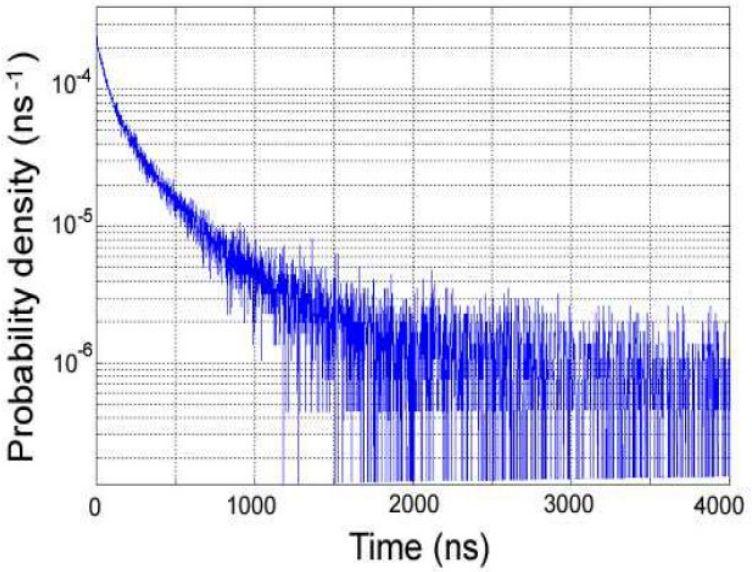
# GM-APD timing: fast and slow components

## 2) Slow component: minor non gaussian tails with time scale O(ns)

Carriers photogenerated in the neutral regions beneath the junction and reaching the electric field region by diffusion

G.Ripamonti, S.Cova Sol.State Electronics (1985)




tail lifetime: $\tau \sim L^2 / \pi^2 D$

L = effective neutral layer thickness

D = diffusion coefficient

S.Cova et al. NIST Workshop on SPD (2003)

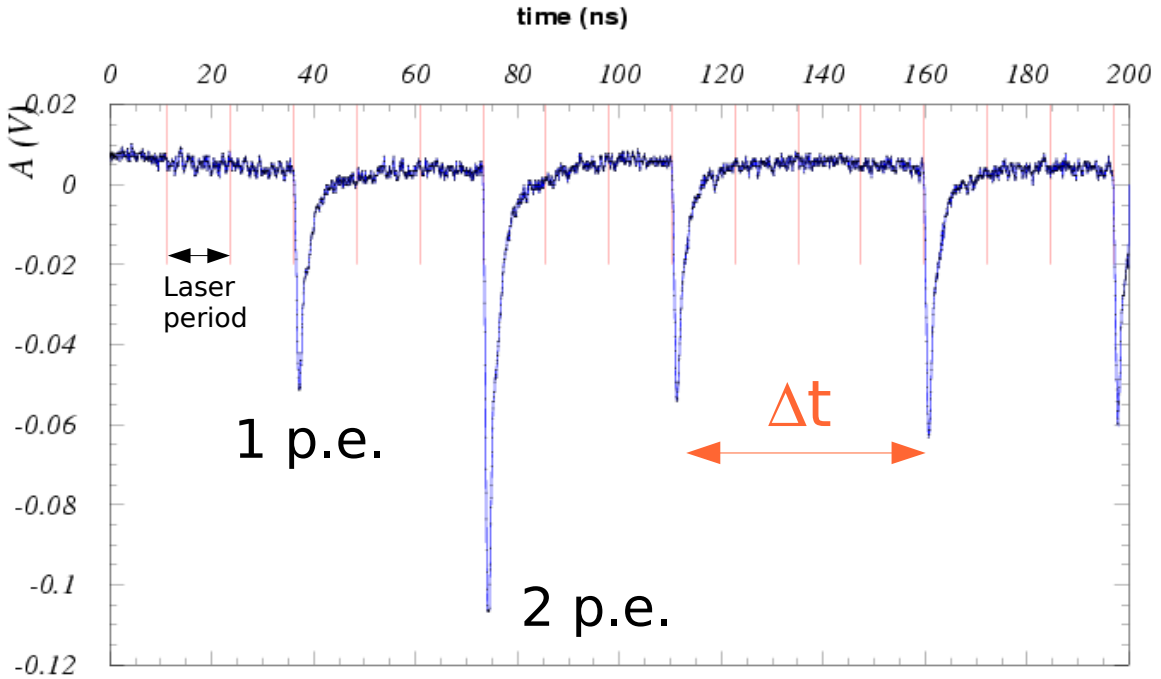
Shorter wavelengths → higher resolution (less tails)

# Waveform analysis: method

(1) Selection of candidate peaks:
- single photon peaks
- proper signal shape
- low instantaneous intensity (no activity before/after within 50ns)
- low noise during the previous 10 ns (typical noise ~ 1mV rms)

(2) Peak reconstruction
- optimum time reconstruction
- amplitude and width (baseline shift correction)

(3) Time difference Δt between consecutive peaks




NOTE: working fine at 20MHz counting rate

# Single Photon Timing Resolution (SPTR)

Analysis of the distributions of the t difference between successive peaks (modulo the laser period $T_{laser}$=12.367ns)

Gaussian + Tails (long λ)

rms~50-100 ps

~ exp (-t / O(ns))

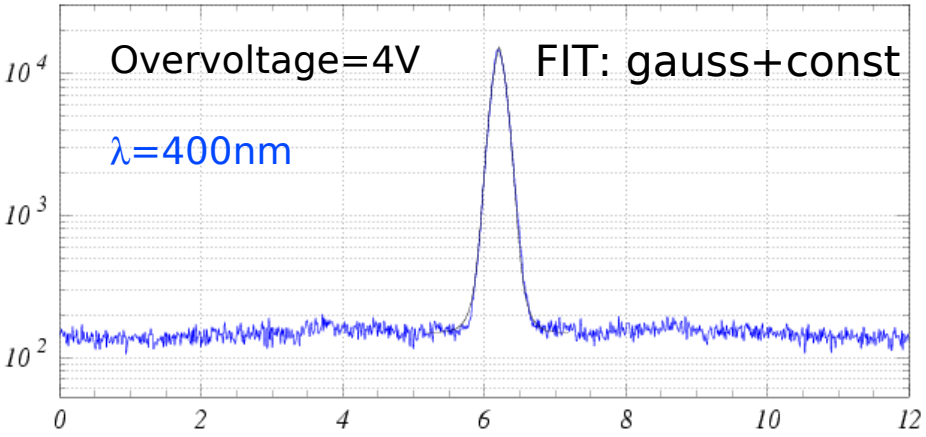
contrib. several % for long wavelengths

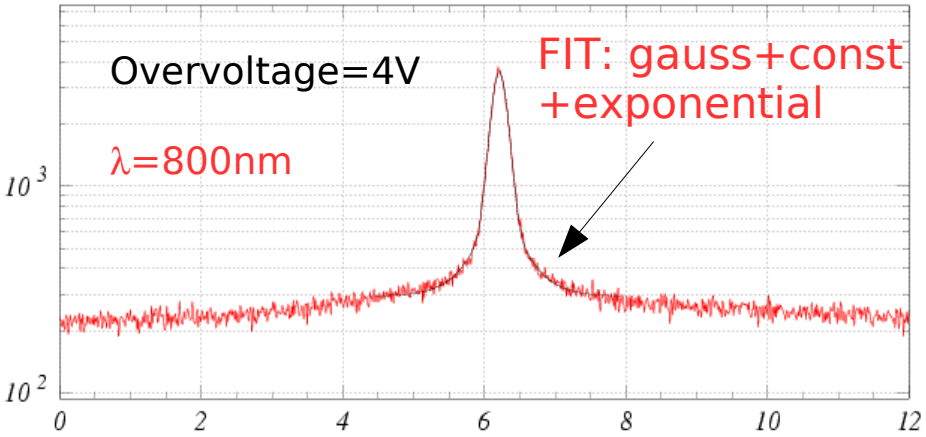
Data at λ=400nm
fit gives reasonable $\chi^2$
with gaussian ($\sigma_t^{fit}$) +
constant term (dark noise contribution)

The detector resolution is obtained by
$\sigma_t^{fit}/\sqrt{2}$

Data at λ=800nm
fit gives reasonable $\chi^2$ with an additional exponential term exp(-Δt/τ)

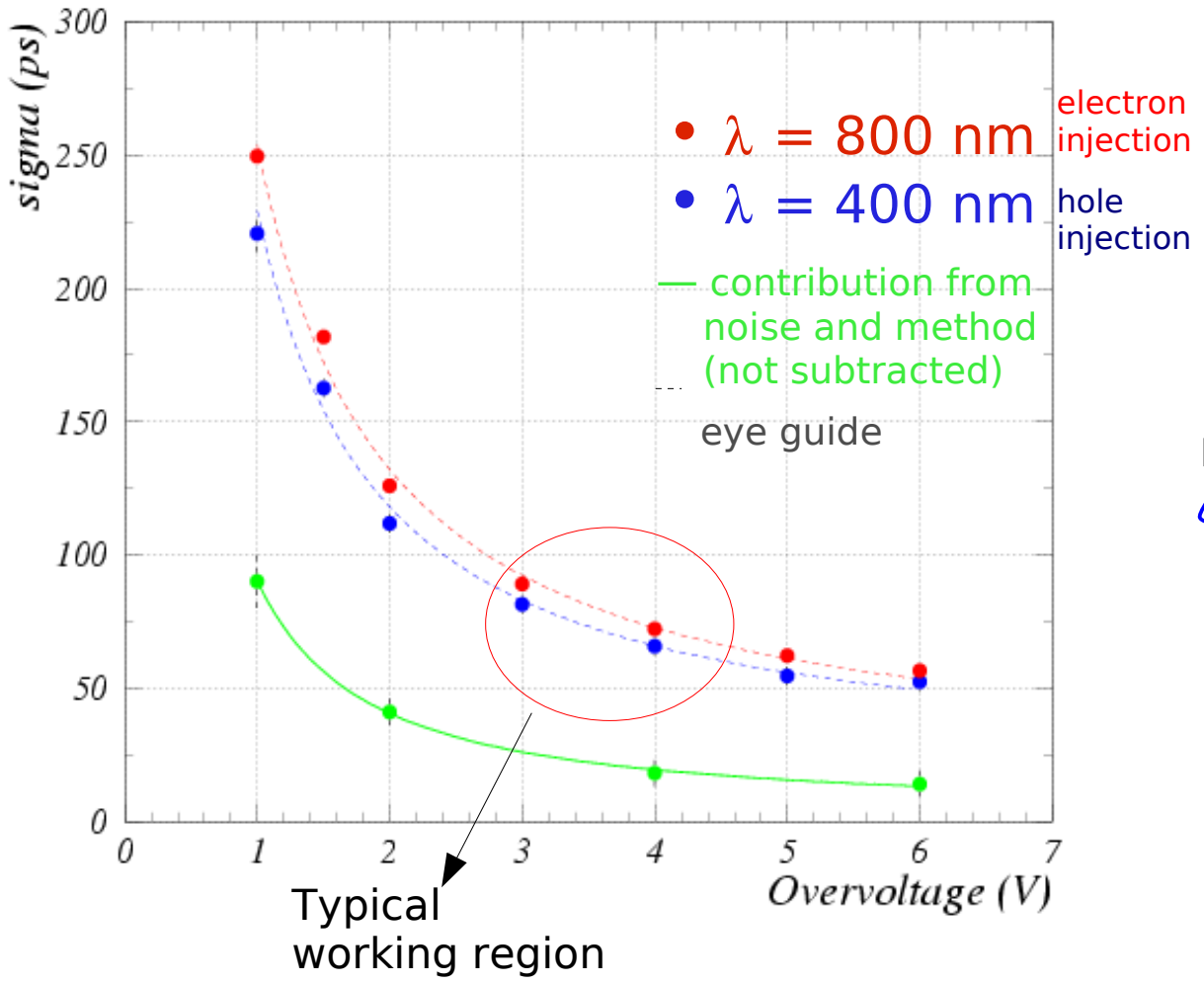
- τ ~ 0.2÷0.8ns in rough agreement with diffusion tail lifetime: $\tau \sim L^2 / \pi^2 D$ if L is taken to be the diffusion length
- Contribution from the tails ~ 10÷30% of the resolution function area



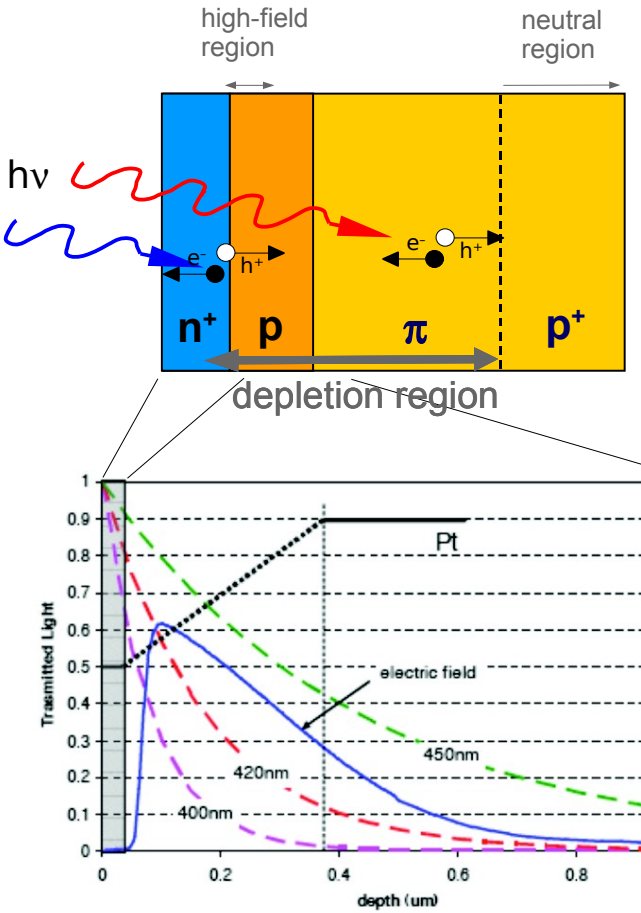




Distributions of the difference in time between successive peaks (modulo the measured laser period $T_{laser}$=12.367ns)

# IRST – single photon timing res. (SPTR)




Better resolution for short wavelengths: carriers generated next to the high E field region

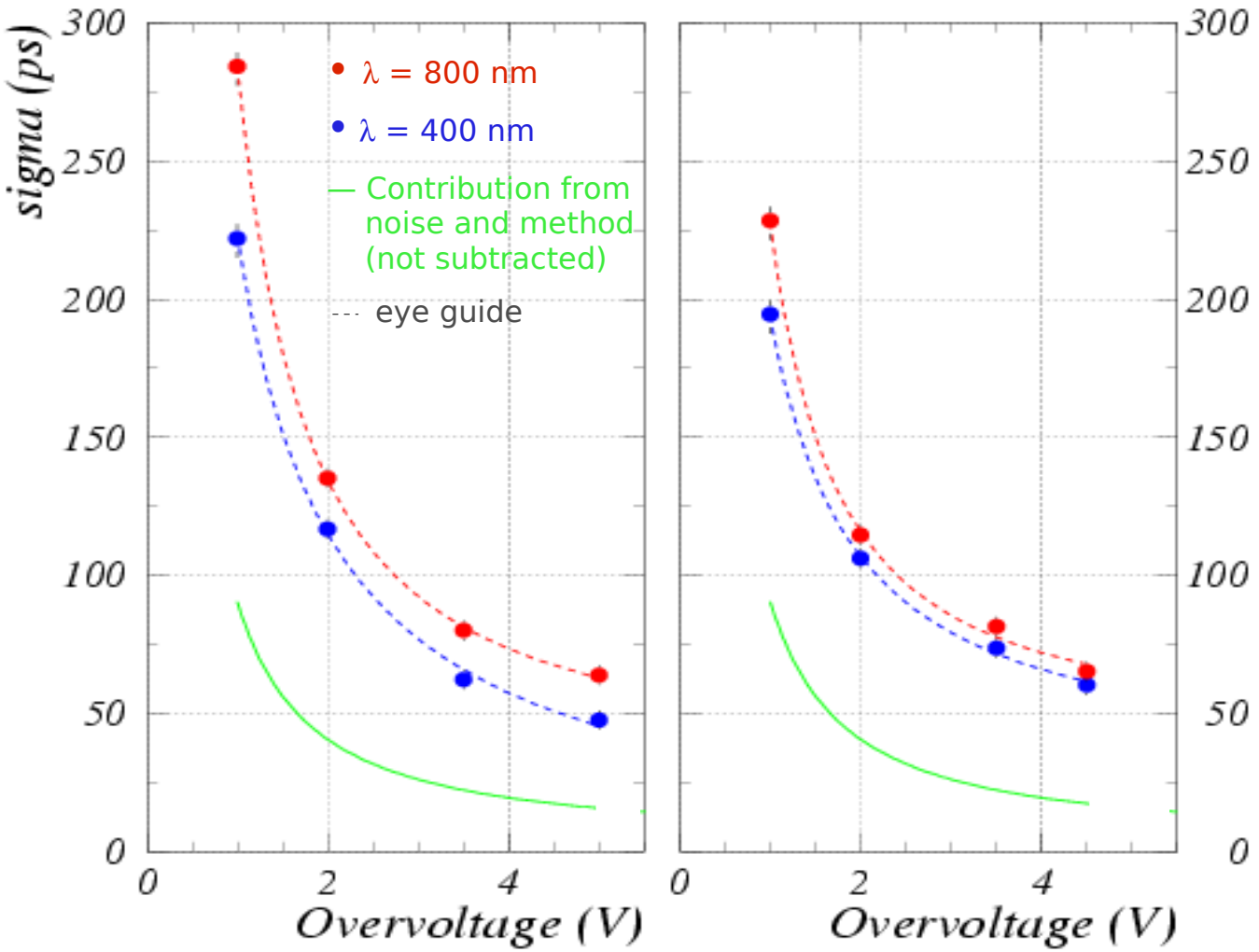


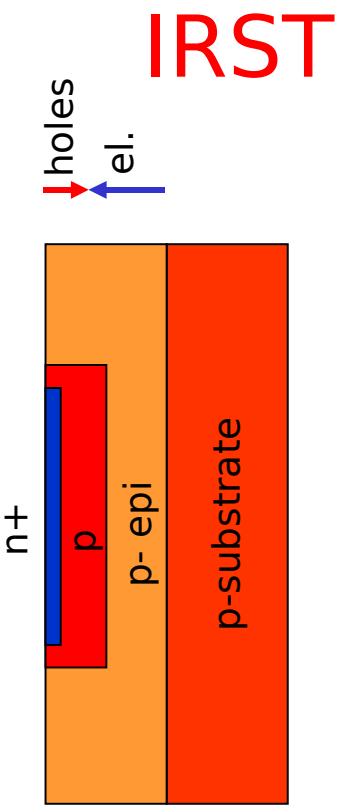

G.Collazuol et al NIMA 581 (2007) 461

# IRST devices (different types)

SiPM type without optical trench

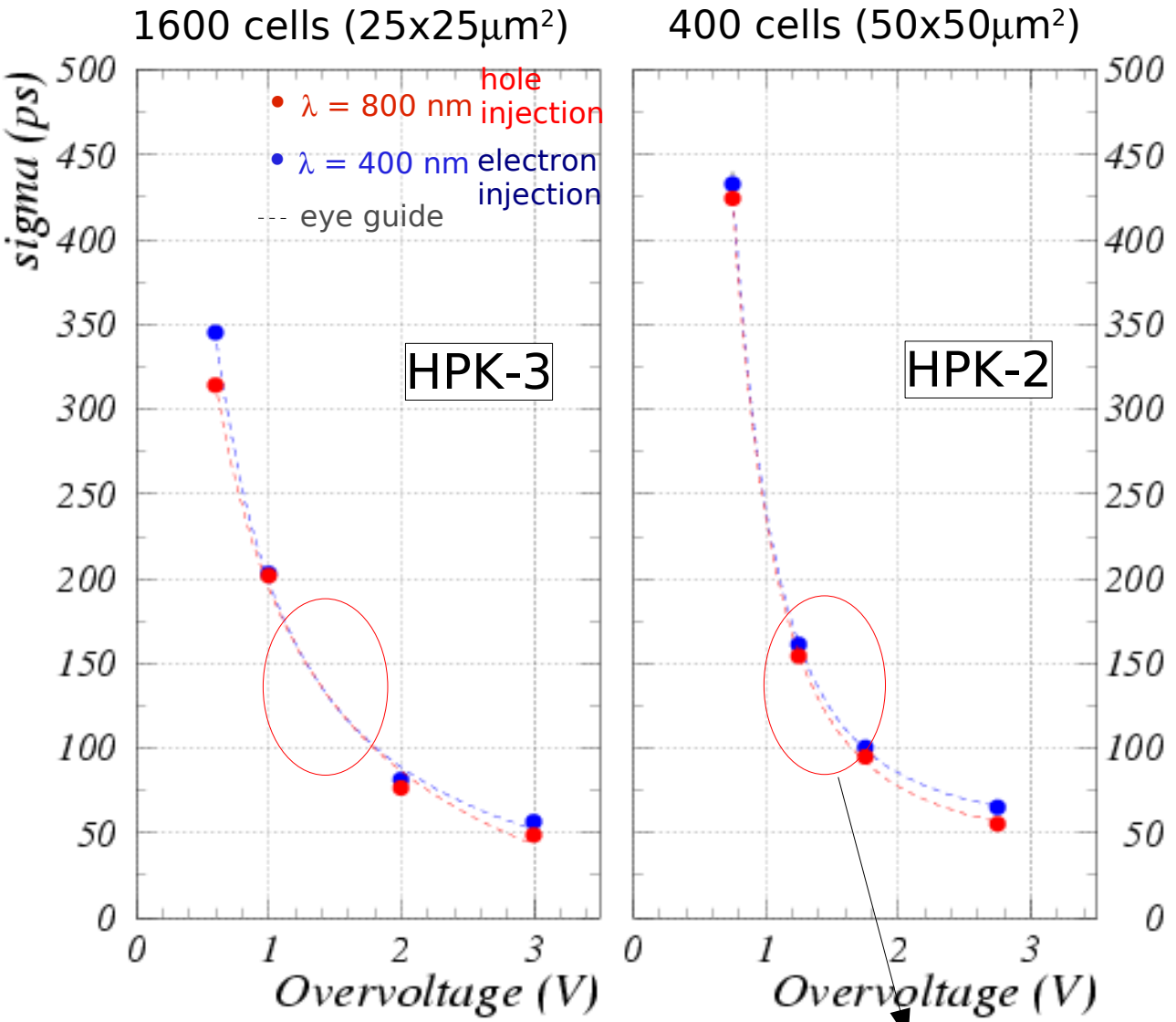
SiPM type with optical trench




Results in fair agreement for devices with the same structure

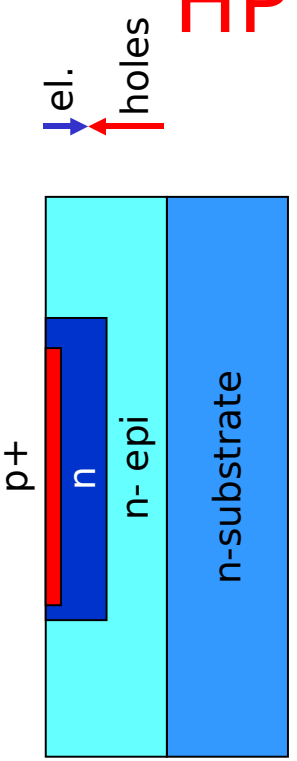
# Hamamatsu – single photon timing res.




G.Collazuol et al (unpublished)

Suggested Operating range

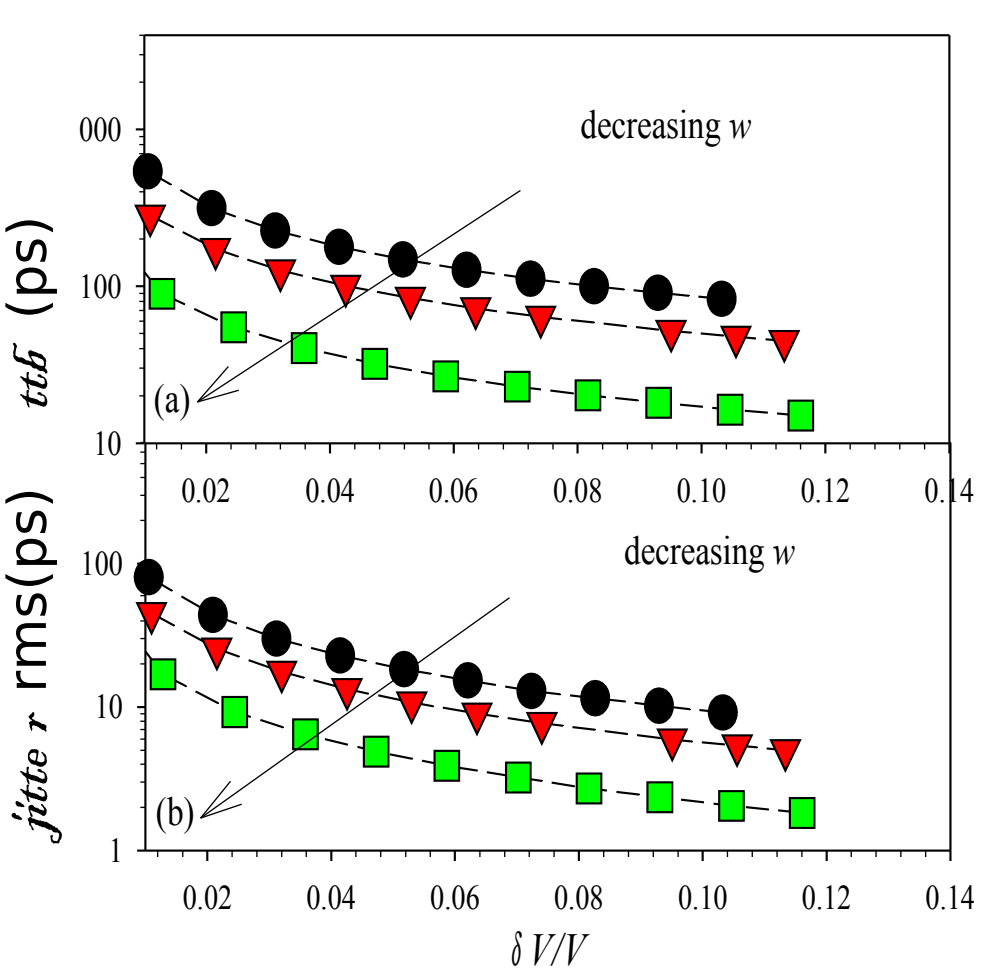
HPK



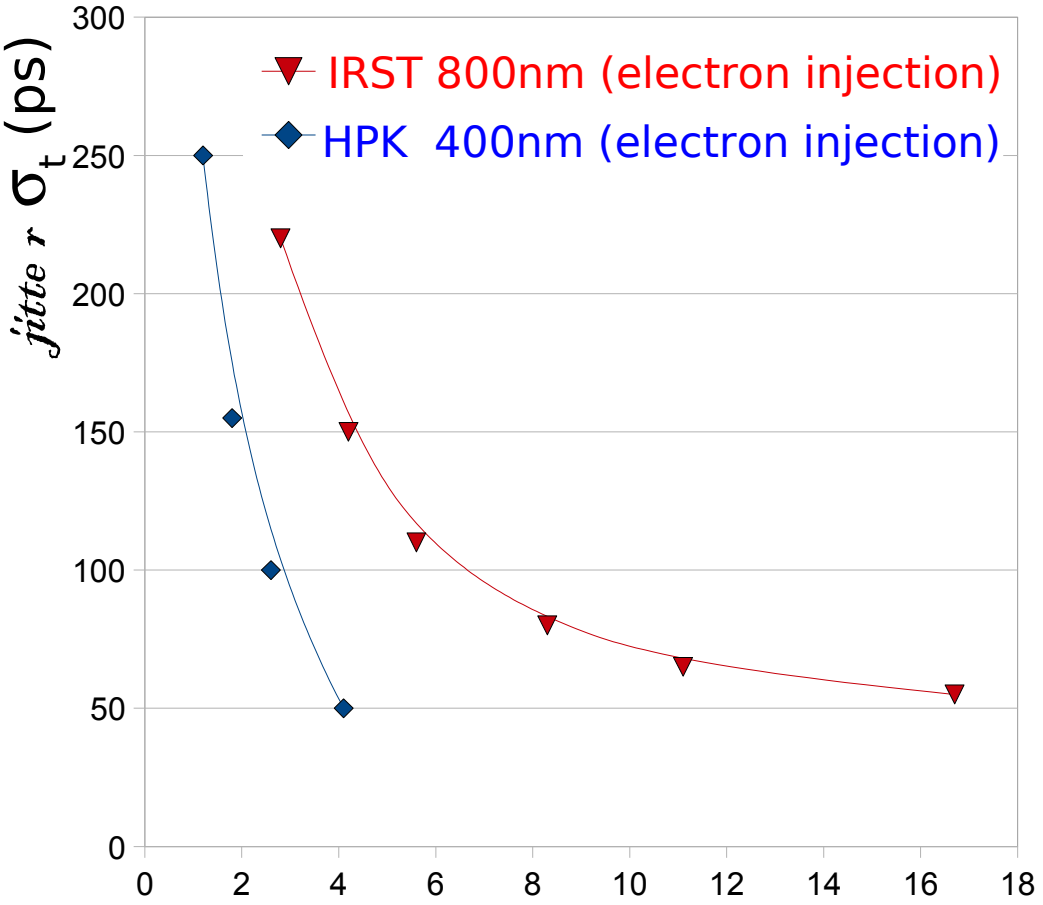
# RPL model vs data: comparison ... not yet

Example of RPL simulation of pure electron injection in Si SPAD

Courtesy of C.H.Tan




DATA
(DASIPM)

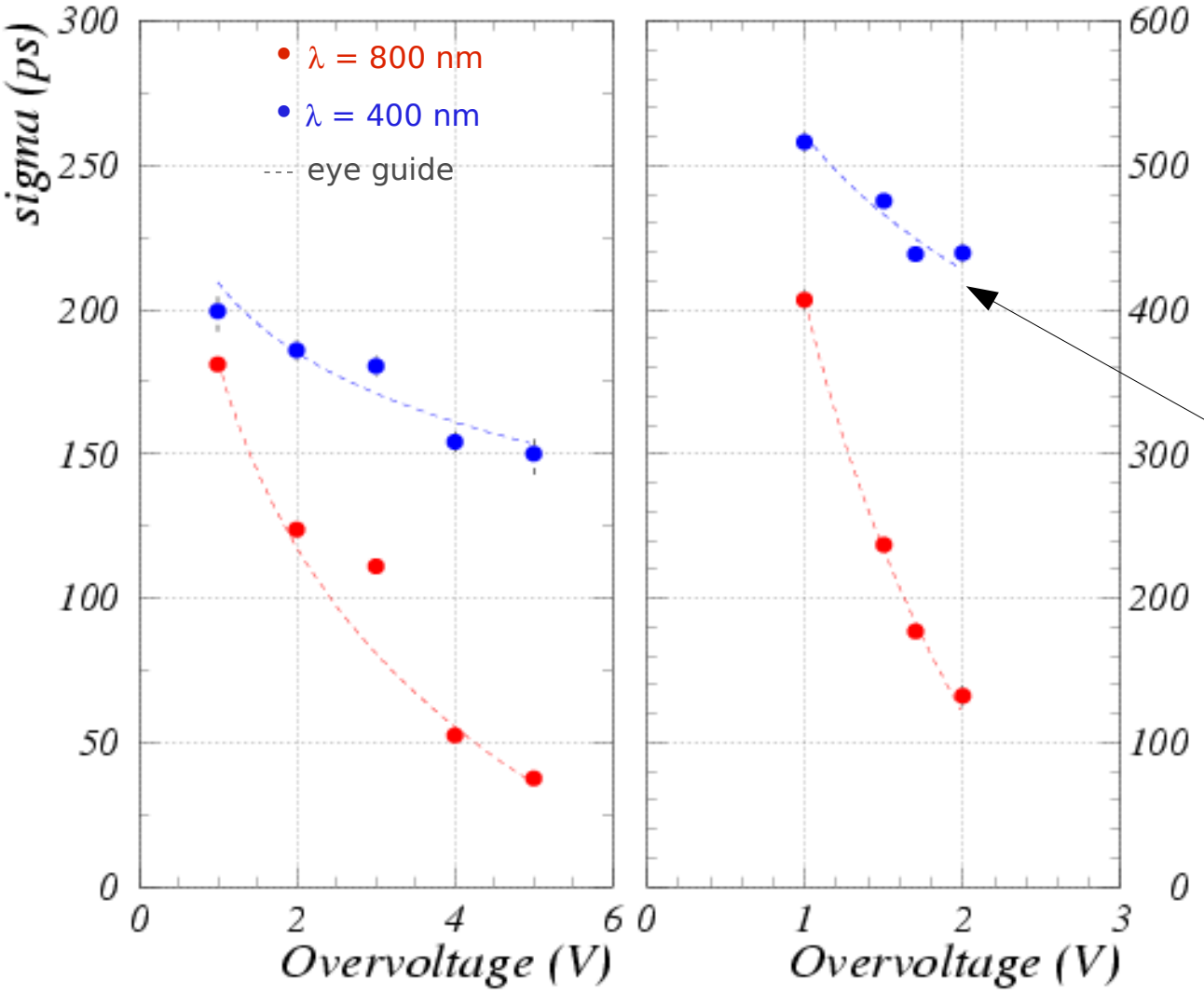



Reminder:
IRST structure is n on p
HPK structure is p on n

# CPTA/Photonique – single photon timing res.

a) Green-Red sensitive
SSPM 050701GR_TO18

b) Blue sensitive
SSPM 050901B_TO18




Two different structures:
a) thick $n^+/p$
b) $p^+/n$ deep junction

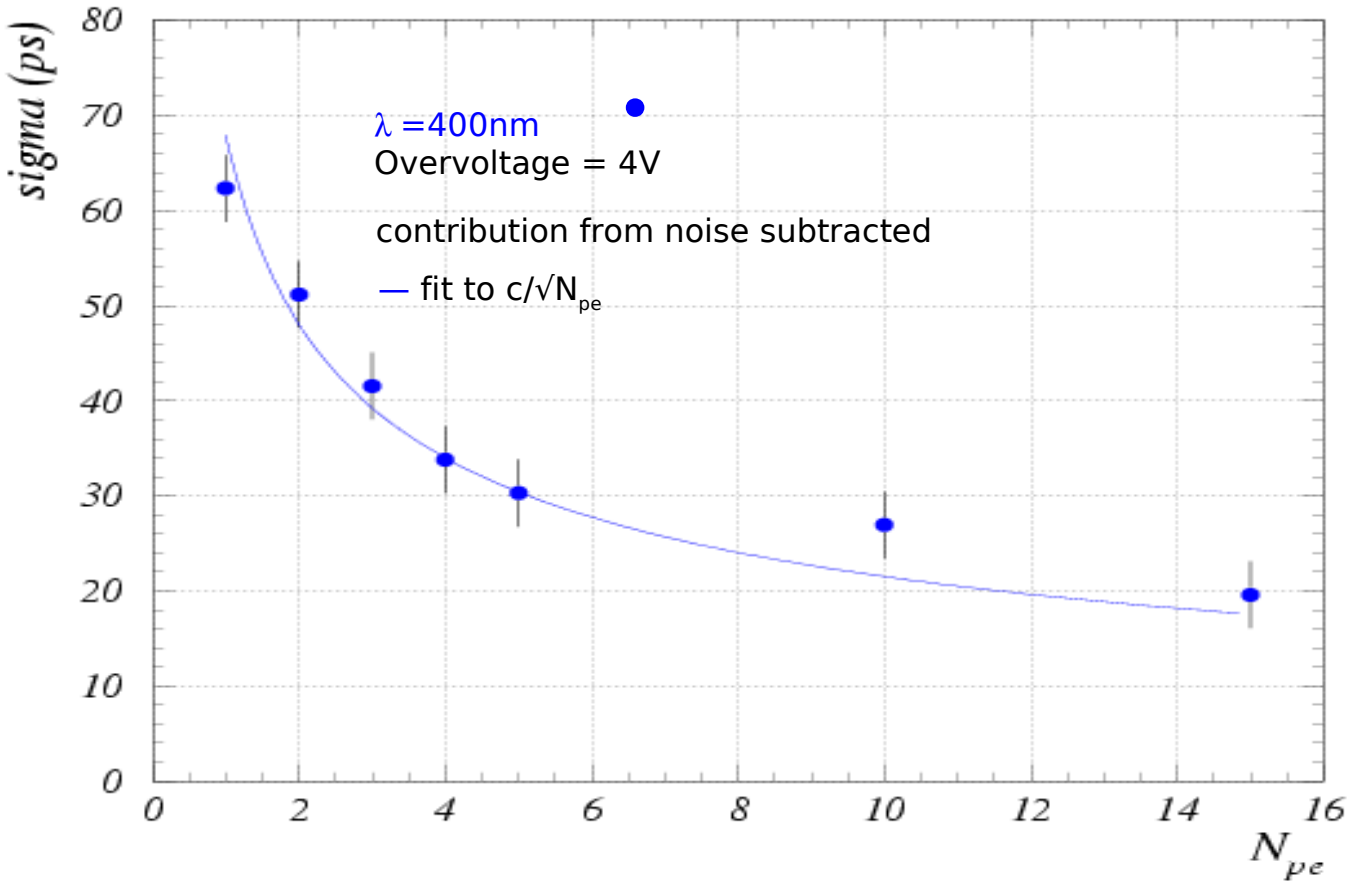
Carrier trapping ?

G.Collazuol et al (unpublished)

# Timing studies

Dependence of SiPM timing on the number of simultaneous photons

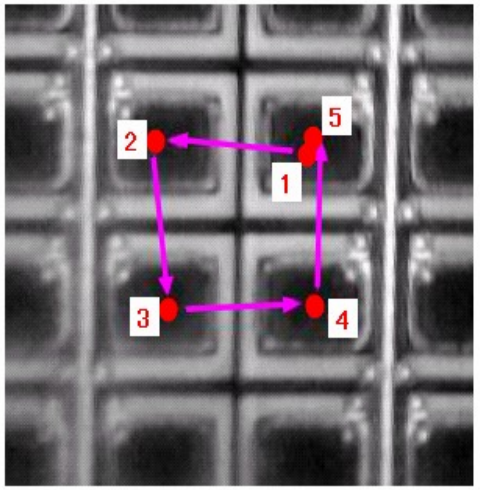
Poisson statistics: $\sigma_t \propto 1/\sqrt{N_{pe}}$



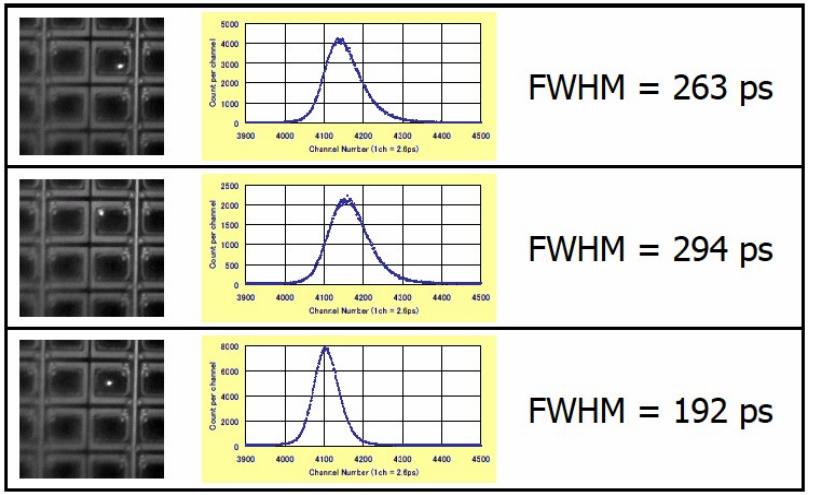

N of simultaneous photo-electrons

# SPTR(*): position dependence

Yamamoto et al (Hamamatsu)

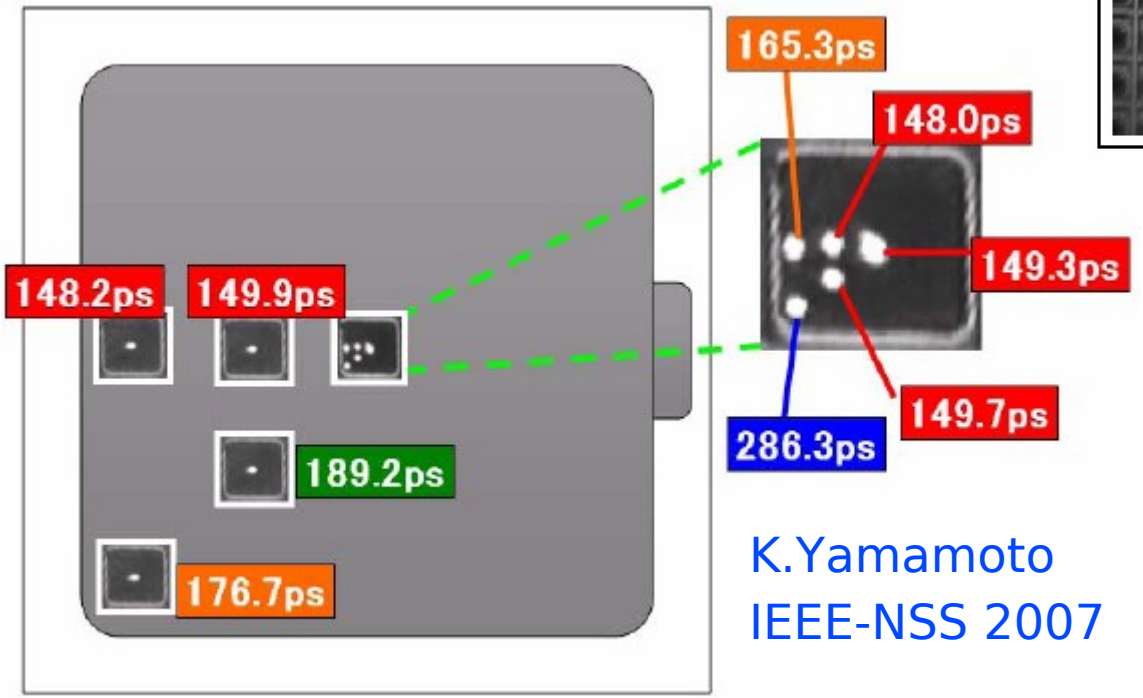



| | FWHM (ps) | FWTM (ps) |
|---|---|---|
| 1 | 199 | 393 |
| 2 | 197 | 389 |
| 3 | 209 | 409 |
| 4 | 201 | 393 |
| 5 | 195 | 383 |




K.Yamamoto PD07

Lower jitter if photoproduction at the center of the cell




K.Yamamoto IEEE-NSS 2007

Data include the system jitter (common offset, not subtracted)

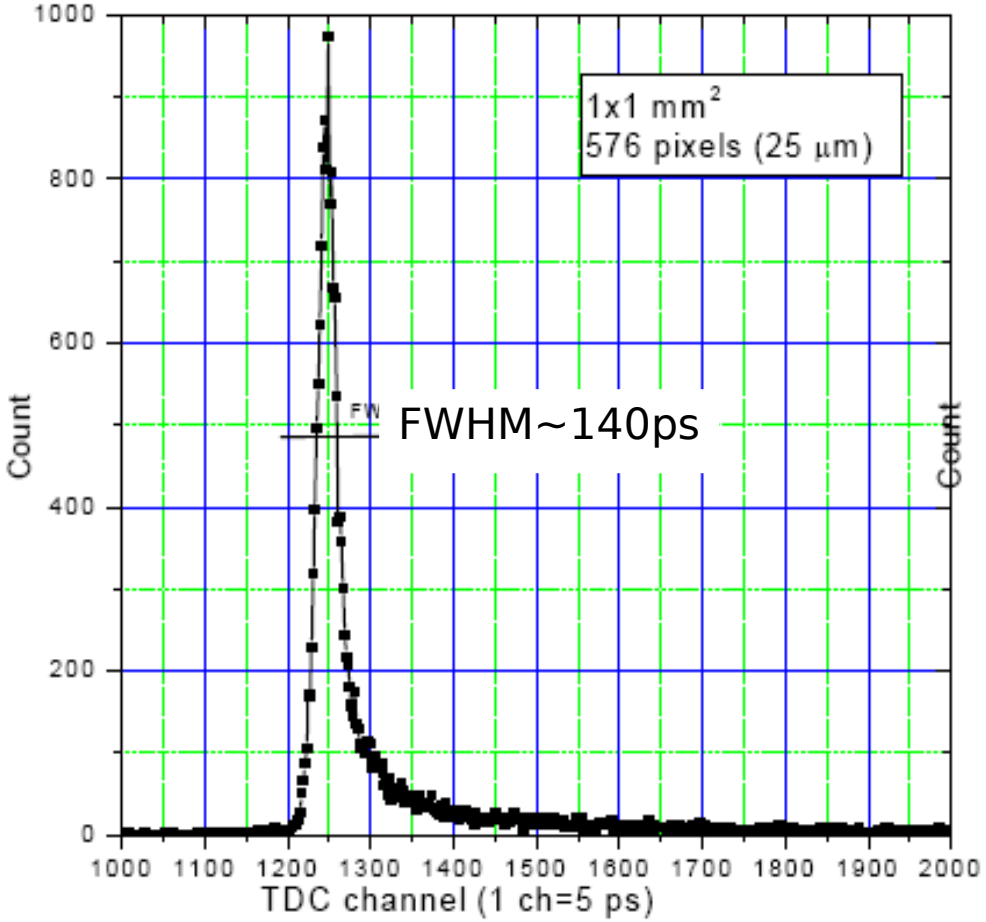
(*) NOTE: when laser is impinging only on one cell it is not easy to be sure that only one photons is in the cell: while the amplitude is invariant (many photons = one photon), timing is much affected by the number of photons (A.Ronzhin)

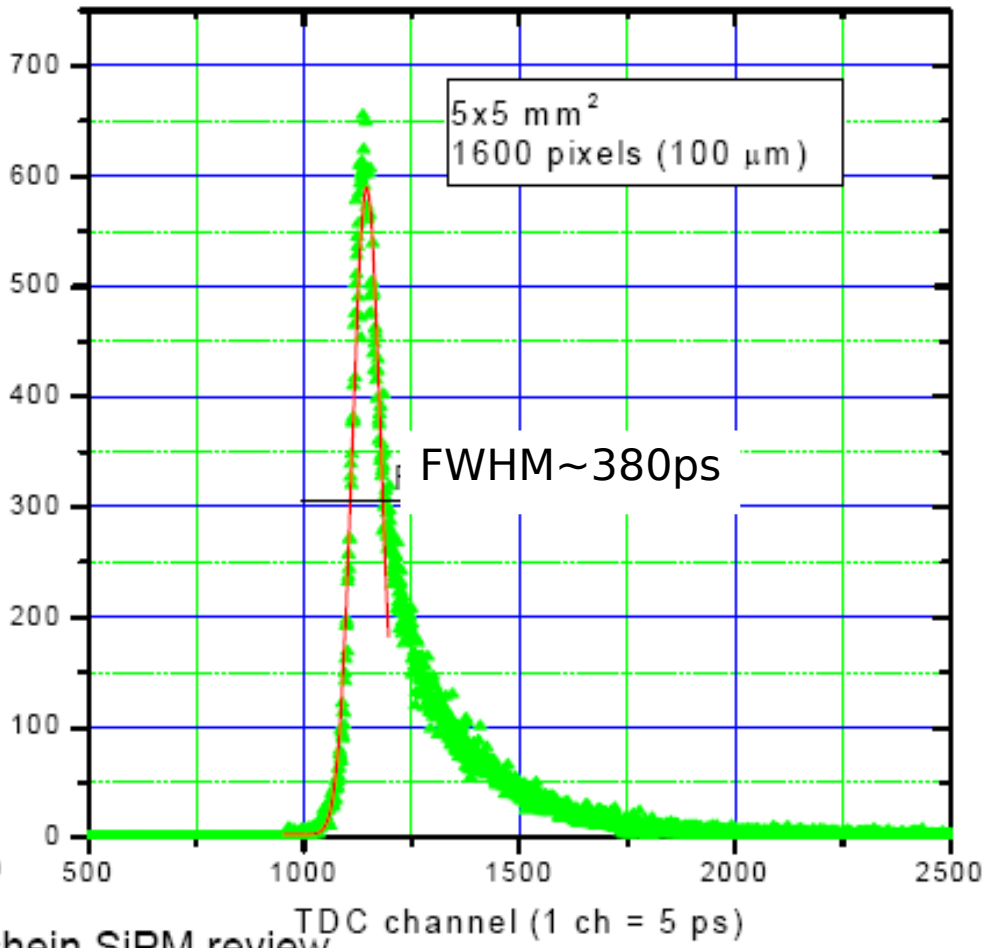
# SPTR: cell and sipm size dependence

B.Dolgoshein – LIGHT07

SiPM – MePhI/Pulsar:
576 cells (25x25μm²)
Area = 1x1 mm²

SiPM – MePhI/Pulsar:
1600 cells (100x100μm²)
Area = 5x5 mm²



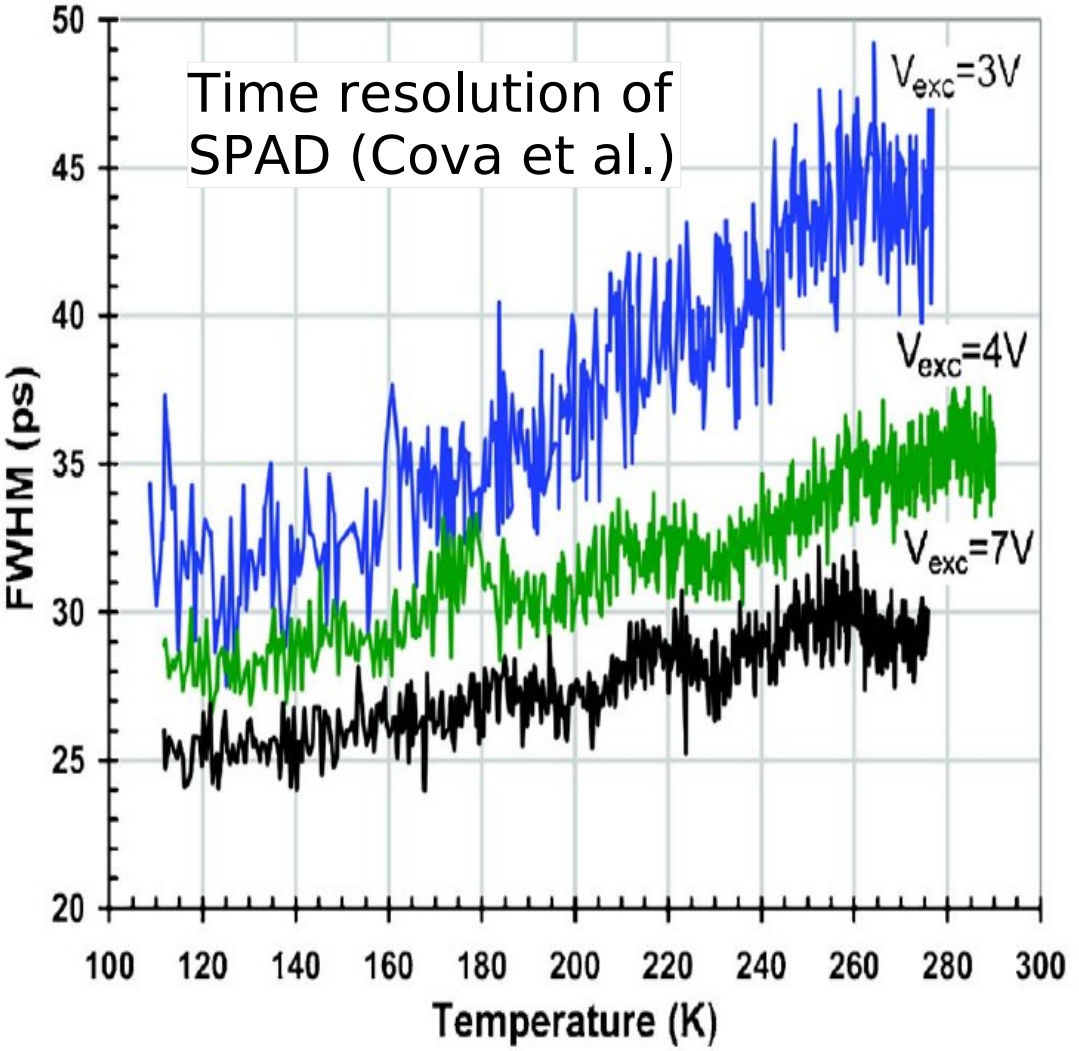



# SPTR: T dependence for SPAD devices

## Timing: better at low T

Lower jiitter at low T due to higher mobility

(Over-voltage fixed)




I.Rech el al, Rev.Sci.Instr. 78 (2007)

# Radiation damage

related to the recharge of the diode capacitance from $V_{BD}$ to $V_{bias}$ during the avalanche quenching time after $I_{latch}$ is reached.

$G=(V_{BIAS}-V_{BD}) * C_D/q$

valid for few volts above $V_{BD}$

Gain

Noise: dark count

pulses triggered by non-photogenerated carriers (thermal / tunneling generation in the bulk or in the surface depleted region around the junction)

afterpulse

carriers can be trapped during an avalanche and then released triggering another avalanche

optical cross-talk

photoegeneration during the avalanche discharge. Some of the photons can be absorbed in the adjacent cell possibly triggering new discharges

**$PDE = QE * P_{01} * \varepsilon$**

QE = quantum efficiency

$P_{01}$ = avalanche triggering prob.

$\varepsilon$ = geometrical fill factor

Photo-detection efficiency

Dynamic Range

Related to the density of cells and recovery time

Related to the photogeneration and to the avalange propagation

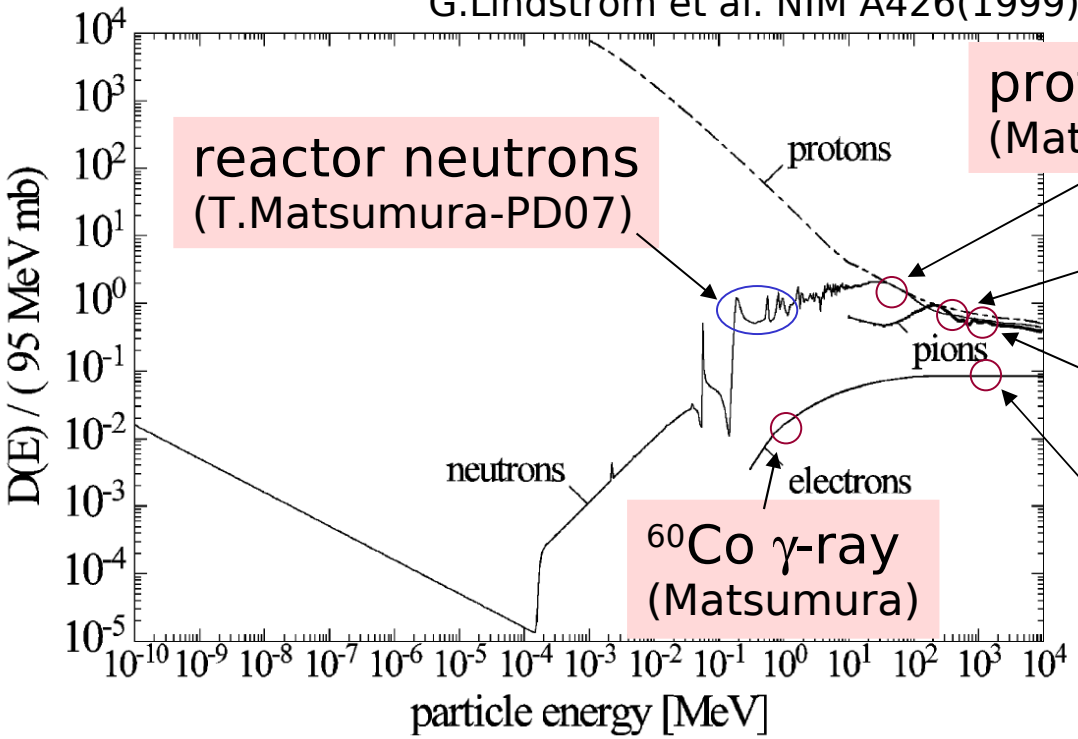
Time resolution

# Radiation damage: two types

- Bulk damage due to Non Ionizing Energy Loss (NIEL) ← neutrons, protons
- Surface damage due to Ionizing Energy Loss (IEL) ← γ rays
  (accumulation of charge in the oxide (SiO2) and the Si/SiO2 interface)

Assumption: damage scales linearly with the amount of Non Ionizing Energy Loss (NIEL hypothesis)

G.Lindstrom et al. NIM A426(1999)1-15




Examples of radiation tolerances for HEP and space physics

ATLAS inner detector ... $3\times10^{14}$ hadrons/cm²/10 year
~ $10^4$ hadrons/mm²/s

General satellites ... ~ 10 Gy/year

Expectations:

protons / γ-ray ~ 100

protons / neutrons ~ 2~10

# Radiation damage: effects on SiPM

## 1) Increase of dark count rate due to introduction of generation centers

Increase ($\Delta R_{DC}$) of the dark rate:

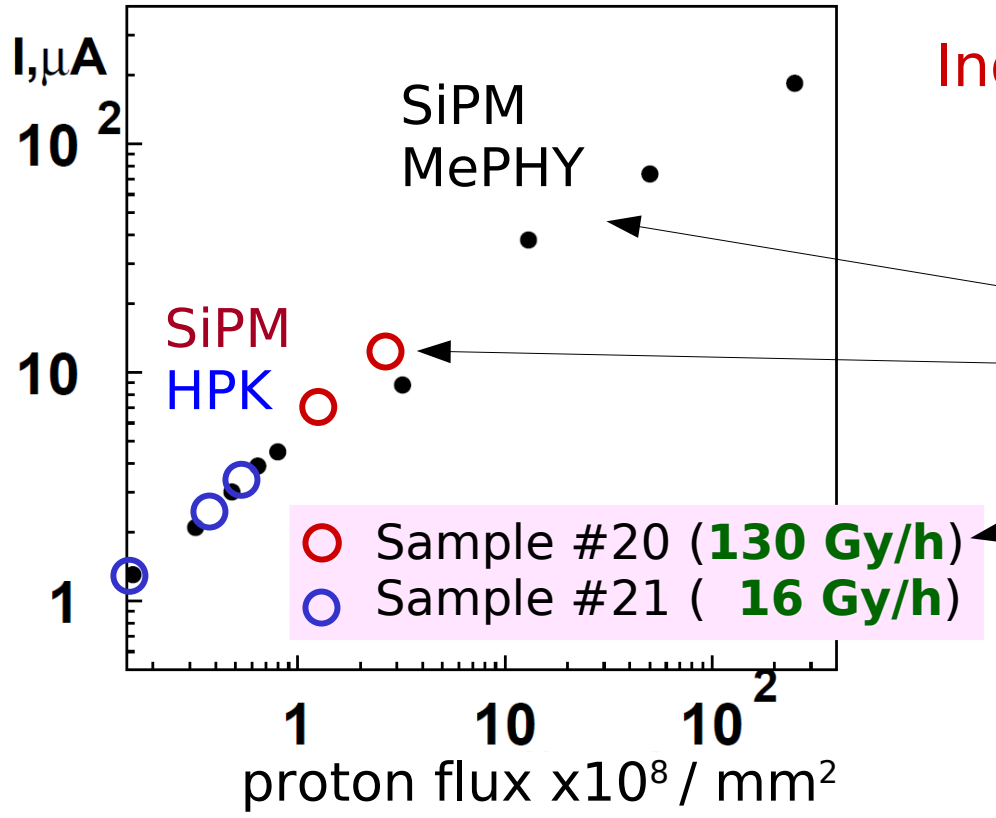
$\Delta R_{DC} \sim P_{01}\, \alpha\, \Phi_{eq}\, Vol_{eff} / q_e$

where $\alpha \sim 3 \times 10^{-17}$ A/cm is a typical value of the radiation damage parameter for low E hadrons and $Vol_{eff} \sim Area_{SiPM} \times \varepsilon_{geom} \times W_{epi}$

NOTE:
The effect is the same as in normal junctions:
- independent of the substrate type
- dependent on particle type and energy (NIEL)
- proportional to fluence

## 2) Increase of after-pulse rate due to introduction of trapping centers

→ loss of single cell resolution → no photon counting capability




## Indications from measurements:

1) no dependence on the device
similar effects found for SiPM from MePHY (Danilov) and HPK (Matsumura) (normaliz. to active volume)

2) no dependence on dose-rate
HPK (Matsumura)

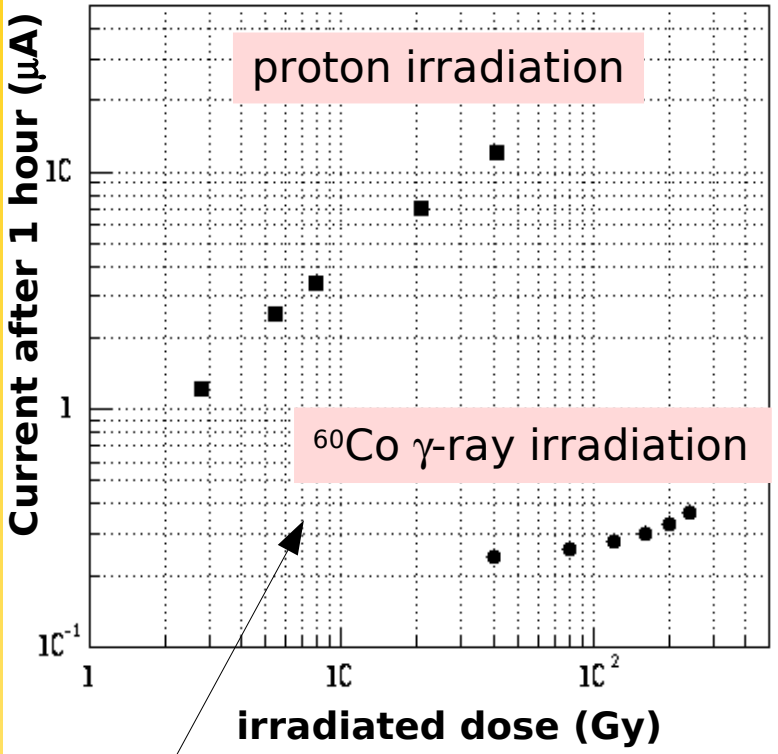
3) n similar damage than p

4) p x$10^1$-$10^2$ more damage than γ

# Damage comparison

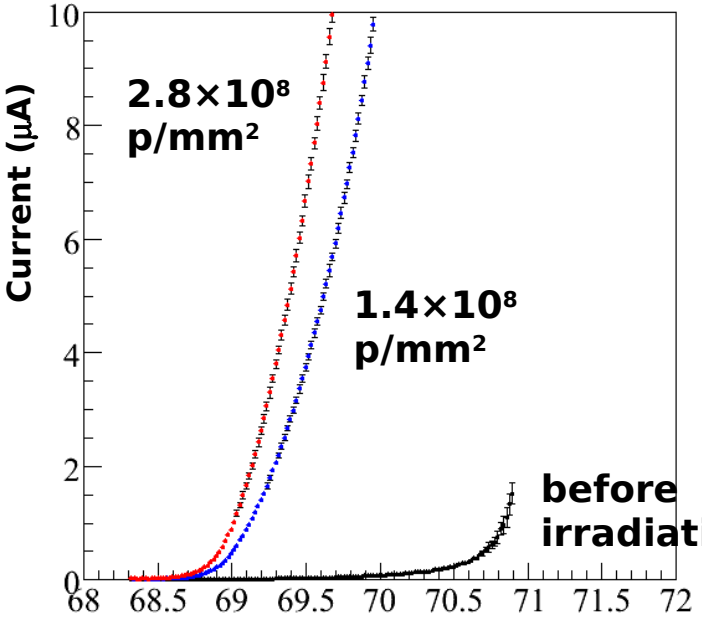
**Damage effect ...**
almost the same for protons and neutrons

HPK devices
T.Matsumura – PD07



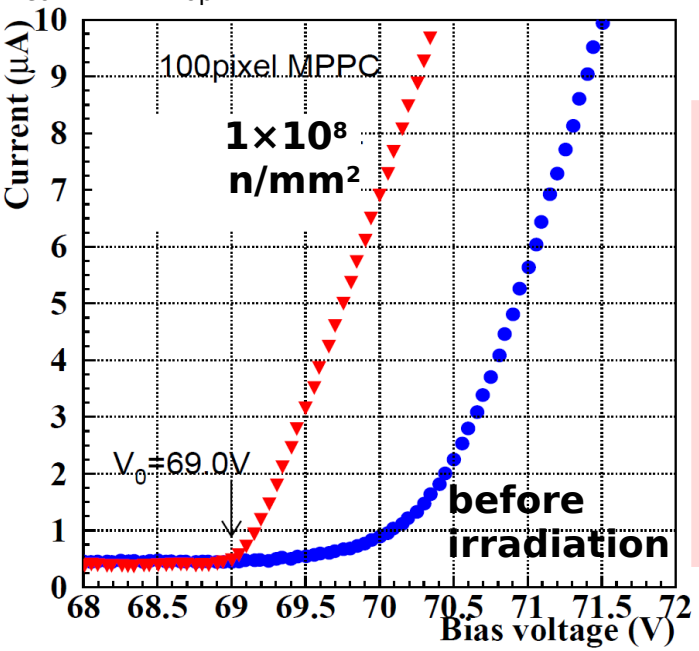

**Damage effect ...**
1~2 orders larger with protons than γ-ray irradiation

$2.3\times10^5$ p/mm²/s (130 Gy/h)
$I_{leak}$ @ ($V_{op}$, $1.4\times10^8$ p/mm²) = 6.7 μA




proton irradiation

$4.2\times10^5$ n/mm²/s
$I_{leak}$ @ ($V_{op}$, $1.0\times10^8$ n/mm²) = 8.5 μA

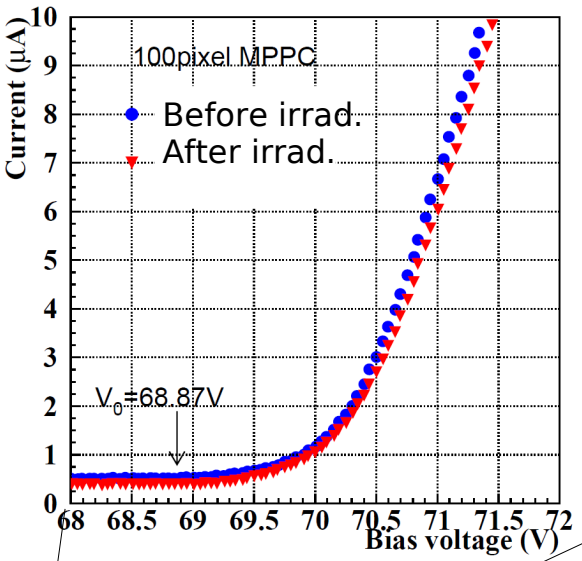


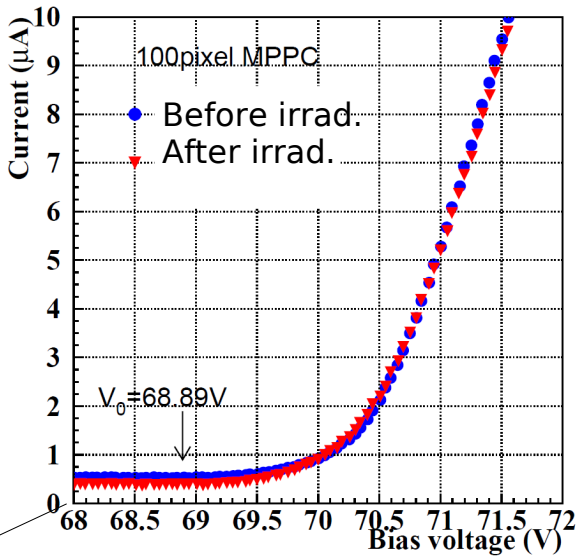

Neutron irradiation

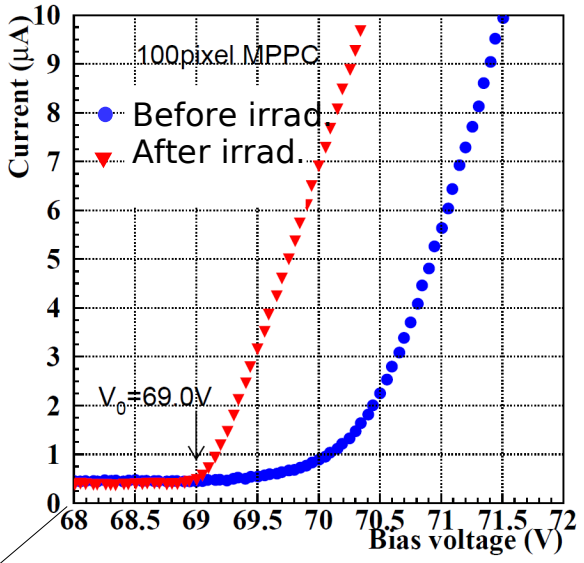
T.Matsumura – PD07

# Radiation damage: neutrons (0.1 -1 MeV)

$8.3\times10^{4}$ n/mm$^2$ $3.3\times10^{5}$ n/mm$^2$ $1.0\times10^{8}$ n/mm$^2$



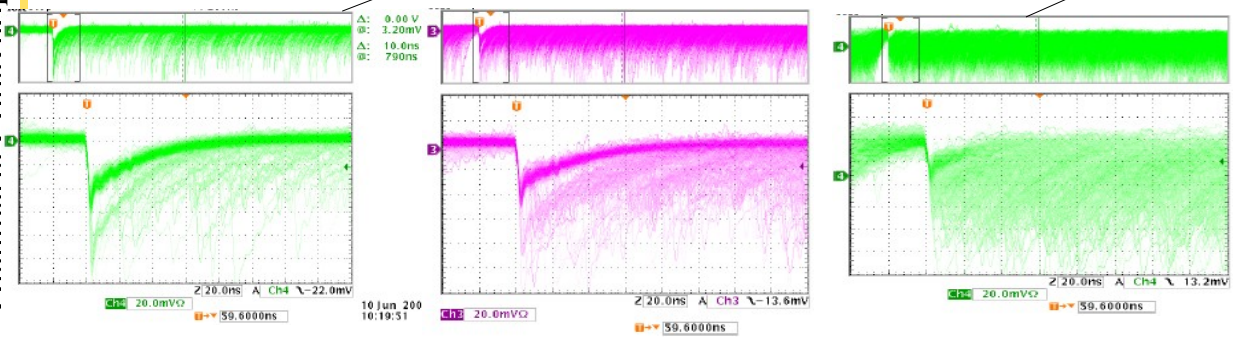




T.Matsumura – PD07

$10^{5}$ n/mm$^2$ $10^{6}$ n/mm$^2$ $10^{7}$ n/mm$^2$ $10^{8}$ n/mm$^2$ $10^{9}$ n/mm$^2$ $10^{10}$ n/mm$^2$

**No significant change**

n dose



I-V drastically change. No signal
Signal pulse is still there,
but continuous pulse height.
(No photon-counting capability)

Nakamura at NDIP08

# Detector performances

related to the recharge of the diode capacitance from $V_{BD}$ to $V_{bias}$ during the avalanche quenching time after $I_{latch}$ is reached.

$G=(V_{BIAS}-V_{BD}) * C_D/q$

valid for few volts above $V_{BD}$

Gain

pulses triggered by non-photogenerated carriers (thermal / tunneling generation in the bulk or in the surface depleted region around the junction)

Noise: dark count
afterpulse
optical cross-talk

carriers can be trapped during an avalanche and then released triggering another avalanche

photoegeneration during the avalanche discharge. Some of the photons can be absorbed in the adjacent cell possibly triggering new discharges

**$PDE = QE * P_{01} * \varepsilon$**

QE = quantum efficiency
$P_{01}$ = avalanche triggering prob.
$\varepsilon$ = geometrical fill factor

Photo-detection efficiency

Dynamic Range

Related to the density of cells and recovery time

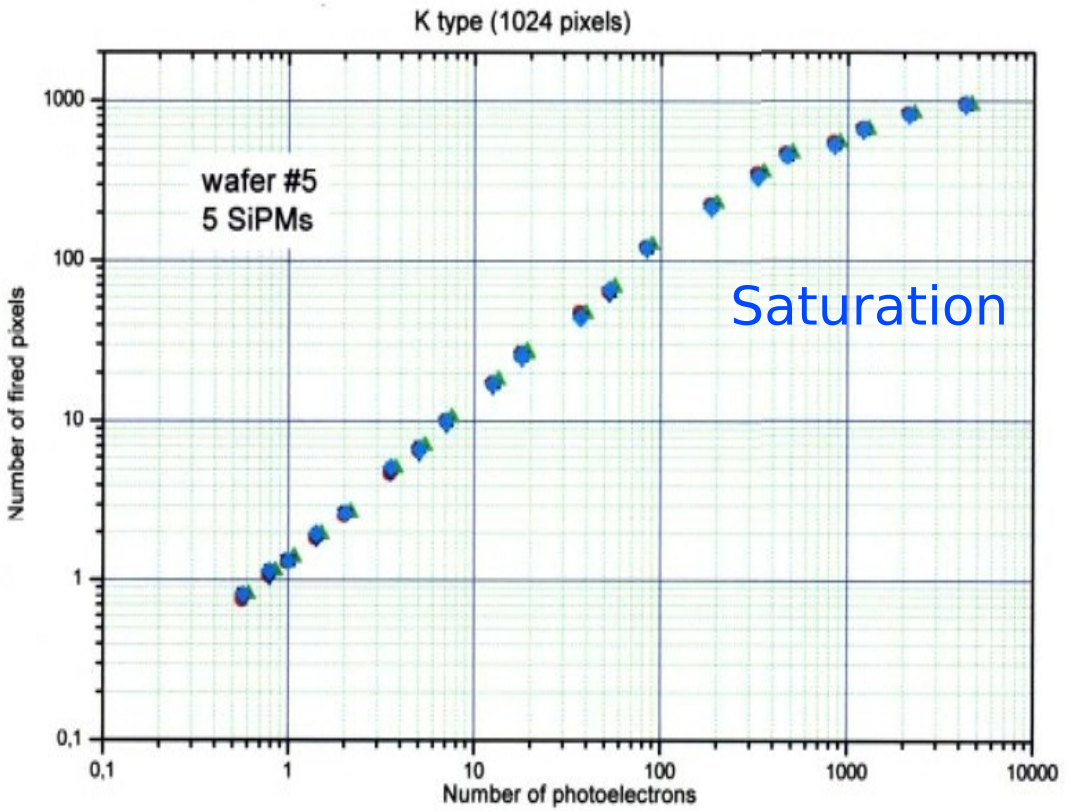
Related to the photogeneration and to the avalange propagation

Time resolution

# Dynamical range

SiPM output = sum of binary cell's output !

The output signal is proportional to the number of fired cells as long as the number of photons in a pulse ($N_{photon}$) times the photodetection efficiency PDE is significant smaller than the number of cells $N_{total}$.




$$A \approx N_{firedcells} = N_{total} \cdot \left(1 - e^{-\frac{N_{photon} \cdot PDE}{N_{total}}}\right)$$

eg: 20% deviation from linearity if 50% of cells respond

→ Best working conditions: $N_{photo\text{-}electrons} < N_{SiPM\ cells}$

NOTE:
1) correction to $N_{fired\text{-}cells}$ due to cross-talk and after-pulse
2) effective dynamic range depends on recovery time and time scale of signal burst
(see discussion about GLD calorimeter)

# Summary of SiPM properties

Most important features of a SiPM are:

- sensitivity to extremely low photon fluxes
  providing proportional information with excellent resolution
  and highest possible photon detection efficiency
- extremely fast response with small fluctuation (jitter)
  sub-ns risetime and <100ps jitter

More features:

- low bias voltage (<100V)
- low power consumption (<50$\mu$W/mm$^2$)
- long term stability, no ageing
- insensitive to magnetic fields (up to 15T) and EM pickup
- small sensitivity to charged particle traversing the device
- tolerate accidental illumination, and illumination with no bias
- robust and compact
- low cost (in the future, now from 100 to 10/mm$^2$) + low peripheral costs

Technology parameters: may be tuned to match the specific application

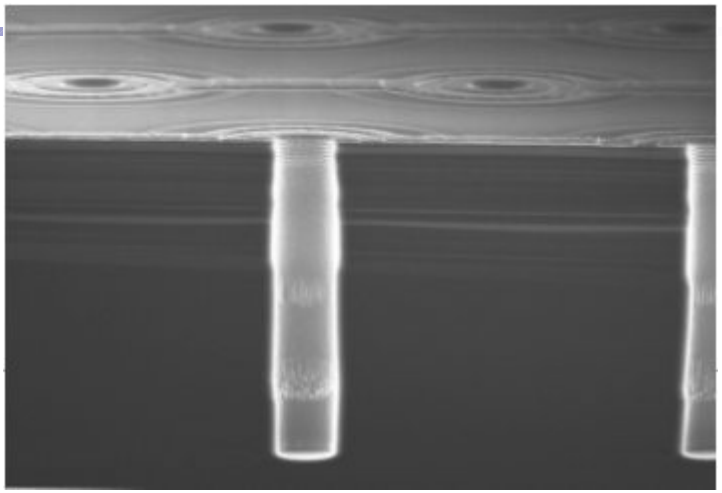
- silicon quality (dark rate, after-pulse)
- doping concentration (operating voltage and its range)
- layer structure and thickness (PDE wavelength range, optical cross-talk)
- optical cell insulation (optical cross-talk)
- effective area of the cells (gain, fill factor, dynamic range, recovery time)
- quenching resistor (recovery time, dynamic range)

# Future development threads

- Matrices to cover wide surfaces
  → imaging

- Optical coupling
  → feed-trough (anodes to the back-side)
  → planar process (CMOS)

E.g. development at RMD
C.J.Stapes, NDIP08



Future IRST technology: vias for electrodes feed-through. To avoid wide bonding on the active surface

- Cooling (moderate, easy) and technological efforts (difficult) to suppress noise
  ... a ''single photon detector'' having MHz noise sounds a bit funny

- Integration with electronics:
  1. CMOS compatibility → mass production / low cost ... but Si quality ?
  2. Exploit the intrinsic high bandwidth possibility of the detector

- Back-side illumination and 'internal, vertical quenching resistor' in study/developement at MPI (see J.Ninkovic et al NDIP08)

# Examples of applications

## Particle High Energy, Space, Astro - Physics

- Calorimetry in magnetic fields
- Active target for neutrino experiments
- Fiber tracking (+ TOF)
- Fast timing, TOF (scintillators, cherenkov)
- Proximity focusing RICH, Cherenkov Differential counter
- Liquid Xenon scintillation
- Telescopes: Cherenkov and Fluorescence light

## Medical Physics

- Medical imaging (PET + TOF + MRI)

## Other applications

- Quantitative biological and molecular recognition: Fluorescence spectroscopy, Confocal microscopy, Flow Cytometry, Micro-Fluidics
- Adaptive optics, Tracking of luminous sources, LIDAR
- X-ray and neutron detection/monitoring, Homeland security
- Non-Invasive Testing of ULSI Circuits

# R&D for future Calorimeters

GREEN/BLUE sensitive

Requirements:
- Operate in high magnetic field (5T)
- High longitudinal and transverse segmentation/granularity

Possible solution: (Sci/Fe sampling) Scintillator tiles + WLS fibers + SiPM
- Green-red sensitive SiPM to match WLS emission spectrum
- tests with MEPhI-Pulsar SiPM (1156 cells on 1mm$^2$), recently with HPK
- mounted directly on scintillator
- no nuclear counter-effect

Key points:
- Noise: MIP ~ 15p.e. $\rightarrow$ MIP efficiency > 95% need low noise at few p.e. threshold (MIP detection is needed x tracks and calibration)
  $\rightarrow$ this requirement fixes over-voltage $\rightarrow$ gain and PDE

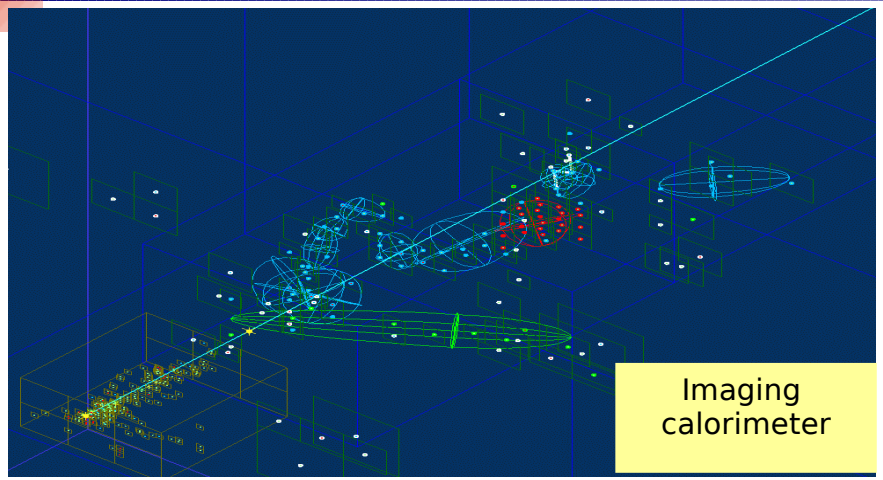
- Dynamic range and recovery time: multi cell signal saturation and fluctuation
  $\rightarrow$ number of cell/MIP is needed for the calibration

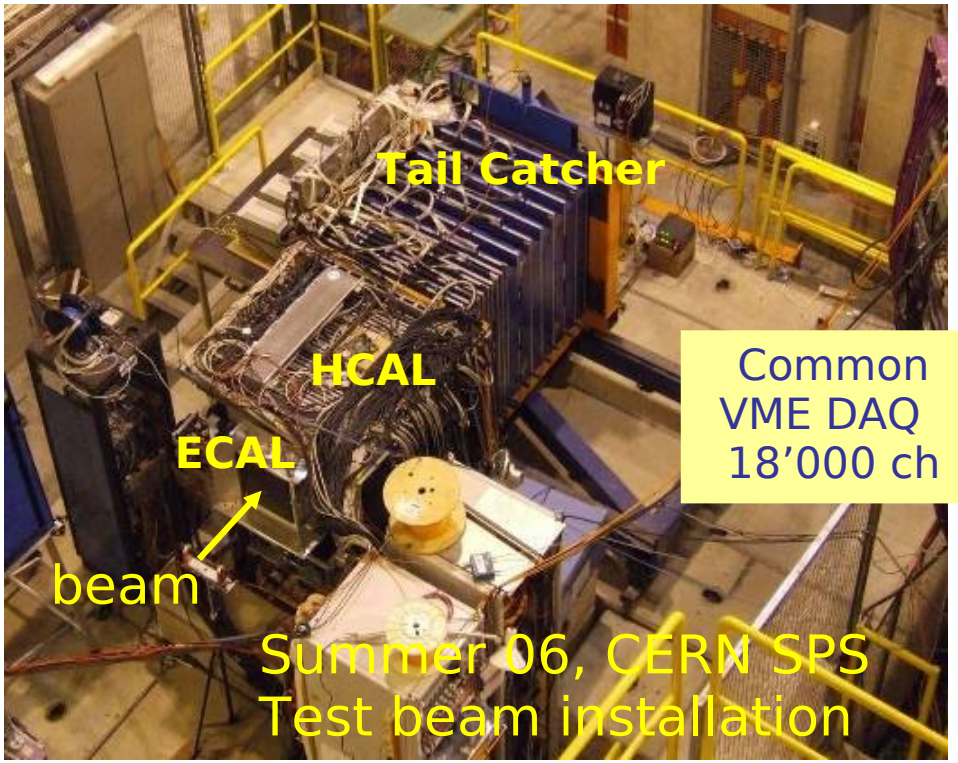
- Gain stability wrt V and T: $\frac{\Delta G}{G \Delta T} \approx 0.04/^{\circ}C$

  $\frac{\Delta G}{G \Delta V} \approx 0.01/20mV$

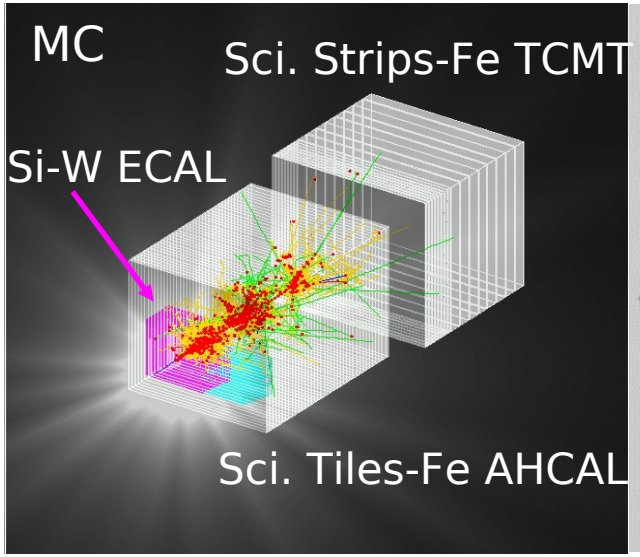
- Two signal ranges: no amplifier for signal + ampli. x20 for measuring SiPM gain
- Large number of channels (10000): gain and PDE spread

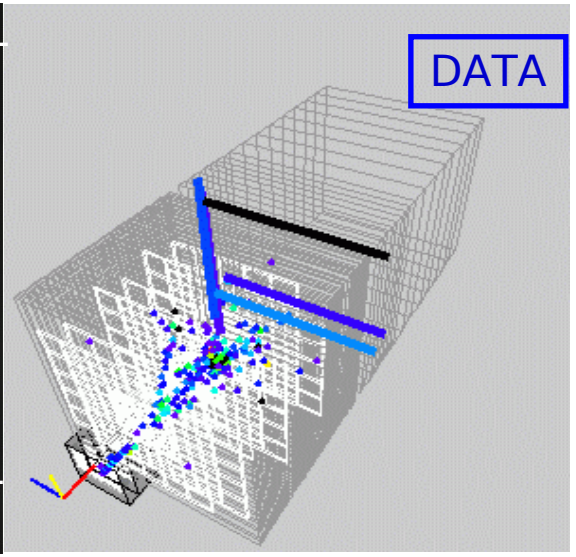
# The prototype calorimeter system for ILC



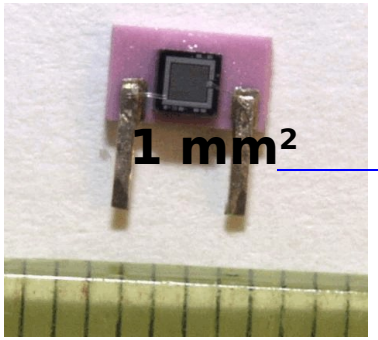



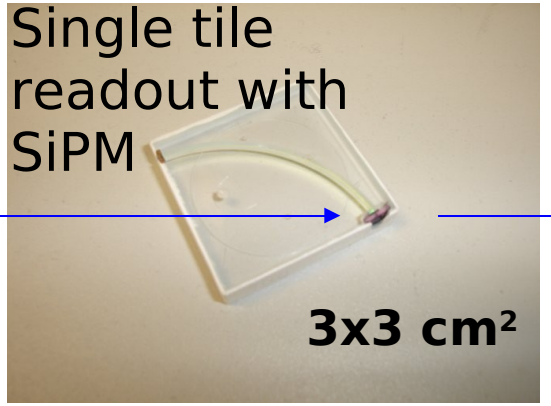

- ILC: exclusive final sates with heavy bosons W, Z, Higgs
  - Multi-jet events
- Need 2x better jet energy precision w.r.t. present calo's
  (jet E resolution at 30% !)
- Particle Flow Approach: Reconstruct each particle individually (1-100 GeV)
- Requires high longitudinal and transverse **granularity**
  - Particle separation
  - Software compensation
- ECAL + HCAL (neutral hadr)
  + Tail Catched Muon Tracker

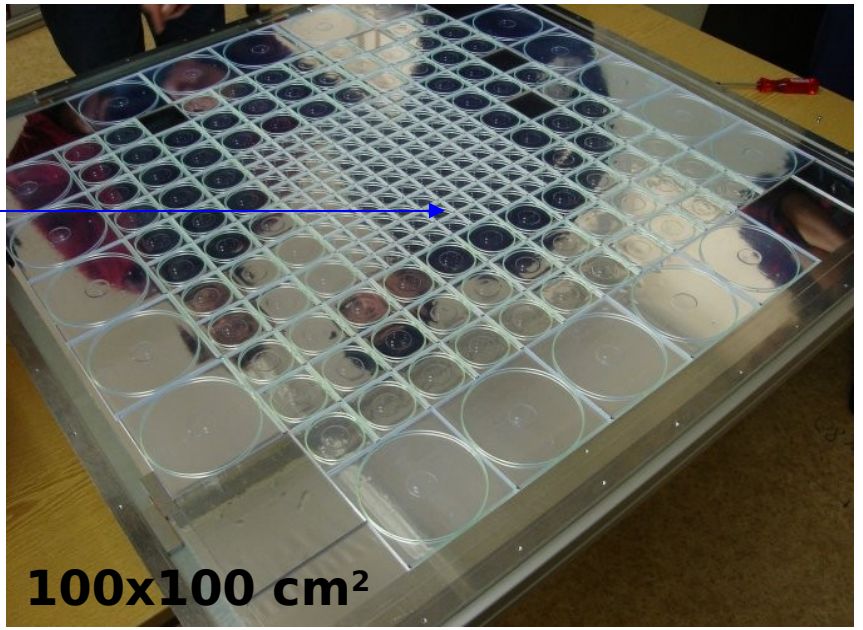


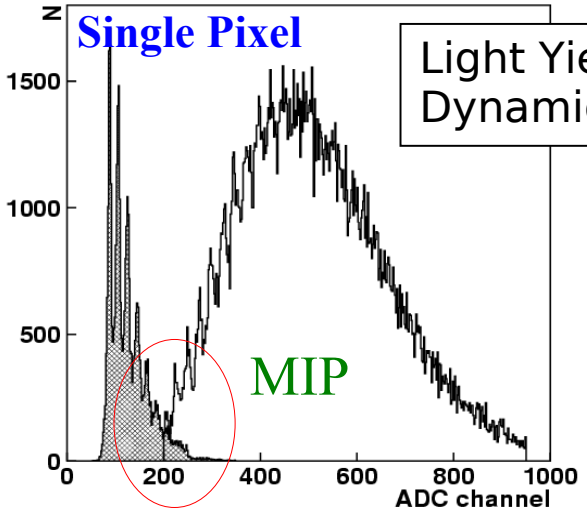
# High granularity hadronic calorimeter




1x1m² prototype calorimeter with 8000 channels readout with SiPM (MePHI/Pulsar)




Light Yield = 15 pixels / MIP
Dynamic range ~ 100 MIPs

NOTE: 2 scales
- auto-calibration of SiPM gain from single-pixel spectra
- MIP spectra for E calibration

Allows unprecedented high granularity

38 layers (~4.5 λ)
Scintillator – Steel sandwich structure (0.5:2cm)

All channels calibrated with muons
Threshold (offline) = ½ MIP for 95% efficiency

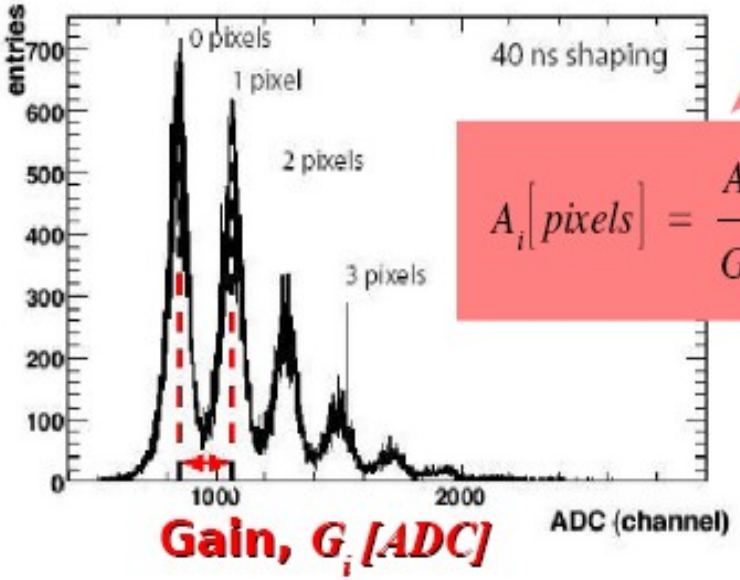
# AHCAL calibration chain

amplitude in ADC channels

$$E_i[MIP] = \frac{A_i[ADC]}{A_i^{MIP}[ADC]} \times f_{sat}\left(A_i[pixels]\right)$$

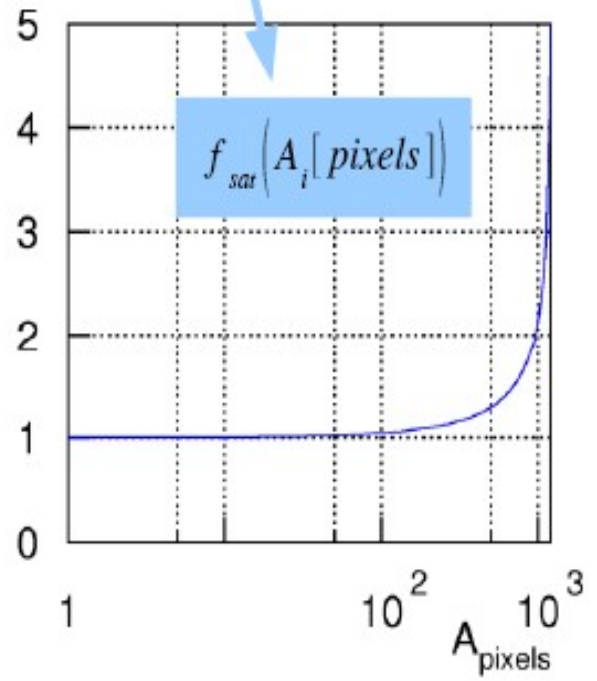
use of muons as minimum ionizing particles

measured amplitude in pixels:




$$A_i[pixels] = \frac{A_i[ADC]}{G_i[ADC]}$$

correction for SiPM non-linearity:



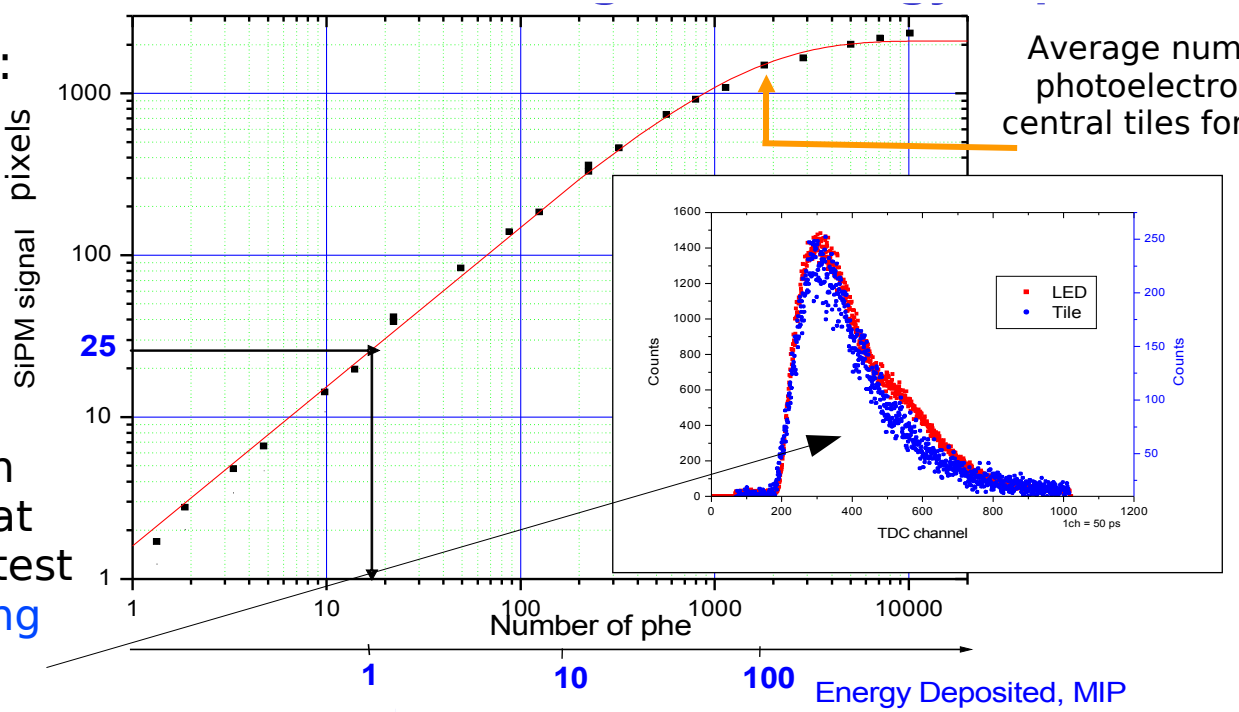

A.Kaplan NDIP08

# Nuisance #1: Dynamic range

Calibration needs 2 scales:
MIP and SiPM pixel

- E[MIP] = $A/A_{MIP}$ * F ($N_{pixel}$)
  $N_{pix}=A/A_{pixel}$

- F = non-linearity correction
  it is not an issue (if saturation is avoided !) It can be made at tile level with LASER/LED on test bench ... with careful matching of signal light pulse shape




NOTE: the effective dynamic range depends on two time scales:
(1) the pixel recovery time ($R_Q$ resistor) and (2) the characteristic scintillator/WLS time

Fast recovery time (eg ~10ns for $R_Q$~0.3 MΩ) → extended dynamic range for large/slow signals: each pixel can fire many times during the same light pulse.
→ Saturation curve depends on $R_Q$ and light pulse duration (WLS → slow/direct coupling → fast)

→ Thread between effective dynamic range and calibration with single p.e. (measurement of F)
CALICE uses large $R_Q$ ~ 2MΩ → slow recovery: OK for WLS, but makes dark noise an issue
Critical also in case of higher light yields: e.g. in ECAL due to compact EM shower signals

# Nuisance #2: Dark rate



1) First limitation comes from requirement to observe single photo-electron signals (for gain monitoring and non-linearity correction).

2) Critical parameter is the noise above threshold (½MIP for 95% efficiency):

- required noise hit detector occupancy < $10^{-4}$

  → noise rate above threshold < 300Hz

  (from FEE bandwidth and need to identify isolated neutrons in 300ns windows)

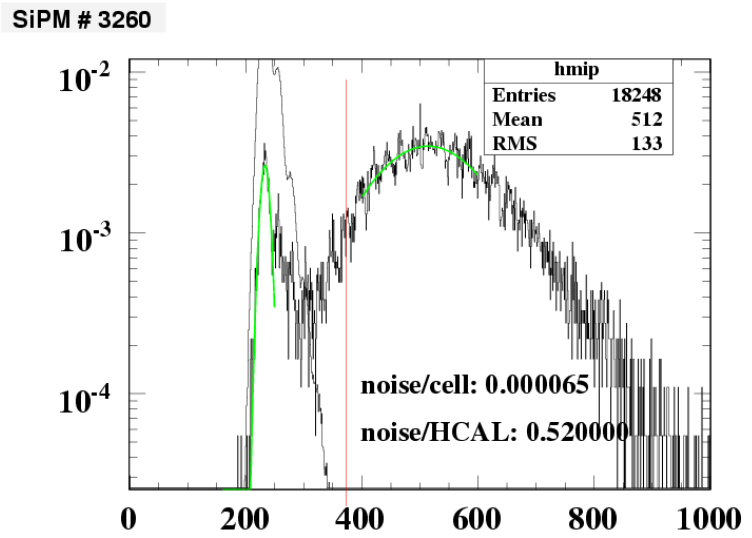
  occupancy measured at test beam ~ $10^{3}$

- rate above threshold depends on dark rate and X-talk probability according to the following rule of thumb:

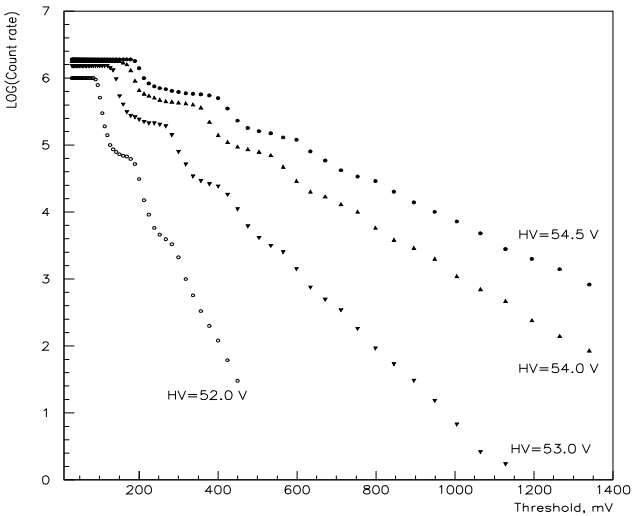
$$R(Thr) = R_{dark} * (P_{X\text{-}talk})^{Thr}$$

$$\rightarrow 2MHz * (0.3)^{7} = 400\ Hz$$

→ New tests with SiPM from Hamamatsu ongoing.
Noise level ~ x 1/10 than MePHY SiPM
and lower cross-talk!

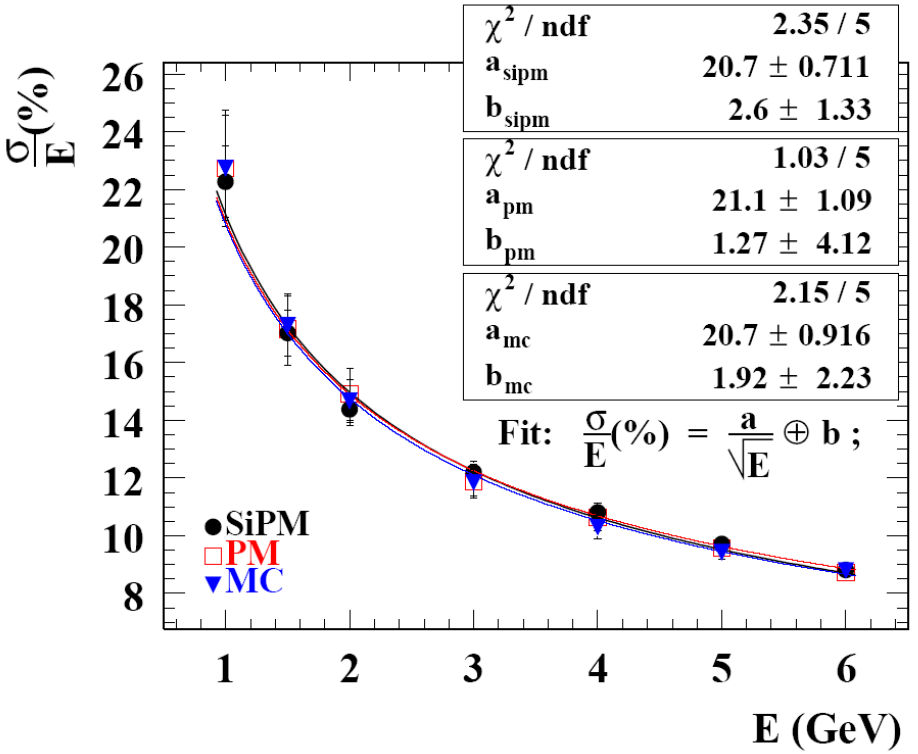


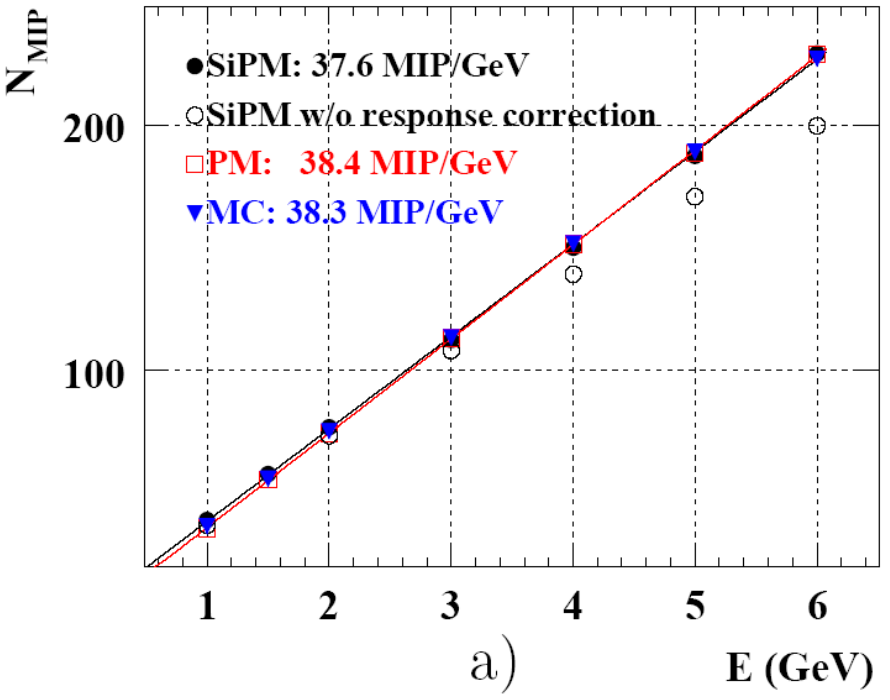

Dark rate vs. Threshold




1p.e. noise rate ~2MHz.
threshold 3.5p.e. ~10kHz
threshold 6p.e. ~1kHz

# Tile HCAL for ILC



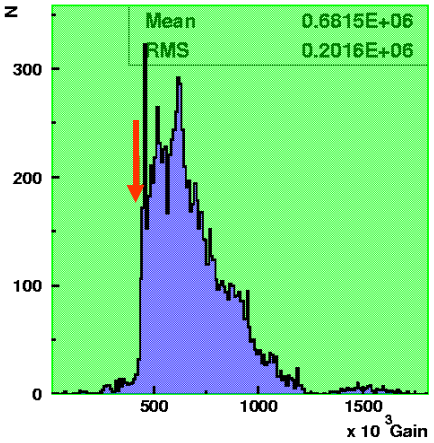




- Very good agreement between SiPM, MAPMT, APD(not shown) and MC in the whole range 1 - 6 GeV
- SIPM non-linearity can be corrected even for dense e/m showers for each tile and does not spoil the resolution
- Possibility to observe peaks for different number of p.e. crucial for calibration

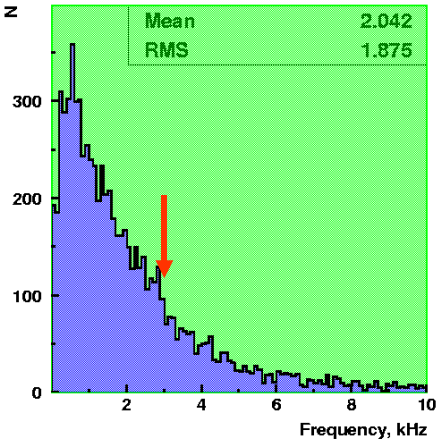
# Tile HCAL for ILC

The first large scale (~$10^4$) successful application of SiPMs

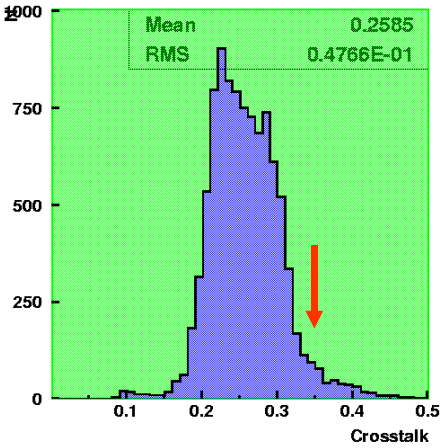
Parameters' spread




GAIN




NOISE AT ½ MIP(7.5 pixels)




CROSS TALK

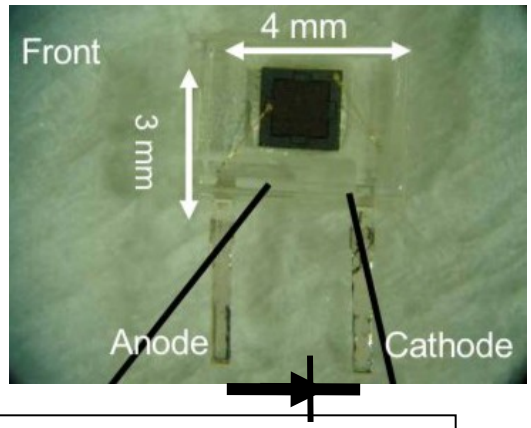
For better resolution (eg EM calorimetry):
- single channel control of T and V
- lower cross-talk

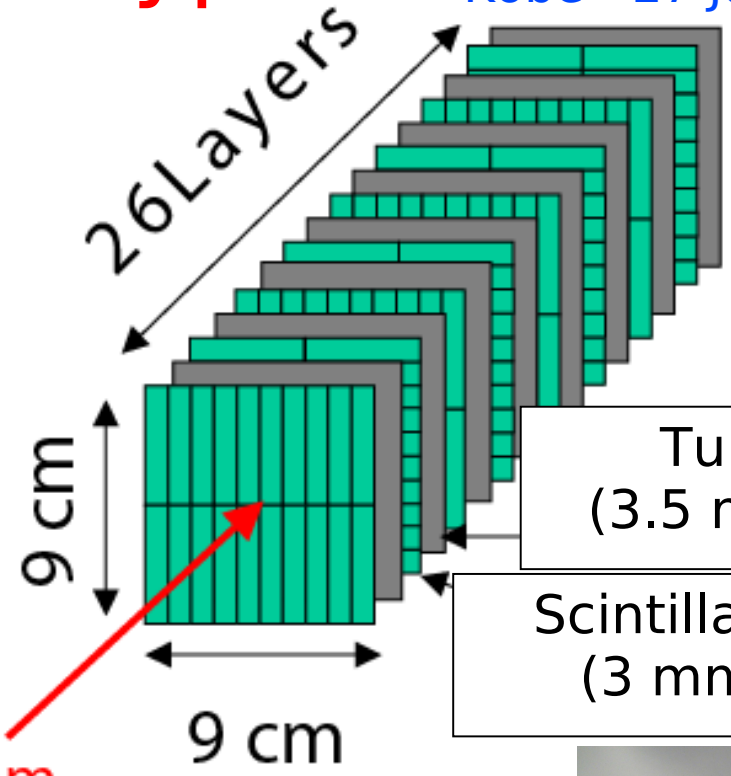
Direct coupling to scintillator still problematic:
- less number of photons (no WLS) + small area devices
  → blue sensitive SiPM with enhanced PDE (like HPK or IRST)
- surfaces coupling

# The GLD ECAL prototype



MPPCs
(HPK)
(1600 pixels)

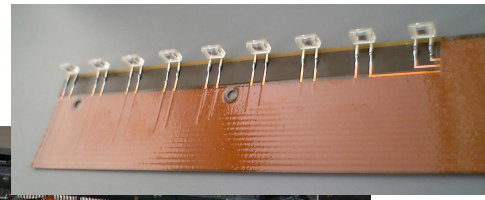


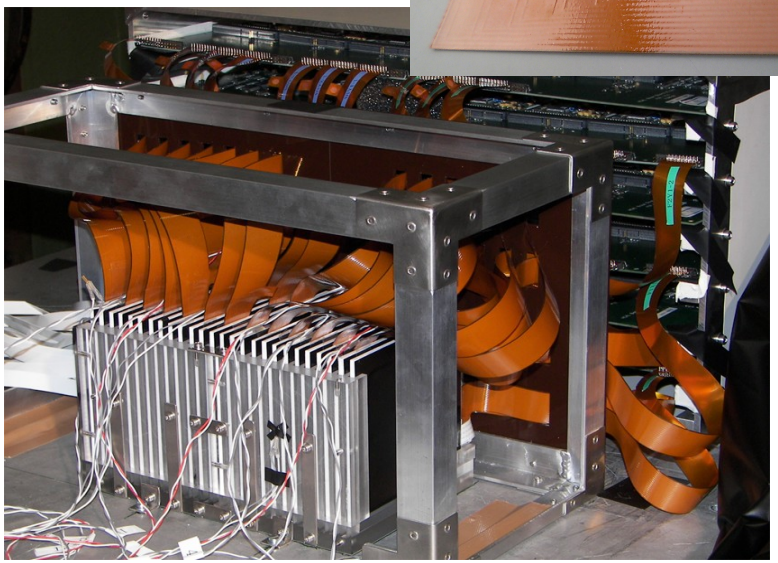



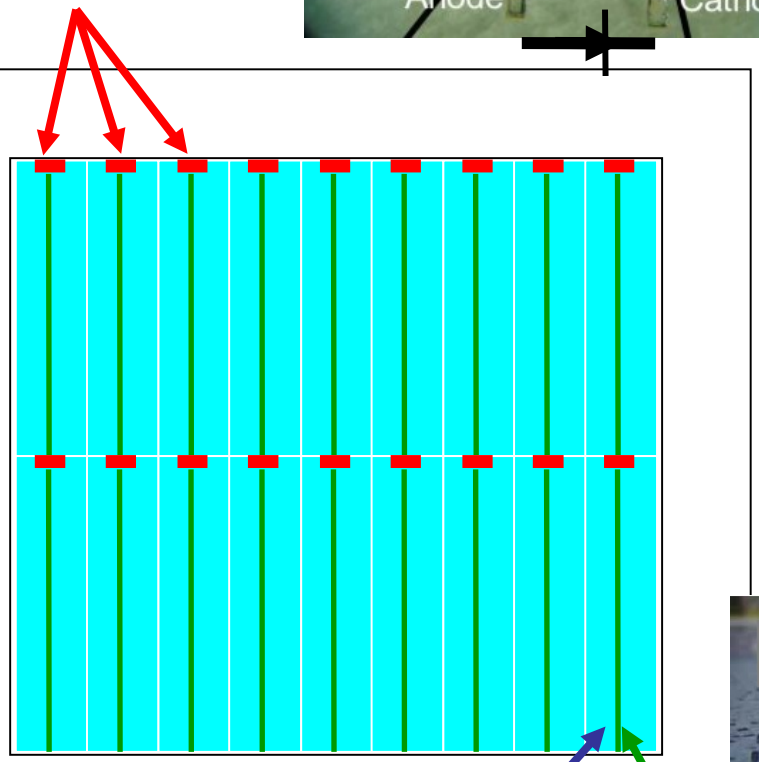

Tungsten
(3.5 mm thick)

Scintillator layer
(3 mm thick)

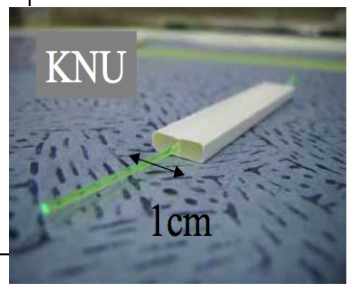
$e^+$ beam
(DESY)







Frame

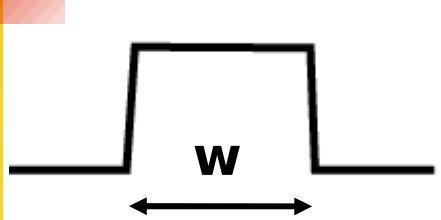


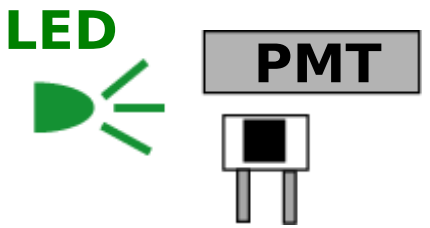

Scintillator strip
(1 x 4.5 x 0.3 cm)

WLS fibre

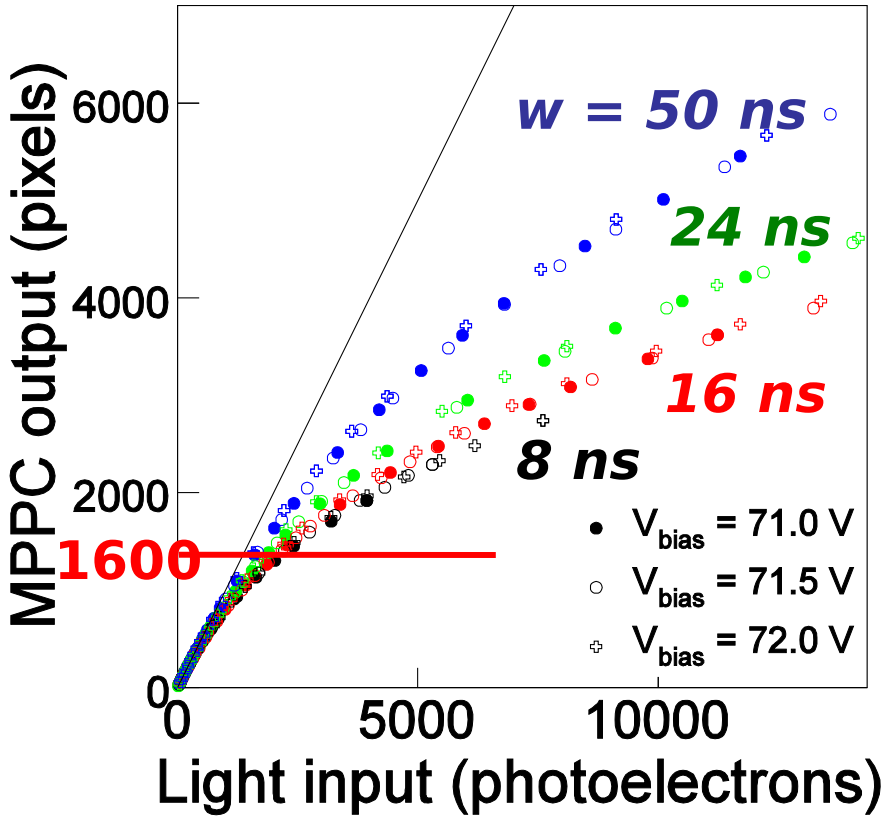
# Response Curve

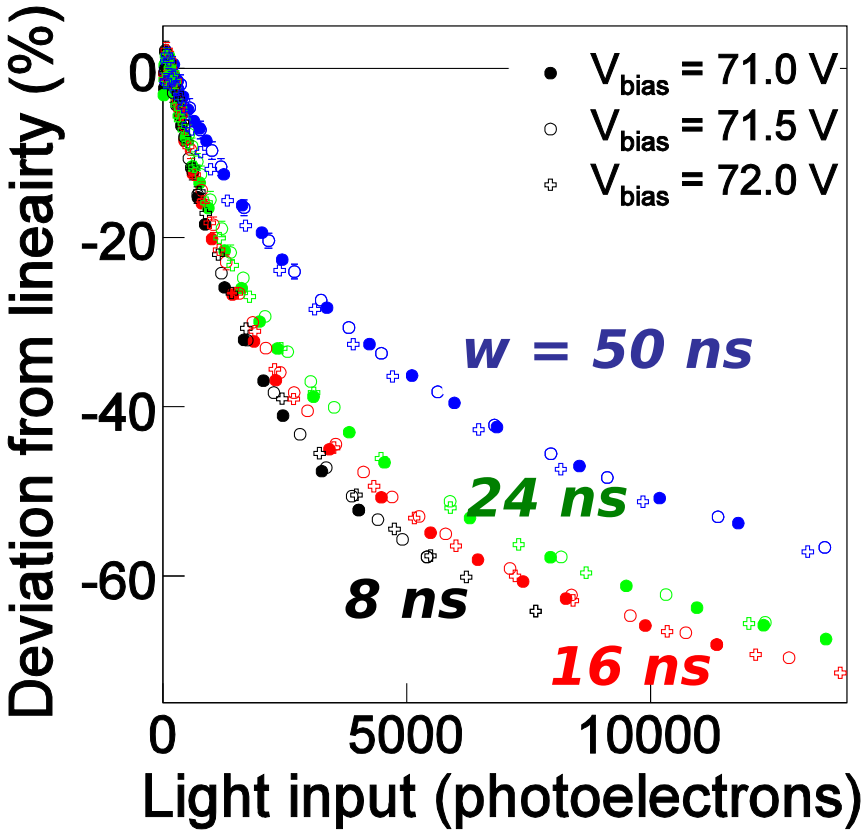





Response curves taken with various width of LED light pulses. (gate width = 100 ns)



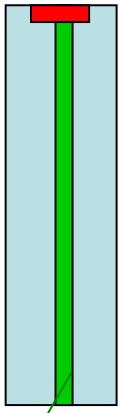




- Dynamic range is enhanced with longer light pulse
- Time structure of the light pulse gives large effects in non-linear region.
- No significant influence with changing bias voltage.
- Knowing time structure of scintillator/WLS light signal is crucial

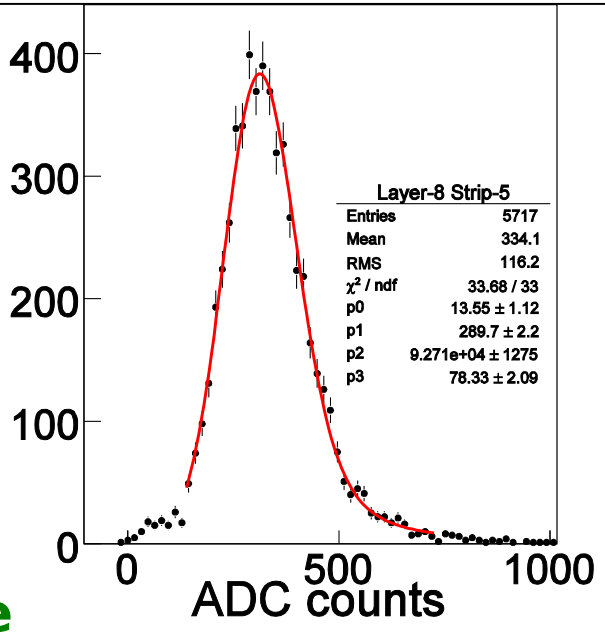
# MIP events (data without absorber)

## WLS Fibre readout strip

MPPC

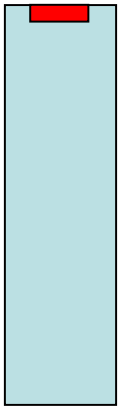


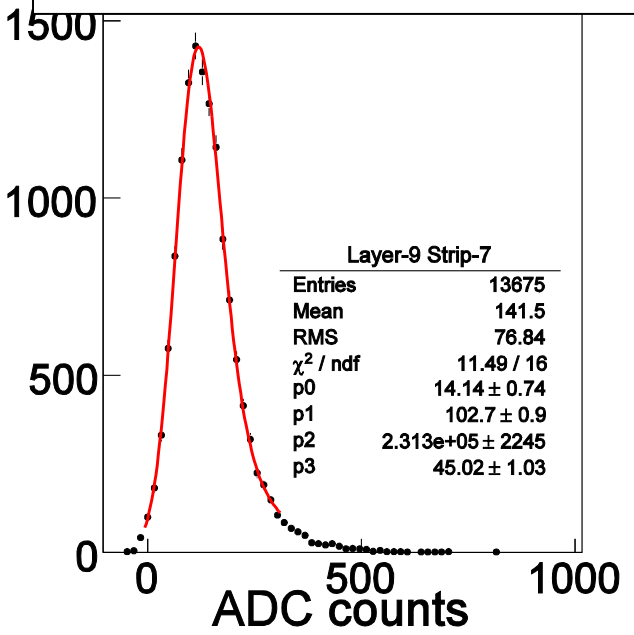
WLS fibre



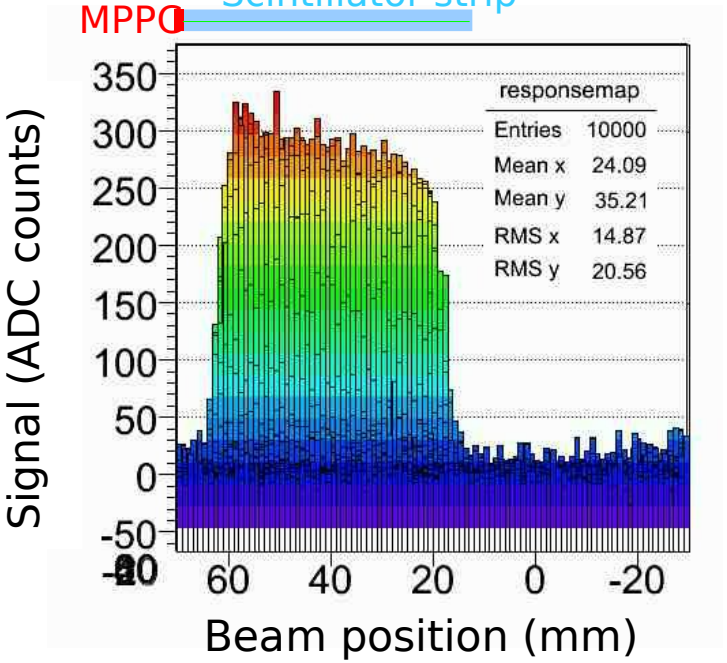

## Direct readout strip

MPPC

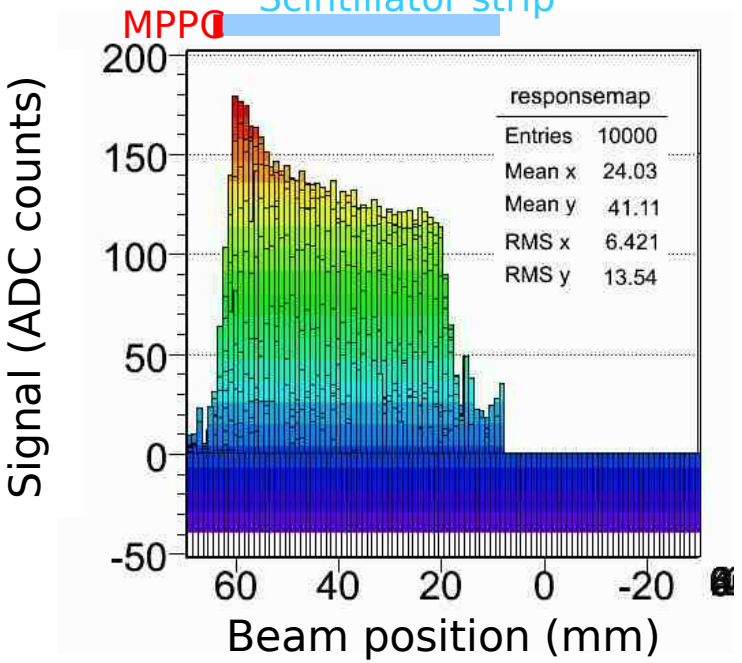




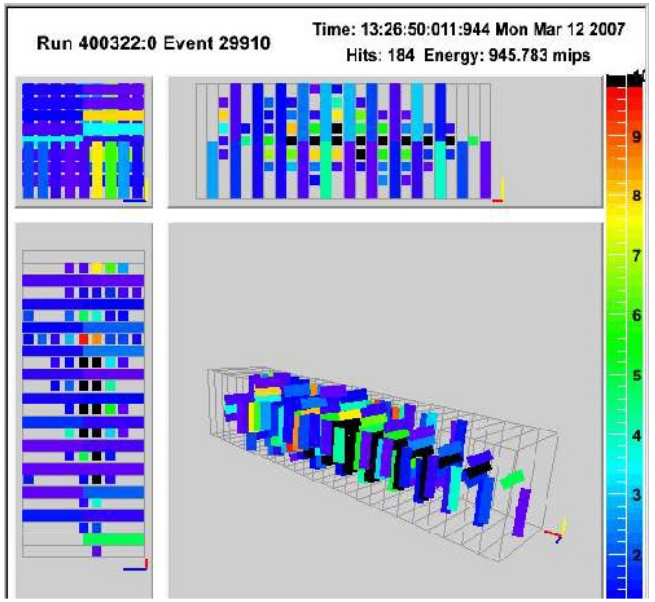

MPPC Scintillator strip



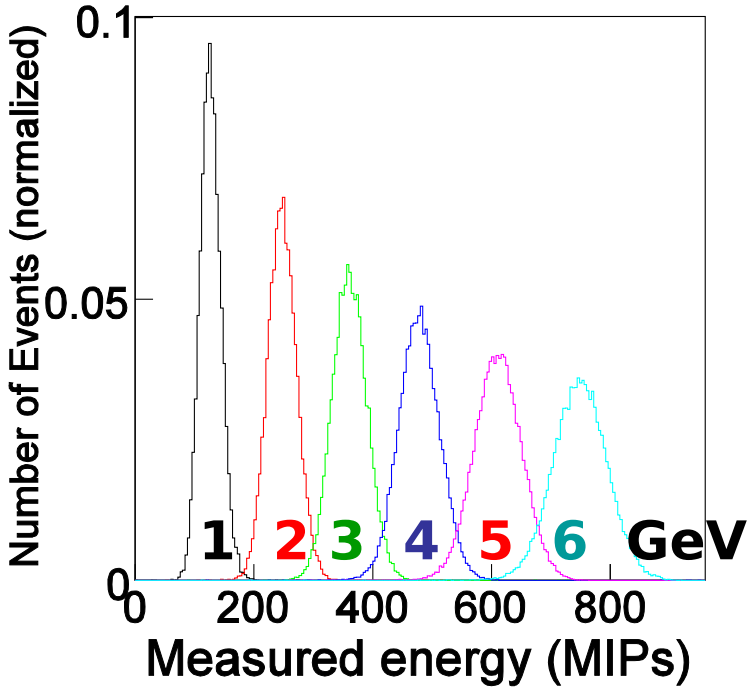

MPPC Scintillator strip

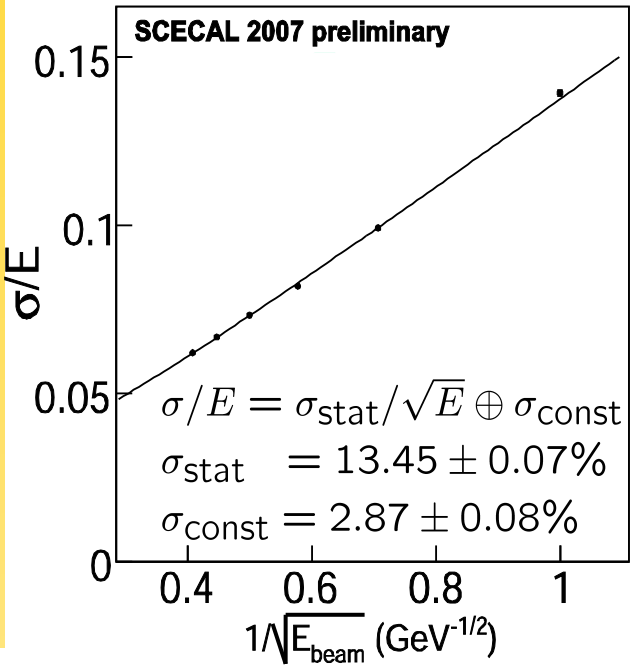


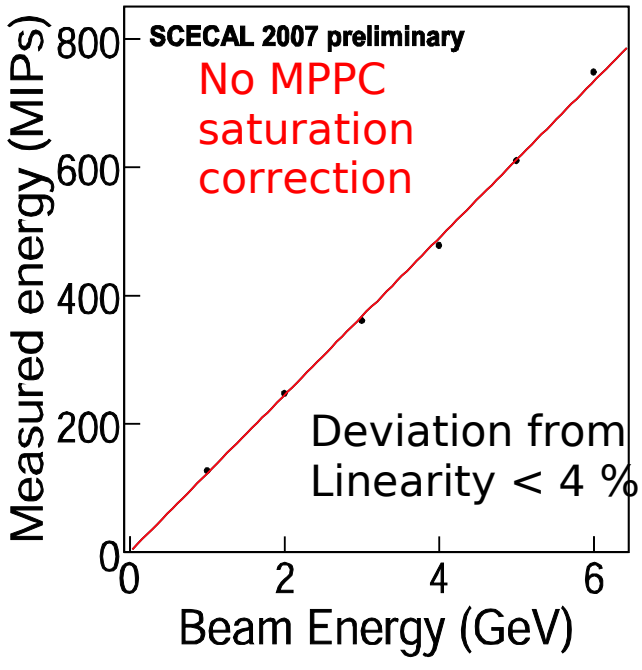
# Results with tungsten absorber




**6 GeV e⁺ center injection**




The calorimeter with MPPC is working!

WLS seems a better options:
1) enhanced dynamic range
2) better uniformity

New results expected from studies at higher beam energy at FNAL (running just now Oct 2008)

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- Adaptive optics, Tracking of luminous sources, LIDAR
- X-ray and neutron detection/monitoring, Homeland security
- Non-Invasive Testing of ULSI Circuits

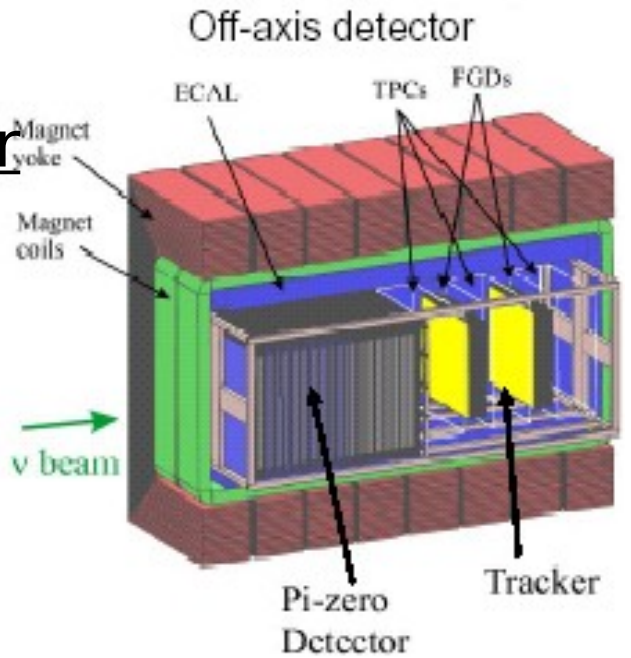
# T2K experiment

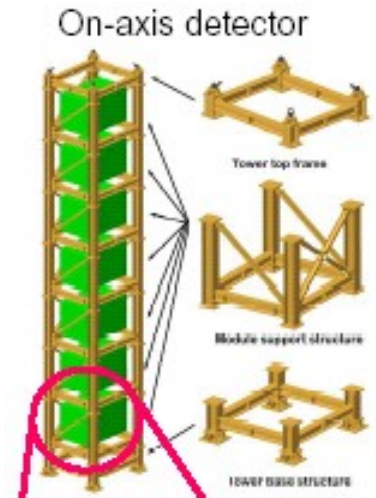
Data taking starting in april 2009




Far detector

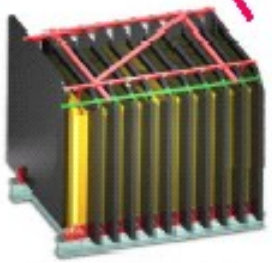
Near detector




Target

Proton beam



Iron + scintillator

**1.** Almost whole near detector composed of segmented scintillator and WLS fiber. The number of total readout channels is ~60000 (*).

**2.** Off-axis detector placed under 0.2T magnetic field (UA1/NOMAD magnet).

(*) price for 60k MPPC is 20$ each (2007)

# T2K Fine Grained Detector

F.Retiere – PD07

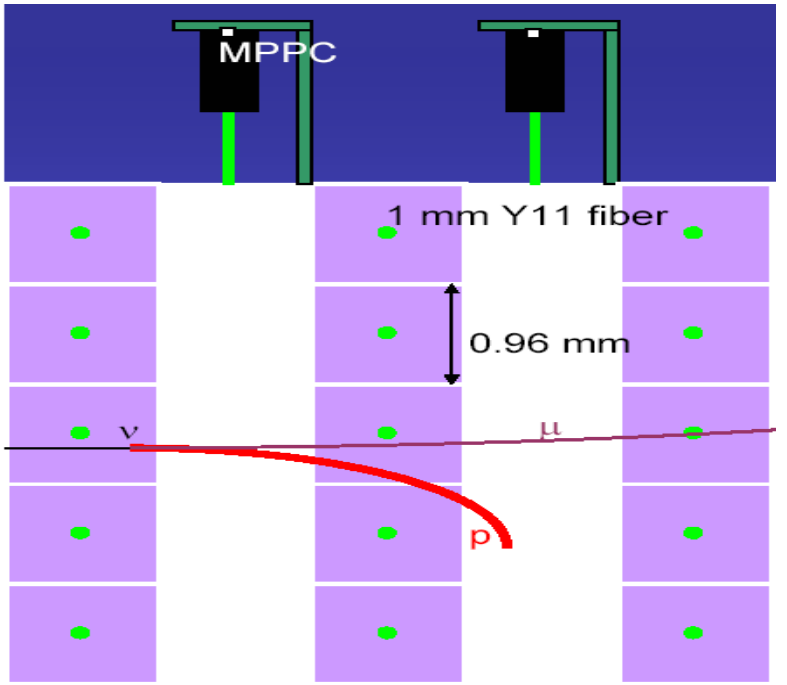
Active target for neutrino interaction:
- Plastic scintillator bar (POPOP) 2m
- Light collection with WLS
- Readout by ~10000 HPK MPPC channels

Requirements:
- 100% efficiency for MIP crossing a bar

- Particle identification:
  - dE/dx for particle crossing the FGD
  - Range (stopping p, large E release ~ 10 MIPs)
  - Detecting Michel $e^+$ for stopping $\pi^+$

- Position resolution:
bar width, no info along the bar

- Timing resolution:
~ 3ns per neutrino interaction
(matching with photons in calorimeter)




Electronics: compromise between Energy and Time measurements
→ Waveform Digitization

Switch Capacitor Array designed for TPC (AFTER ASIC):
- Slow shaper (100 ns rise time)
- 50 MHz sampling frequency
- 10 μs total integration time

# T2K Fine Grained Detector

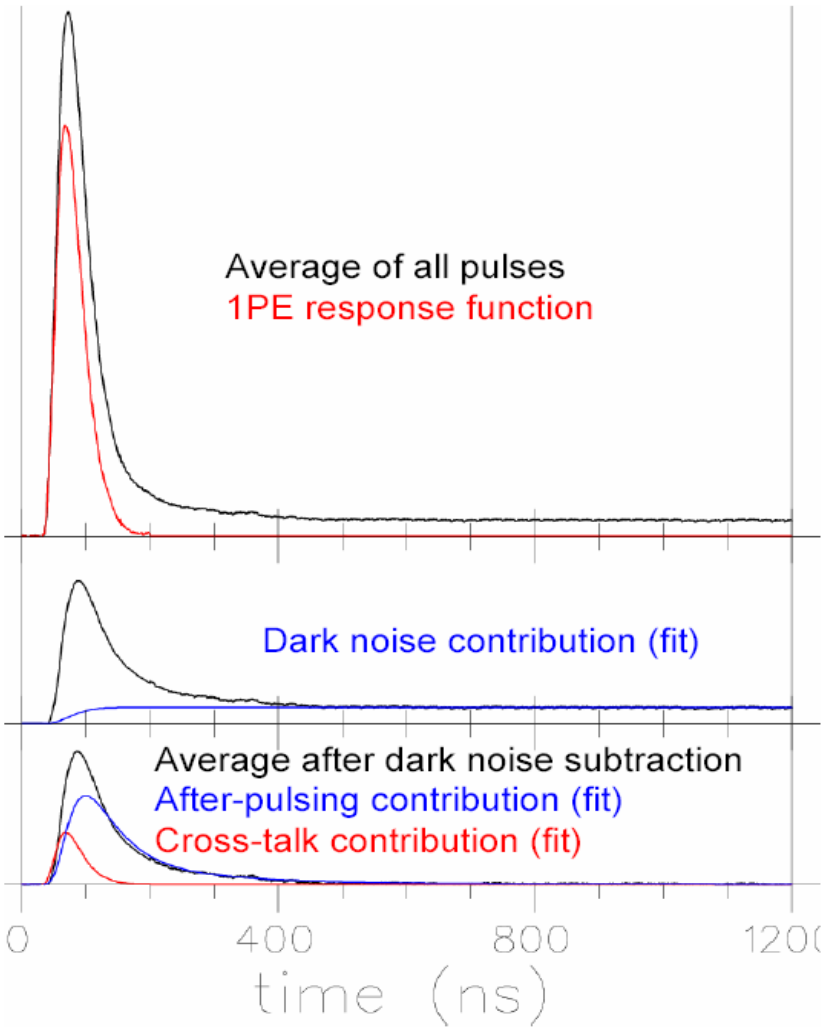
SiPM is OK for:

- High PDE → 30 photons/MIP is OK for MIP efficiency
  → 25% with 15 pe
- Dynamic range:
  400 pixels OK → up to 50 MIPs
  (nees also 1 pixel for calibration)
- Dark counts: for HPK < 500kHz:
  not an issue for both timing and E

Nuisances:

- Cross-talk: worse energy resolution + skew timing res.
- After-pulsing: worse timing resolution when fitting full waveform

→ lower over-voltage, to keep cross-talk and afterpulse prob. < 10% which is OK for FGD (alternative: use higher $R_Q$ to quench afterpulses ...)



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## Other applications

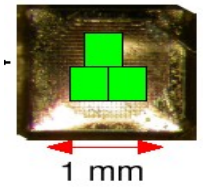
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- Non-Invasive Testing of ULSI Circuits

# PEBS balloon experiment

H.Gast – R.Yearwood
VCI07

Design: Tracker = SciFi (250μm) 55000 channels on 1770 SiPM arrays
ECAL = Pb + Scintillator Stripa + WLS + 2000 SiPM
TOF = scintillator (2x2x3mm³) + SiPM

- 2 types of SiPM Photonique tested
- 2 fibre ribbons (square fibers 300μm)
  - 3x10 fiber each
  - 3 fibers/SiPM
- Cooling with Peltier (+dry $N_2$)

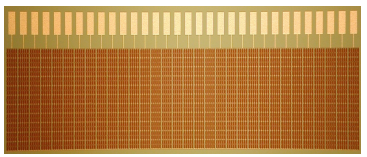


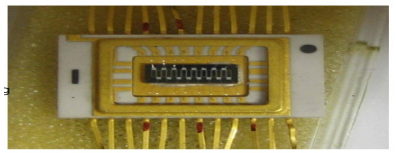

- signal: 6p.e. / MIP (w. refl. foil)
- efficiency > 95% @ S/N ~20
- position resolution ~ 90μm

- Future:

CPTA arrays of 32x1 SiPM's (0.25 x 1mm² each chan.)

and IRST arrays of 32x1 SiPM's (0.25 x 1.1 mm² each chan.)





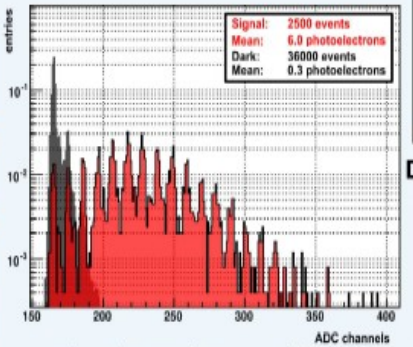
- TOF capabilities ???

## Beamtest of prototype in 10GeV proton beam

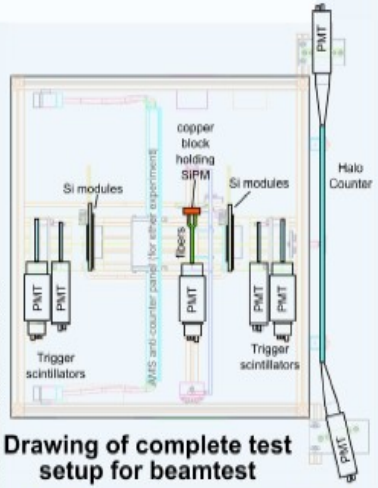
- prototype in a closed box, flooded with dry nitrogen
- 4 CMS Tracker modules as beamtelescope
- 2 fiber ribbons tested, with and without reflective foil
- tracker prototype rotated by up to 40° w.r.t. beam axis
- SiPM cooled to -10°C



T9 beamline of PS, CERN




signal spectrum and dark spectrum of single SiPM in 0° configuration, both normalized to 1




Drawing of complete test setup for beamtest

A prototype of two 3x10 fiber ribbons read out by 10 SiPM of type SSPM 0606EXP with 35% photodetection efficiency and 9 SiPM of type SSPM 050701GR with approximately 20% photodetection efficiency for scintillation light from the fibers.

The copper block around the SiPM was cooled to up to -16°C. The complete fiber prototype could be rotated in 10° steps to up to 40° tilt with respect to the beam axis.

A beam telescope was built from 4 silicon modules from the CMS tracker project. The Silicon modules were read out using the ARC system and provided a spatial resolution for incident particles better than 40μm.

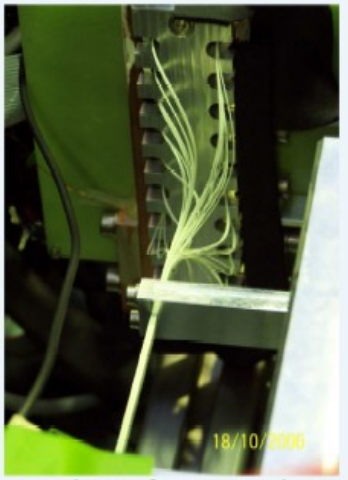


photo of prototype in beamtest setup

# PEBS balloon experiment



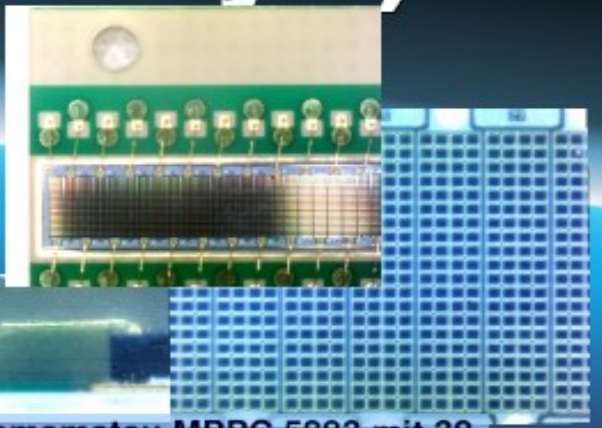
11

## SiPM(-Arrays) im Test





Hamamatsu S10362-11-100C MPPC



Hamamatsu MPPC 5883 mit 32 Kanälen



IRST SiPM-Array auf Hybrid gebondet



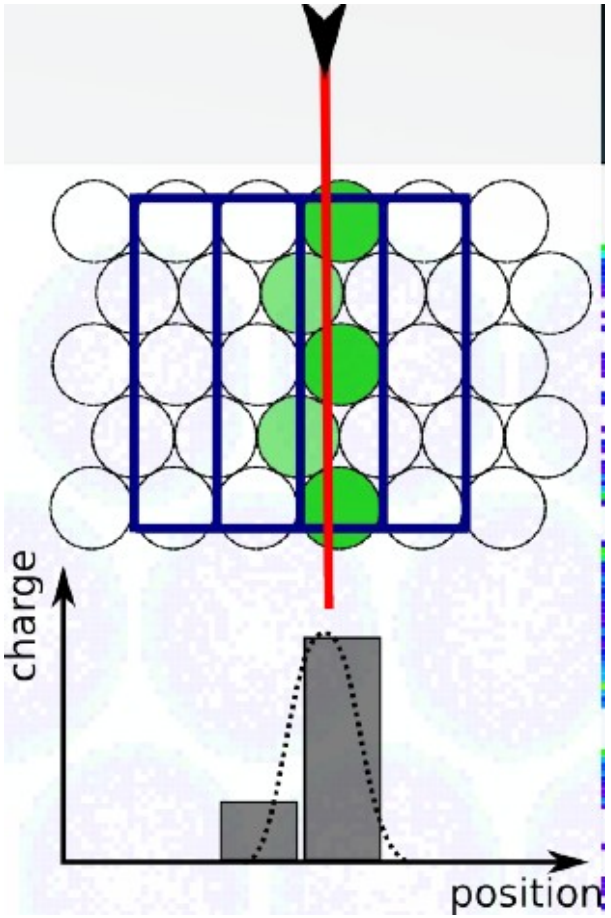
IRST SiPM-Array

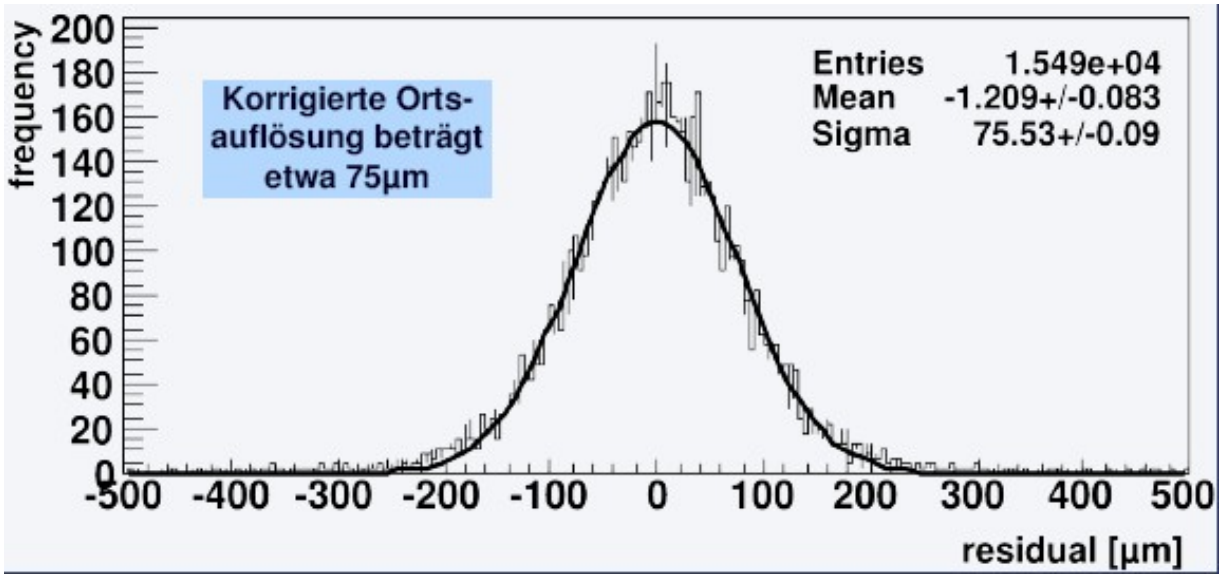
| Produzent | Hamamatsu | Hamamatsu | IRST-ITC |
|---|---|---|---|
| Typ | MPPC S10362-11-100C | MPPC 5883 | |
| Kanäle | 1 | 32 | 32 |
| Pixel | 100 | 32 x 80 | 32x110 |
| Aktive Fläche | 1mm x 1mm | 32 x 0.22mm x 1.1mm | 32 x 0.23mm x 1.1mm |
| Betriebsspannung | 70V | 70V | 32V |
| | | | |
| Photoneffizienz | 40% @ 500nm; 60% @ 430nm | ? | 20% @ 500nm |
| 1-Photon-equivalentes Rauschen | 600kHz | 300kHz | 1MHz |
| Pixelcrosstalk Wahrscheinlichkeit | 0,58 | 0,3 | 0,1 |

# PEBS balloon experiment



Yield of ~10p.e./MIP obtained with
0.25mm Kuraray fiber & Hamamatsu SiPM array




Position resolution
of 75 micron achieved

Latest results at this conference by R.Yearwood

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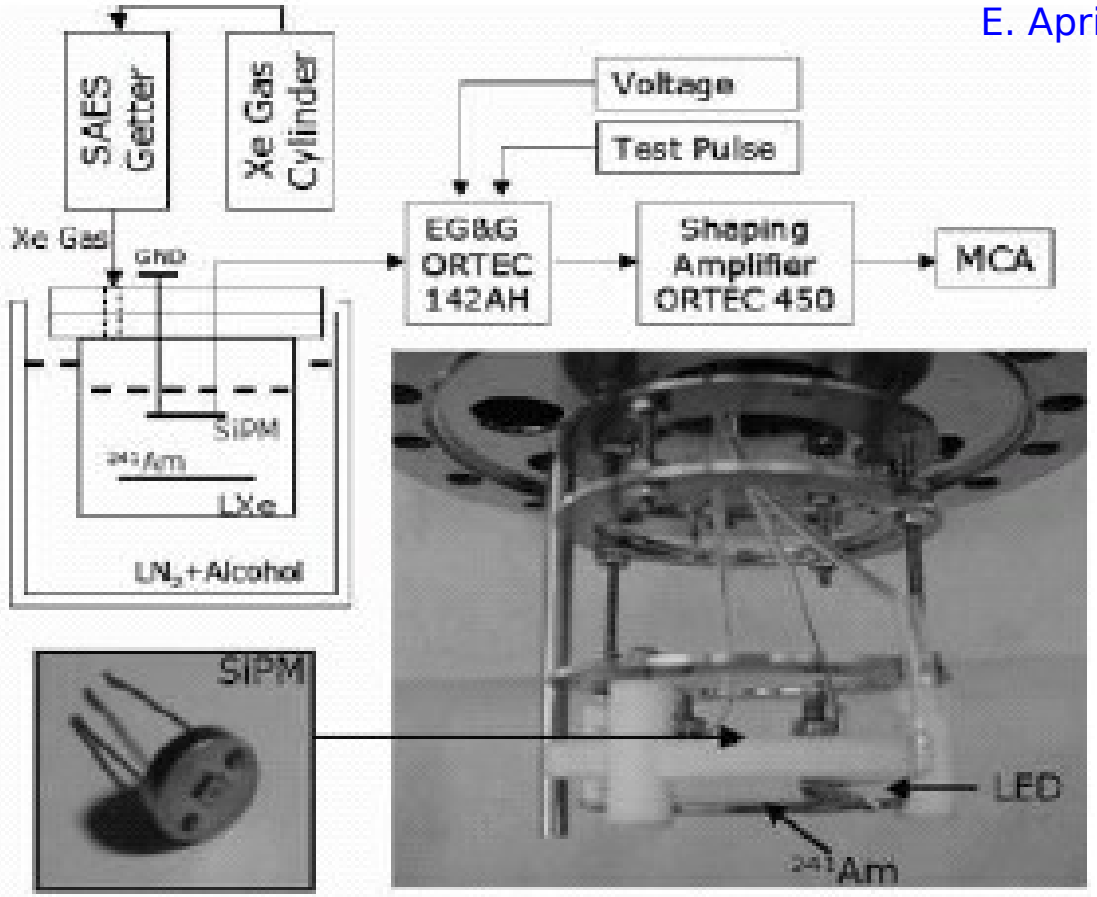
## Medical Physics

- Medical imaging (PET + TOF + MRI)

## Other applications

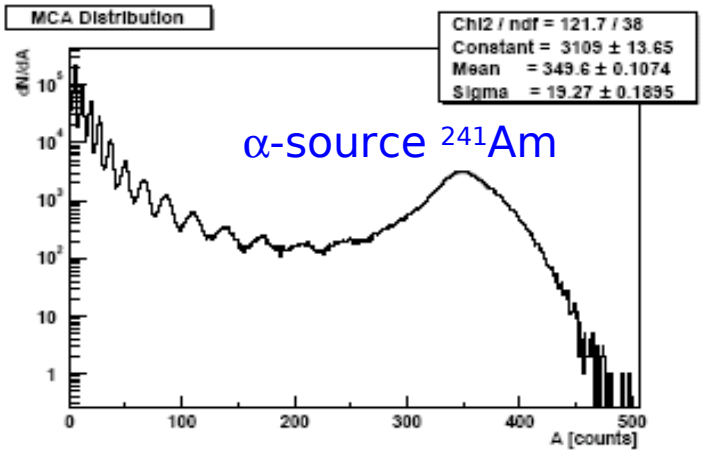
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# Deep UV detection: Liquid Xenon detector




E. Aprile, P. Cushman, K. Ni, P. Shagin

Liquid Xenon: T = -95°C
$\lambda$ = 178 nm




PDE ~ 5.5% ($\lambda$ = 178 nm)
typical PMT applied in TPC for WIMP searches PDE ~20%

576 pixels SiPM, MePHI/Pulsar

- Unexpectedly high PDE to be understood !!!
- Attractive alternative to PMT for UV photon detection at low energy detection threshold
- Would be great for Dark Matter searches !

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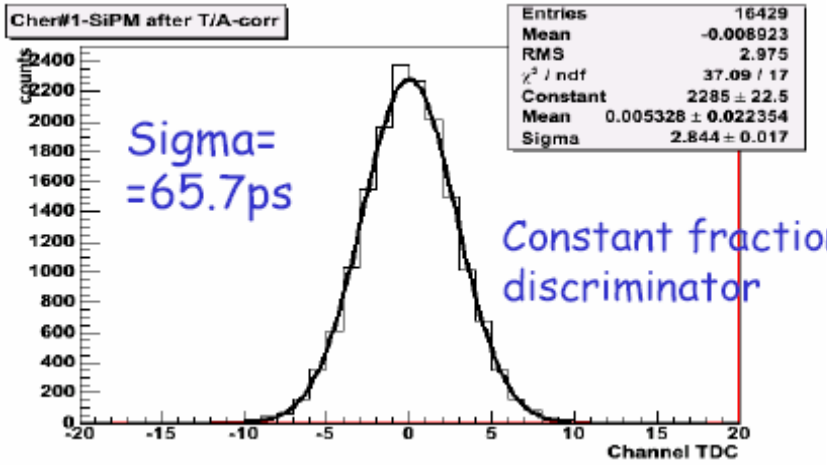
# Timing with fast scintillator and SiPM

Dolgoshein
Beaune05

SiPM 3x3 mm$^2$ (5600 cells) glued directly to BICRON-418 scintillator 3x3x40 mm$^3$
Signal is readout directly from SiPM without any preamp and shaper !

Timing between:
- PMT(FEU 187) + Cherenkov radiator
- SiPM + BC418




Results: $\sigma_{(PMT+Ch.rad)}$=49ps ; $\sigma_{(electronincs)}$=32ps

$\rightarrow \sigma_{(SiPM+BC418)}$=33ps

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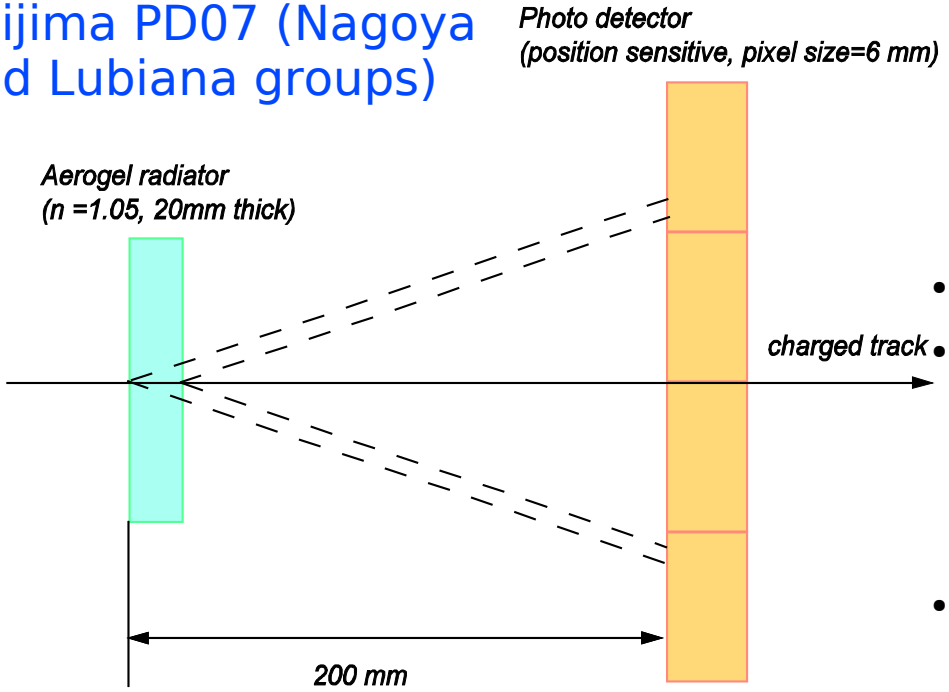
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## Other applications

# Proximity Focusing Aerogel RICH + TOF

T.Iijima PD07 (Nagoya and Lubiana groups)

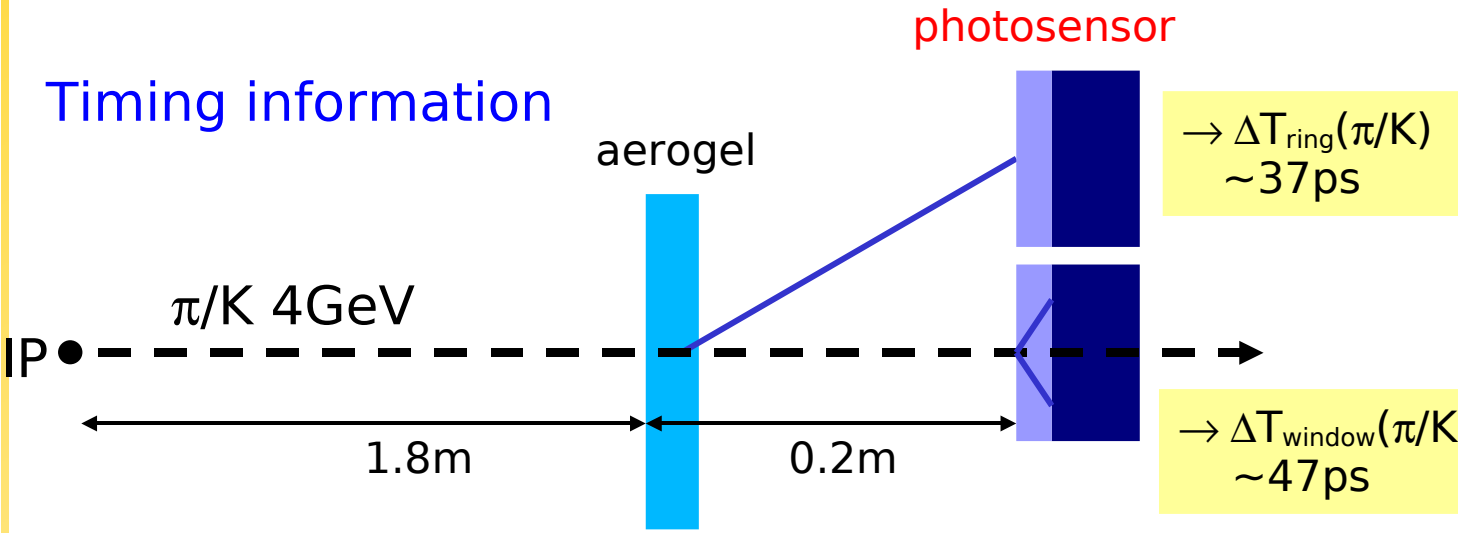


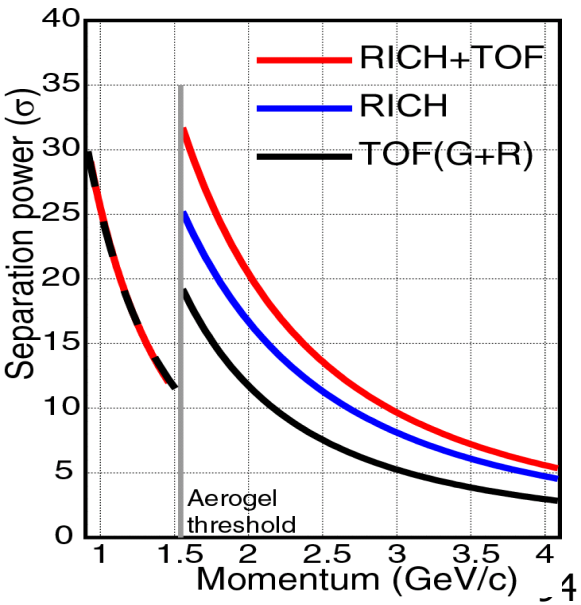

- Proximity focusing geometry
  - aerogel radiator (n~1.05, ~2cm)
  - no mirror complex.
  - suitable for collider and space experiments
- $>4\sigma$ K/$\pi$ for $0.7 < p < 4.5$ GeV/c
- Rayleigh scattering dominates in aerogel.

  → position measurement of single photons in the visible wave length region.
- Timing measurement

  → TOF and noise rejection by coincidences

Timing information



# Possibility of SiPM for RICH

Korpar et al (Lubiana and Nagoya groups)

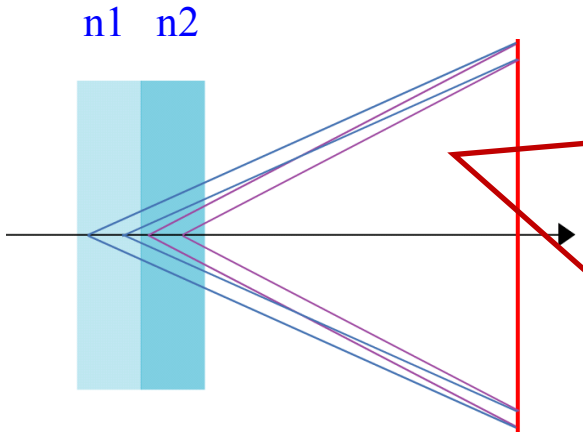
Addressing the problems of noise and size (*):

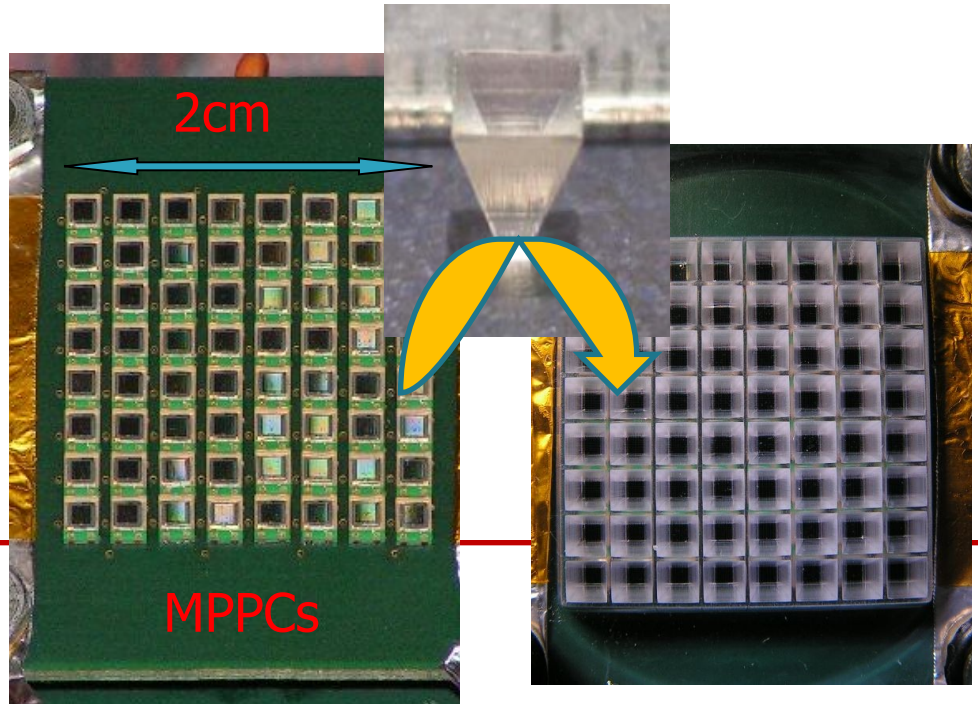
- light collectors (with optimization of pad size to the ring thickness)
- light collectors bonus: serves as a light emitter for TOF measurement
- time coincidences by a narrow time window (<10ns).

SiPM + Optics (Light-guide. Lens)

MPPCs: array of 8x8 SMD mount Hamamatsu S10362-11-100P with 0.3mm protective layer

Aerogel-RICH



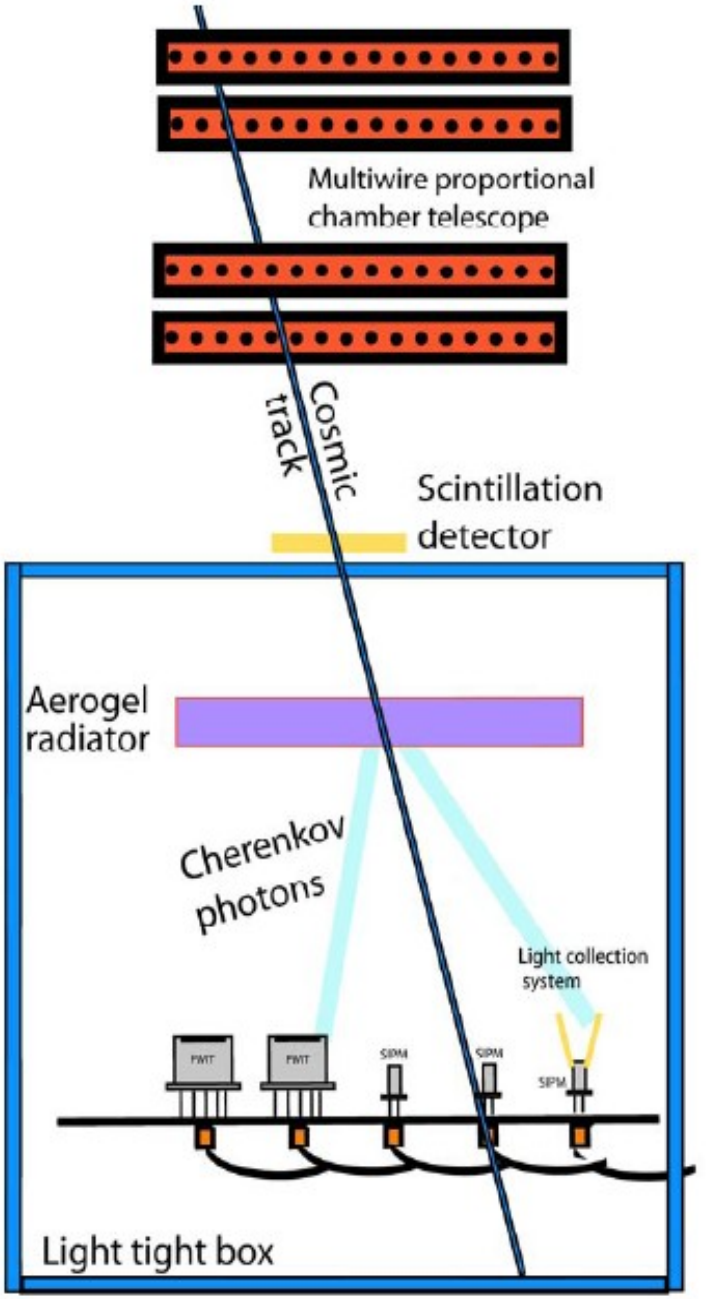



# Cherenkov tests with cosmic rays

P.Krizan et al (Lubiana and Nagoya groups)

Photon detector:

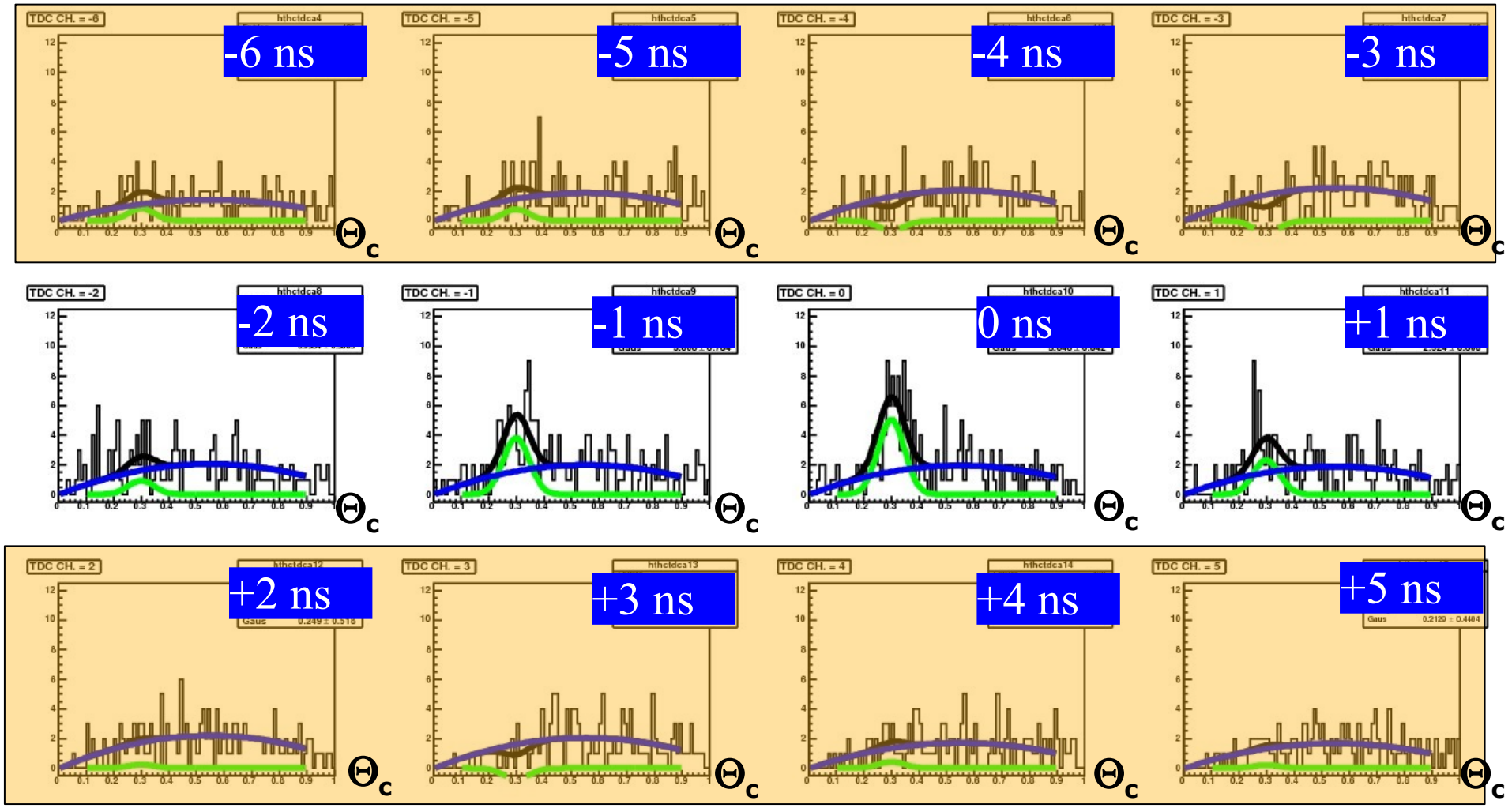
- Array of 6 SiPMs
- Array of 12 R5900-M16 PMTs as reference



# Cherenkov tests with cosmic rays

## SiPM: Cherenkov angle distributions for 1ns time windows

P.Krizan (INSTR08 - Novosibirsk 2008)




Cherenkov photons appear in the expected time windows !!!

See also detection of Cherenkov light by M.Kim at this conference

# Conclusions

- Device physics is understood and under control
- Device technology is in continuous evolution in many countries
- Detector now is stable and reliable, ready for many applications
- Already at work in Calorimetry and scintillator counters (TOF and tracking). Recently first results on Cherenkov light

## Particle High Energy, Space, Astro - Physics

- Calorimetry in magnetic fields
- Active target for neutrino experiments
- Fiber tracking (+ TOF)
- Fast timing, TOF (scintillators, cherenkov)
- Proximity focusing RICH, Cherenkov Differential counter
- Liquid Xenon scintillation
- Telescopes: Cherenkov and Fluorescence light

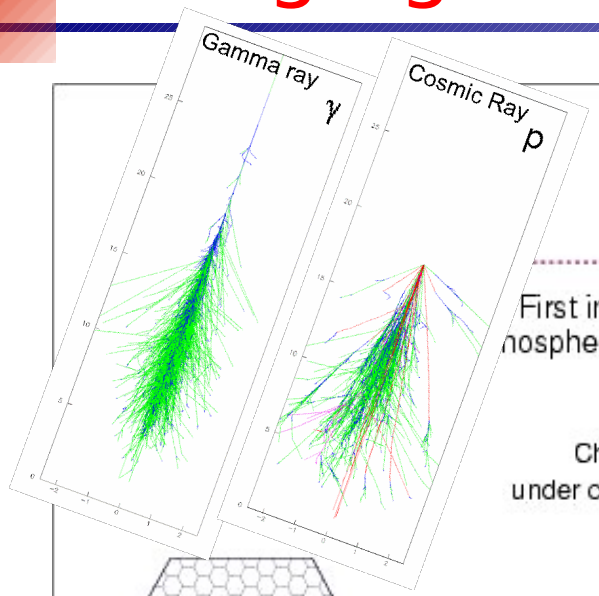
## Medical Physics

- Medical imaging (PET + TOF + MRI)

## Other applications

- Quantitative biological and molecular recognition: Fluorescence spectroscopy, Confocal microscopy, Flow Cytometry, Micro-Fluidics
- Adaptive optics, Tracking of luminous sources, LIDAR
- X-ray and neutron detection/monitoring, Homeland security
- Non-Invasive Testing of ULSI Circuits

# Imaging Air Cherenkov Telescopes (IACT)




Primary particle:

First interaction with nuclei of
nosphere at about 20 km height

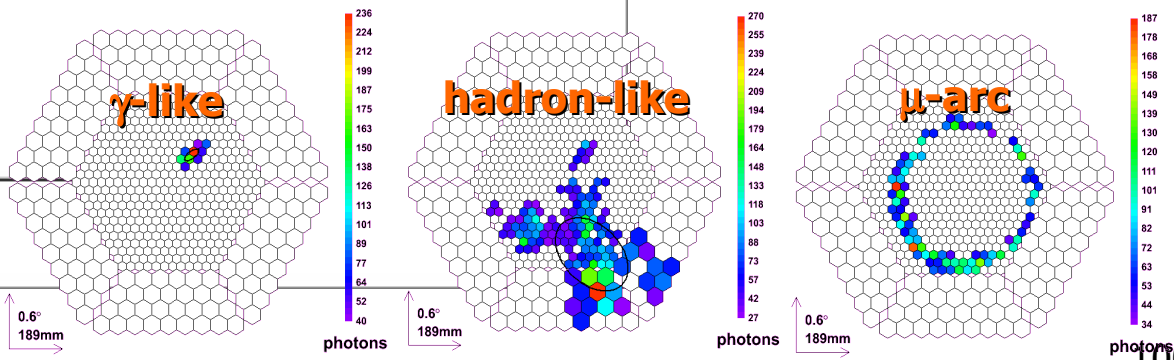
Cherenkov light emission
under characteristic angle $\theta_C$

~ 5 km

~ 20 m

Camera

Cherenkov
Telescope




Pixelization is important !

## Single photon applic.
## UV sensitive SiPM

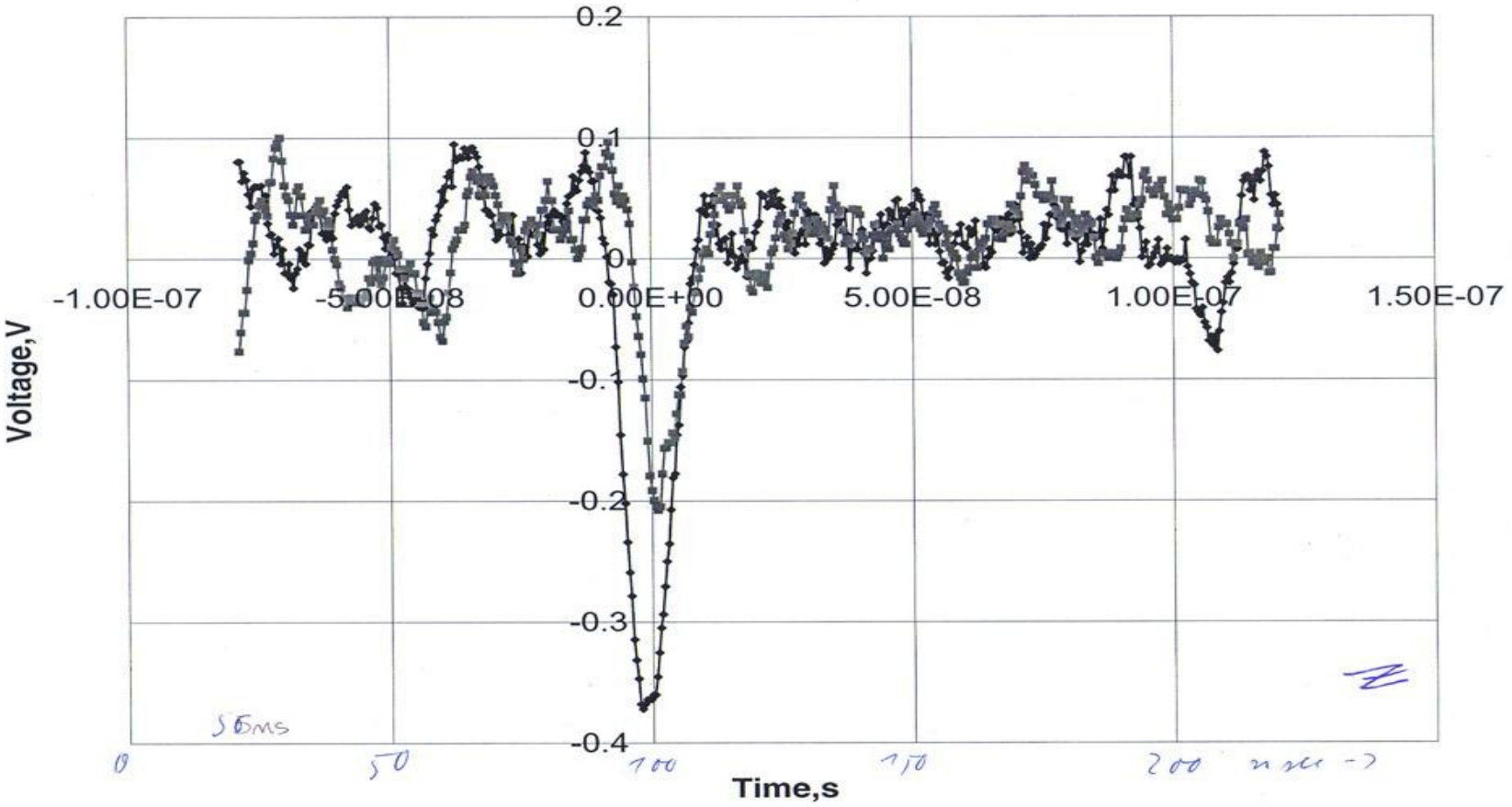


SiPM for the MAGIC camera

- better PDE (wrt PMT)
  lower E threshold (30 GeV !)
- can be exposed to light
  (extend data taking to
  moonlight nights)
- dark NOISE lower than
  dark sky noise: no cooling

# IACT, test #1: SiPM on MAGIC camera

16 G-APDs (Photonique SSPM_0606BG4MM, area 4.4 mm$^2$ each, peak sensitivity at 580 nm) were mounted in 4 groups of 4 G-APDs each in the focal plane of the MAGIC telescope on the side of the camera. The signals of the 4 G-APDs in each of the groups were added and gave thus in total only 4 output signals. Each group covered an area of 18 mm$^2$. With one of the signals a digital oscilloscope was triggered and the output of two other groups were recorded. The fourth group was excluded because of amplifier instability. The well defined timing and the narrowness of the signals indicate that Cherenkov light has been recorded.



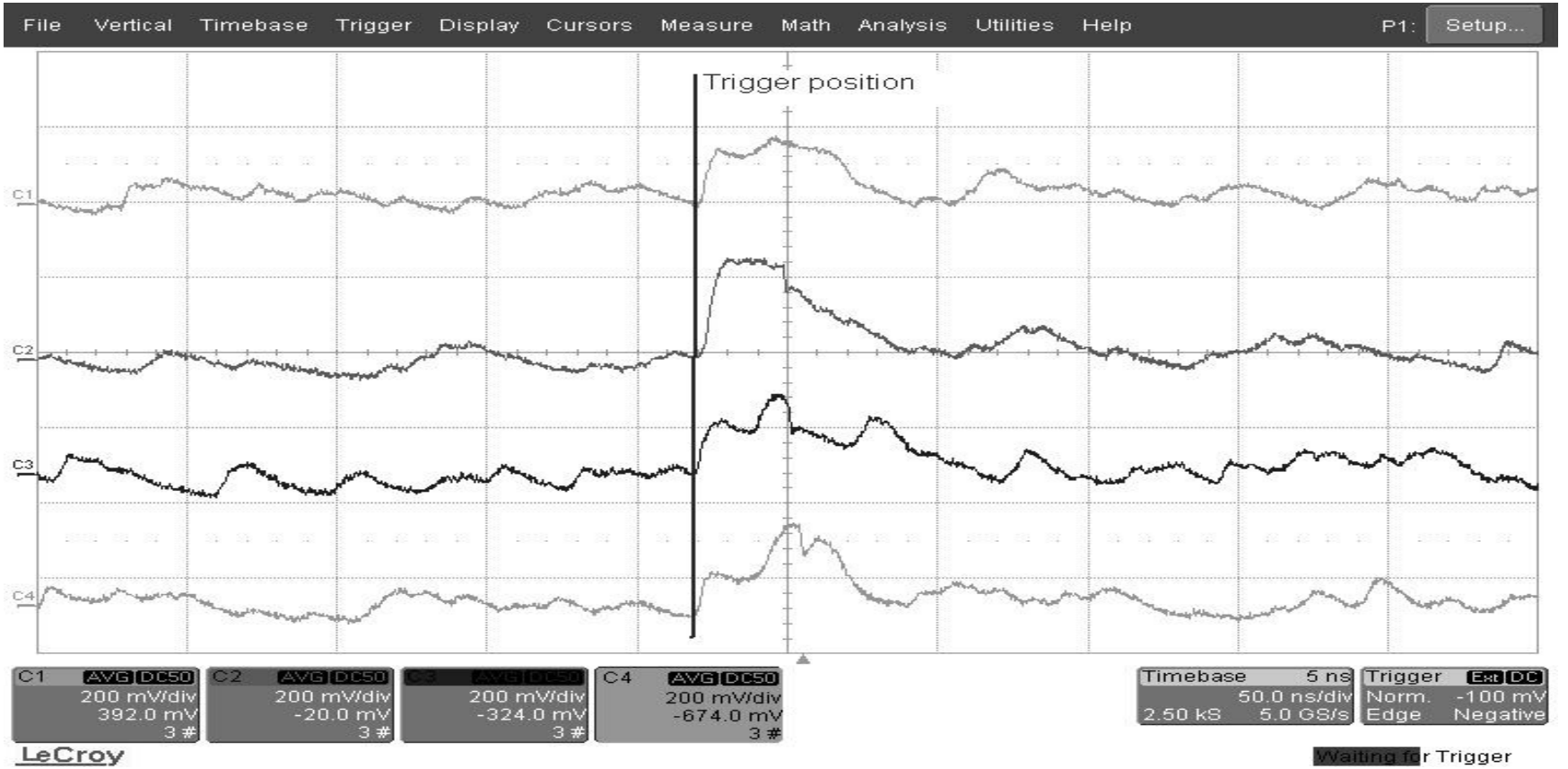

D.Renker et al - VCI07

# IACT, test #2: direct sky observation

Hamamatsu SiPMs (better blue sensitivity) were grouped in an array of 4 detectors ( 9 $mm^2$ each).
Two PMT (+Winston cones) were mounted at two sides of the array to trigger (coincidence) a scope.
The assembly viewed the night sky around the zenith.

The threshold for the PMT signals was set to 8 photo electrons within < 5 nsec.
Thus selected cosmic showers with a photo density of 10^26 photons/cm2,
from ultrahigh energy (greater than 500 TeV) cosmic events within 1 sterad.

The trigger rate was about 1 event per 5 min
(~ 50% accidentals + 50% genuine events from air showers)
which is in accordance with the measurement of the former Airobicc experiment.



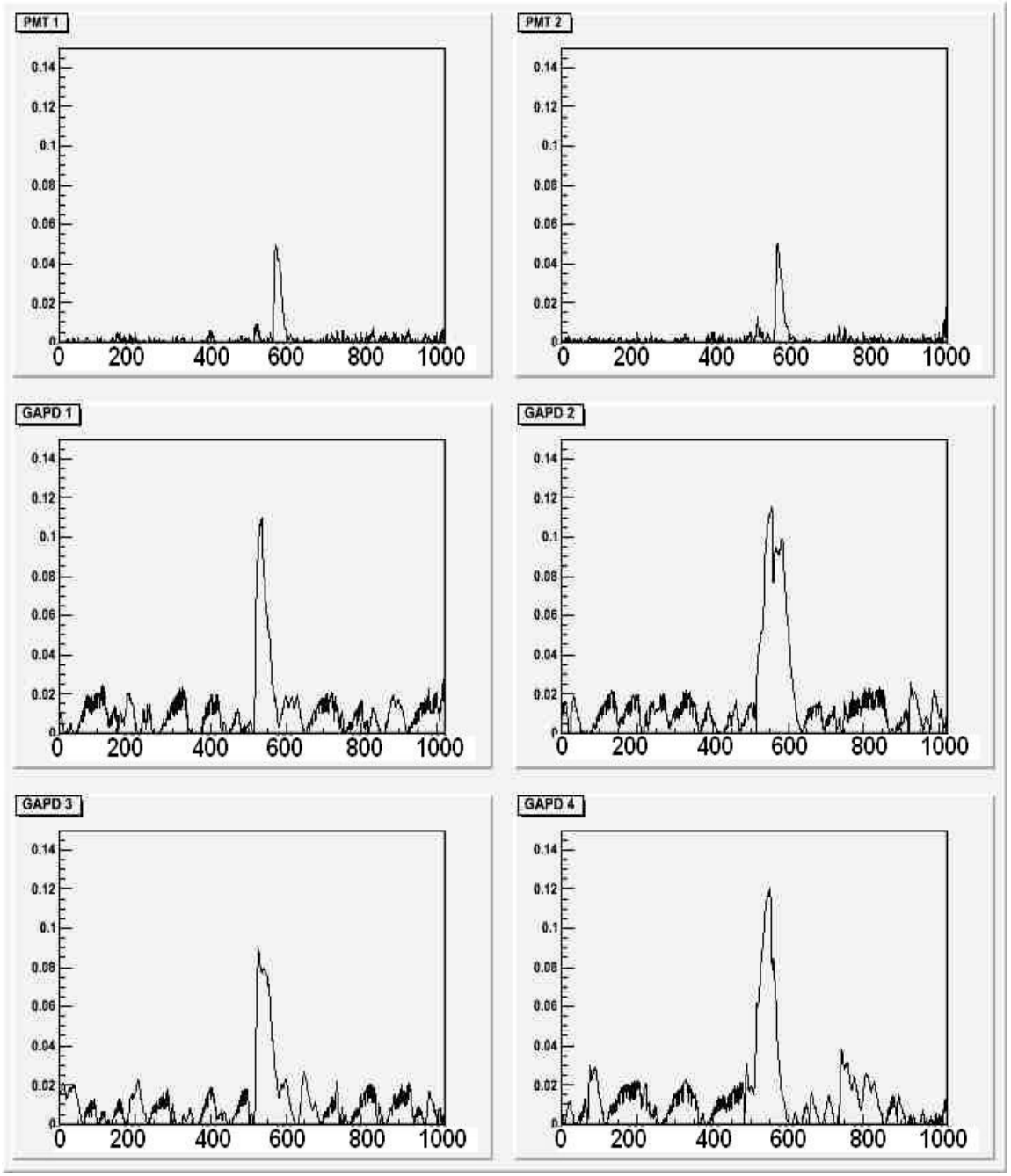
# IACT, test #3: solar concentrator at PSI

The assembly of test #2 was mounted in the focal plane of the solar concentrator at the PSI. (a planar (120 m2) and a parabolic mirror with 8.5 m diameter).

The Domino Ring Sampler with a 2 GHz sampling frequency was used to readout the 6 channels. The trigger threshold was lower (6 p.e.)

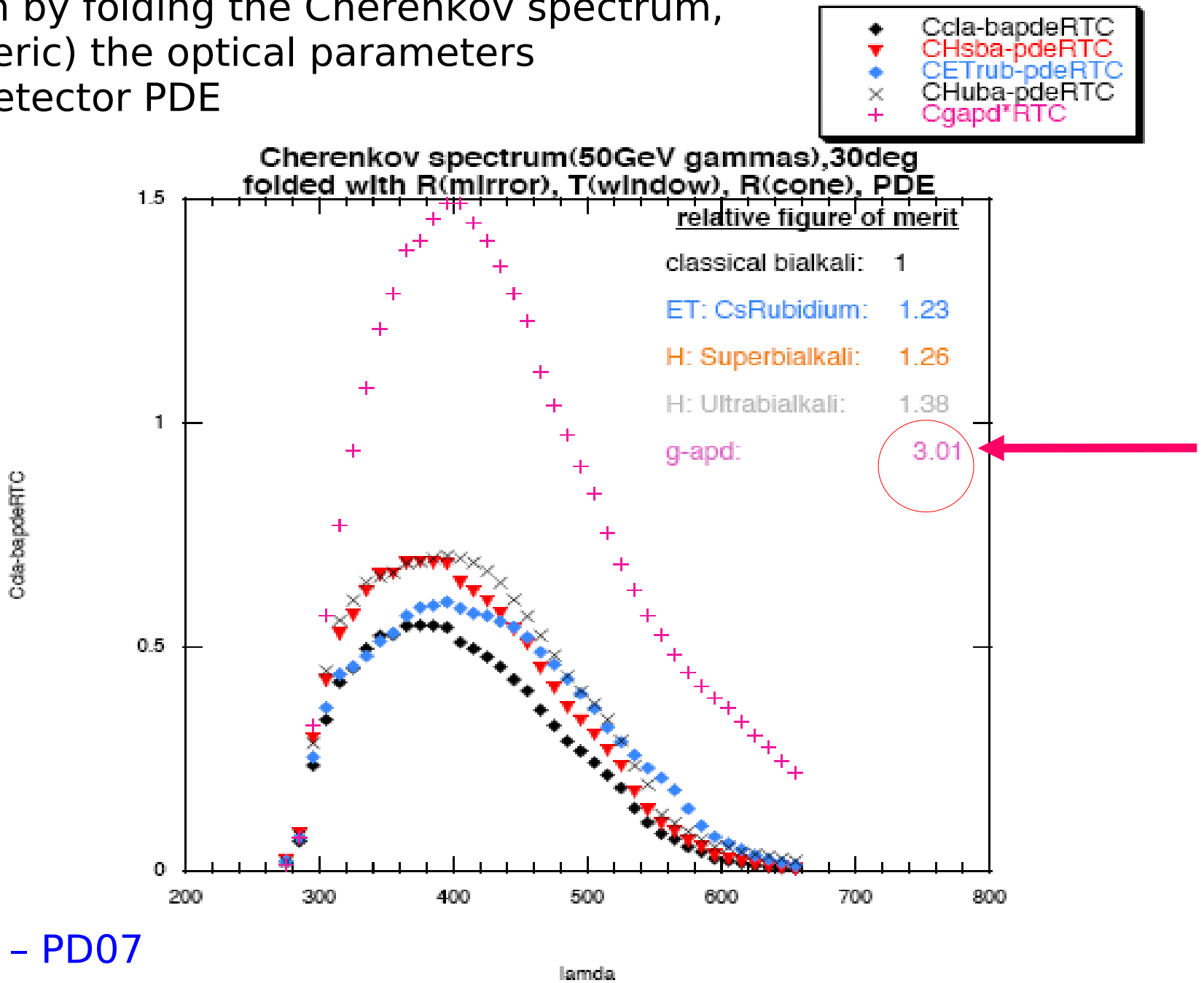
For all large light flashes seen in the PMTs always genuine signals were observed in all SiPM groups.

D.Renker et al - VCI07



# Figure of merit of different sensors

Evaluation by folding the Cherenkov spectrum, (atmospheric) the optical parameters and the detector PDE




E.Lorentz – PD07

# Improvements of SiPMs for γ-ray astronomy

Cherenkov light from air showers can be detected with SiPM with > x2 gain wrt flat window, standard bialkali PMTs.

Further improvements to come:

- Widening of high PDE spectral range
- Adding WLS in plastic coating to enhance UV sensitivity
- Optical filters with transmission between 300 and 700 nm to get rid of IR night sky light
- Micro-lenses or micro-light-catchers to overcome dead area between cells → higher PDE
- Wide area SiPMs ! 10x10 $mm^2$ MPPC with 100x100 $\mu m^2$ cell size (w/o degradation in rise time)

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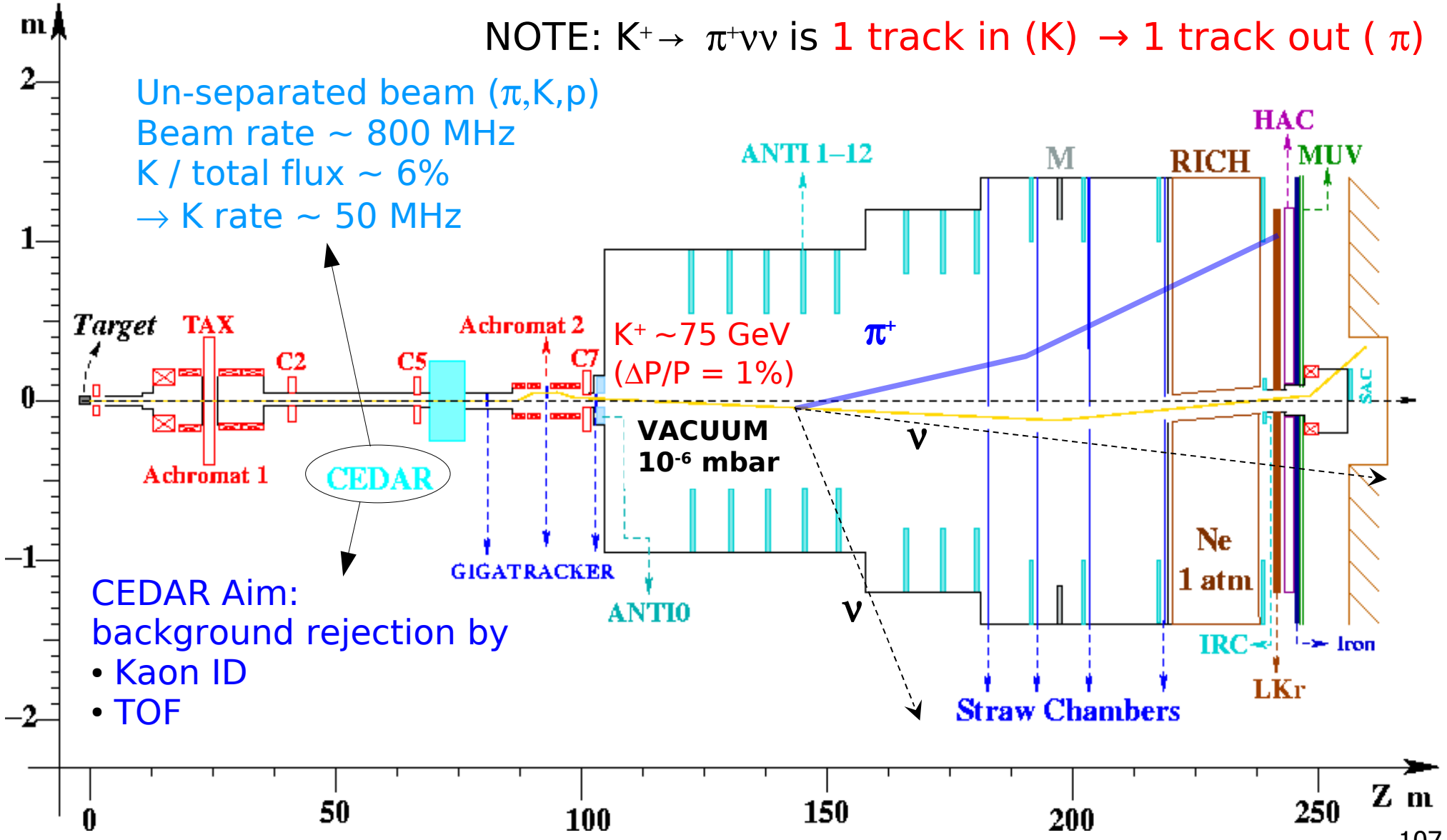
# Experiment NA62 at CERN

Goal: to measure O(100) ultra rare decays $K^+ \to \pi^+\nu\nu$ at the CERN SPS

Experimental method: K decay in flight in vacuum

Beam: High energy, monochromatic, high intensity

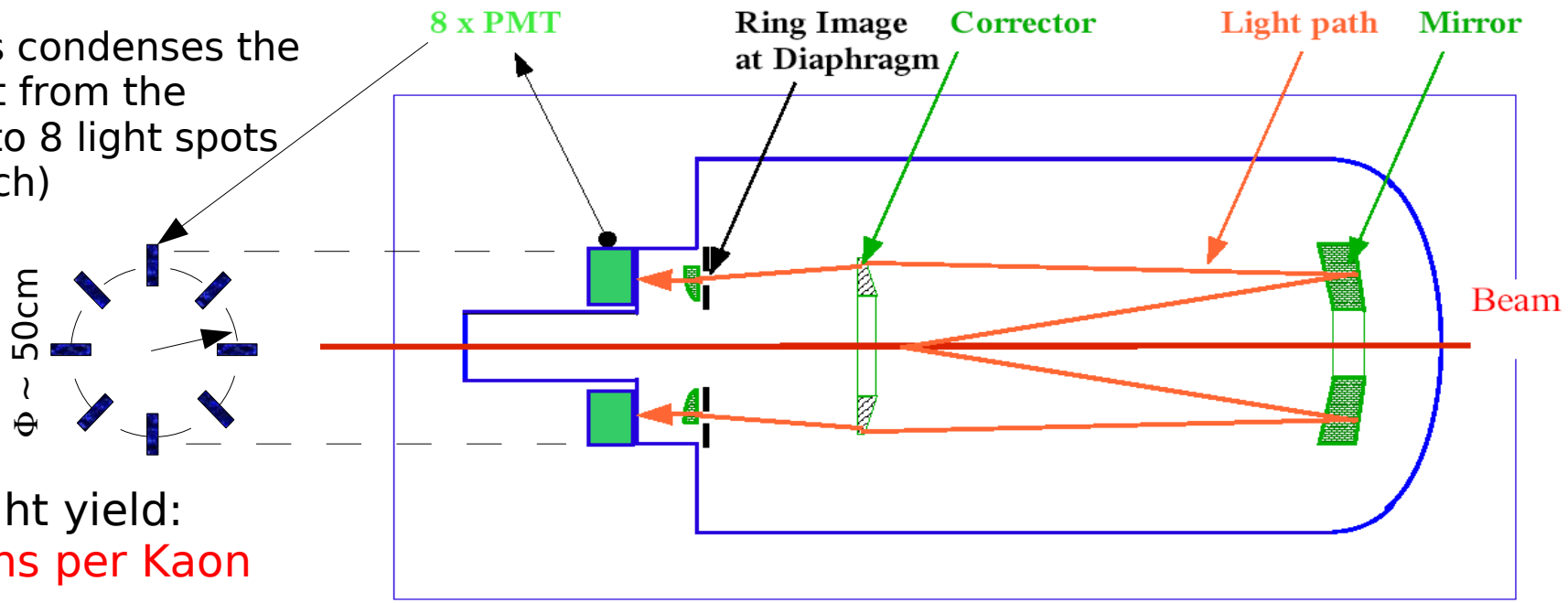
NOTE: $K^+ \to \pi^+\nu\nu$ is 1 track in (K) → 1 track out ( $\pi$)



# The CEDAR detector at CERN



Current optics condenses the cerenkov light from the diaphragm into 8 light spots (~1x3 cm$^2$ each)



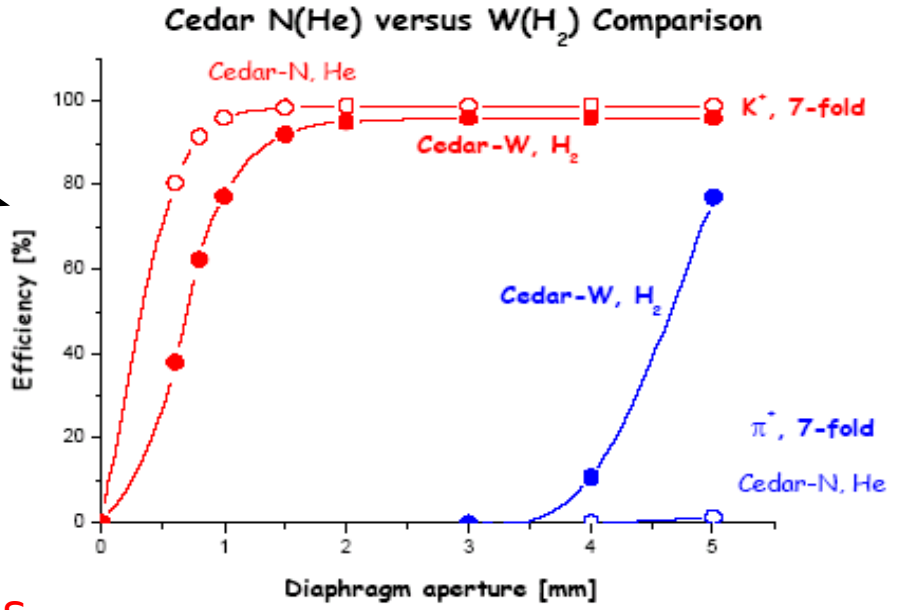

Cerenkov light yield:
~100 photons per Kaon

NA62 requirements:
Minimal material budget: use $H_2$ (2.6 bar)
Kaon tag efficiency > 95%
Contamination < $10^{-6}$
Pion rate ~ 800MHz
Kaon rate ~ 50MHz

→ photon rate ~ 2 MHz / mm$^2$

~ x10 the limit of the old readout

Need new read-out detector and electronics



## Particle High Energy, Space, Astro - Physics

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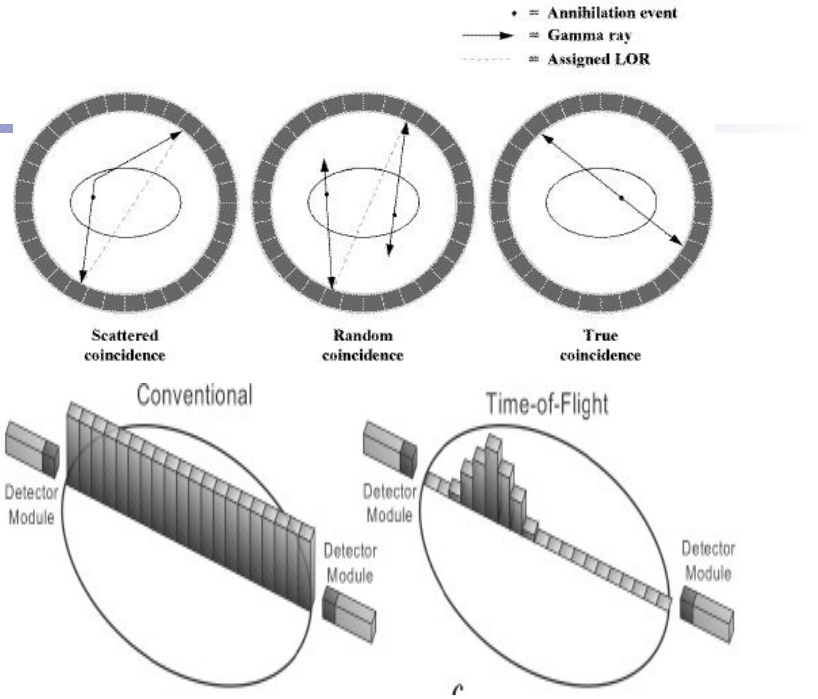
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- Medical imaging (PET + TOF + MRI)

## Other applications

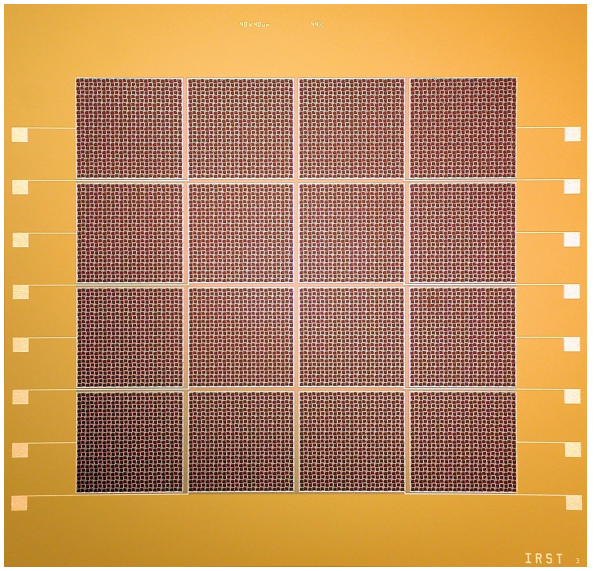
- Quantitative biological and molecular recognition: Fluorescence spectroscopy, Confocal microscopy, Flow Cytometry, Micro-Fluidics
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- X-ray and neutron detection/monitoring, Homeland security
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# Medical imaging (PET)




Aims: PET detectors with
- high spatial resolution (sub-millimeter)
- high sensitivity
  (low dose and high signal/background ratio)
- high time resolution
  (TOFPET → background rejection)
- DOI capability (no or → less parallax)
- no sensitivity to magnetic fields, EM pickup and RF (simultaneous NMR scan)

Key issues:
- Granularity: matrices of SiPM

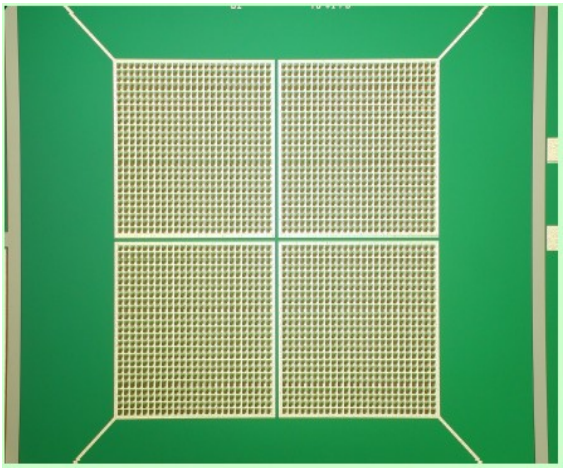


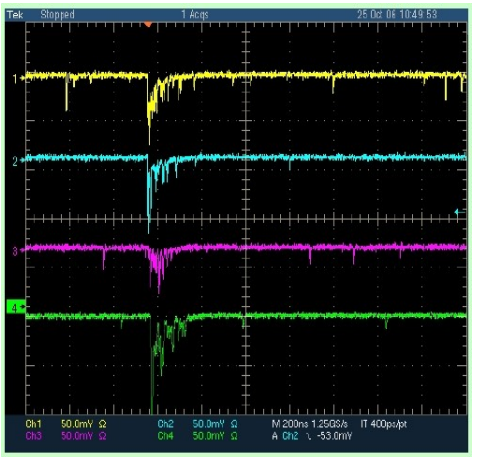
- High PDE for short wavelengths (420nm):
  - for coupling to high light yield crystals (scintillators)
  - max E resolution @ high efficiency to reject background Compton scattering (threshold)

- Optical coupling with scintillator

- Dynamic range and recovery time: multi cell signal saturation and fluctuation
- Gain stability with V and T: individual control of O(10000) channels

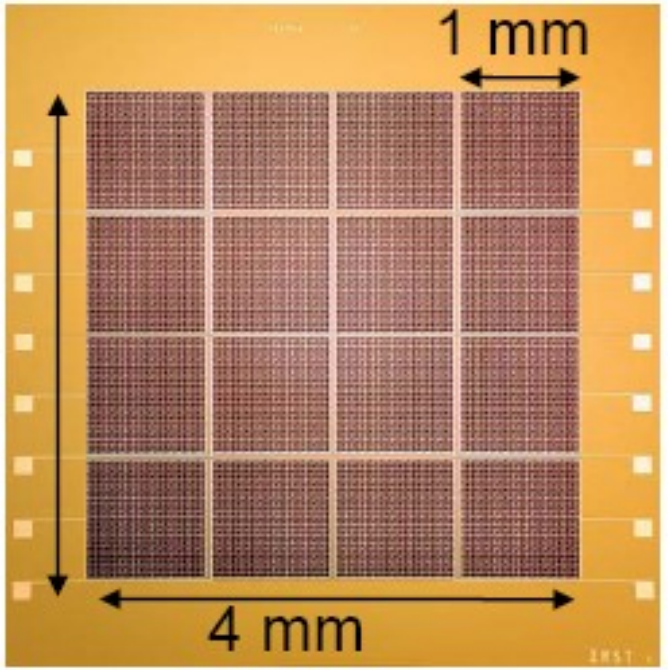
# Matrices of SiPM - IRST

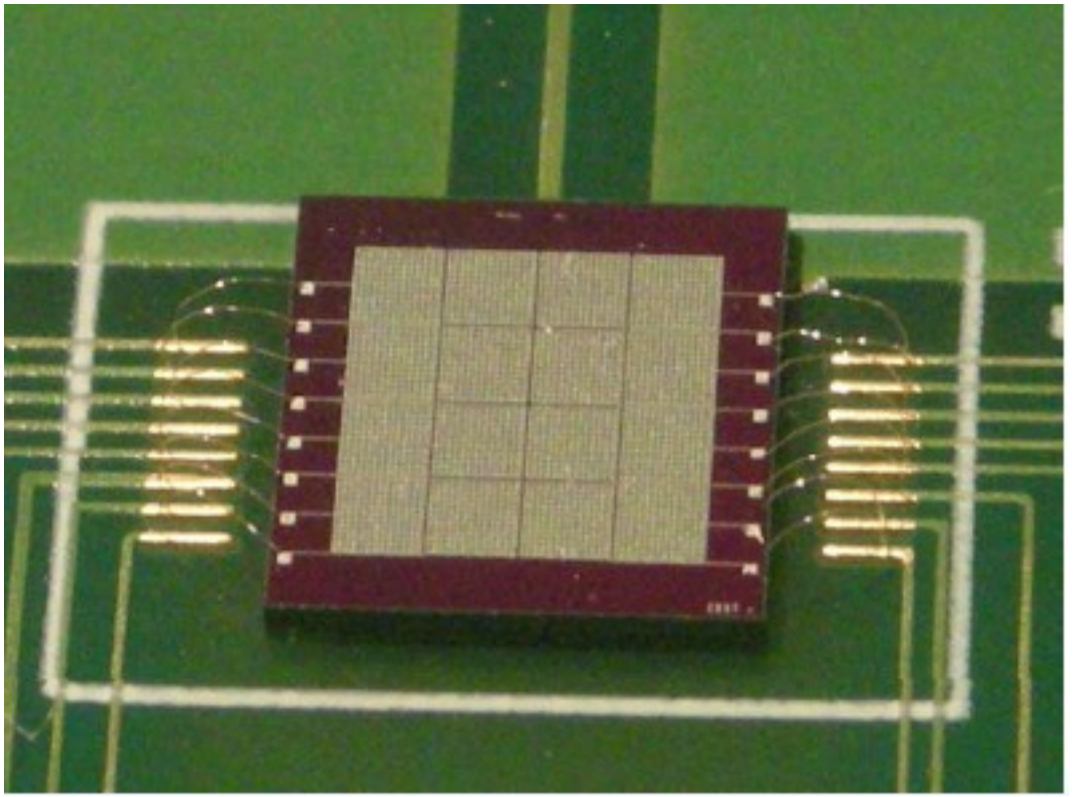




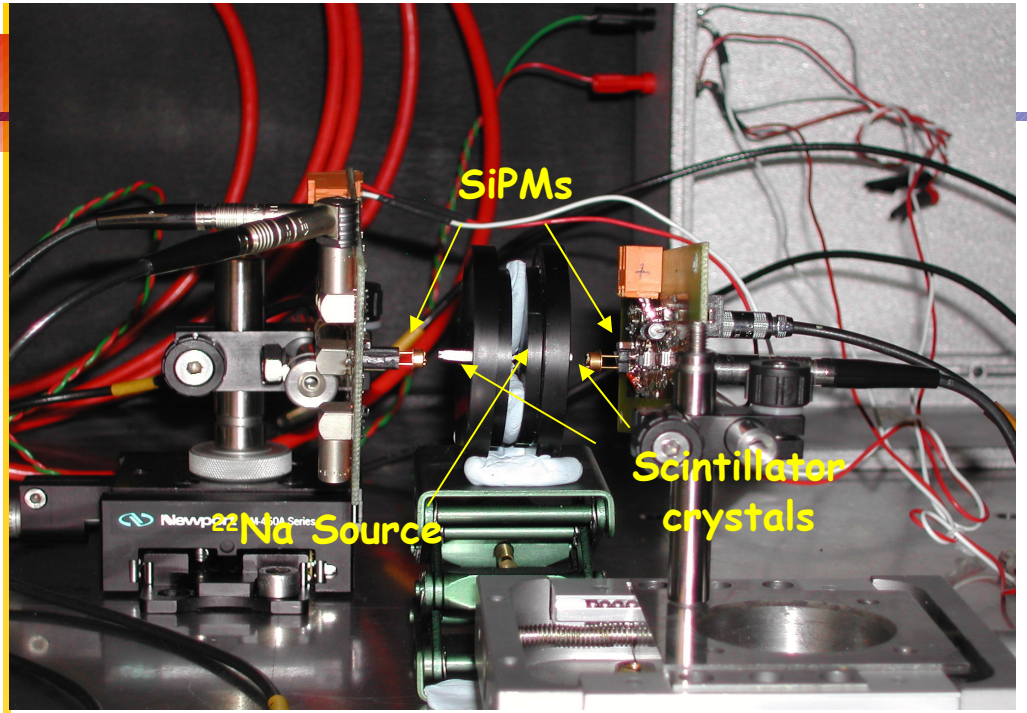
The first matrices of 2x2 blue sensitive SiPMs have been developed (IRST) since Y2006

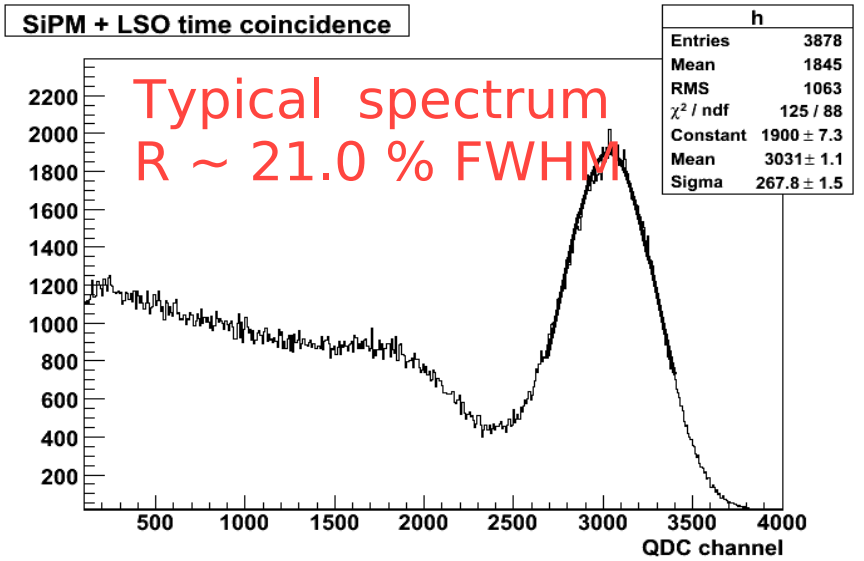
G.LLosa et al. IEEE NSS 2006 CD record M06-88



G.LLosa et al. NDIP08



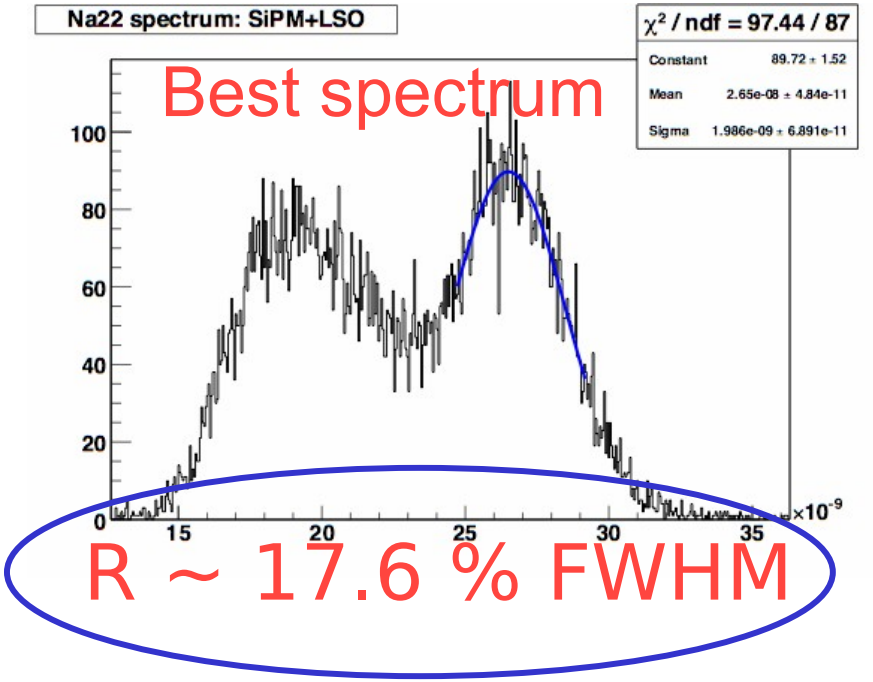




A $^{22}$Na spectrum was obtained with a 1 mm x 1 mm x 10 mm LSO crystal coupled to a SiPM ($\varepsilon_{geom}$~30.9%)

Two devices were operated in time coincidence. A typical energy resolution of 21% FWHM was obtained.

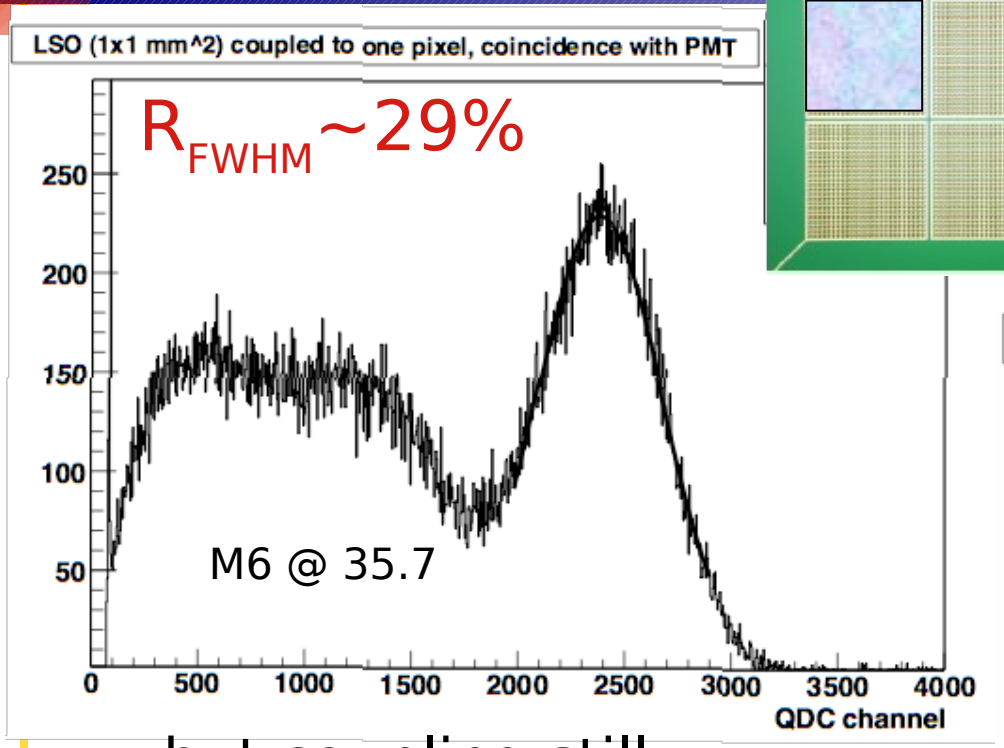
World best resolution w/ LSO (3x3x20) and PT XP2020 10%(FWHM) [intrinsic 8.9% at 511 keV]

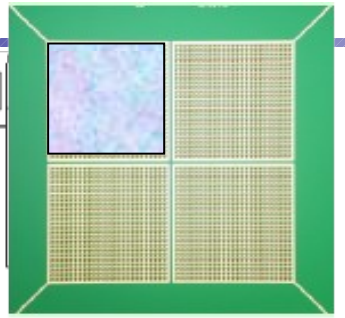
Balcerzyk et al., IEEE TNS 47(2000)1319



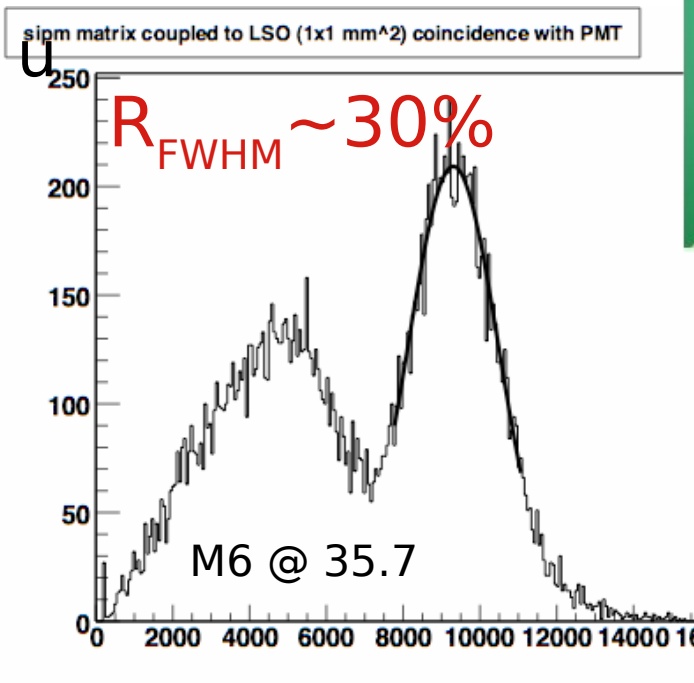

DASIPM-2 collaboration

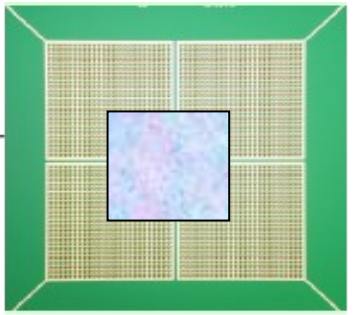
# Scintillator readout with SiPM matrices



- LSO crystal (1x1 mm²)
- time coincidence with a PMT



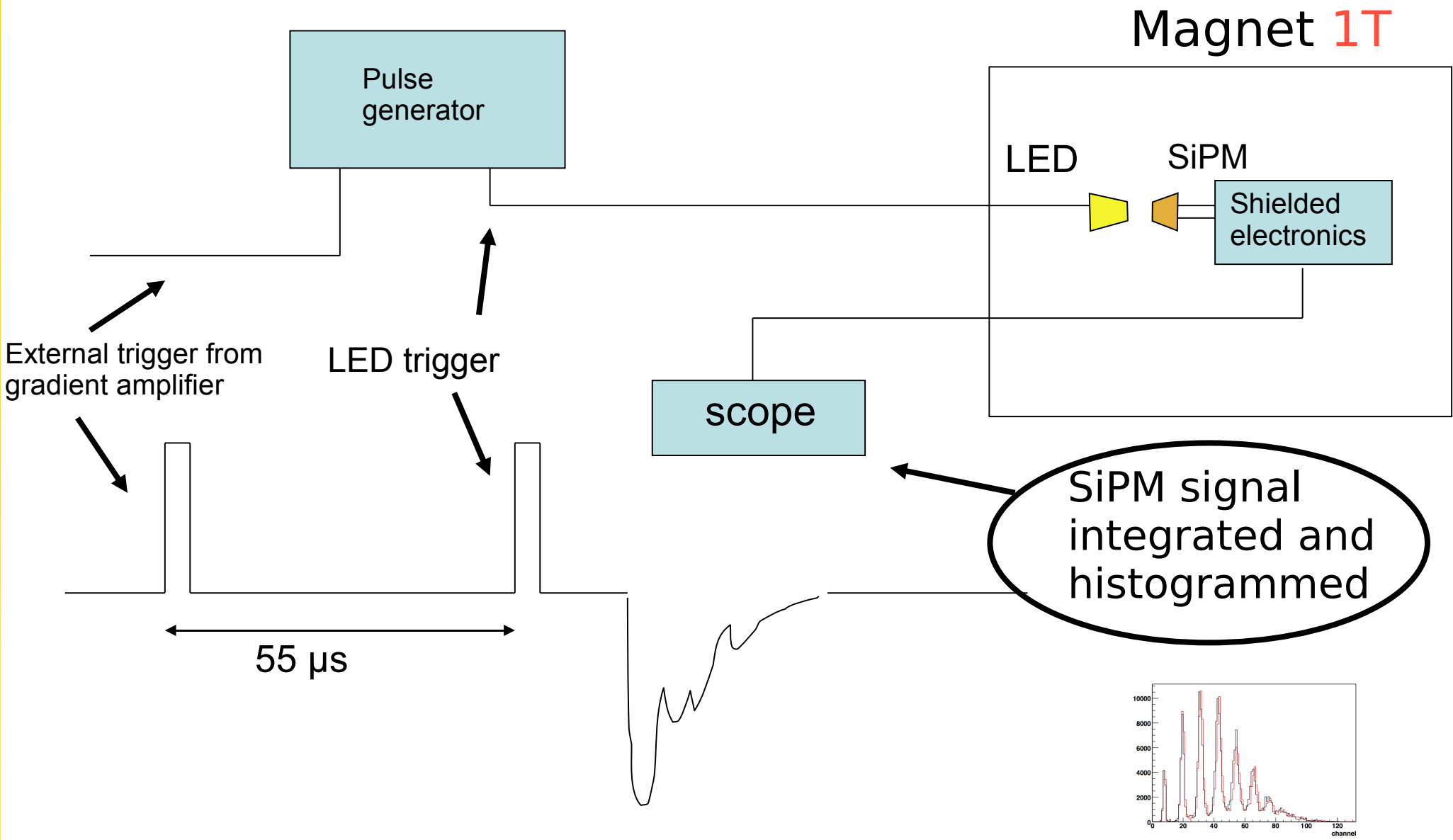



- ...but coupling still to be improved

A.Del Guerra et al, workshop on "Photon Detection" - Perugia 13-14/6/2007

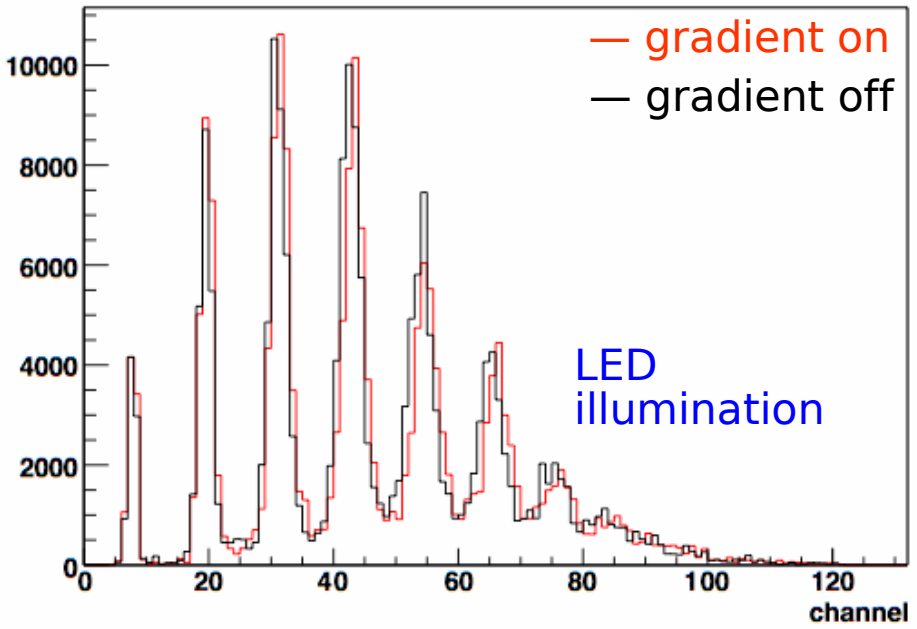
New results with 4x4 matrices with both pixellated and continuous crystals expected at IEEE-NSS-MIC 2008 (Dresda)

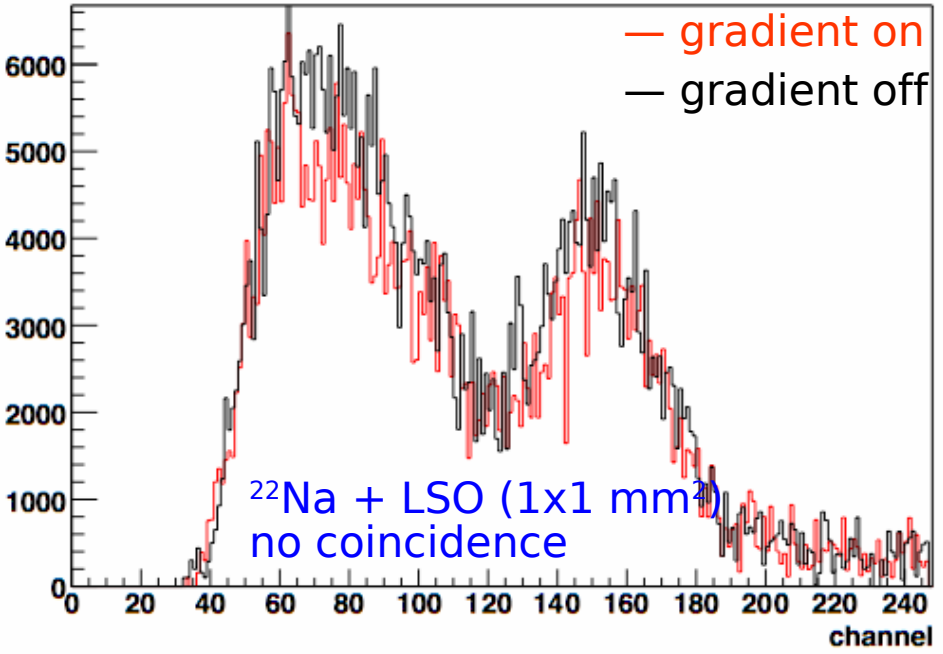
# SiPM in static magnetic field + gradient

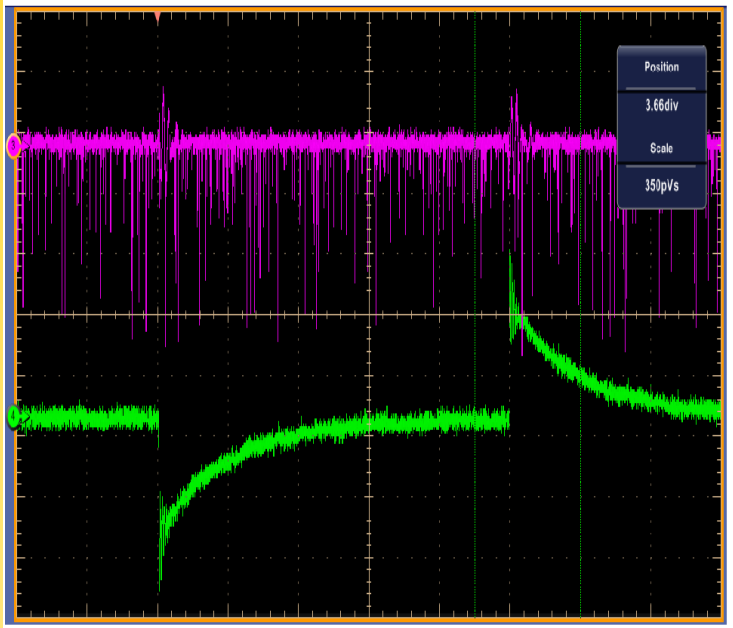



SiPM signal is acquired while the gradient is increasing

# SiPM compatibility with MRI: Z gradient on

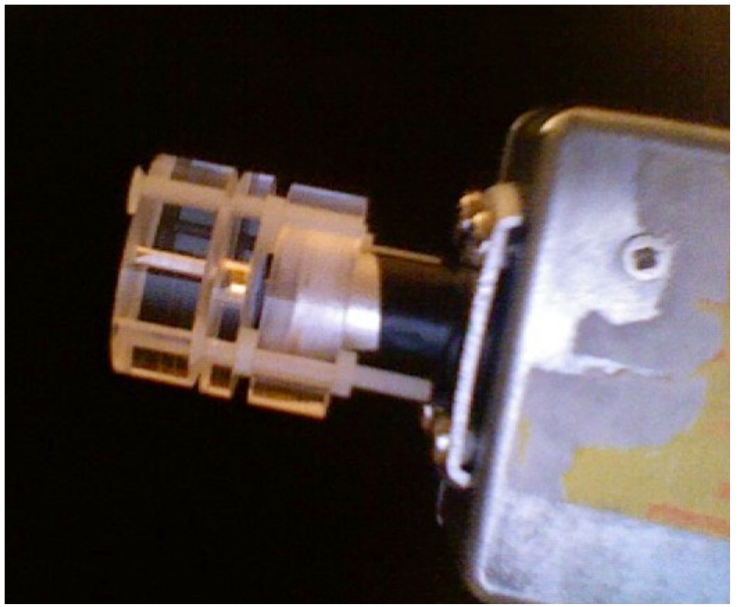


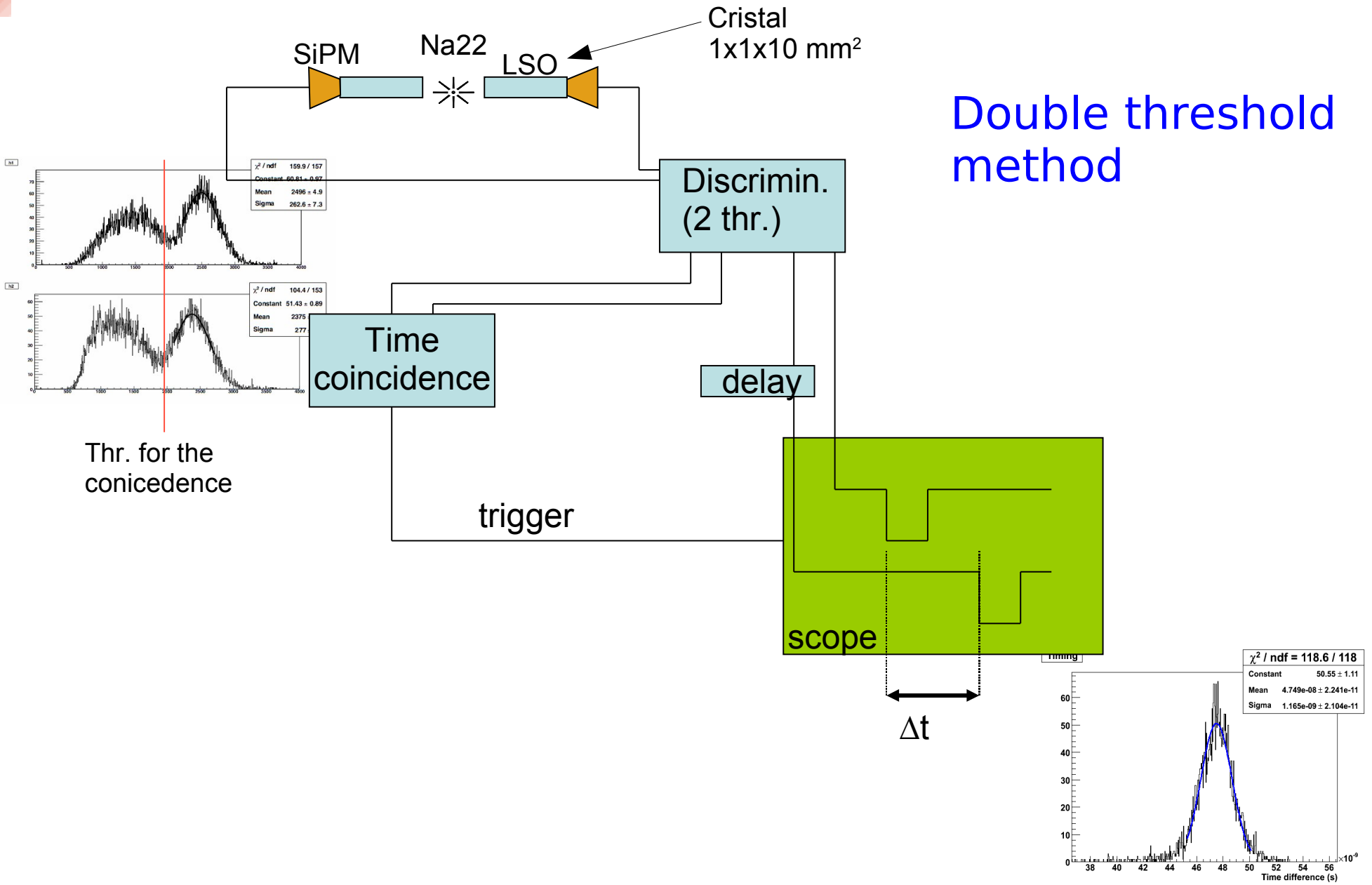




SiPM dark signal

Pickup coil signal

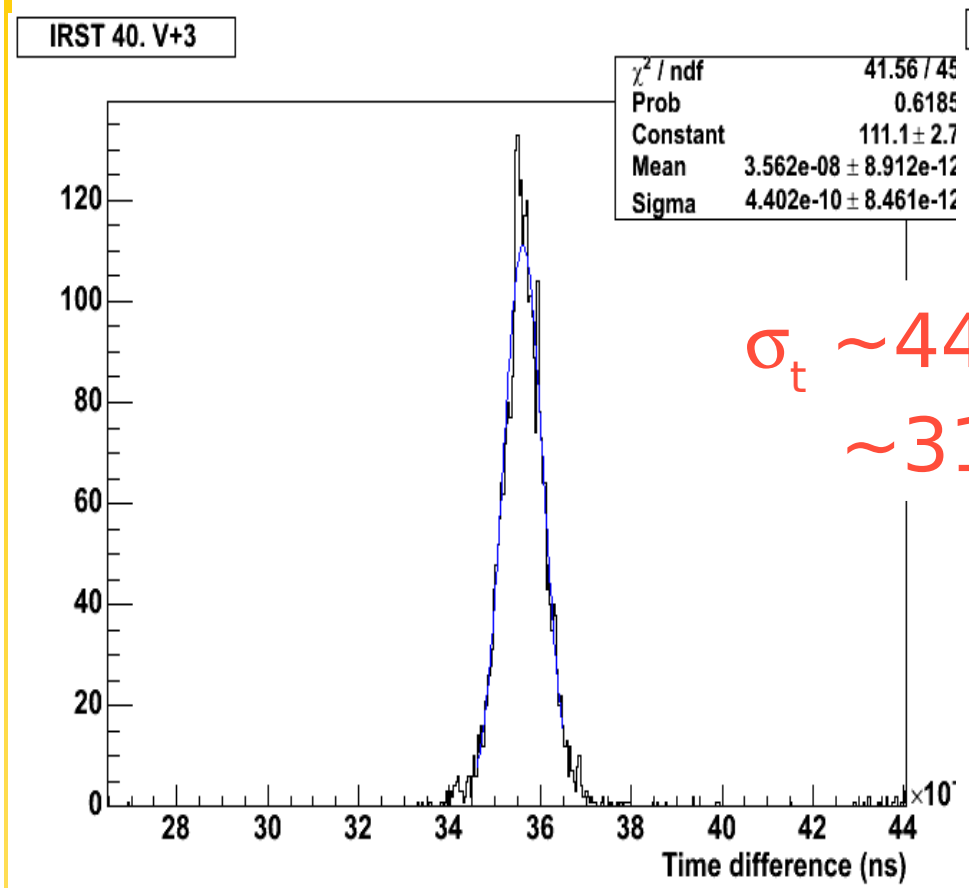


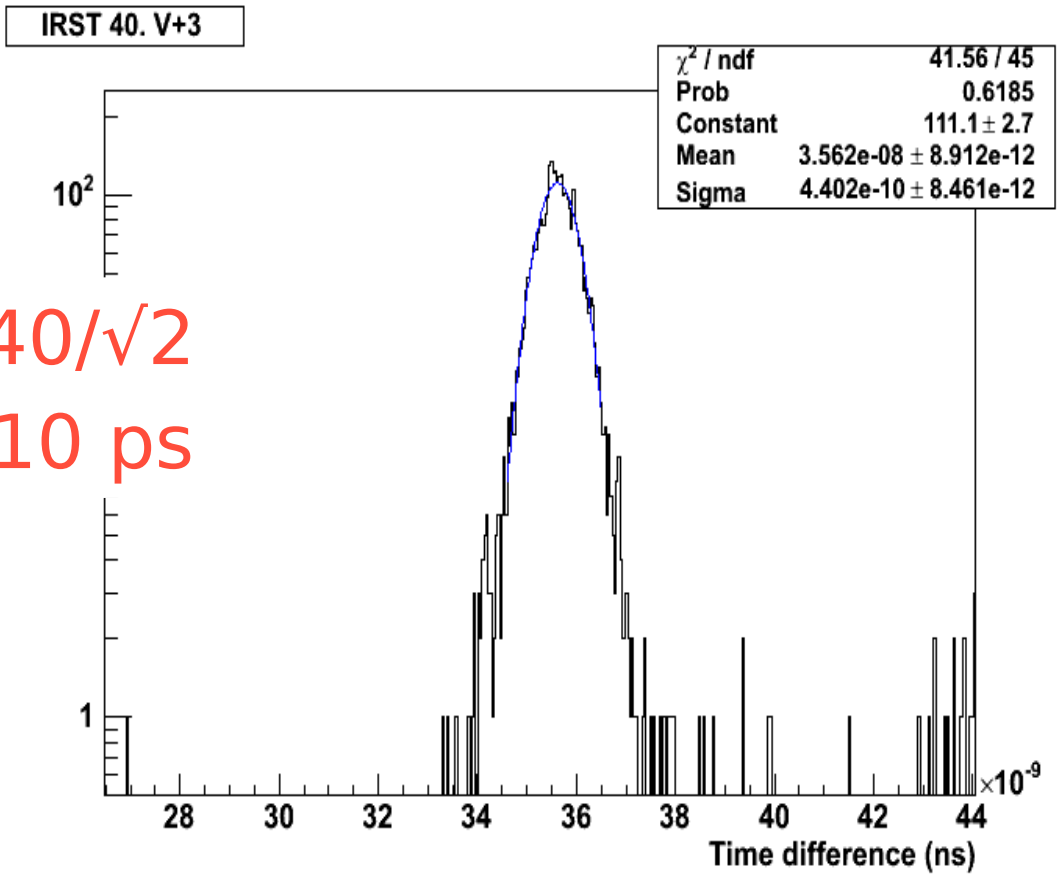
# Timing with LSO



# Timing with LSO

SiPM - IRST
1x1 mm² (50x50μm²)




$\sigma_t$ ~440/√2
~310 ps

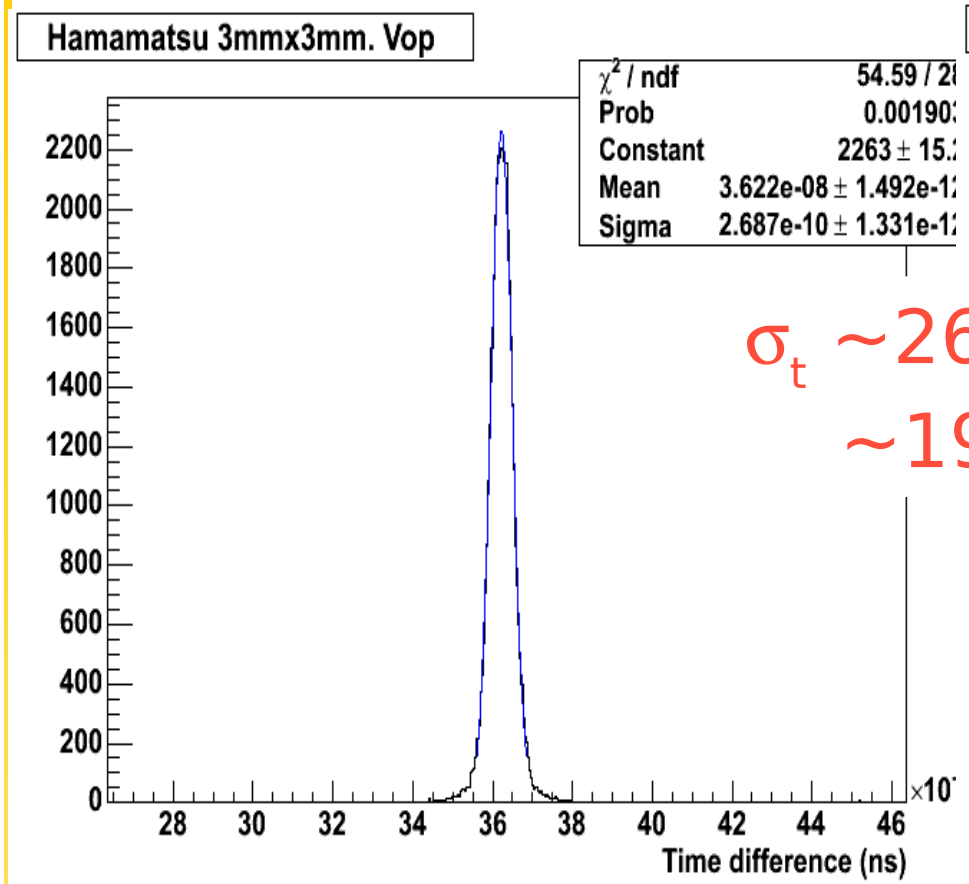
Note: $\sigma_t \approx \frac{\sqrt{N_{thr.[p.e.]}}}{N_{p.e.}} \cdot \tau_{decay}$ Post P.R. 80 (1950) 1113

E.g.
$\tau_{decay}$ ~ 40ns (decay time for LSO)
$N_{p.e.}$ ~ 100 p.e. (photopeak)
$N_{thr.}$ ~ 1 p.e. (trigger on 1st p.e.)

Expected: $\sigma_t$ ~ 400 ps

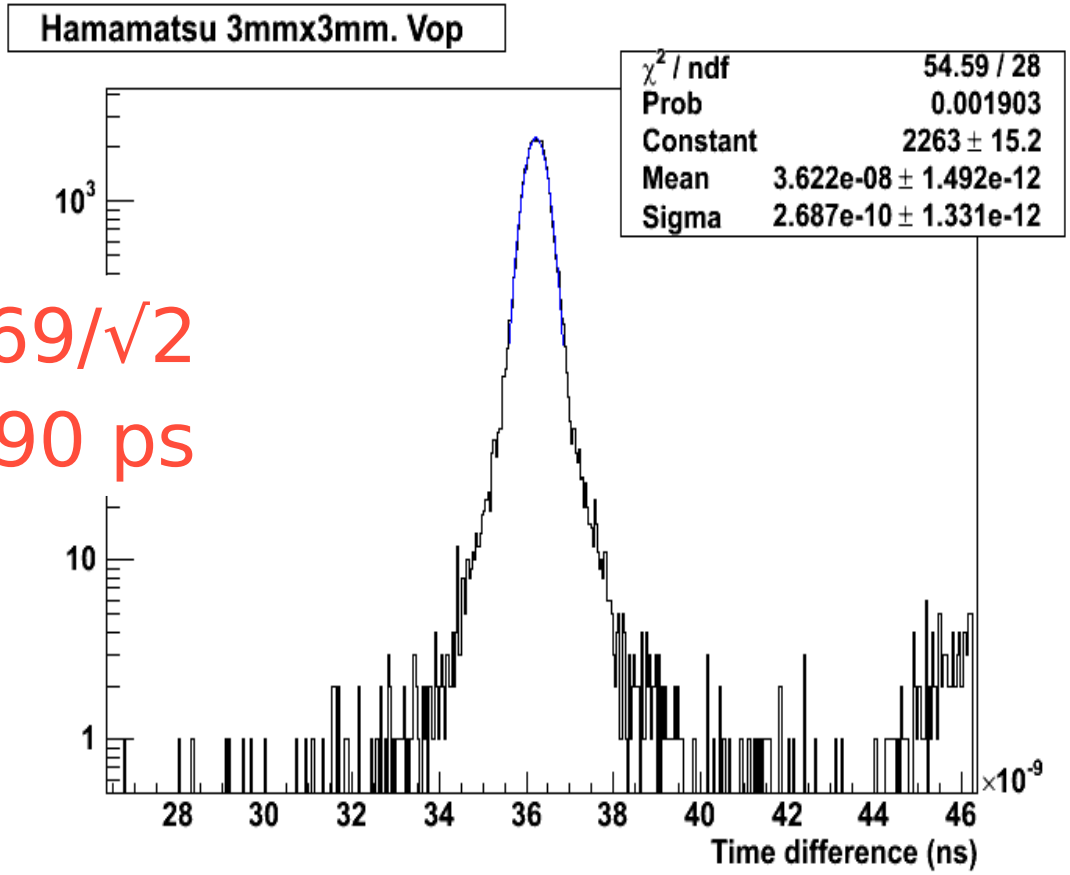
# Timing with LSO

SiPM – HPK
3x3 mm² (50x50μm²)




$\sigma_t$ ~269/√2

~190 ps




More photoelectrons

G.Llosa, DASIPM (unpublished)

# Conclusions

- Device physics is understood and under control
- Device technology is in continuous evolution in many countries
- Detector now is stable and reliable, ready for many applications
- Already at work in Calorimetry and scintillator counters (TOF and tracking). Recently first results on Cherenkov light

# Additional information

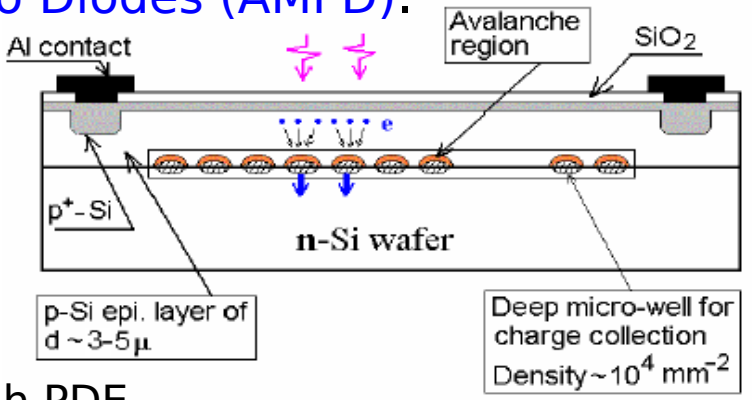
# A bit of history

Pioneering work since late 80-ies at Russian institutes

Investigations of various multi-layer silicon structures with local micro-plasma suppression effect to develop low-cost APD devices

Sadygov - JINR (Dubna)

Avalanche Micro-channel/pixel Photo Diodes (AMPD).

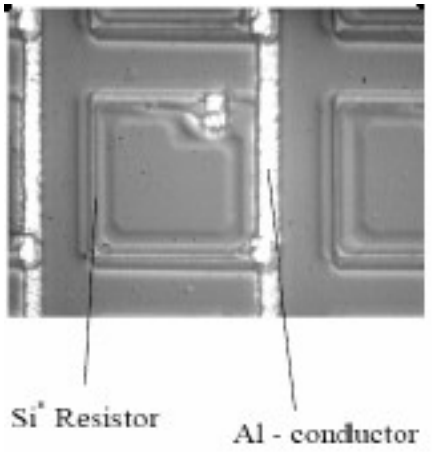



- high PDE
- very high density of microcells

Sadygov, NIMA 567 (2006)

Dolgoshein - Mephi/PULSAR (Moscow)

Polysilicon resistor

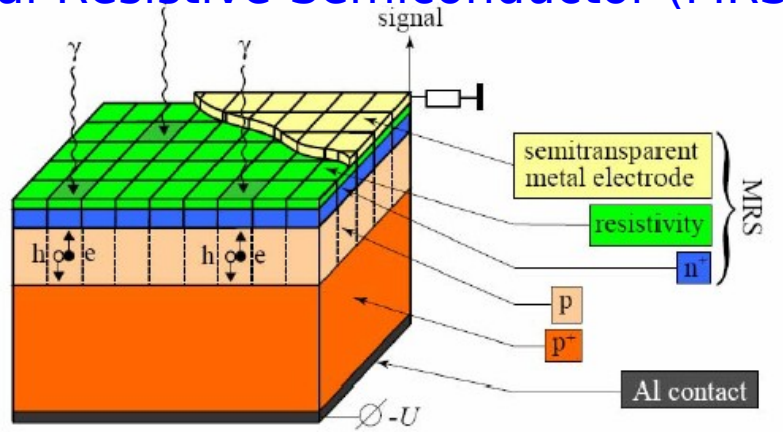



- Low area efficiency
- Standard fabrication tech.

e.g., Dolgoshein, NIMA 563 (2006)

Golovin - Obninsk/CPTA (Moscow)

Metal-Resistive-Semiconductor (MRS)

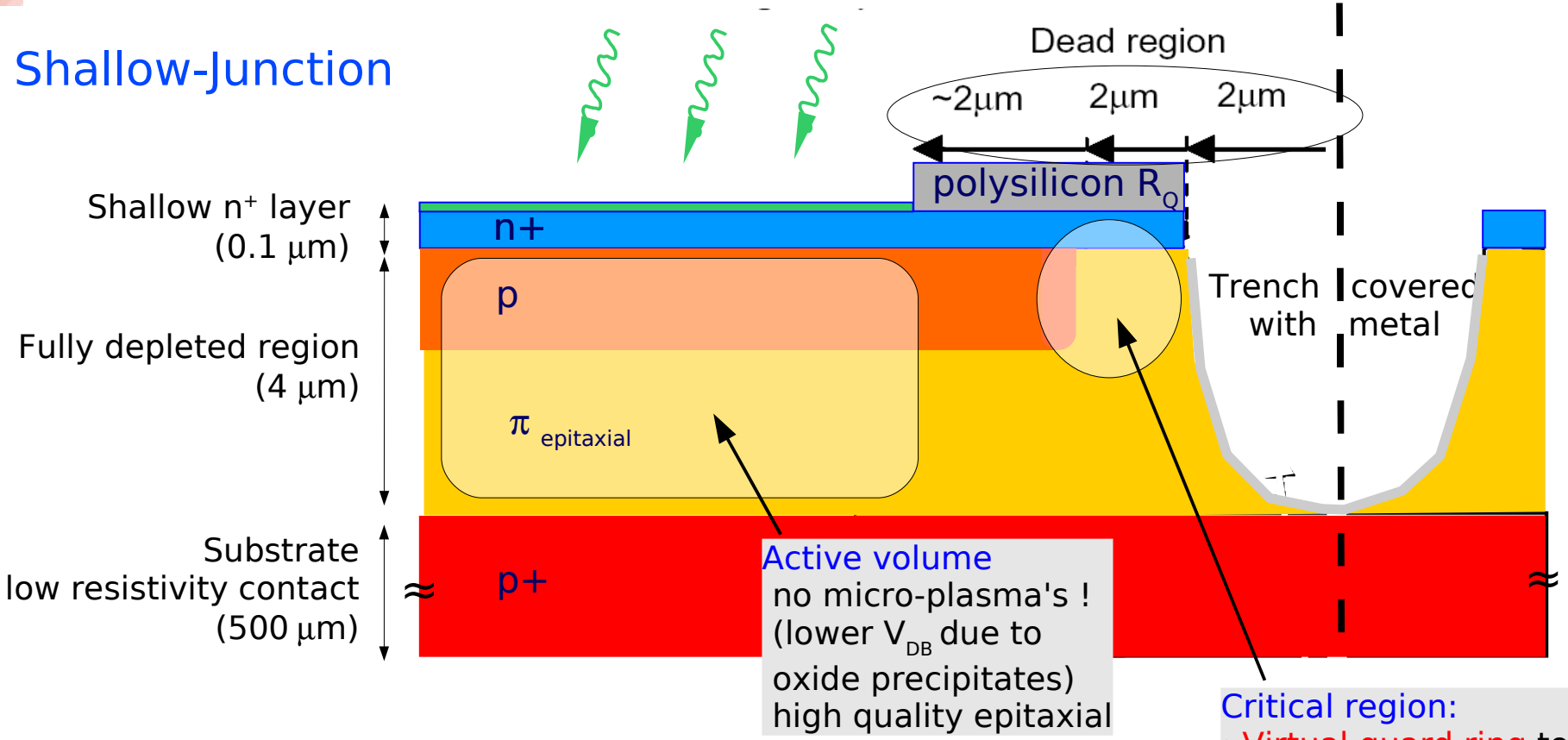



- High Area efficiency
- high density of microcells

e.g., Golovin NIMA 539 (2005)

# Close up of a cell (IRST technology)

## Shallow-Junction




## Optimization for the blue light (~400nm)

- n⁺ on p abrupt junction structure
- Anti-reflective coating (ARC) optimized for λ~420nm
- Very thin (100nm) n⁺ layer: "low" doping n⁺ layer
  → minimize Auger and SHR recombination
- Thin high-field region: "high" doping p layer (limited by tunneling breakdown)
  → fixes $V_{BD}$ junction well below $V_{BD}$ at edge
- $R_Q$ by doped polysilicon
- Trenches for optical insulation (low cross-talk)
- Fill factor: 20% - 80%

C.Piemonte NIM A 568 (2006) 224

# Summary about timing

SiPM are intrinsically very fast:
- core (gaussian) fluctuations below 100ps, depending on $\Delta V$
- Non gaussian tails up to O(ns), depending on wavelength

The conditions to optimize the device timing (high impact ionization ratio $\beta/\alpha$) are opposite of those to optimial (fast) rise PDE vs $\Delta V$

High bandwidth current amplifier + fast sampling electronics best choice for timing applications (robust against noise and pile-up)

Best (gaussian) resolutions in applications:
- Cherenkov: O(10ps)
- Fast plastic scintillators O(20ps)
- Fast crystal scintillators O(150ps)

Non-gaussian tails to be considered (for long wavelengths)

# Introduction: from vacuum tubes to semiconductor photo-detectors

Vacuum Photo-multiplier Tube: first PMT by Elster and Geiter 1913

- high price, bulky shape, sensitive to magnetic fields
- Single photons can be detected (since years 40's)

PIN photo-diode: successfully used, e.g. in HEP experiments

- No internal gain: necessary Q sensitive amplifier (noise, slow)
  $\rightarrow$ minimum of several 100 photo-electrons (p.e.) detectable
- Nuclear counter-effect

Avalanche photo-diode: used in big experiments (CMS at LHC)

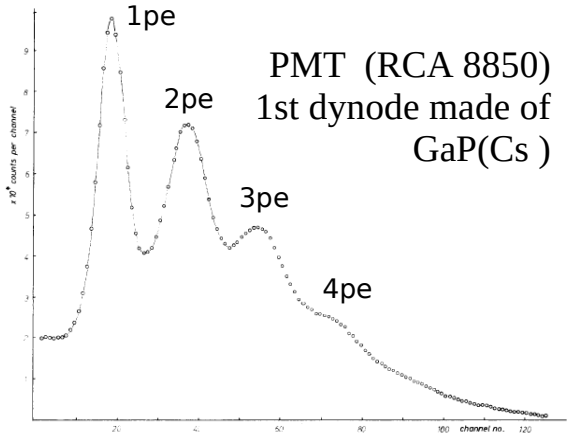
- Internal multiplication: S/N improved $\rightarrow$ still >20 p.e. detectable
- Gain limited by the excess noise (fluctuations of avalanche multiplication)

Geiger mode avalanche photo-diode:

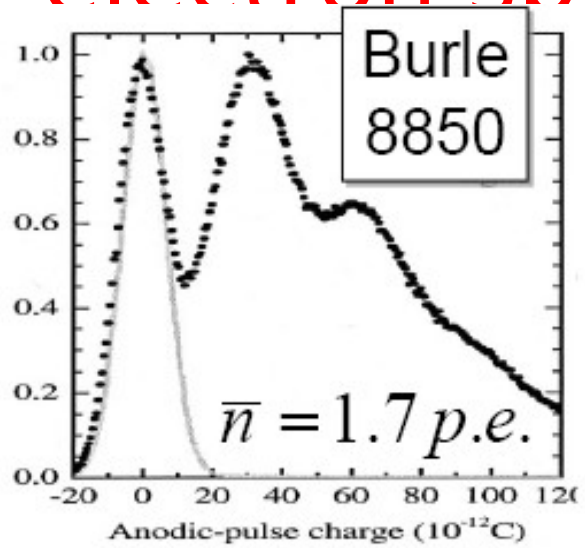
- High multiplication (x$10^6$) $\rightarrow$ single photons detected
  $\rightarrow$ Silicon photo-multiplier (SiPM)
- Limited excess noise factor (cross-talk and after-pulses)
- Impressive charge and timing resolution with simple electronics at room T

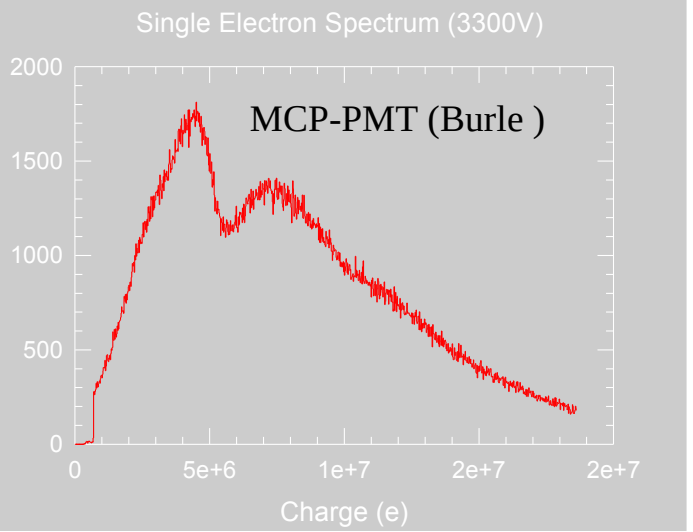
# Single photo-electron spectra

NIM 112 (1974) 121



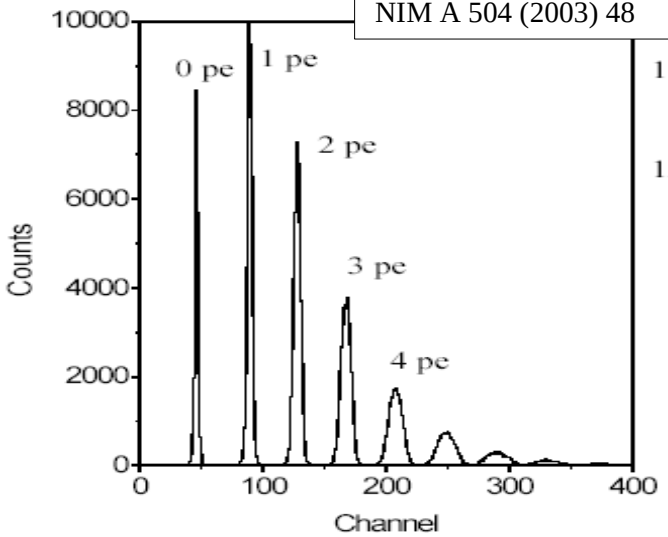

PMT (RCA 8850)
1st dynode made of GaP(Cs )



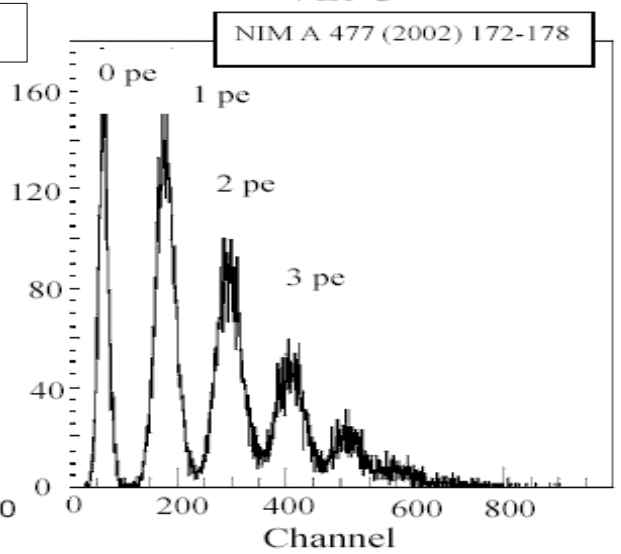

SiPM

NIM A 504 (2003) 48

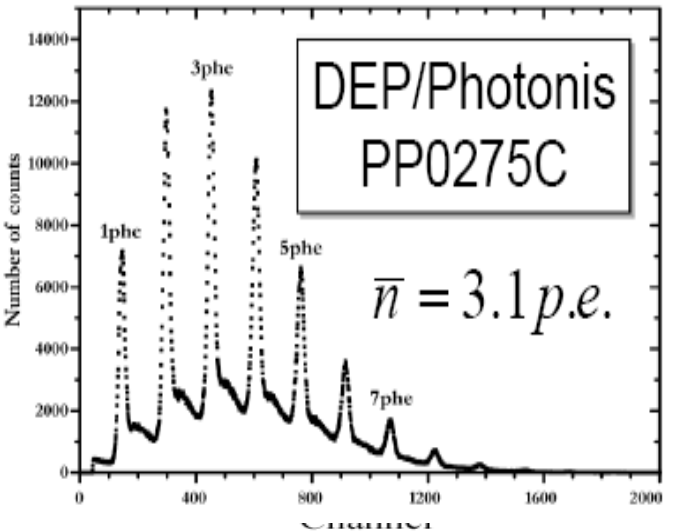



VLPC

NIM A 477 (2002) 172-178




HPD




Silicon Photo-Multiplier
- room T
- V~50V

Visible Light Photon Counter
- T=6.5K
- V~5V

Hybrid PhotoDiode
- room T
- V~20000V

# Operation principle of a GM-APD

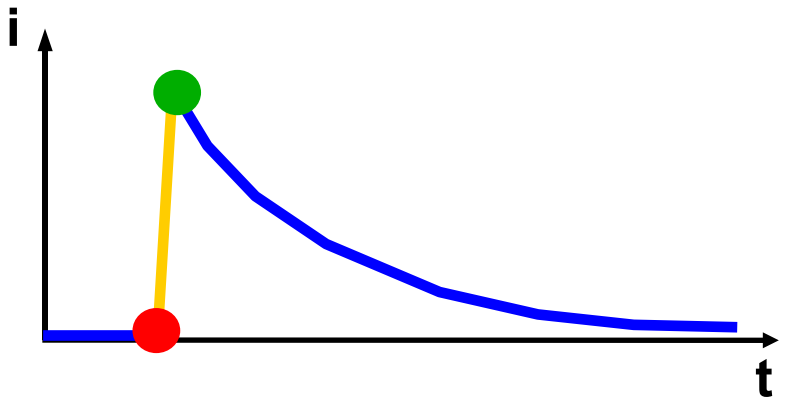
In GM-APD the detection of a photon essentially triggers the full discharge of a condenser !

Gain = charge on the condenser / electron charge.

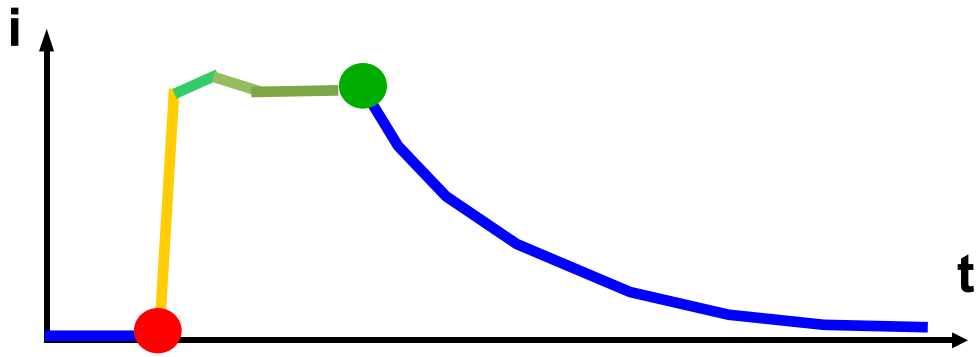
The Gain fluctuations in GM-APD are very small and different in nature compared to APD where the statistical process of internal amplification shows peculiar fluctuations (multiplication noise)

Note: effective quenching ($R_Q$) is crucial to have a well defined gain

Good quenching

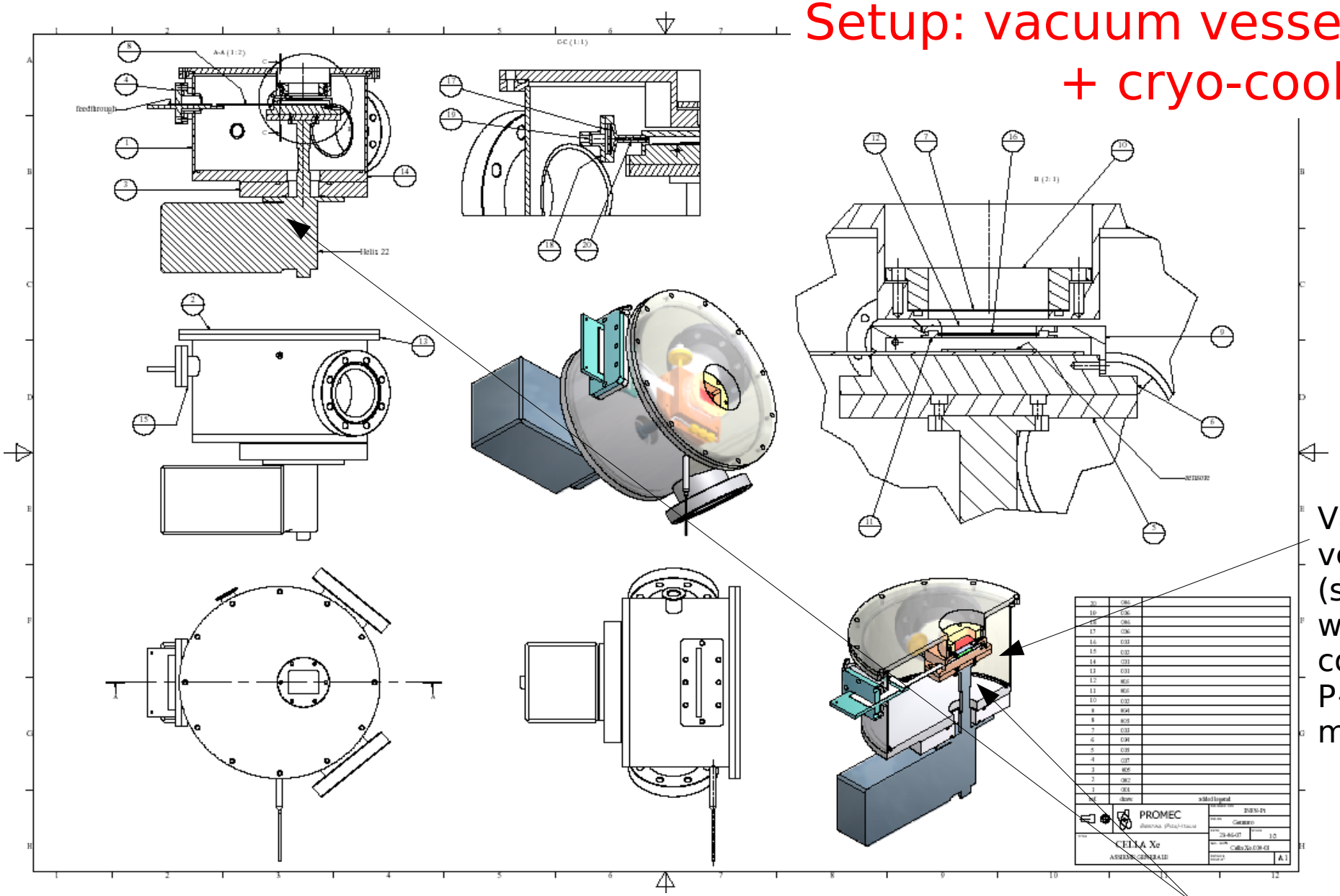



Bad quenching



# Temperature characterization

## Setup: vacuum vessel + cryo-cooler




Vacuum vessel (std working cond. $P \sim 10^{-4}$ mbar)

cryo-cooler

Acknowledgments: A.Baldini, A.Brez and M.Minuti (INFN Pisa)

SiPM's are in thermal contact with a cooled Cu rod

# RPL model: fast simulation

"Statistics of Avalanche Current Buildup Time in Single-Photon Avalanche diodes"
C.H.Tan, J.S.Ng, G.J.Rees, J.P.R.David (Sheffield U.)
IEEE J.Quantum Electronics 13 (4) (2007) 906

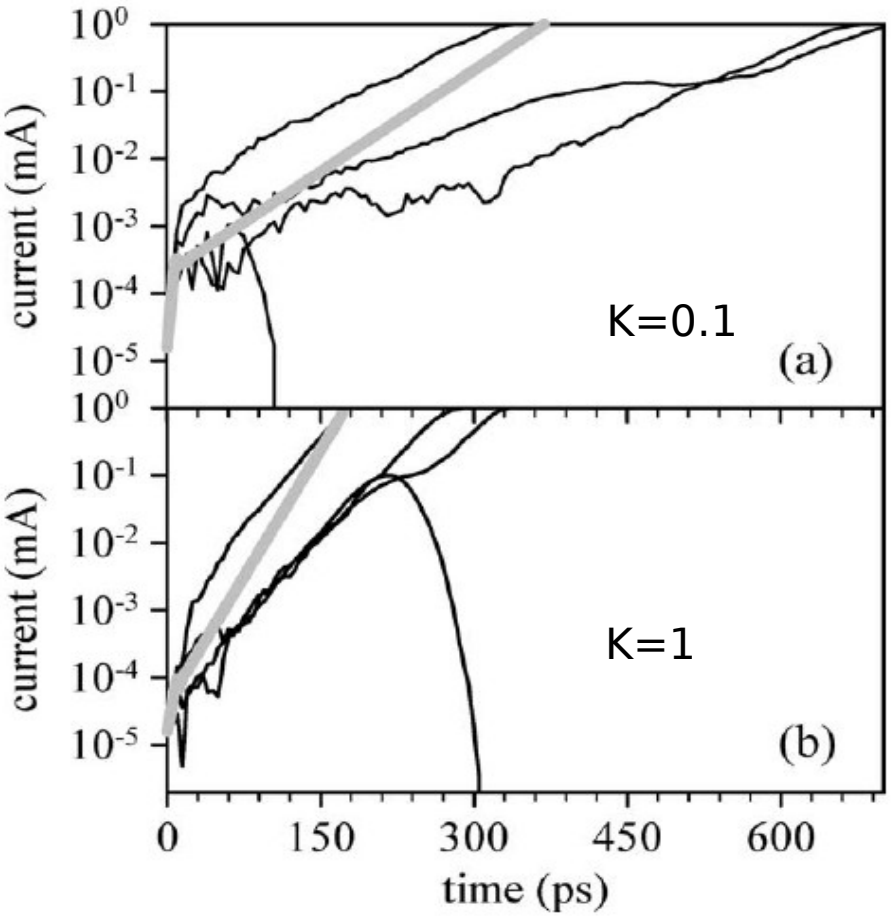
Numerical model (MC): Random distribution of impact ionization Path Length (RPL)

Analysis of breakdown probability, breakdown time and timing jitter as functions of avalanche region width (w), ionization coefficient ratio ($k=\beta_{hole}/\alpha_{electron}$) and dead space parameter (d) (uniform E field, constant carrier velocity)

1) increasing k:
- improves timing performances
- but breakdown probability $P_{br}$ increases slowly with overvoltage

1a) hole injection results in better timing than electron injection (in Si devices)

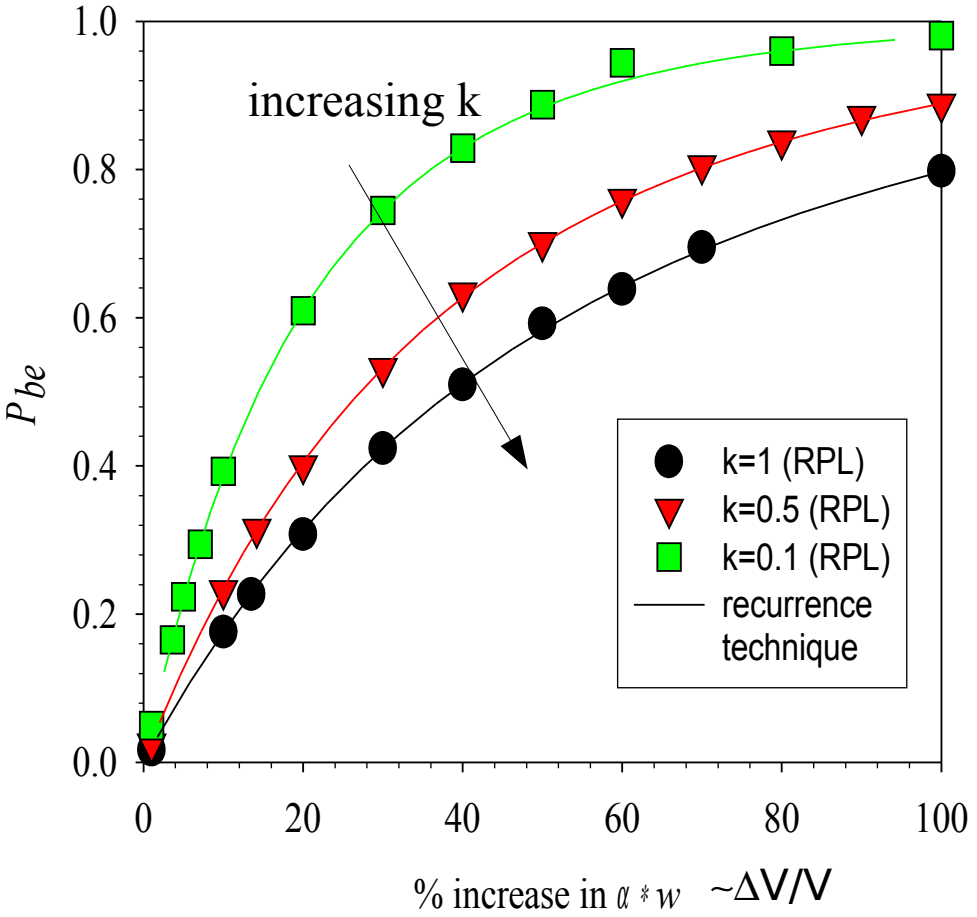
2) dead space effects worsen timing performances (the more at small k)
Important for devices with small w



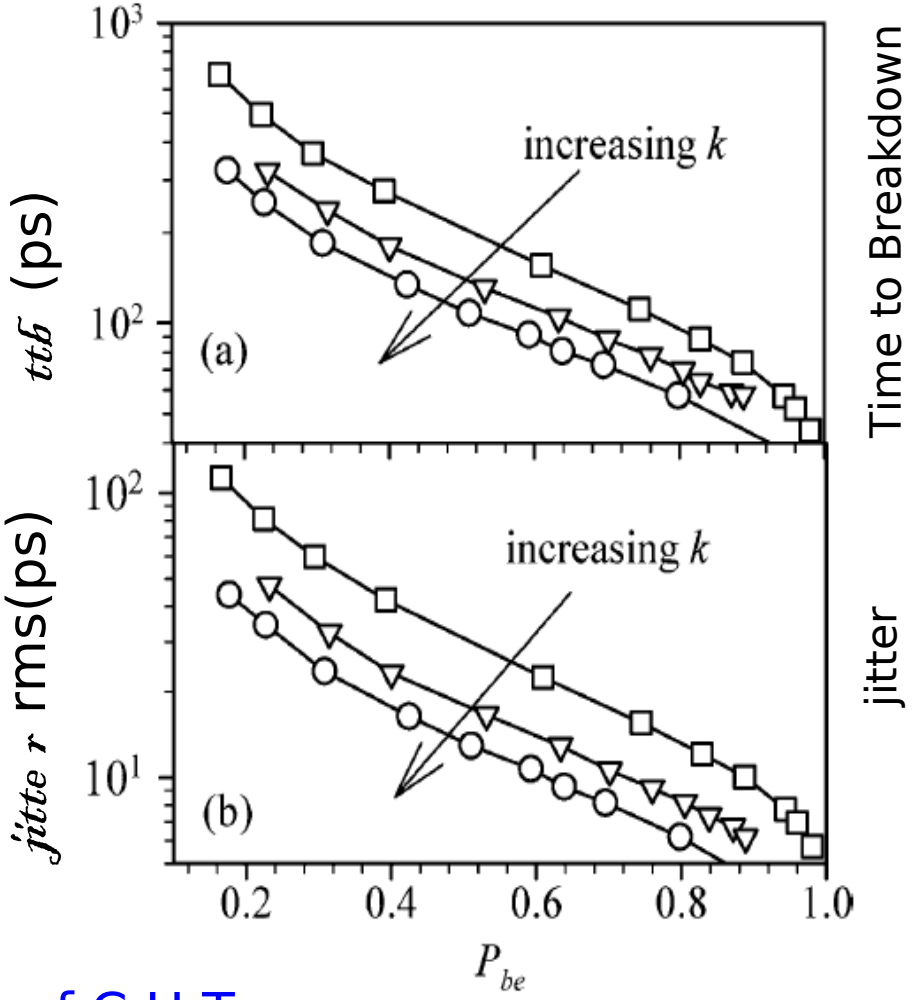
# RPL model

Tan, Rees, Davis et al

Increasing k
→ slower slope $P_{br}$ vs ΔV




Increasing k
→ optimized timing



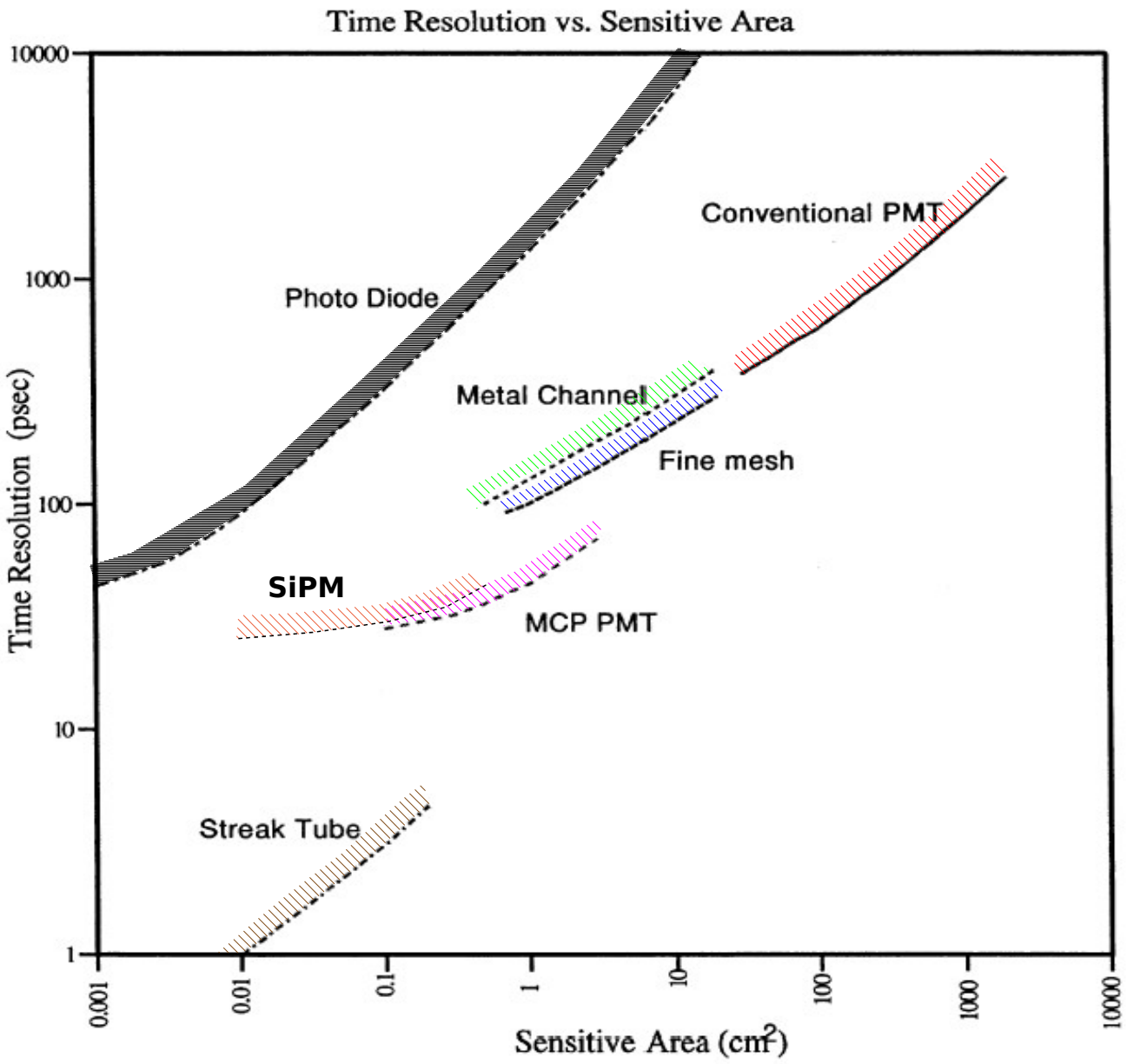

Courtesy of C.H.Tan

The conditions to optimize the device timing (high impact ionization ratio β/α) are opposite of those to optimial (fast) rise PDE vs ΔV

# Photo-detectors comparison

| **Photon detector** | **Timing** | | **Intensity / Energy** | | **Position** | |
|---|---|---|---|---|---|---|
| | sensitivity | dead time | sensitivity | dyn.range | sensitivity | dyn.range |
| | | | #photons per unit area | | | |
| Human eye | 100 ms | | $10^{4}$ | $10^{10}$ | 0.1 mm | 10 cm |
| | | | | | | |
| **Vacuum** | | | | | | |
| PMT | 200 ps | 10 ns | 5 | $10^{7}$ | 50 mm | 50 cm |
| Pos. sens. PMT | 200 ps | 10 ns | 5 | $10^{7}$ | 1 mm | 0.5 cm |
| MCP PMT | 30 ps | 10 ns | 5 | $10^{4}$ | 1 mm | 3 cm |
| Image Intensifier | 100 ps | | 10 | $10^{4}$ | 0.01 mm | 0.2 cm |
| Streak Tube | 1 ps | | 10 | $10^{4}$ | 0.1 mm | 0.2 cm |
| | | | | | | |
| **Solid State** | | | | | | |
| PIN diode | 1 ns | | $10^{3}$ | $10^{15}$ | 1 mm | 5 cm |
| APD | 100 ps – 1 ns | 10 ns | 10 | $10^{12}$ | 0.1 mm | 5 cm |
| CCD | 10 ms | | 10 | $10^{5}$ | 0.01 mm | 0.1 cm |
| VLPC | 300 ps – 1 ns | | 1 | | | |
| SiPM | 50 ps | < ns | 1 | $10^{3}$ | 0.1 mm | 1 cm |
| | | | | | | |
| **Hybrid** | | | | | | |
| PIN diode | 1 ns | | 5 | $10^{10}$ | 2 mm | 2 cm |
| APD | 1 ns | | 5 | $10^{8}$ | 0.5 mm | 0.5cm |
| CCD | 10 ms | | 5 | $10^{4}$ | 0.01 mm | 0.1 cm |
| Silicon Strip | 1 ns | | | $10^{5}$ | 0.01 mm | 0.1 cm |
| Silicon pixel | 1 ns | | | $10^{4}$ | 0.01 mm | 0.1 cm |

# Timing resolution of photo-detectors (PD)




Note resolution improves:
- by decreasing active area
- by decreasing the intrinsic
- detector capacitance
- as $\sigma_t/t \sim 1/\sqrt{Npe} \sim 1/\sqrt{QE}$

Note tails matter:
- MCP
- APD < 2ns at $1/10^5$ (best)
- SiPM< 1ns at $1/10^2$ (only large$\lambda$)

modified from K.Arisaka NIMA 422 (2000)

# PD comparison: Energy resolution

| Detector | QE | δ1 | ENF | Gain | $\sigma_E/E$ |
|---|---|---|---|---|---|
| **Vacuum** | | | | | |
| PMT | 30% | 5 – 10 | 1 | $10^5$–$10^8$ | $\sqrt{(4/N_{pe})}$ |
| Fine Mesh | 30% | 2 | 2 | $10^6$ | $\sqrt{(7/N_{pe})}$ |
| MCP | 20% | 2 | 1.5 | $10^5$ | $\sqrt{(7/N_{pe})}$ |
| **Solid state** | | | | | |
| PIN | 90% | | 1 | 1 | $\sqrt{(1.2/N_{pe})+(10^3/N_{pe})^2}$ |
| APD | 60% - 90% | | >2 | <100 | $\sqrt{(2.5/N_{pe})+(14/N_{pe})^2}$ |
| SiPM | 20% - 80% | | 1.1 | $10^5$ – $10^6$ | $\sqrt{(2.5/N_{pe})}$ |
| VLPC | 80% | | 1.1 | $10^5$ | $\sqrt{(1.6/N_{pe})}$ |
| **Hybrid** | | | | | |
| HPD | 30% | $10^3$ | 1 | $10^3$ | $\sqrt{(3/N_{pe})+(3/N_{pe})^2}$ |
| HAPD | 30% | $10^3$ | 1 | $10^5$ | $\sqrt{(3/N_{pe})}$ |

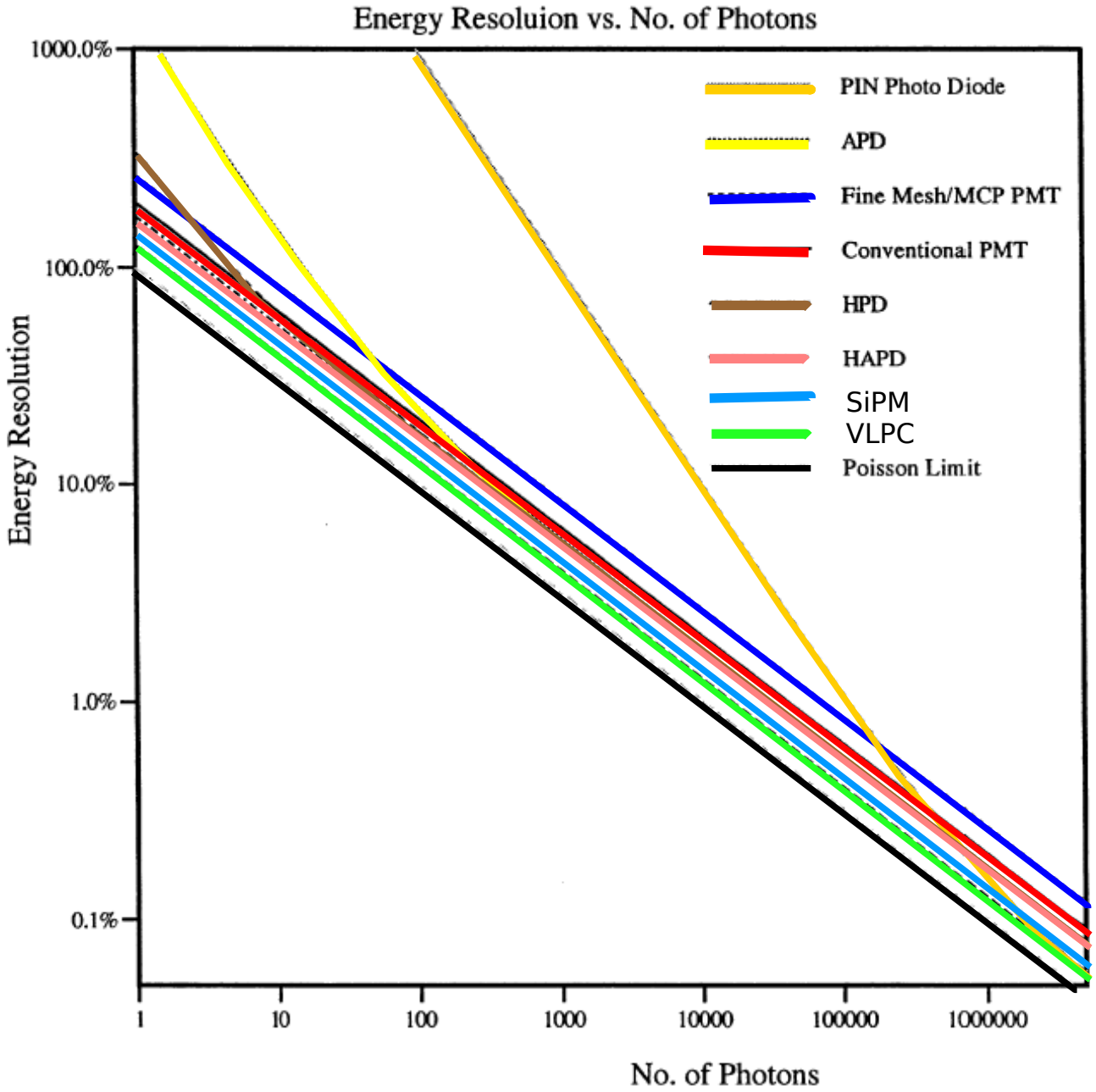
NOTE:
1) PMT and HPD with enhanced cathode QE (>35%) are going to be come standard

2) after-pulsing (at 1p.e. threshold)
PMT (HPK) ~ few $10^{-2}$
HPD (HPK) ~ few $10^{-3}$

- δ is the multiplication factor of a dynode
- ENF is the excess noise
  (eg ENF~ 1+ 1/d for PMT and ENF ~ kM + (1-k)(2-1/M)) for APD)
- ENC is the Equivalent charge noise (typically $10^3$) of the amplifier

$$\sigma_E/E \sim \sqrt{\frac{ENF}{N_{pe}} + \left(\frac{ENC}{N_{pe}G}\right)^2}$$

# PD comparison: Energy resolution

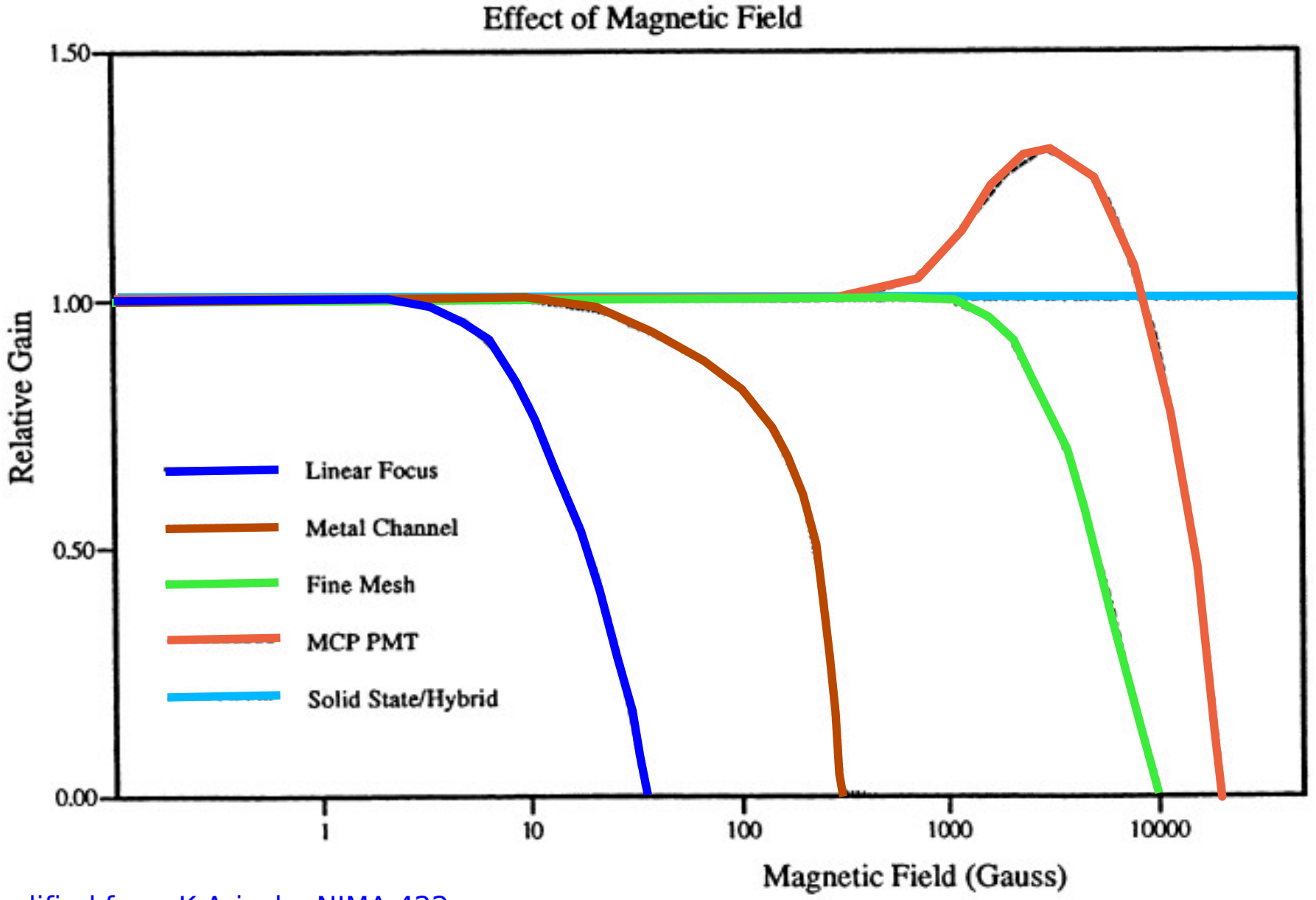



modified from K.Arisaka NIMA 422 (2000)

# Gain vs T and V dependence

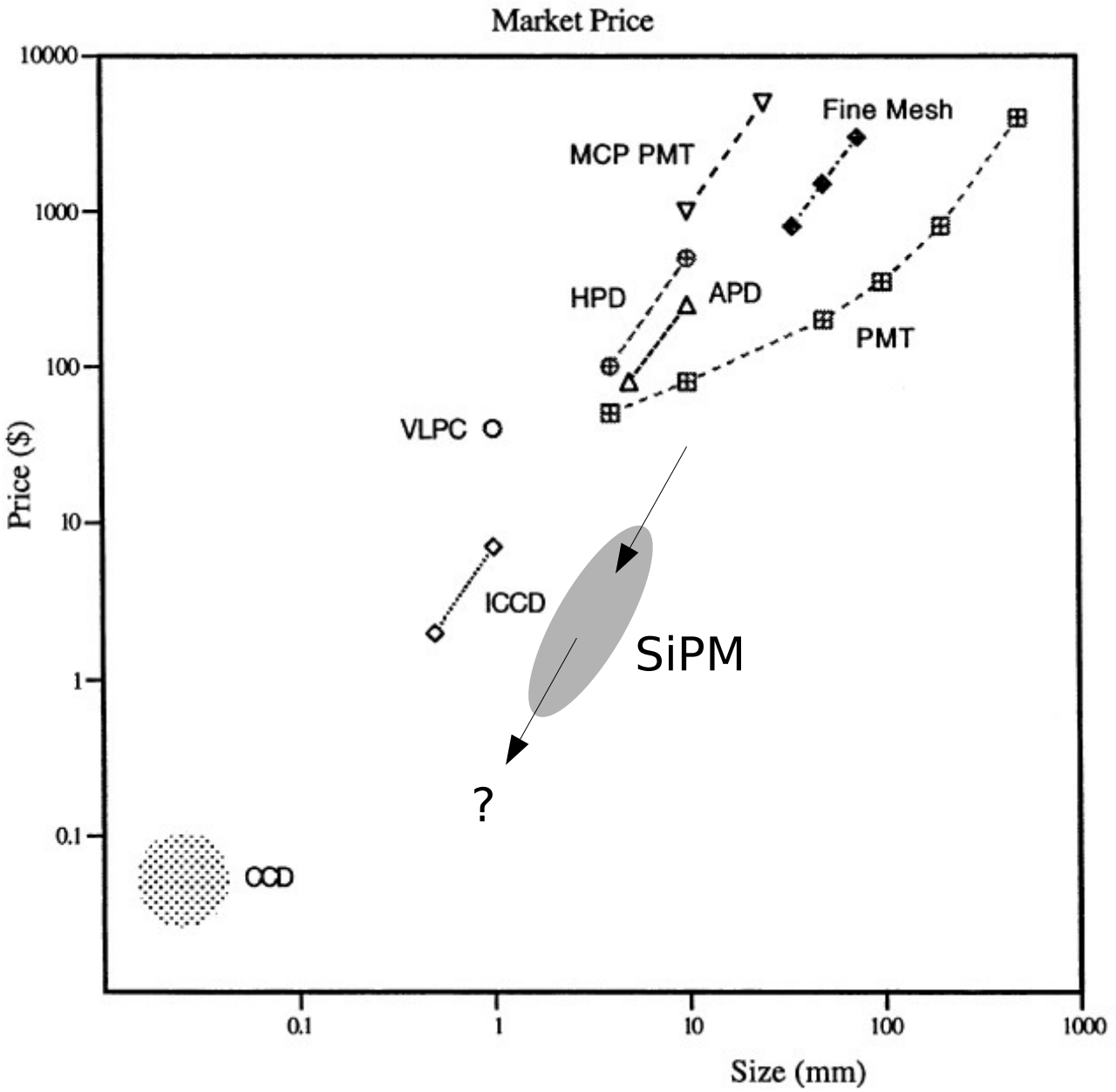
| Photo detector | $\Delta T$ for $\Delta G/G = 1\%$ | $\Delta V / V$ for $\Delta G/G = 1\%$ |
|---|---|---|
| PMT | ~5°C | 0.5V / 1500V = $3 \cdot 10^{-4}$ |
| APD ($G=10^2$) EG&G | ~0.15°C | 0.4V / 400V = $10 \cdot 10^{-4}$ |
| APD ($G=10^2$) Hamamatsu (High C) | ~0.3°C | 0.04V / 300V = $1.5 \cdot 10^{-4}$ |
| HPD ($G=10^5$) Hamamatsu R9792U-40 | ~0.5°C | 0.04V / 400V = $1.0 \cdot 10^{-4}$ (APD dominates) |
| SiPM ($G=10^6$) IRST (C ~ 80 fF) | ~0.3°C | 0.02V / 35V = $6 \cdot 10^{-4}$ |

# PD comparison: Effect of Magnetic field



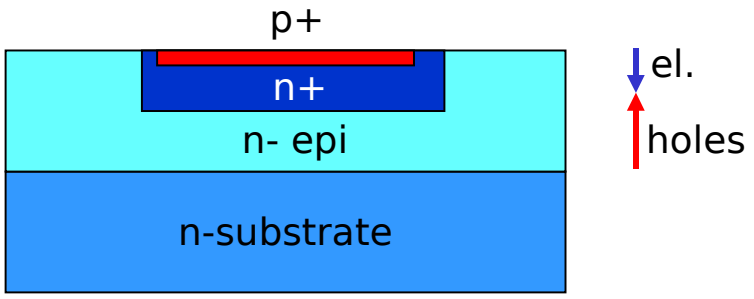
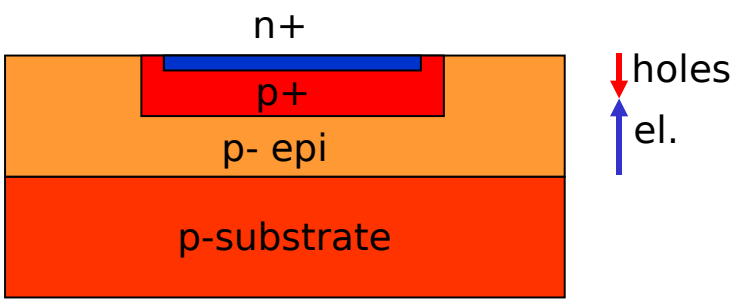

modified from K.Arisaka NIMA 422 (2000)

# PD comparison: Price




modified from K.Arisaka NIMA 422 (2000)

# Comparison HPK-IRST

| Product. | Hamamatsu | IRST |
|---|---|---|
| Type |   |   |
| Gain | $10^5$ - $10^6$ | $10^5$ - $10^6$ |
| PDE | 30-70%<br>(UV)-blue-green | 30-70%<br>(blue)-green-IR |
| Noise | 200kHz - 1MHz | ~ HPK x 2 |
| After-pulse | ~ 10% | ~ 1% |
| Cross-talk | ~ 10% | ~ 1% |
| Timing | ~ 100 ps | ~ 60 ps |

NOTE: in the standard working $V_{bias}$ range

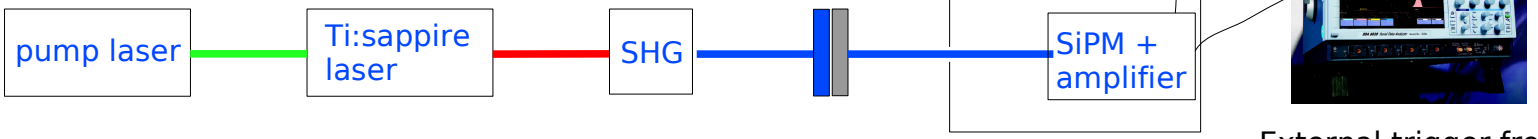
# Experimental Setup

## Pump Laser
*Millenia V (Spectra-physics)*
solid state CW visible laser

## Crystal for Second Harmonic Generation (SHG)
conversion 800 nm → 400 nm
efficiency at % level

Low noise LV suppliers

Dark box






## LeCroy SDA 6020

Analog bandwidth: 6GHz
Sampling rate: 20GS/s
Vertical resolution: 8 bits

(Aknowledments:
E.Marcon, LeCroy)

External trigger from Ti:sappire laser signal

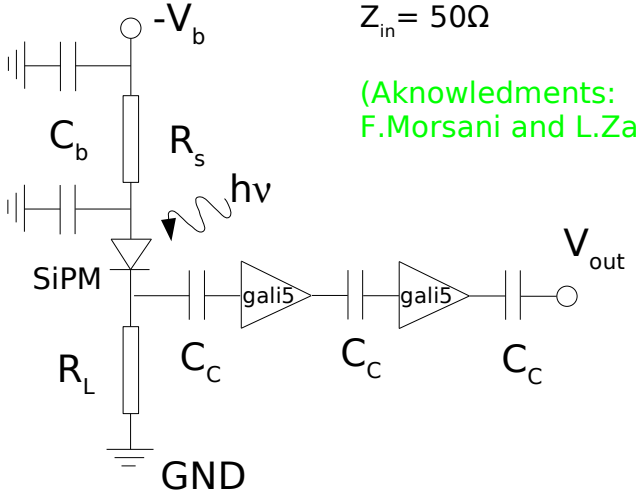
## Mode-locked Ti:sapphire Laser
*Tsunami (Spectra-physics)*
femtosecond pulsed laser

## Filters
blue + neutral
for rejecting IR light and tune intensity

wavelength: tuned at 800±15 nm
pulse width: ~ 60 fs FWHM
pulse period: ~ 12 ns
pulse timing jitter < 100 fs

## Electronics
I → V conversion via $R_L$ (500Ω)
Two stage voltage amplification (= x50) based on high-bandwidth low-noise RF amplifier: gali-5 (MiniCircuit)
$Z_{in}$ = 50Ω

(Aknowledments:
F.Morsani and L.Zaccarelli, INFN-Pisa)

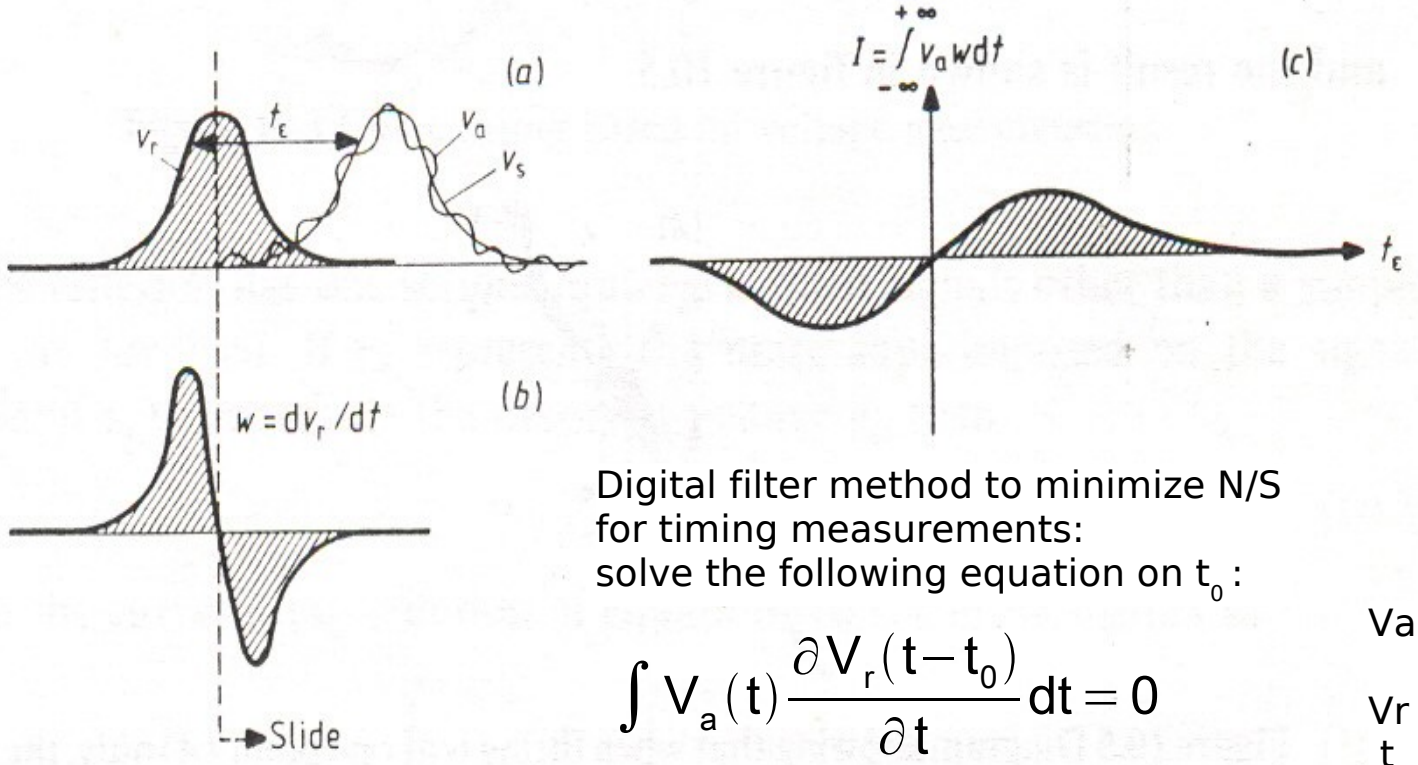



## Data taking conditions:
- different $V_{bias}$
- both at 800 nm and 400 nm
- with different light intensities (counting rates in the range 10÷20 Mhz ie 15÷30 KHz per single cell)

# Waveform analysis: time reconstruction

Different methods to reconstruct the time of a peak:

✗ parabolic fit to find the peak maximum
✗ average of time samples weighted by the waveform derivative
✔ digital filter: weighting by the derivative of a reference signal
→ best against noise (signal shape known)




Digital filter method to minimize N/S for timing measurements:
solve the following equation on $t_0$ :

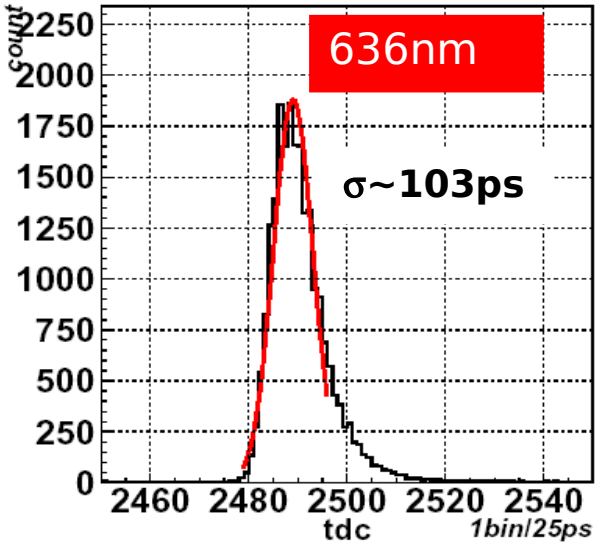
$$\int V_a(t) \frac{\partial V_r(t-t_0)}{\partial t} dt = 0$$

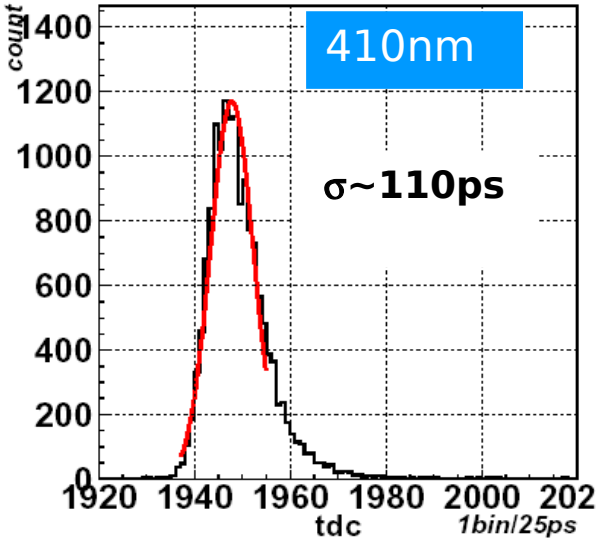
Va = measured signal (includes noise)
Vr = reference signal
$t_0$ = reference time

# SPTR: HPK/CPTA comparison

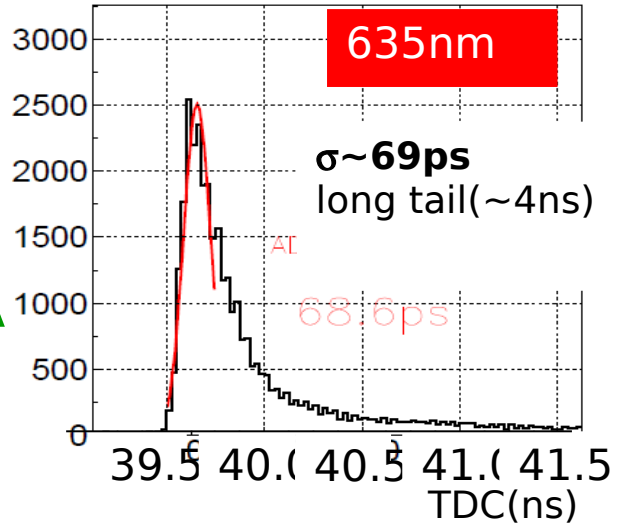
T.Iijima – PD07
Nagoya and Lubiana groups

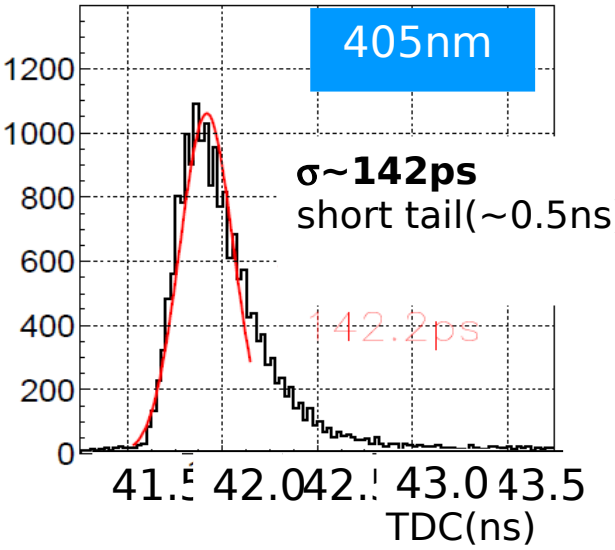
SiPM - HPK (MPPC)




SIPM - CPTA (MRS-APD)



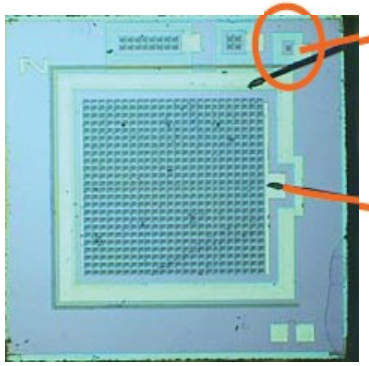

Method: CFD + TDC + Time walk corrections

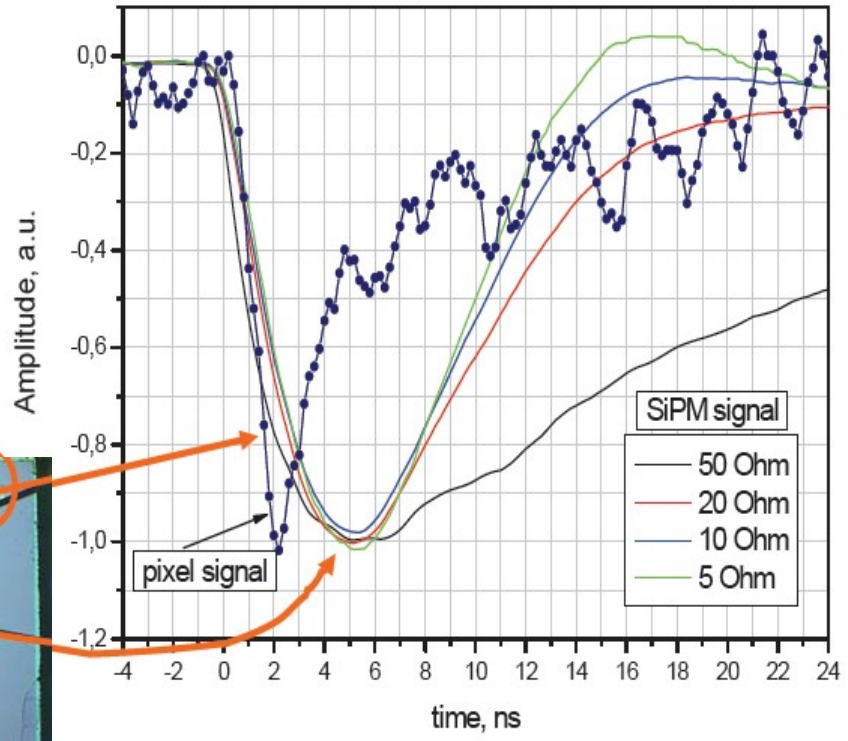
Compatible with DASIPM measurements

# Electronics for timing applications

# SiPM signal: effect of $C_{tot}$ and $Z_{load}$

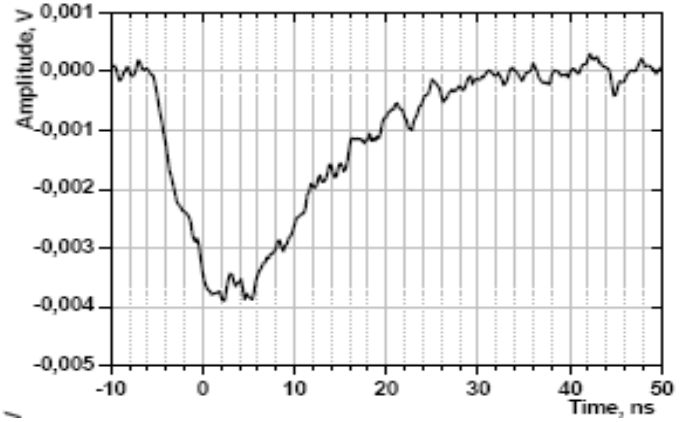
SiPM – MePhI/Pulsar:
1600 cells (100x100μm²)
Area = 5x5 mm²
$C_{tot}$~ 160pF



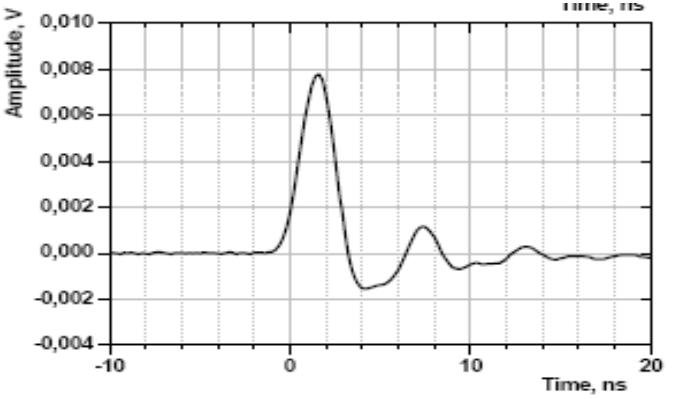



*B.Dolgoshein and E.Popova LIGHT07*

Zin~50Ω
*FWHM~15ns*




Zin~7Ω *+ shaper*
*FWHM~2.5ns*




*Trans-impedance amplifier*

# Ideal FE Electronics for timing (and high rate)

1) current amplifier or I-V converter:
   - exploit detector fast component → smallest $Z_{in}$ (and smallest $C_{parasitic}$)
   - detector need low external Gain ~ x20 → fast amplifier
   - detector rise time ~ 100ps → *match with* 3GHz amplifier BW

2) RC shaper
   → minimize signal occupancy (pile-up) per channel
   → maximize the double pulse resolution (DPR)

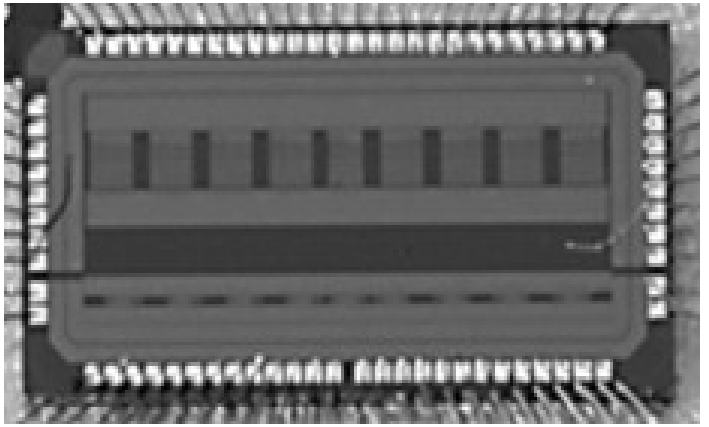
3) sampling with FADC
   → sampling at 1GHz time resolution better than 20 ps rms
   and DPR better than 5ps are easy to obtain
   → very robust to noise
   → cost/channel ~ O(100 $)

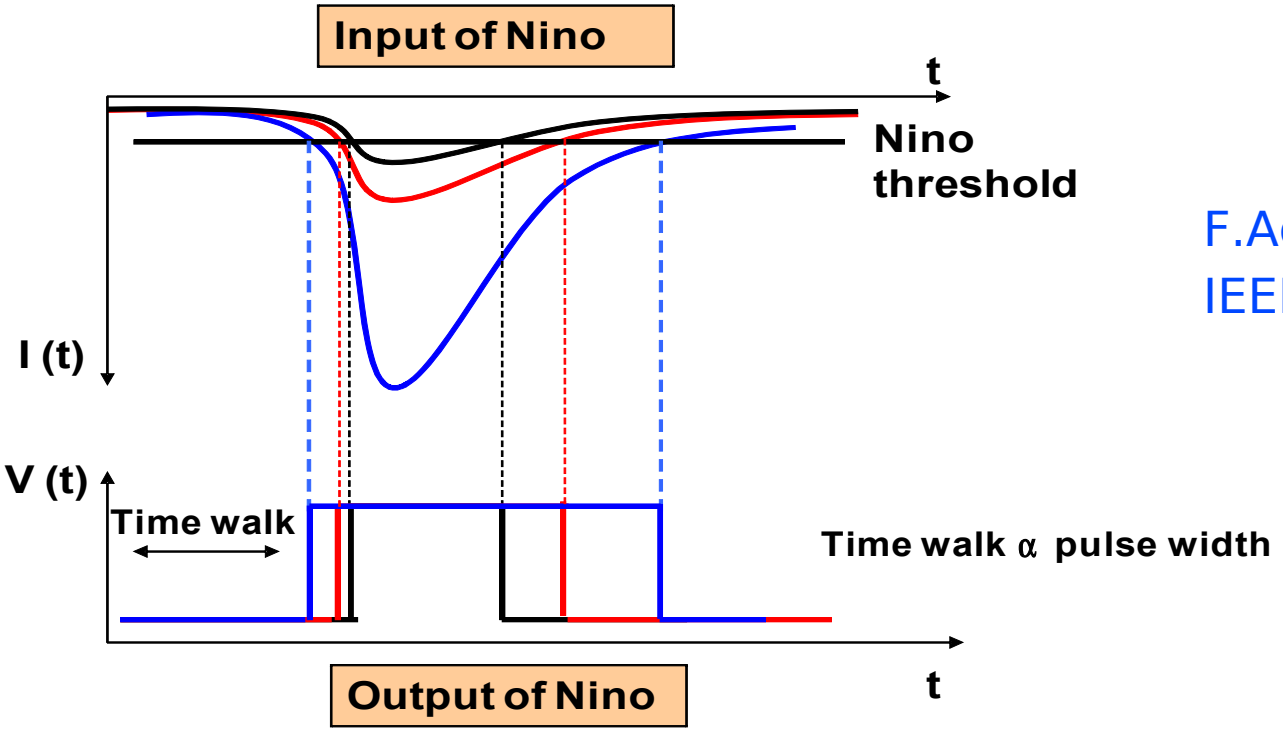
3') Time over Threshold (ToT) Discriminator (Slew correction)
   → time resolution better than 40 ps rms and DPR better than 10ps
   (depending on noise and signal shape)
   → cost/channel ~ O(10 $)

# Time over Threshold technique (ToT)

NINO chip

- developed for Alice TOF
- Ultra-fast low-power discriminator.
- 8 channels (250 nm CMOS)
- 1 ns peaking time
- 25 ps rms time jitter
- Time over threshold technique

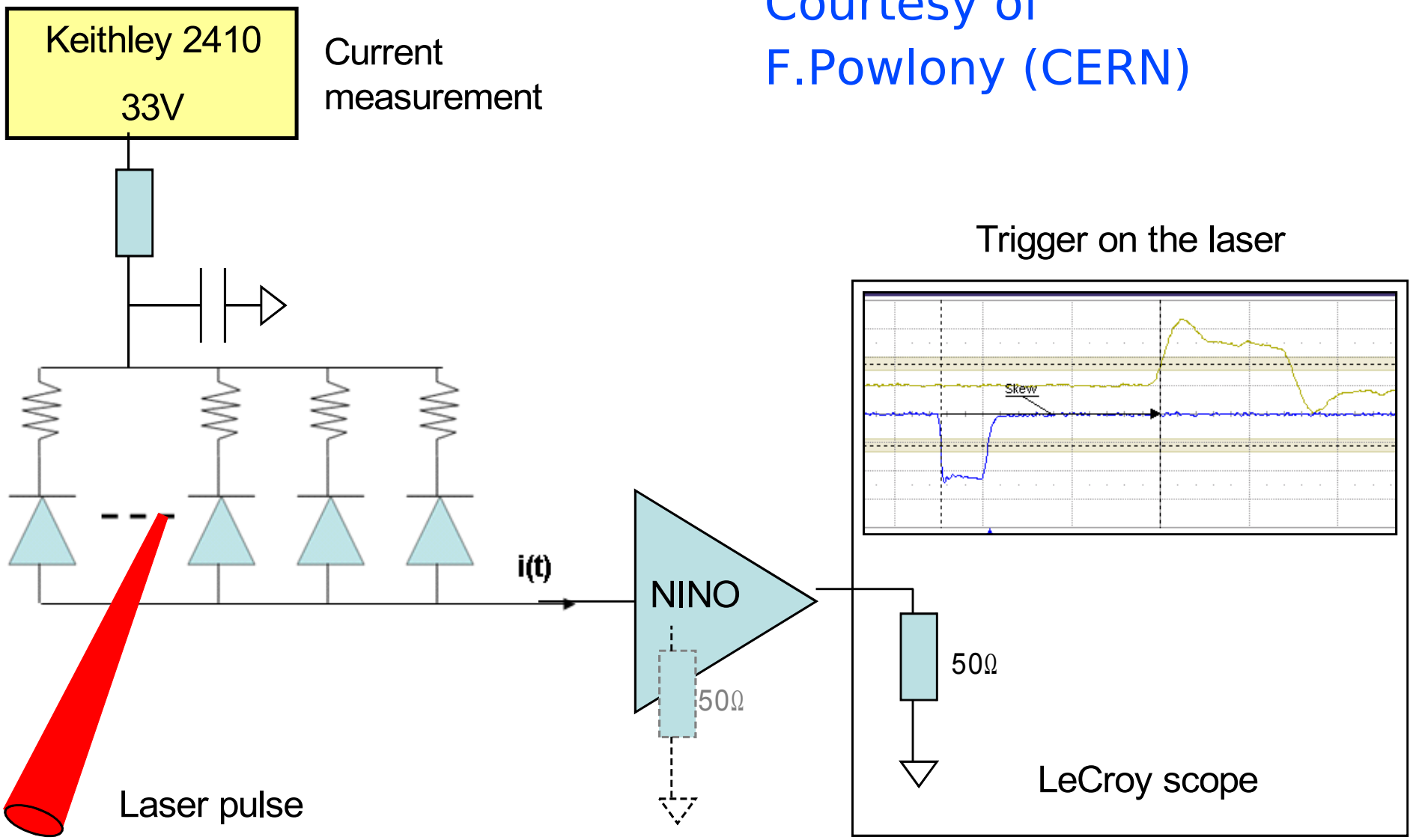




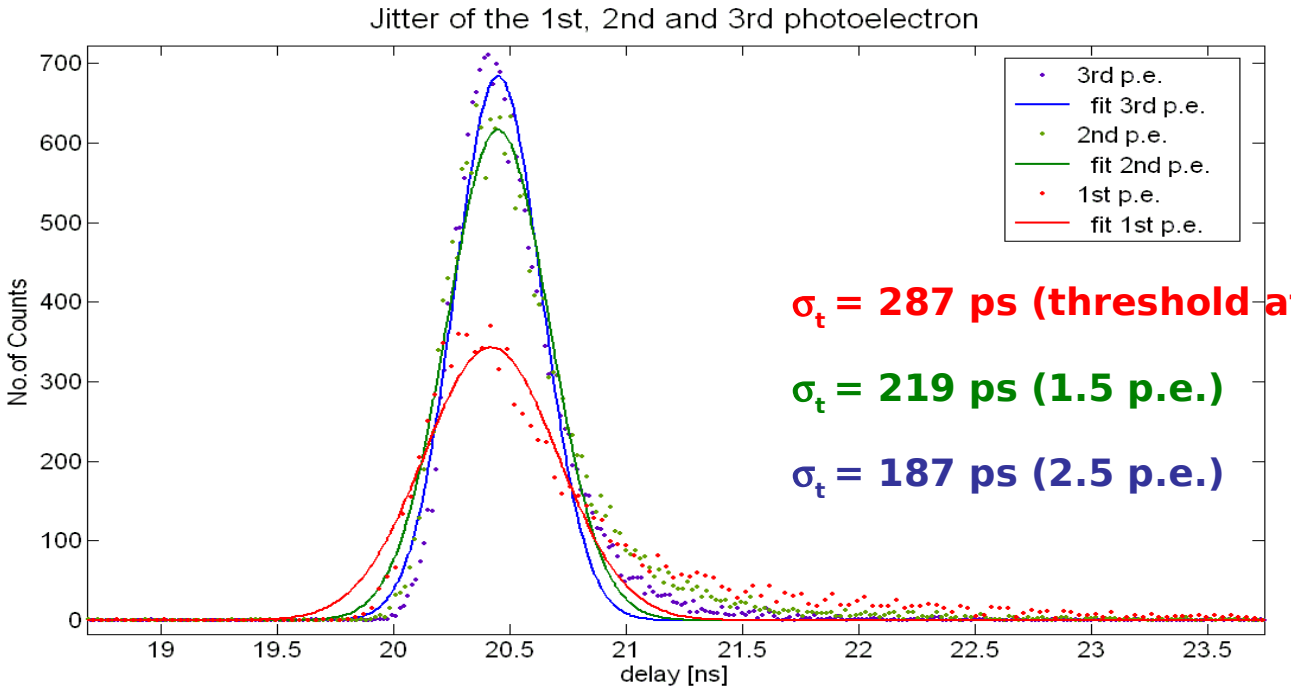

F.Aghinolfi et al
IEEE TNS 51 (2004) 1974

# Time resolution with NINO

Courtesy of
F.Powlony (CERN)

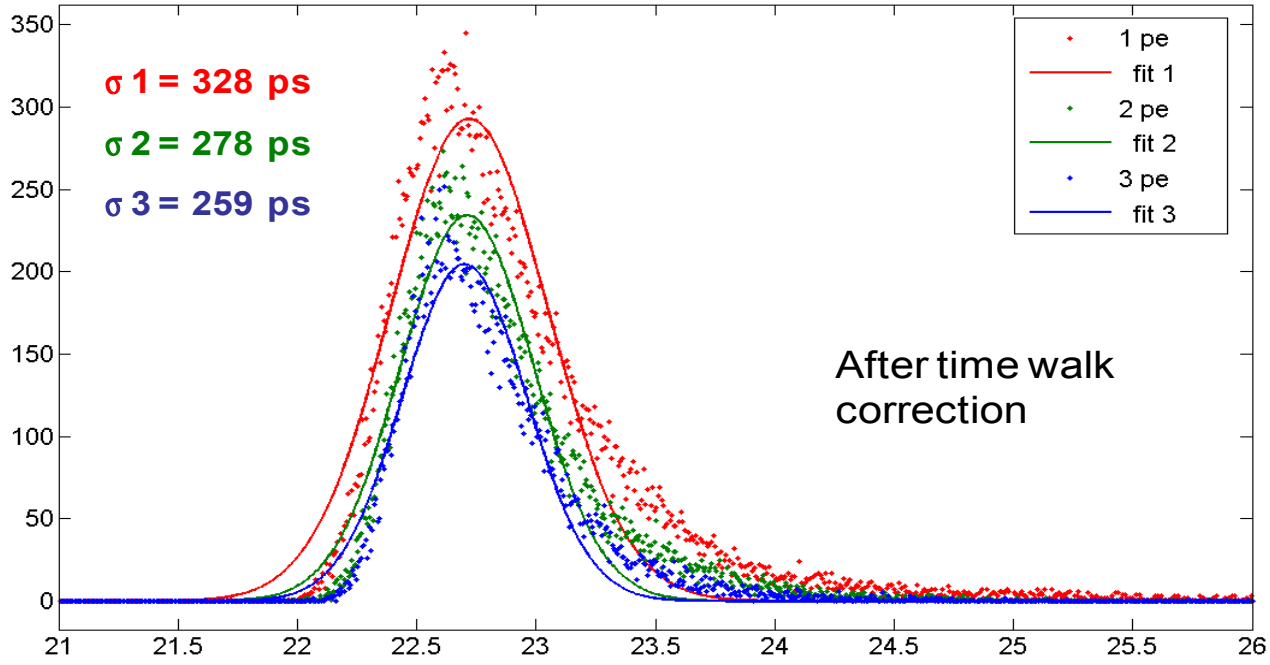


# Time resolution with NINO




SiPM directly to scope

Preliminary results

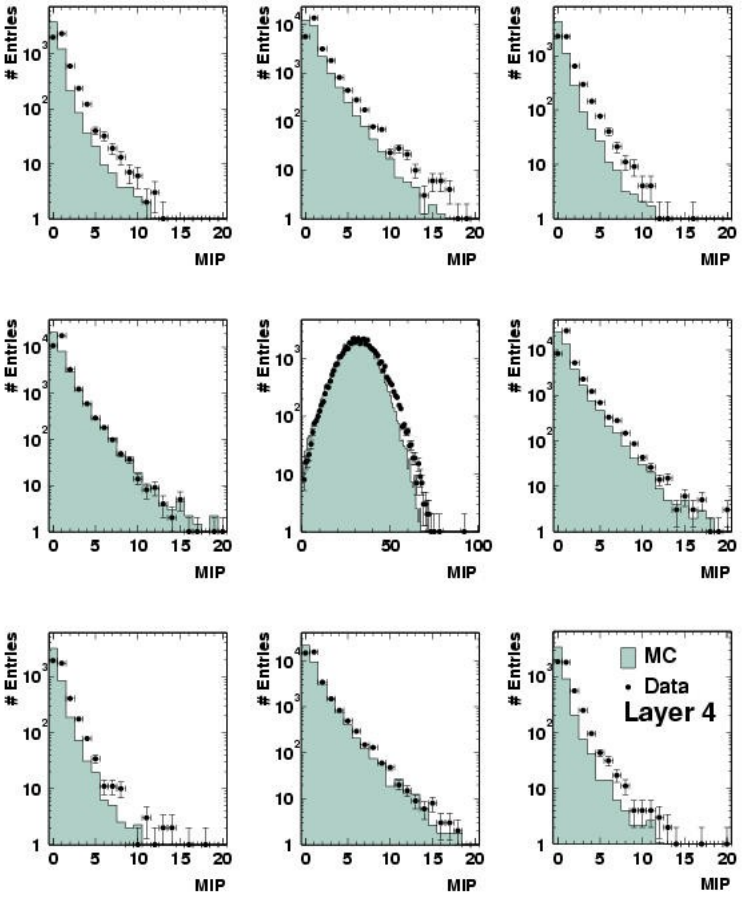



SiPM + NINO

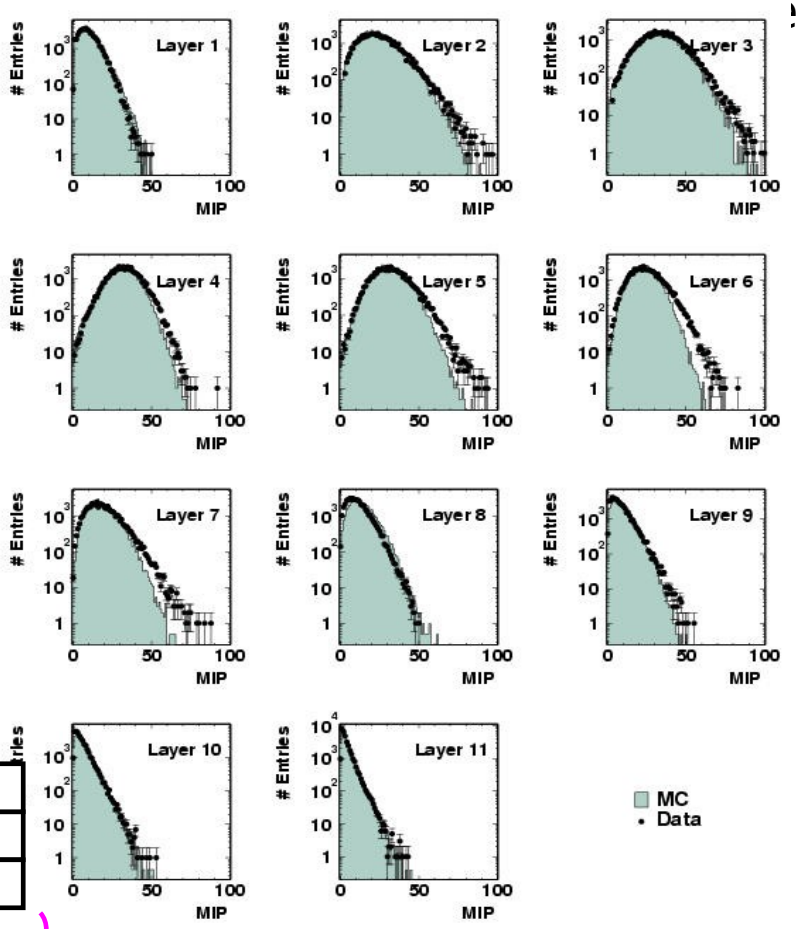
Courtesy of F.Powlony (CERN)

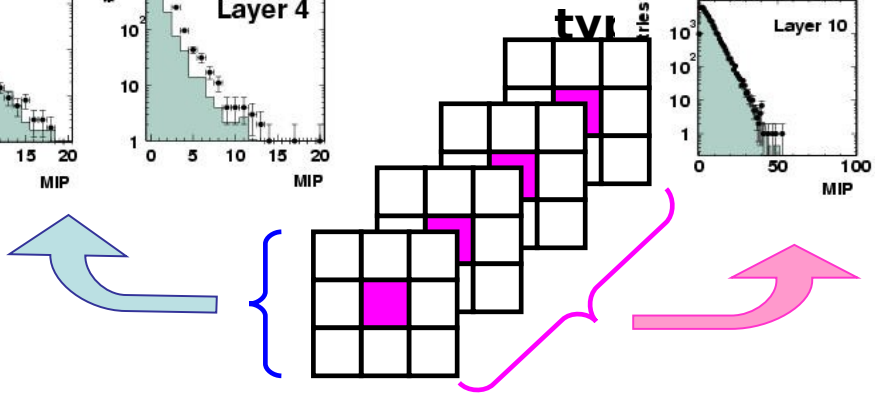
# First prototype calorimeter with SiPM

## Lateral shower shape




## Longitudinal shower shape



Courtesy of E.Garutti - DESY

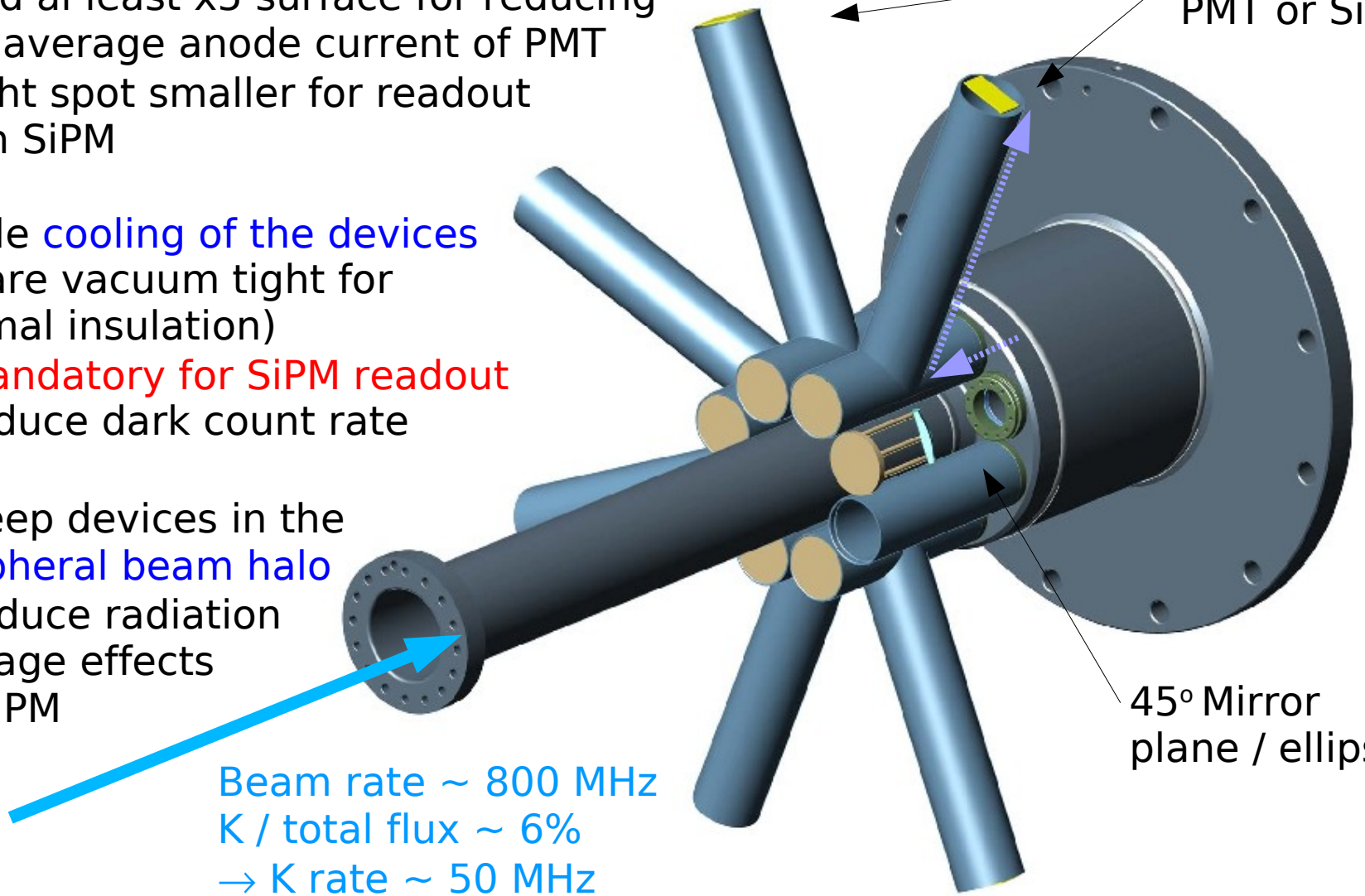
# The new mechanics of the CEDAR “nose”

CEDAR ''nose'' new mechanics (additional T's) for allowing:

1. to tune of optics of the 8 Cerenkov light spots out of the quartz windows
   → light spot widening is mandatory for the PMT readout:
   need al least x5 surface for reducing the average anode current of PMT
   → light spot smaller for readout with SiPM

2. simple cooling of the devices (T's are vacuum tight for thermal insulation)
   → mandatory for SiPM readout to reduce dark count rate

3. to keep devices in the peripheral beam halo
   → reduce radiation damage effects on SiPM



# Comparison SiPM vs PMT

## for applications of photon counting and timing at high rates

| | PMT | SiPM |
|---|---|---|
| Reference device (eff. area) | HPK R7600 (18x18 $mm^2$) | HPK S10362-11-50C (1x1 $mm^2$) |
| Gain (G) | $\geq 10^6$ | $\geq 10^6$ |
| $\delta V/V$ for $\delta G/G=1\%$ | $3 \times 10^{-4}$ | $6 \times 10^{-4}$ |
| $\delta T$ for $\delta G/G=1\%$ | 5° C | 0.3° C |
| Max average anode current | 100 µA (350 $mm^2$) | 3 µA (1 $mm^2$) |
| Efficiency (on active area) | ~25% @ 400nm ~40% (UBA) | ~95% @ 400nm |
| Collection efficiency | 70% | 40% to 80% |
| Time resolution | ~300ps | 50ps to 100ps |
| Dark noise (1 p.e.) | few kHz | 0.5 MHz @room T |
| After-pulse (thr. @ 1 p.e.) | 1 % level | 10 % level |
| B-field immunity | No | Yes |
| Radiation damg. | No (also at single photon level ?) | Yes |

Gain = $C_{cell} \times \Delta V$

20MHz limit to avoid signal pileup. Not mandatory: can use proper shaping !!! SiPM can stand at least x10 more rate per unit area than PMT

Approx. $\propto \Delta V$

Fill factor (cell geometry)

Approx. $\propto \Delta V^{-1}$

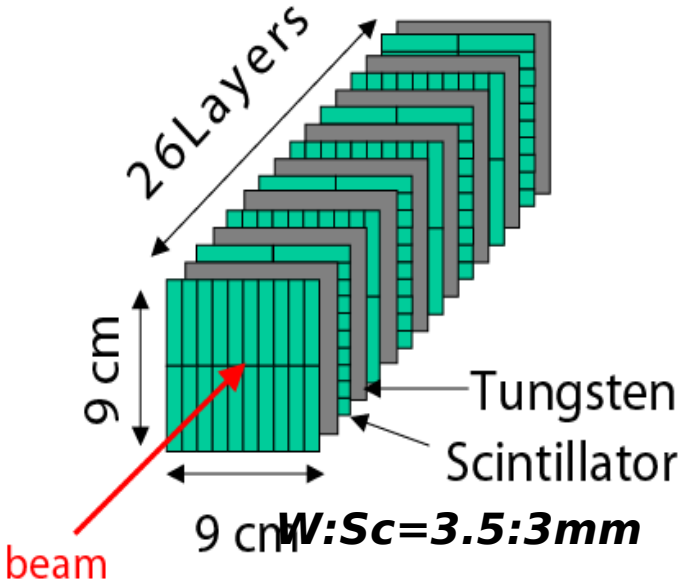
Approx. $\propto \Delta V$

Approx. $\propto \Delta V^2$

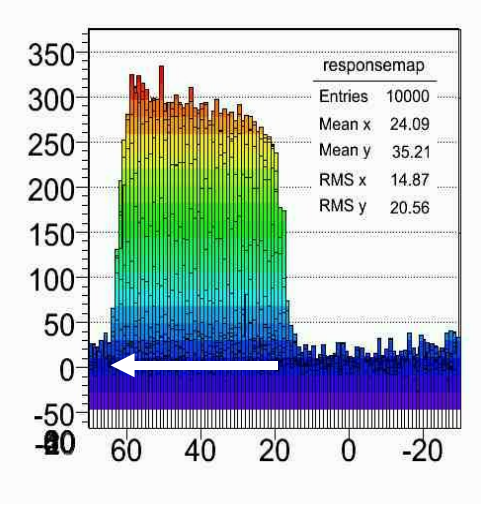
$\Delta V$ is the over-voltage $= V_{bias} - V_{breakdown}$

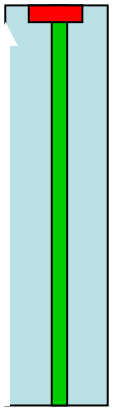
**Result of the beam test demonstrates that the MPPC is feasible for the GLD calorimeter readout.**

| | Photomultiplier | MPPC |
|---|---|---|
| Gain | ~$10^6$ | $10^5$~$10^6$ |
| Photon Detection Eff. | 0.1 ~ 0.2 | ~0.2 for 1600 pix. MPPC |
| Response | fast | fast |
| Photon counting | Yes | Great |
| Bias voltage | ~ 1000 V | ~ 70 V |
| Size | Small | Compact |
| B field | Sensitive | Insensitive |
| Cost | Very expensive ! | Not very expensive |
| Dynamic range | Good | Determined by # of pixels |
| Long-term Stability | Good | Unknown |
| Robustness | decent | **Unknown, presumably good** |
| Noise (fake signal by thermions) | Quiet | Noisy (order of 100 kHz) |

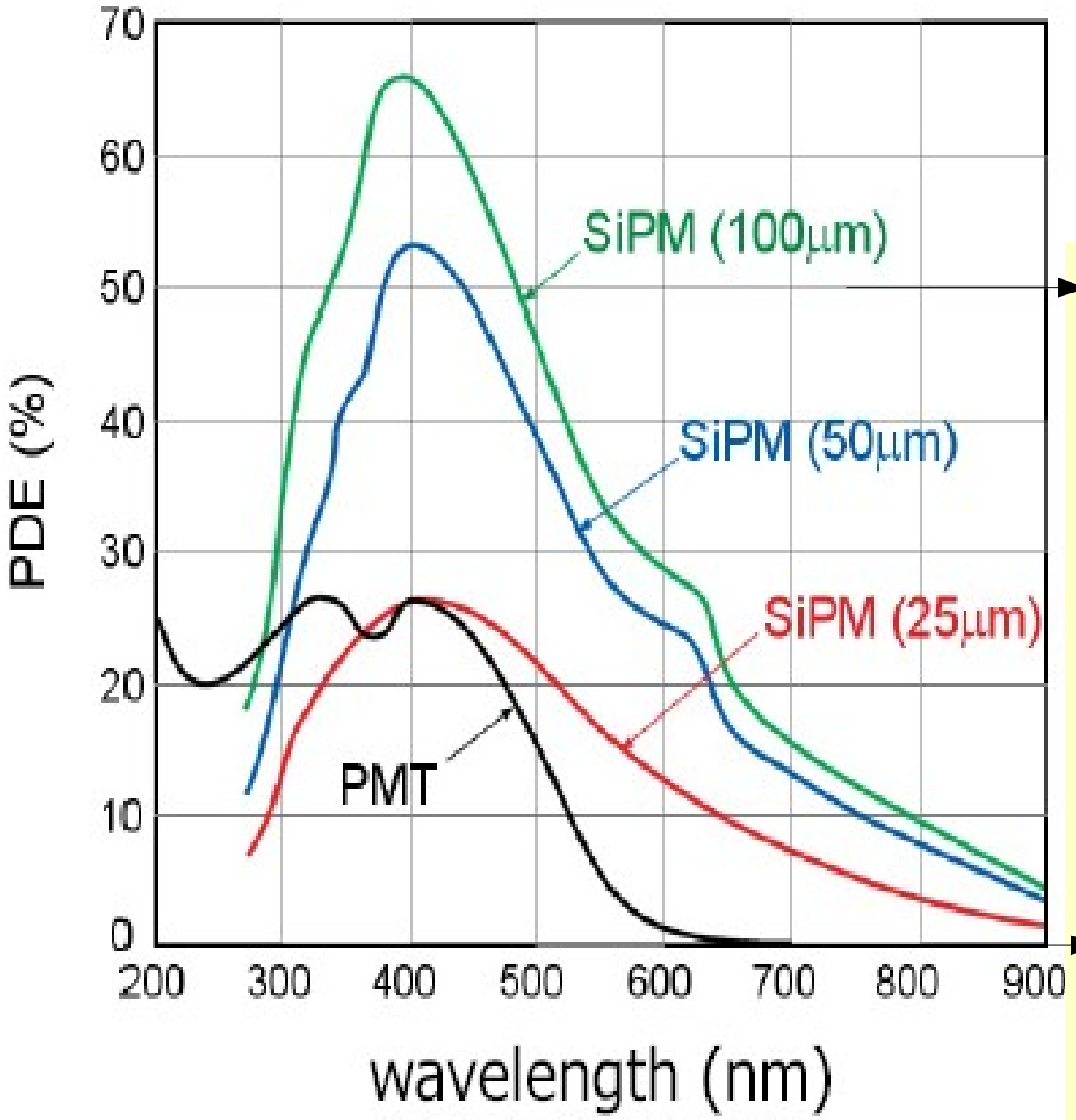


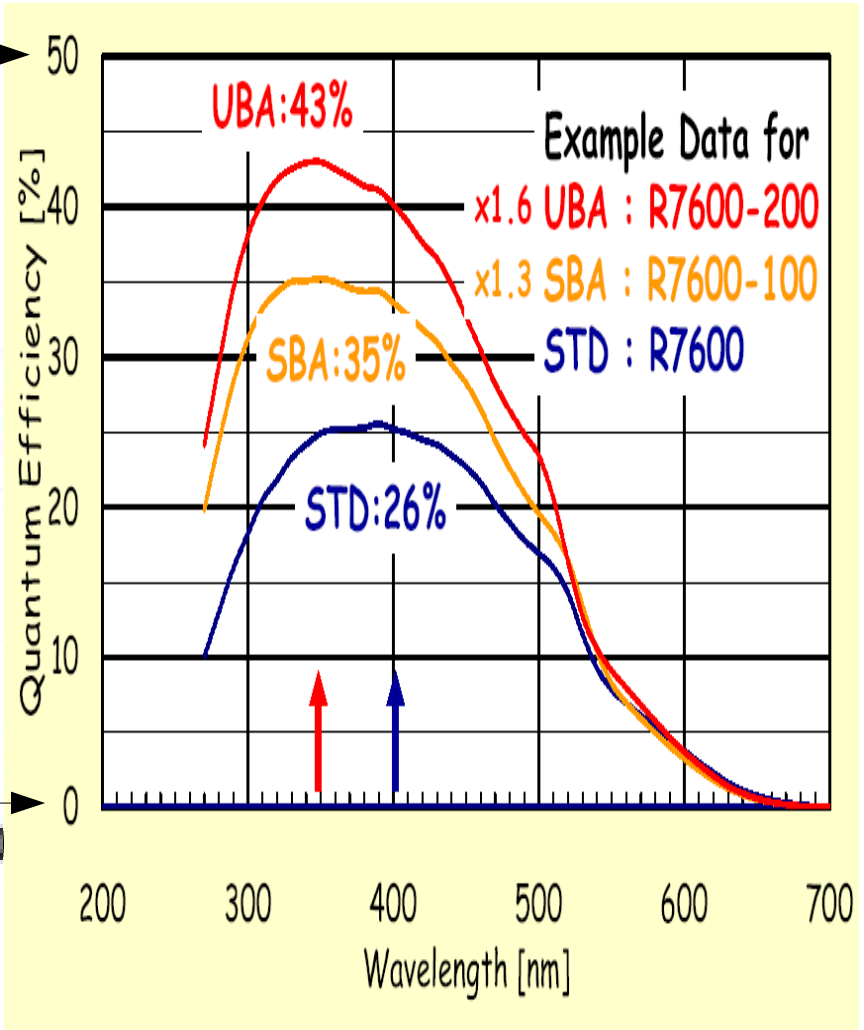

**MPPC**



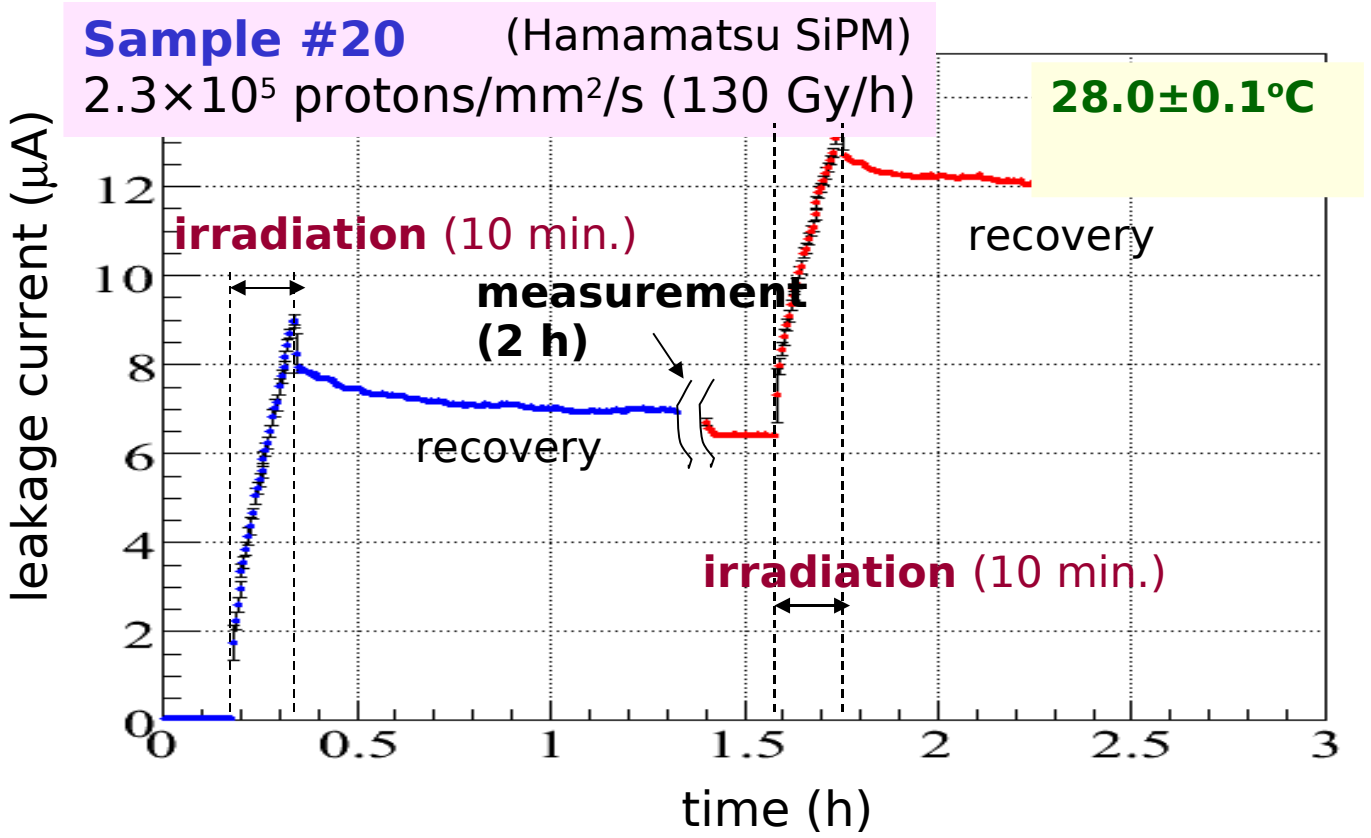



# Hamamatsu: PDE of SiPM vs QE of PMT



# Radiation damage: protons (53.3 MeV)



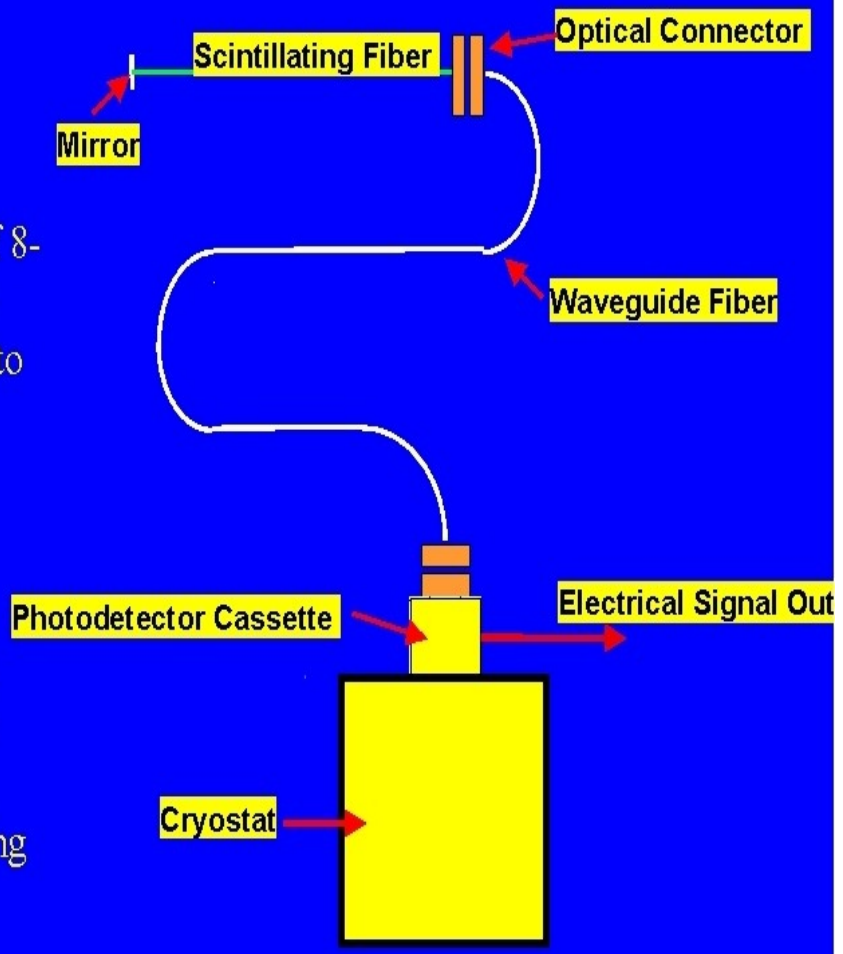

- The leakage current lineally increases with irradiated doses.
- Annealing effects are seen. But the radiation damage is not completely recovered within a few hour.

T.Matsumura – PD07

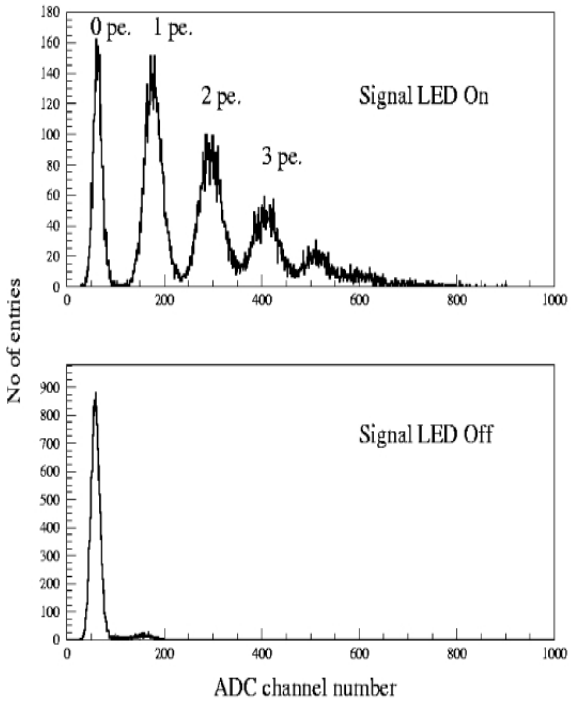
# D0 tracker with VLPD: much easier today !

## DØ Scintillating Fiber Tracker: Operational Principles

- Charged particles cross a scintillating fiber, where it causes a 'blink' of light.
- The light is transported via optical fiber over a distance of 8-11 meters to a device called a VLPC which converts light into electricity.
- VLPC are solid state devices which run at cryogenic temperatures.
- A 'cassette' of VLPC devices contains 1024 channels and is housed in a cryostat, which carefully regulates the operating temperature.




- 80000 fibers 2.5m and 1.7m
- single photon detection
- 80% QE
- operated at 6°K
- gain ~50.000
- position resolution 120 μm



# Summary of SiPM properties

## Energy
- Gain
- Single photon resolution
- Dynamic range

## Noise
- Dark noise
- After-pulse
- Cross-talk

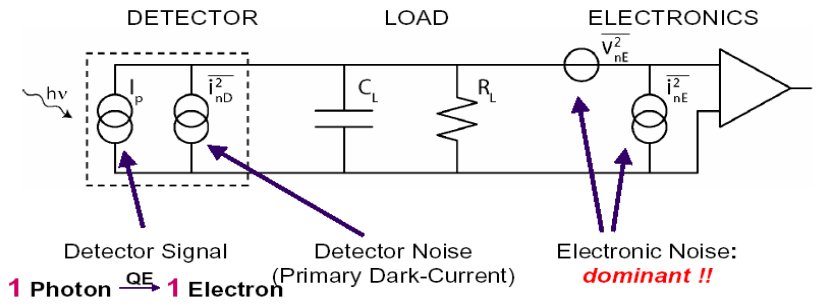
## Coupling :
- scintillator
- wavelength shifters
- direct radiation (eg Cherenkov)

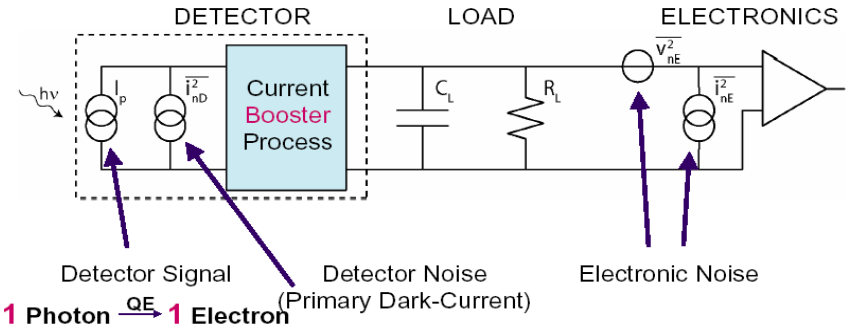
## Issues:
- MHz dark noise at 1 p.e. level (but ∃ improvements HPK, IRST)
- wide area (max 5x5 mm² up to now)
- T and V depend. of dark count and gain
- radiation damage

## Time
- Time resolution
- Rise time
- Recovery time

## Control parameters
- Over-voltage
- Temperature

## Electronics :
- No need for low noise/high gain ele.



# Integration with electronics / CMOS

## High-Density SSPM

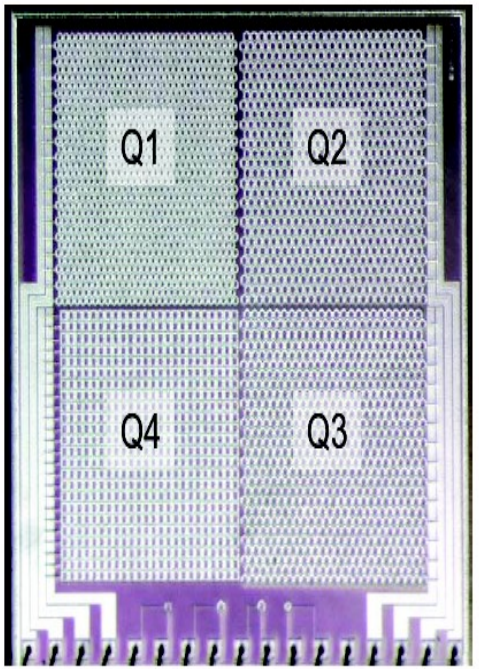
Instrument Research & Development

Q1: Type 12
Fill Factor = 19%
1020 pixels

Q2: Type 12
Fill Factor = 29%
700 pixels

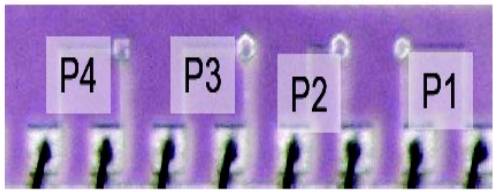
Q3: Type 4
Fill Factor = 29%
700 pixels

Q4: Type 12
Fill Factor = 29%
576 pixels




~3.3 mm

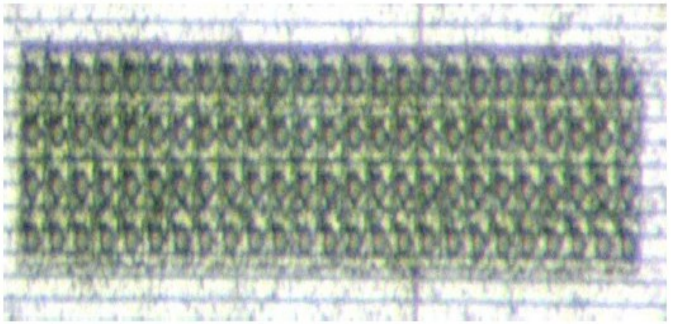
- P1: Q1 pixels (12) P2: Q2 pixels (12)
- P3: Q3 pixels (4) P4: Hyper-dense square pixel (12)




➢ Better fill factor, better SSPM DE

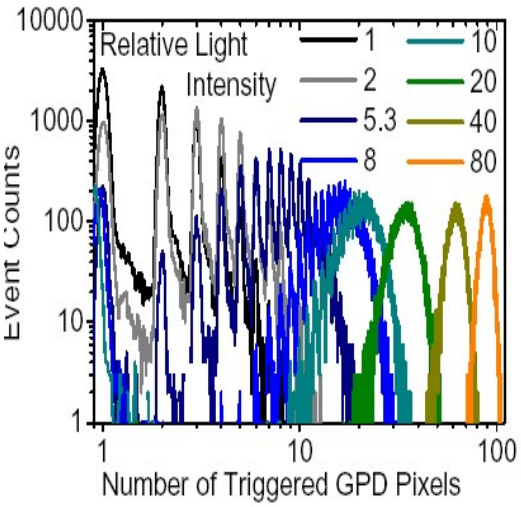
## CMOS SSPM Prototype

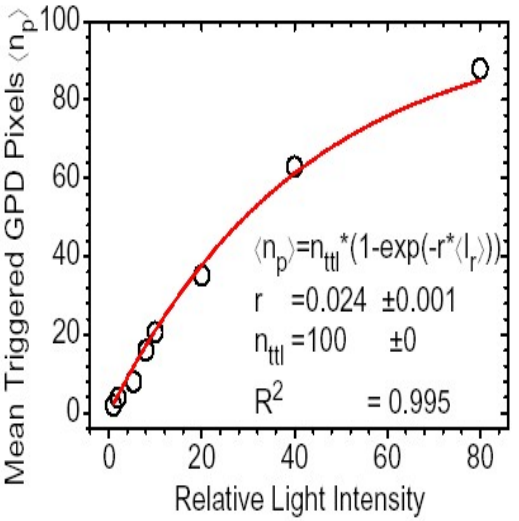
Instrument Research & Development



4 x 25, 20-μm dia. CMOS GPD pixel array, 43-μm pitch

- Signal from pulsed diode laser

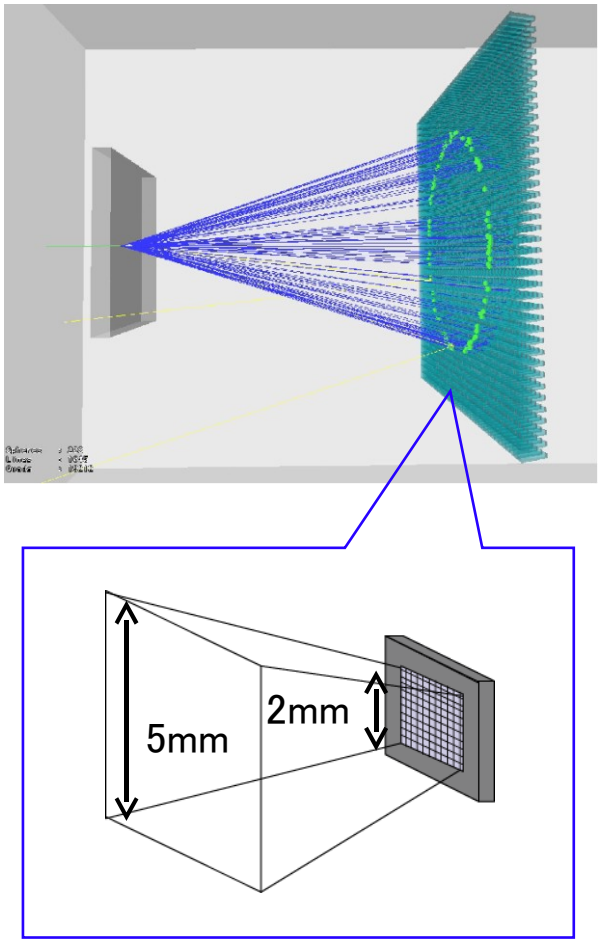


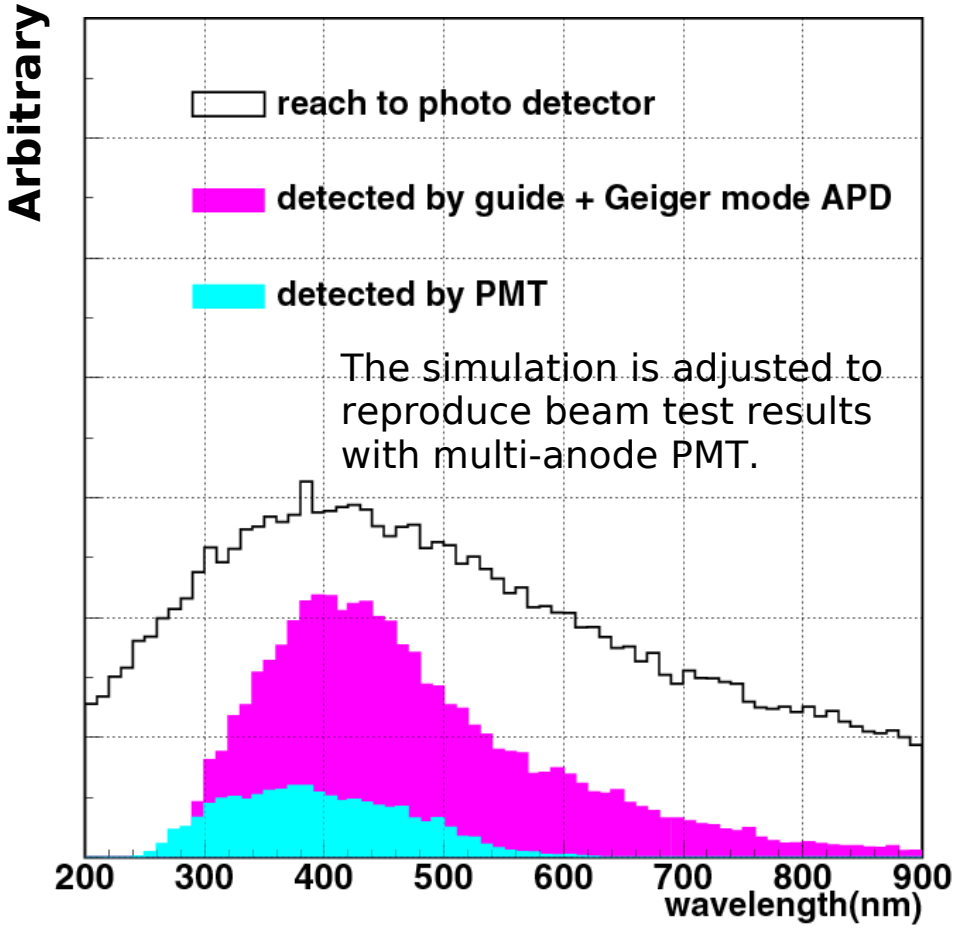




➢ Increased intensity triggers more GPD pixels
➢ Energy calibration

*J.F.Christian, LIGHT 07 workshop, Sept 2007, Germany*

# Expected RICH Performance



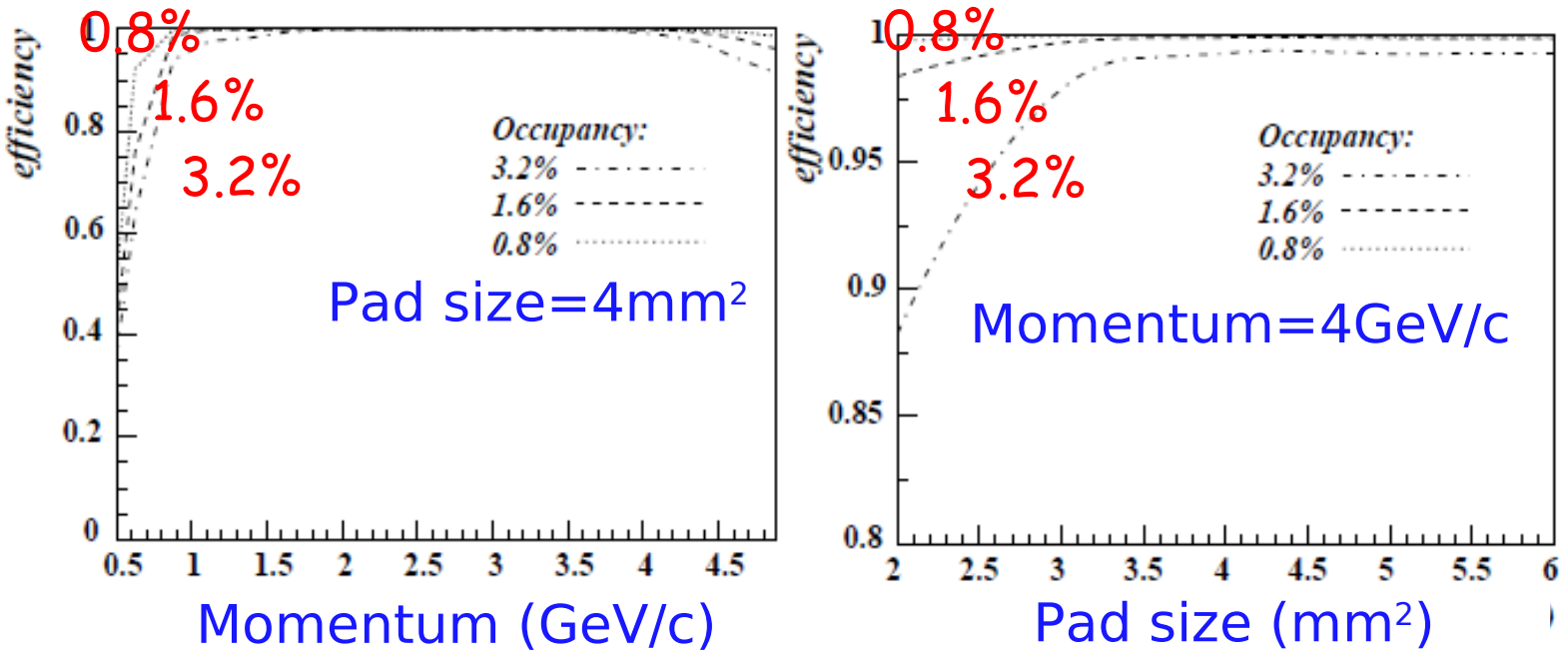




A photosensor based on Light-guide + Geiger-mode APD can give x 4Npe (number of detected photons).

# Effect of Noise

- Simulation including random background
- Assume $N_{pe}$ = 20 / ring (conservatively)

Kaon ID efficiency at 1% pion mis-ID probability

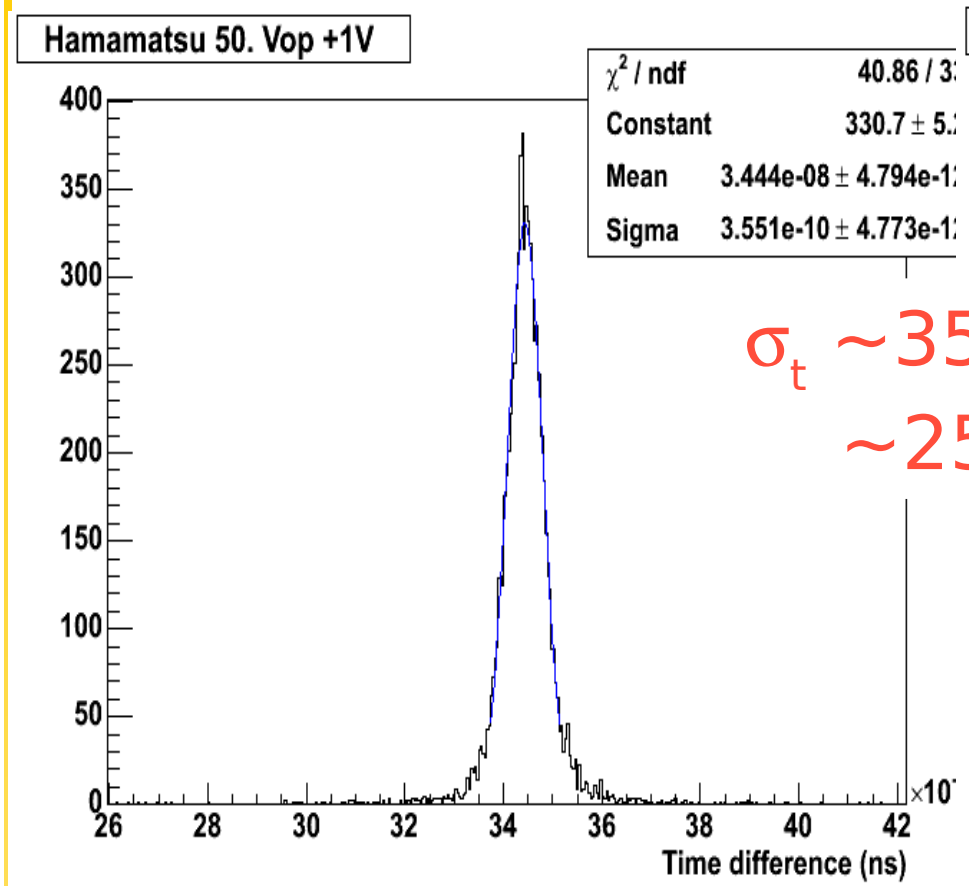



Noise rate (HPK-100) = 200kHz/mm² @ $10^6$ gain.

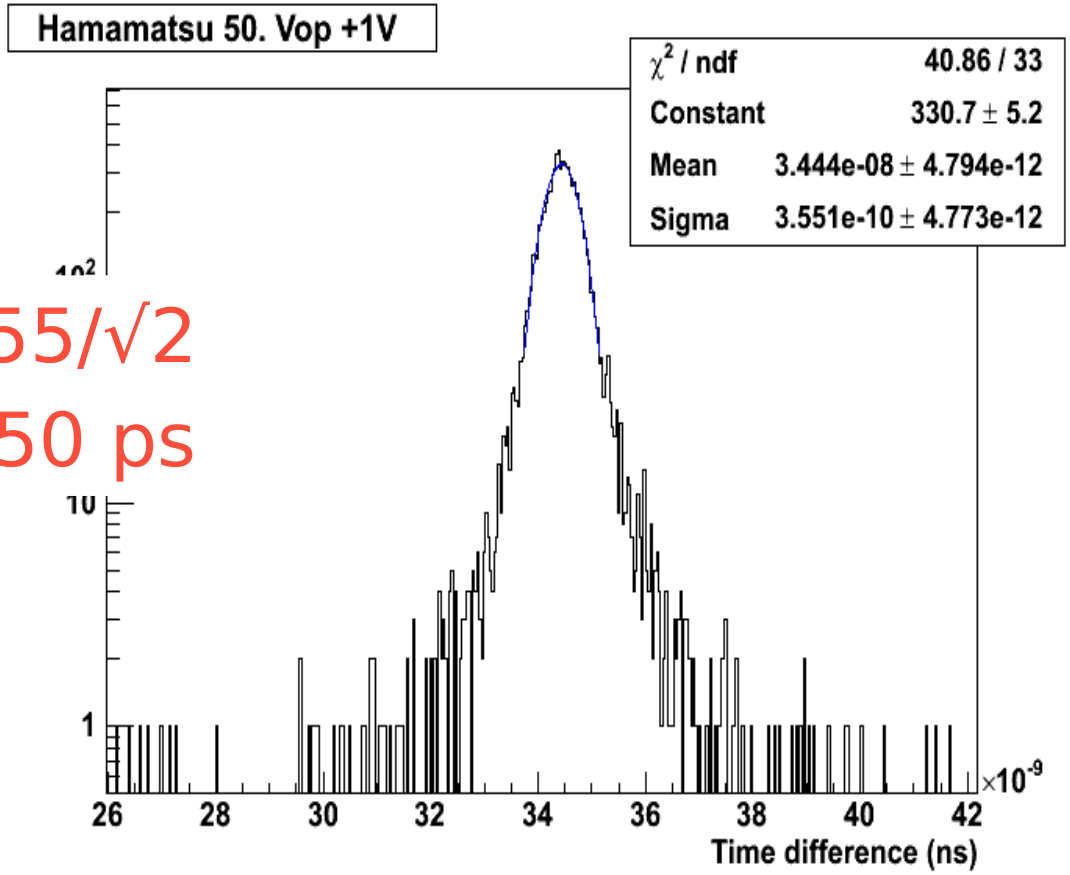
➡ 0.8% occupancy for 4mm² pad and 10ns time window

# Timing with LSO

SiPM – HPK
1x1 mm$^2$ (50x50μm$^2$)




$\sigma_t$ ~355/√2
~250 ps

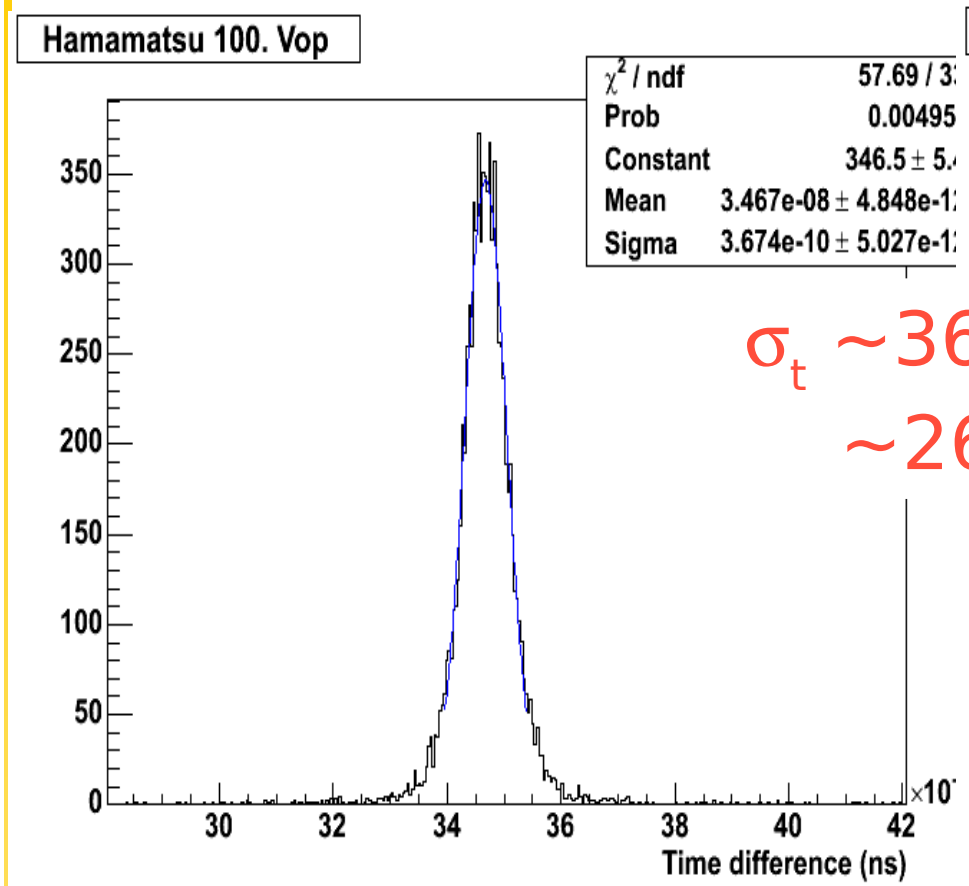


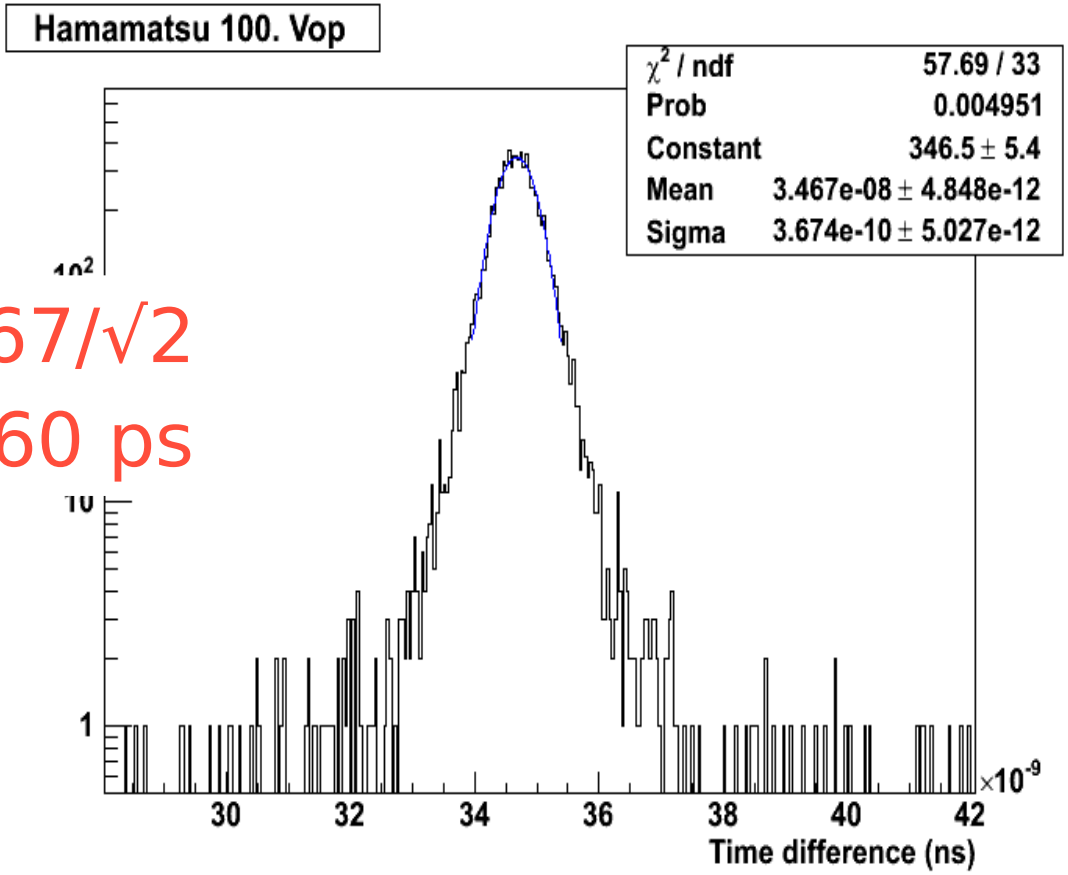

More photoelectrons
(higher PDE)

G.Llosa, DASIPM (unpublished)

# Timing with LSO

SiPM - HPK
1x1 mm$^2$ (100x100μm$^2$)




$\sigma_t$ ~367/√2
~260 ps

G.Llosa, DASIPM (unpublished)